FINITELY GENERATED UNIT LAURENT CRYSTALS

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Abstract. Let p be an odd prime, let k be a perfect field of characteristic p, and let X be a smooth qcqs p-adic formal scheme over $\mathrm{W}(k)$ equipped with the log structure $\mathcal{M}$ induced by a horizontal normal crossings divisor D. We show that there is an equivalence of categories between constructible p-Kummer-étale $\mathbf{Z}_p$-sheaves on the log adic generic fiber $(\mathrm{X}, \mathcal{M})_\eta$ for the logarithmic stratification induced by D and certain finitely generated unit Laurent crystals on the log prismatic site. We verify p^+-perverse t-exactness, give a local description of finitely generated unit crystals on a framed log affine in terms of (φ, Γ)-modules, and identify at the level of hearts the objects corresponding to ordinary constructible sheaves on the adic generic fiber of X.

Contents

1. Introduction

Let k be a perfect field of characteristic p. For smooth k-schemes, Emerton and Kisin gave a contravariant Riemann-Hilbert correspondence describing constructible $\mathbf{F}_p$ sheaves using finitely generated unit Frobenius modules in [EK04b]. Their result is compatible with Katz's correspondence in [Kat73], which describes the full subcategory of étale local systems.

In mixed characteristic, Katz's theorem plays an important role in describing local systems on the adic generic fiber of a p-adic formal scheme using (φ, Γ) modules; importantly, this can also be done over a deperfected base ring. This is originally due to Fontaine in [Fon90], and later generalized in [AI08]. Due to Kedlaya-Liu in [KL15], this theory also admits a logarithmic variant describing what are now called p-Kummer-étale local systems on the log adic generic fiber (see Definition 4.19). Unlike the situation in positive characteristic, there is not a known constructible variant of these theories of (φ, Γ)-modules.

In this article, we will show such a result in logarithmic setting using a new notion of finitely generated unit (φ, Γ) modules over a deperfected base. Our correspondence allows one to study p-Kummer-étale constructible sheaves on a fixed normal crossings stratification on the log adic generic fiber. We use a log formal scheme $(\mathrm{X}, \mathcal{M})$ where $\mathrm{X}/\operatorname{Spf}\mathrm{W}(k)$ is smooth and qcqs and $\mathcal{M}$ is the log structure attached to a horizontal normal crossings divisor D. By this we mean that $\mathrm{D} \cup \mathrm{X}_s$ is a normal crossings divisor and no irreducible component of D is contained in the special fiber; equivalently, strict-étale locally the pair has the form of Definition 3.2.[1] Since any description using (φ, Γ)-modules is necessarily local and coordinate dependent, similarly to [BS23] and [IKY26] we globalize local results about (φ, Γ)-modules using crystals on the absolute log prismatic category $(\mathrm{X}, \mathcal{M})_{\Delta}$. This is done in Definition 4.3, where we introduce the category $\mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ of finitely generated unit Laurent crystals.

We now explain the target of the functor in more detail. We write $\mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$ for the category of Kummer-étale sheaves which are constructible for the stratification induced by D on the log adic generic fiber $(\mathrm{X}, \mathcal{M})_\eta$. For a geometric point $\bar{x}$ of X_η, put

$$\rho(\bar{x}) = \mathrm{rk}_{\mathbf{Z}}\, \overline{\mathcal{M}}^{\mathrm{gp}}_{\bar{x}}.$$

For $a \geq 0$, let $\mathrm{X}^{(a)}_\eta$ be the locus where $\rho = a$. The logarithmic strata are the connected components of the locally closed loci $\mathrm{X}^{(a)}_\eta$. Strict-étale locally, after labeling the branches $\mathrm{D}_1, \ldots, \mathrm{D}_r$, they are the connected components of the coordinate strata

$$\mathrm{X}_{\eta,\Lambda} = \left(\bigcap_{i \in \Lambda} (\mathrm{D}_i)_\eta\right) \setminus \left(\bigcup_{j \notin \Lambda} (\mathrm{D}_j)_\eta\right) \text{ for } \Lambda \subseteq \{1, \ldots, r\}.$$

[1] Note this is not the same as simply asking D to have no vertical components.

As with the correspondence of [IKY26] for Kummer-étale local systems, there is a condition on objects $\mathscr{F}$ in the essential image which they call *p-Kummer* (see Definition 4.19). In a normal crossings setting, at a log geometric point $\bar{x}$ the log inertia group is given by $\mathrm{I}_{\overline{x}}^{\log} = \mathrm{Hom}(\overline{\mathcal{M}}_{\overline{x}}^{\mathrm{gp}}, \widehat{\mathbf{Z}}(1))$ which will look like $\widehat{\mathbf{Z}}(1)^r$. The p-Kummer condition simply asks that the prime-to-p component $\mathrm{I}_{\overline{x}}^{\log,(p')}$ acts trivially on $\mathscr{F}_{\overline{x}}$ for all $\overline{x}$. This condition is not so restrictive, as it still allows all ordinary constructible sheaves.

The main result of this article is then the following. We let $(\mathrm{X}, \mathcal{M})_{\mathbb{\Delta}}$ denote the category of log prisms, opposite to the log prismatic site of [Kos22], and $\mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]_p^{\wedge}$ the coefficient system assigning a log prism $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}})^a$ to $\mathrm{A}[1/\mathrm{I}]_p^{\wedge}$ as in [IKY26]. We assume p is odd in everything that follows.

Theorem A (Theorem 5.25, Corollary 5.29). Let $\mathrm{X}/\operatorname{Spf} \mathrm{W}(k)$ be smooth and qcqs, and let $\mathcal{M} = \mathcal{M}_{\mathrm{D}}$ be induced by a horizontal normal crossings divisor. There is a fully faithful symmetric monoidal functor

$$\mathrm{T}_{\text{ét}} : \mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]_p^{\wedge})^{\mathrm{op}} \longrightarrow \mathrm{D}^{(b)}_{\mathrm{cons,D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$$

with essential image the p-Kummer complexes of Definition 4.19. If X_η is pure of dimension d, then $\mathrm{T}_{\text{ét}}[d]$ identifies the standard t-structure on the source with the p^+-perverse t-structure on its essential image.

We emphasize that it is not too difficult to check the finitely generated unit condition for a Laurent crystal. A Laurent crystal assigns to every log prism (A, I) a complex of $\mathrm{A}[1/\mathrm{I}]_p^{\wedge}$-modules, compatibly with pullback. It is unit if it is equipped with an isomorphism $\varphi^*\mathcal{E} \simeq \mathcal{E}$. A unit Laurent crystal is further *finitely generated unit* when for any log prism $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}})^a \in (\mathrm{X}, \mathcal{M})_{\mathbb{\Delta}}$ one has $\mathcal{E}(\mathrm{A}, \mathrm{I}, \mathrm{M}) \in \mathsf{Perf}(\mathrm{A}[1/\mathrm{I}]_p^{\wedge}\langle \mathrm{F}\rangle)$ (here we mean the thick subcategory generated by the unit). The noncommutative ring $\mathrm{A}[1/\mathrm{I}]_p^{\wedge}\langle \mathrm{F}\rangle$ is obtained by adjoining F with $\mathrm{F}a = \varphi(a)\mathrm{F}$.[2] One does not need to check all prisms to verify the condition. It in fact suffices to check on a covering family of logarithmic q-de Rham prisms, as in Remark 4.12.

Because the equivalence is derived, it also immediately implies a cohomological comparison. In Corollary 5.29 we explain how straightforward modifications allow one to deduce all results over the base $\mathrm{W}(k)[\zeta_{p^\ell}]$ for $\ell \geq 0$. We primarily work with $\mathrm{W}(k)[\zeta_p]$ in the paper for expository reasons, and later descend the result to $\mathrm{W}(k)$. It is also worth pointing out that one can treat ordinary constructible sheaves within this correspondence. There is a morphism of sites

$$\epsilon : (\mathrm{X}, \mathcal{M})_{\eta,\mathrm{qpkét}} \longrightarrow \mathrm{X}_{\eta,\mathrm{proét}}$$

whose pullback ϵ^* is fully faithful on constructible sheaves and lands in the p-Kummer sheaves in the heart of $\mathrm{D}^{(b)}_{\mathrm{cons,D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$. Thus ordinary perverse sheaves form a full subcategory at the level of hearts. See Remark 5.28 for the corresponding condition on the crystal side, which is easily checked locally.

As mentioned earlier, finitely generated unit Laurent crystals admit an explicit local description in terms of (φ, Γ)-modules. We use this to give an explicit example in Example 5.31. We state the variant over $\mathrm{W}(k)$. In what follows a framed log affine is a $\mathrm{W}(k)$-algebra R that

[2] Strictly speaking we must do this all in a solid setting, but we ignore this for the introduction.

is p-completely étale over $\mathrm{W}(k)\langle \mathrm{T}_1, \dots, \mathrm{T}_r, \mathrm{T}_{r+1}^{\pm}, \dots, \mathrm{T}_n^{\pm}\rangle$ with log structure $\mathcal{M}$ induced by $\mathrm{D} = \mathrm{V}(\mathrm{T}_1 \dots \mathrm{T}_r)$.

Theorem B (Variant of Theorem 4.6). Let $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ be a framed log affine, with framing denoted by $\square$. Evaluation on the cyclotomic log q-de Rham prism $(\mathrm{R}_0^{\square}[\![q-1]\!], [p]_q, \mathbf{N}^r)^a$ over R induces an equivalence

$$\mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}) \simeq \mathrm{D}_{\mathrm{fgu}}(\mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge})^{h\Gamma},$$

where $(a, b_1, \dots, b_n) \in \Gamma = \mathbf{Z}_p^{\times} \ltimes \mathbf{Z}_p(1)^n$ acts by $q \mapsto q^a$ and $\mathrm{T}_i \mapsto q^{pb_i}\mathrm{T}_i$, and "fgu" means that the complex has a unit Frobenius and is perfect over $\mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}\langle \mathrm{F}\rangle$.

We also use this to make $\mathrm{T}_{\text{ét}}$ completely explicit locally in Lemma 4.26 (although in principle the definition in Definition 4.17 is already computable).

Our third main result is pushforward compatibility. The required functor is a shift and twist of the site-theoretic pushforward.

Theorem C (Theorem 6.16). Let $f : (\mathrm{X}, \mathcal{M}) \to (\mathrm{Y}, \mathcal{N})$ be a strict log smooth proper morphism of pure relative dimension d between smooth qcqs p-adic formal schemes over $\mathrm{W}(k)[\zeta_{p^{\ell}}]$ for some $\ell \geq 0$, and let D_{X} and D_{Y} be horizontal normal crossings divisors inducing $\mathcal{M}$ and $\mathcal{N}$. Suppose the adic generic fibers have pure dimensions d_{X} and d_{Y}. Thus $d = d_{\mathrm{X}} - d_{\mathrm{Y}}$. There is a commutative diagram

$$\begin{tikzcd} \mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\mathrm{op}} \arrow[r, "\mathrm{T}_{\text{ét},\mathrm{X}}[d_{\mathrm{X}}]"] \arrow[d, "\mathrm{R}f_*\{d\}[d]"] & \mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}_{\mathrm{X}}}((\mathrm{X}, \mathcal{M})_{\eta}, \mathbf{Z}_p) \arrow[d, "\mathrm{R}f_{\eta,*}"] \\ \mathrm{D}_{\mathrm{fgu}}((\mathrm{Y}, \mathcal{N})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\mathrm{op}} \arrow[r, "\mathrm{T}_{\text{ét},\mathrm{Y}}[d_{\mathrm{Y}}]"] & \mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}_{\mathrm{Y}}}((\mathrm{Y}, \mathcal{N})_{\eta}, \mathbf{Z}_p). \end{tikzcd}$$

The shift and twist arise from a logarithmic Poincaré duality result for perfect Laurent crystals proved in Corollary 6.6.

It is worth pointing out here that actually for the Laurent F-crystal equivalences in either [BS23] or [IKY26], the analogous pushforward compatibility theorem does not seem to appear in the literature (to the best of the author's knowledge). The problem is that it is not obvious that the site-theoretic pushforward $\mathrm{R}f_*$ preserves $\mathsf{Perf}(\mathrm{X}_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\varphi=1}$ (once this is known the claim is more or less immediate). For this reason existing pushforward compatibility results (such as the result stated in [GL25]) assume that the Laurent F-crystal arises from a prismatic F-crystal, as in that situation perfectness can be deduced from the Hodge-Tate comparison. So it appears that even in the lisse case, the pushforward compatibility result is new.

Relation to the work of Bhatt and Lurie. There is a related Riemann-Hilbert correspondence due to Bhatt-Lurie in [Bha25]. Let $\mathrm{X}^{\Delta,\mathrm{pfd}}$ denote the perfected prismatic stack attached to a p-adic formal scheme X as defined in [Bha25]. In *loc. cit.*, Bhatt and Lurie construct a fully faithful p-adic Riemann-Hilbert functor

$$\mathrm{RH}_{\Delta} : \mathrm{D}((\mathrm{X}_{\eta})_{\text{ét}}, \mathbf{Z}_p)^{\mathrm{oc}} \to \mathrm{D}(\mathrm{X}^{\Delta,\mathrm{pfd}})^{a,\varphi=1}$$

which is compatible with their mod p Riemann-Hilbert correspondence in positive characteristic (constructed in [BL19]; note they also prove their functor is dual to the Emerton-Kisin functor in a precise sense). When X arises as a formal completion of a finite type $\mathcal{O}_K$ scheme this admits a variant without using almost mathematics. The functor also has good properties when applied to Zariski-constructible sheaves on X_η, for example compatibility with the perverse t-structure constructed in [BH22]; they also prove pushforward compatibility results.

Our correspondence is much less general, but there is a reasonable tradeoff which makes both correspondences valuable. First, our crystals are deperfected. This allows us to locally formulate the crystal side of the correspondence as a generalization of (φ, Γ) modules over a deperfected base, and also allows us to write them down explicitly. The second is that the functor is an *equivalence* onto the full subcategory of p-Kummer constructible sheaves – in [Bha25] there is not a clear (non-tautological) condition describing constructible sheaves.

Overview of the proofs. All arguments work first over $\mathrm{W}(k)[\zeta_p]$, and then we extend to other bases. The first main result we work towards is strict-étale descent for finitely generated unit Laurent crystals, since most arguments reduce to a framed log affine. We work in the derived p-complete ordinary solid category $\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)$ of Definition A.10. The appendix proves that the logarithmic Breuil-Kisin atlases are completely descendable in the sense of [Kip26b, Definition 2.4.1.7]. Consequently, ordinary solid Laurent crystals are computed by the Čech nerve of the log q-de Rham prism. We also show strict-étale descent using similar methods.

The next step is to show that the finitely generated unit condition itself is strict-étale local. We do this by using *roots*. A coherent root of a unit Frobenius module M is a coherent submodule $\mathrm{M}_0 \subset \mathrm{M}$ equipped with an injection $\mathrm{M}_0 \to \varphi^*\mathrm{M}_0$ so $\mathrm{M} \simeq \operatorname{colim} \varphi^{*n}\mathrm{M}_0$; it is minimal if it contains no proper root. Modulo p, [Bli08] gives a unique minimal coherent root in the sense of Definition 4.5 for a finitely generated unit module over a regular F-finite $\mathbf{F}_p$-algebra, for example $(\mathrm{R}_0^{\square}/p)(\!(q-1)\!)$. The main point is to promote these minimal roots *as modules* from individual log q-de Rham evaluations to minimal roots in *coherent Laurent crystals* on framed log affines. Once this is done, their uniqueness as subcrystals makes them glue and proves the required descent (since ordinary coherent Laurent crystals do form a stack, using [Mat22]).

The section on coherent Laurent crystals provides the technical ingredients to carry this out. The group $\Gamma = (1+p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$ acts on the log q-de Rham prism by $q \mapsto q^a$ and $\mathrm{T}_i \mapsto q^{pb_i}\mathrm{T}_i$. Proposition 3.13 shows that evaluation on this prism is fully faithful when the evaluated object is p-completely nuclear; this includes coherent and finitely generated unit crystals. Theorem 3.5 carries a Γ-equivariant root over $\mathrm{R}_0^{\square}(\!(q-1)\!)^{\wedge}_p$ to a coherent Laurent crystal. Together they give the local (φ, Γ)-module description in Theorem B. Applying Theorem 3.5 to the distinguished minimal roots of the algebraic evaluations produces distinguished minimal roots as crystals. In the process of strict-étale descent we also characterize finitely generated unit Laurent crystals in terms of admitting a global coherent root modulo p.

Once strict-étale descent is known the arguments are more elementary and module-theoretic, as we can localize to a situation where we deal only with finitely generated unit (φ, Γ) modules as in Theorem B. The main equivalence in Theorem A is built first on the open stratum $\mathrm{U}_\eta = \mathrm{X}_\eta \setminus \mathrm{D}_\eta$. Theorem B reduces the open-stratum equivalence locally to a statement about (φ, Γ)-modules. For essential surjectivity on the open stratum, one uses Lütkebohmert's theorem to extend a local system on U_η to a Kummer-étale local system $\mathbb{L}$ on X_η. Letting $j : \mathrm{U}_\eta \to \mathrm{X}_\eta$ be the inclusion, we construct a crystal $\mathcal{O}_{\Delta}(*\mathrm{D})$ corresponding to $j_!\mathbf{Z}_p$. We show that after a finite prime-to-p Kummer cover $(\mathrm{X}_\mathrm{N})_\eta$ we may make $\mathbb{L}$ a p-Kummer local system, and thus [IKY26] provides a logarithmic Laurent crystal $\mathcal{E}$. Keeping track of the descent data in the form of a μ_N-action, one may descend $\mathcal{E}(*\mathrm{D})$ to construct a preimage. Full faithfulness uses as its main component a more delicate decompletion statement that base change to the perfection is fully faithful; Remark 5.13 explains why this does not follow from Frobenius alone. Closed strata are far easier and are handled by a variant of Kashiwara's lemma, which reduces to the corresponding result of [EK04b].

A dévissage argument using Kashiwara's lemma reduces the main theorem to the open and closed comparisons on strata, along with some (derived) full faithfulness results about maps between strata. The essential image is identified because extensions by zero of p-Kummer local systems on the strata generate the p-Kummer constructible subcategory. One must also verify that $\mathrm{T}_{\text{ét}}$ is constructible for the stratification along D; to show this strict-étale descent allows one to use the local formulation in terms of finitely generated unit (φ, Γ) modules where a Fitting ideal argument controls the possible roots (and proves constructibility along D). Once all results are shown over $\mathrm{W}(k)[\zeta_p]$, with trivial modifications we can work over $\mathrm{W}(k)[\zeta_{p^\ell}]$.

Finally, Theorem C first uses the duality of [Kip26b] to prove Poincaré duality for perfect Laurent crystals. Using this along with some modifications of the results in [GR24, Proposition 8.6, Lemma 8.8] then shows that $\mathrm{R}f_+ := (\mathrm{R}f_*(-))\{d\}[d]$ preserves perfect unit crystals. Combining these two with a dévissage we reduce to this case, we show $\mathrm{R}f_+$ also preserves finitely generated unit Laurent crystals. Once this is known we construct a comparison map to witness commutativity of the diagram in Theorem C, and knowing the source and target are constructible we check it is an equivalence by checking on a suitable conservative family of classical log points.

Notation. Unless otherwise stated, all categories are ∞-categories, $(\mathrm{X}, \mathcal{M})$ denotes a smooth qcqs log p-adic formal scheme over $\operatorname{Spf} \mathrm{W}(k)[\zeta_p]$ equipped with the log structure coming from a horizontal normal crossings divisor D. We use the logarithmic stratification by the rank of $\overline{\mathcal{M}}^{\mathrm{gp}}$. On a framed strict-étale chart, $\mathrm{X}_{\eta,\Lambda}$ denotes the coordinate stratum on which exactly the boundary coordinates indexed by Λ vanish. We write $\mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$ for the quasi-pro-Kummer-étale complexes which are derived lisse on each logarithmic stratum. The p-Kummer sheaves form a full subcategory; see Definition 4.19.

We use $(\mathrm{X}, \mathcal{M})^{\Delta}$ for Olsson's log prismatic stack (see Appendix A). We reserve $(\mathrm{X}, \mathcal{M})^{\Delta,\blacksquare}$ and $(\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}$ for the prismatic and Laurent analytic stacks derived from the construction of [Kip26b] and used only for perfect-coefficient duality.

On a framed log affine $(\operatorname{Spf} \mathrm{R}, \mathcal{M})$ with framing $\square$ and boundary coordinates $\mathrm{T}_1, \dots, \mathrm{T}_r$, let R_0 be the $\mathrm{W}(k)$-algebra in the framing, so that $\mathrm{R} \simeq \mathrm{R}_0 \widehat{\otimes}_{\mathrm{W}(k)} \mathrm{W}(k)[\zeta_p]$. The attached log q-de Rham prism is $(\mathrm{R}_0^\square[\![q-1]\!], [p]_q, \mathbf{N}^r)^a$, where $\mathrm{R}_0^\square[\![q-1]\!] = \mathrm{R}_0[\![q-1]\!]$. Its quotient by $[p]_q$ is R. The associated coefficient ring is $\mathrm{R}_0^\square(\!(q-1)\!)_p^\wedge = \mathrm{R}_0^\square[\![q-1]\!][1/[p]_q]_p^\wedge$. The superscript $\square$ records the framing. The group $\Gamma = (1 + p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$ acts on the prism by $q \mapsto q^a$ and $\mathrm{T}_i \mapsto q^{pb_i}\mathrm{T}_i$. We write A for the coefficient ring when the framing is fixed, and $\overline{\mathrm{A}}$ for its reduction modulo p.

We write $\mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ for the derived p-complete category of Laurent crystals with ordinary solid coefficients. Derived p-completion is built into Definition A.10. We write $\mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ for its subcategory of finitely generated unit Laurent crystals as in Definition 4.3. The functor $\mathrm{T}_{\text{ét}}$ is the (Kummer)-étale realization functor, while RSol denotes the crystalline Emerton-Kisin solution functor in characteristic p.

We write $\mathrm{RHom}(-,-)$ for mapping complexes and $\underline{\mathrm{RHom}}(-,-)$ for internal derived Hom. Underived internal Hom is denoted by $\underline{\mathrm{Hom}}(-,-)$.

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2. The Emerton-Kisin correspondence

In this section, we review the Emerton-Kisin correspondence from [EK04b].

The story naturally begins with Katz's correspondence, which the Emerton-Kisin correspondence generalizes to constructible sheaves. The original statement of this result is [Kat73, Proposition 4.1.1], allowing the generality of a regular $\mathbf{F}_p$-algebra R, but for our purposes we will be most interested in the case where R is a perfect $\mathbf{F}_p$-algebra. In this setting the theorem appears in [KL15, Proposition 3.2.7] and [KL15, Theorem 8.5.3] (once we pass to the limit). However we will also need a derived variant, which appears to not have a direct reference but is a formal extension of existing results. We also take the opportunity to introduce a dévissage method for idempotent complete categories which we will use frequently in what follows.

We let $\mathrm{D}^{(b)}_{\mathrm{lisse}}(\operatorname{Spec}\mathrm{R},\mathbf{Z}_p)$ denote locally bounded derived p-complete complexes in $\mathrm{D}((\operatorname{Spec}\mathrm{R})_{\text{proét}},\mathbf{Z}_p)$ such that each cohomology sheaf of the derived mod p reduction is lisse in the usual sense.

Theorem 2.1 (Katz). Let R be a perfect $\mathbf{F}_p$-algebra. Then there is an equivalence of categories
$$\mathsf{Perf}(\mathrm{W}(\mathrm{R}))^{\varphi=1}\simeq \mathrm{D}^{(b)}_{\mathrm{lisse}}(\operatorname{Spec}\mathrm{R},\mathbf{Z}_p),$$
induced by the solution functor
$$\mathrm{M}\longmapsto \left(\mathrm{M}\otimes_{\mathrm{W}(\mathrm{R})}\mathrm{W}(\mathcal{O}_{\operatorname{Spec}\mathrm{R},\text{ét}})\right)^{\varphi=1}.$$

Proof. We apply Lemma 2.3. We first verify the hypotheses. The solution functor is easily checked to be exact and $\mathsf{Perf}(\mathbf{Z}_p)$-linear, as well as compatible with derived reduction modulo p^n (using perfectness). Both categories are idempotent complete, since their defining conditions are preserved under retracts. The source is presented as $\lim_n \mathsf{Perf}(\mathrm{W}_n(\mathrm{R}))^{\varphi=1}$ and the target as a limit of categories of lisse complexes whose stalks have finite Tor-dimension over $\mathbf{Z}/p^n$, i.e. $\mathrm{D}^{(b)}_{\mathrm{lisse}}(\operatorname{Spec}\mathrm{R},\mathbf{Z}/p^n)$. The mod p functor is the equivalence of [BS23, Proposition 3.4], so Lemma 2.3 proves the result. $\square$

Remark 2.2. The Emerton-Kisin correspondence is contravariant. Since Perf is self-dual, the previous correspondence also admits a dual or contravariant version. The solution functor is $\mathrm{M} \mapsto \mathrm{RHom}_{\mathsf{Perf}^{\varphi}(\mathrm{W(R)}_{\text{ét}})}(\mathrm{M}_{\text{ét}}, \mathrm{W(R)}_{\text{ét}})$. Similarly other correspondences reducing to Katz's theorem, such as [BS23, Corollary 3.7] or [IKY26] admit contravariant versions.

The dévissage method we have just applied is a categorified version of Nakayama's lemma. For an object X of a $\mathsf{Perf}(\mathbf{Z}_p)$-linear stable ∞-category, write $\mathrm{X}/p^n := \mathrm{cofib}(p^n : \mathrm{X} \to \mathrm{X})$. Recall a stable ∞-category $\mathcal{C}$ is idempotent complete when $\mathcal{C}$ is closed under retracts in $\mathsf{Ind}(\mathcal{C})$.

Setup. Let $\mathcal{C}$ be an idempotent complete $\mathsf{Perf}(\mathbf{Z}_p)$-linear stable ∞-category. All tensor products of categories below are taken among idempotent complete stable ∞-categories. Put

$$\mathcal{C}_n := \mathcal{C} \otimes_{\mathsf{Perf}(\mathbf{Z}_p)} \mathsf{Perf}(\mathbf{Z}/p^n).$$

We say that $\mathcal{C}$ is derived p-complete if the natural functor $\mathcal{C} \to \varprojlim_n \mathcal{C}_n$ is an equivalence. Write $\mathrm{red}_n : \mathcal{C} \to \mathcal{C}_n$ for reduction and $\mathrm{X}_n = \mathrm{red}_n\mathrm{X}$. Since $\mathsf{Perf}(\mathbf{Z}/p^n)$ is thickly generated by its unit, $\mathcal{C}_n$ is thickly generated by the objects X_n, and

$$\mathrm{Map}_{\mathcal{C}_n}(\mathrm{X}_n, \mathrm{Y}_n) \simeq \mathrm{Map}_{\mathcal{C}}(\mathrm{X}, \mathrm{Y}/p^n).$$

We ask that functors between derived p-complete categories $\mathrm{F} : \mathcal{C} \to \mathcal{D}$ in this setup induce compatible functors $\mathrm{F}_n : \mathcal{C}_n \to \mathcal{D}_n$.

Lemma 2.3. Let $\mathcal{C}$ and $\mathcal{D}$ be derived p-complete idempotent complete $\mathsf{Perf}(\mathbf{Z}_p)$-linear stable ∞-categories. Let $\mathrm{F} : \mathcal{C} \to \mathcal{D}$ be an exact $\mathsf{Perf}(\mathbf{Z}_p)$-linear functor in the setup above. If F_1 is an equivalence, then F is an equivalence.

Proof. Full faithfulness is clear, using derived Nakayama on mapping complexes. Indeed,

$$\mathrm{Map}_{\mathcal{C}_1}(\mathrm{X}_1, \mathrm{Y}_1) \simeq \mathrm{Map}_{\mathcal{C}}(\mathrm{X}, \mathrm{Y}/p) \simeq \mathrm{Map}_{\mathcal{C}}(\mathrm{X}, \mathrm{Y}) \otimes^{\mathrm{L}}_{\mathbf{Z}_p} \mathbf{F}_p.$$

Note that by our setup $\mathrm{Map}_{\mathcal{C}}(\mathrm{X}, \mathrm{Y})$ is derived p-complete. Note that similarly one gets full faithfulness of F_n as we will need later in the argument.

For essential surjectivity, it suffices to show the F_n are essentially surjective; we know this for $n = 1$. Let $\mathrm{Y}_n \in \mathcal{D}_n$, and consider the fiber sequence

$$\mathrm{Y}_n/p \longrightarrow \mathrm{Y}_n/p^n \longrightarrow \mathrm{Y}_n/p^{n-1}.$$

This shows Y_n/p^n belongs to the thick subcategory generated by Y_n/p; we know Y_n/p is in the essential image of F_n because F_1 is essentially surjective, and the essential image is a thick subcategory. Thus Y_n/p^n is in the essential image of F_n, so denote a choice of preimage by X_n.

Because we take the derived quotient by p^n, since $p^n = 0$ in $\mathcal{D}_n$ one has $\mathrm{Y}_n/p^n = \mathrm{cofib}(0 : \mathrm{Y}_n \to \mathrm{Y}_n) = \mathrm{Y}_n \oplus \mathrm{Y}_n[1]$. So in particular Y_n is a retract of Y_n/p^n. But then idempotent completeness implies that Y_n itself is in the essential image of F_n. Indeed we have

$$\mathrm{Y}_n \xrightarrow{\ i\ } \mathrm{Y}_n/p^n \xrightarrow{\ r\ } \mathrm{Y}_n$$

where ri is the identity; in particular ir is an idempotent in $\mathrm{End}_{\mathcal{D}_n}(\mathrm{F}_n(\mathrm{X}_n))$. By derived full faithfulness, it is an image of an idempotent e in $\mathrm{End}_{\mathcal{C}_n}(\mathrm{X}_n)$ which splits as

$$\mathrm{X}'_n \xrightarrow{i'} \mathrm{X}_n \xrightarrow{r'} \mathrm{X}'_n$$

where $r'i' = \mathrm{id}_{\mathrm{X}'_n}$ and $i'r' = e$. Then necessarily $\mathrm{F}_n(\mathrm{X}'_n) \simeq \mathrm{Y}_n$, and the F_n are all essentially surjective. Thus every F_n is an equivalence, and derived p-completeness implies that F is an equivalence. □

Katz's theorem is crucial in proving the equivalence

$$\mathsf{Perf}(\mathrm{X}_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})^{\varphi=1} \simeq \mathrm{D}^{(b)}_{\mathrm{lisse}}(\mathrm{X}^{\diamond}_{\eta})$$

for bounded p-adic formal schemes X in [BS23]. Precisely, this means that when $\mathrm{X} = \mathrm{Spf}\,\mathrm{R}$ is perfectoid, the result reduces to Katz's equivalence; their theorem is deduced by a reduction to this case. Our result for finitely generated unit Laurent crystals will be compatible in a similar way with the Emerton-Kisin correspondence for constructible sheaves.

We now extend the Emerton-Kisin correspondence of [EK04b] to $\mathbf{Z}_p$ coefficients, deduced from the mod p theorem by Lemma 2.3. We fix a perfect field k of characteristic p. If $\mathscr{X} / \mathrm{Spf}\,\mathrm{W}(k)$ is a smooth formal scheme, we let $\mathrm{D}_{\mathscr{X}}$ denote the sheaf of infinitesimal differential operators. Locally, if $\mathrm{X} \to \mathbf{A}^n_k$ is étale, this can be pulled back from the sheaf associated to $\mathrm{D}_{\mathrm{Spf}\,\mathrm{W}(k)\langle \mathrm{T}_1,\ldots,\mathrm{T}_n\rangle} = \mathrm{W}(k)\langle \mathrm{T}_1, \ldots, \mathrm{T}_n, \partial_1^{[i_1]}, \ldots, \partial_n^{[i_n]}\rangle$, where all ∂_i are given full divided powers. We put

$$\mathrm{D}_{\mathrm{F},\mathscr{X}} := \widehat{\bigoplus}_{i\geq 0} \mathrm{F}^{*i}\mathrm{D}_{\mathscr{X}}.$$

Modules over this noncommutative sheaf are infinitesimal crystals $\mathcal{E}$ with semilinear Frobenius $\varphi_{\mathcal{E}}$. A certain Taylor formula comparing Frobenius lifts makes $\mathrm{D}_{\mathrm{F},\mathscr{X}}$ canonical and global; see [EK04a, §§6.1-6.2].

The Emerton-Kisin correspondence is compatible with a perverse t-structure on the étale side, which we now introduce.

Definition 2.4. Write $({}^p\mathrm{D}^{\leq 0}, {}^p\mathrm{D}^{\geq 0})$ for Gabber's perverse t-structure on $\mathrm{D}^{(b)}_{\mathrm{cons}}(\mathrm{X}, \mathbf{Z}_p)$ [Gab04], and write ${}^p\mathrm{H}^i$ for perverse cohomology. Its dual tilt is the p^+-perverse t-structure

$$^{p^+}\mathrm{D}^{\leq 0} = \{\mathrm{K} \in {}^p\mathrm{D}^{\leq 1} \mid {}^p\mathrm{H}^1(\mathrm{K}) \text{ is locally bounded } p\text{-power torsion}\},$$

$$^{p^+}\mathrm{D}^{\geq 0} = \{\mathrm{K} \mid \mathrm{K}/p \in {}^p\mathrm{D}^{\geq 0}(\mathrm{X}, \mathbf{F}_p)\}.$$

Here locally bounded means that the torsion is killed by a fixed power of p after restriction to any quasicompact open. For every $n \geq 1$, derived reduction modulo p^n is right t-exact for p-perversity and left t-exact for p^+-perversity. Moreover, $\mathrm{K} \in {}^p\mathrm{D}^{\leq 0}$ if and only if $\mathrm{K}/p \in {}^p\mathrm{D}^{\leq 0}(\mathrm{X}, \mathbf{F}_p)$, while $\mathrm{K} \in {}^{p^+}\mathrm{D}^{\geq 0}$ if and only if $\mathrm{K}/p \in {}^p\mathrm{D}^{\geq 0}(\mathrm{X}, \mathbf{F}_p)$. This is similar in spirit to [BH22, Construction 4.3 and Proposition 4.6].

One defines $\mathrm{D}^b_{\mathrm{fgu}}(\mathrm{D}_{\mathrm{F},\mathscr{X}})$ as the full subcategory of $\mathrm{D}_{\mathrm{unit}}(\mathrm{D}_{\mathrm{F},\mathscr{X}})$ (i.e. $\mathrm{F}^*\mathrm{M} \simeq \mathrm{M}$) which are locally bounded and have cohomology sheaves finitely generated over $\mathrm{D}_{\mathrm{F},\mathscr{X}}$. We will assume X is quasicompact in this section for convenience (it is not strictly needed), so this amounts to being bounded.

Theorem 2.5. Let $\mathscr{X} / \operatorname{Spf} \mathrm{W}(k)$ be smooth and quasicompact. Then there is an equivalence

$$\mathrm{RSol} : \mathrm{D}^b_{\mathrm{fgu}}(\mathrm{D}_{\mathrm{F},\mathscr{X}})^{\mathrm{op}} \simeq \mathrm{D}^b_{\mathrm{cons}}(\mathrm{X}, \mathbf{Z}_p),$$

where X/k is the smooth special fiber of $\mathscr{X}$. Explicitly, the functor is given by $\mathrm{RSol}_{\mathrm{X}}(\mathrm{M}) = \mathrm{RHom}_{\mathrm{D}(\mathrm{D}_{\mathrm{F},\mathscr{X}_{\mathrm{pro\acute{e}t}}})}(\mathrm{M}_{\mathrm{pro\acute{e}t}}, \mathcal{O}_{\mathscr{X},\mathrm{pro\acute{e}t}}) \in \mathrm{D}(\mathrm{X}_{\mathrm{pro\acute{e}t}}, \mathbf{Z}_p)$. If X is pure of dimension d, then $\mathrm{RSol}_{\mathrm{X}}[d]$ identifies the standard t-structure on the source with the p^+-perverse t-structure on the target.

Proof. We first note that one has

$$\mathrm{D}^b_{\mathrm{fgu}}(\mathrm{D}_{\mathrm{F},\mathscr{X}}) \simeq \varprojlim_n \mathrm{D}^b_{\mathrm{fgu,tf}}(\mathrm{D}_{\mathrm{F},\mathscr{X}_{p^n=0}})$$

and similarly

$$\mathrm{D}^b_{\mathrm{cons}}(\mathrm{X}, \mathbf{Z}_p) \simeq \varprojlim_n \mathrm{D}^b_{\mathrm{ctf}}(\mathrm{X}_{\mathrm{pro\acute{e}t}}, \mathbf{Z}/p^n).$$

Here tf and ctf mean finite Tor dimension and constructible finite Tor dimension, respectively. These categories are closed under retracts in their ambient idempotent complete categories. They therefore satisfy the hypotheses of Lemma 2.3. Moreover, [EK04b, Proposition 15.4.1 and Equation (16.1.3)] give

$$\mathrm{RSol}_{\mathrm{X}}(\mathrm{M})/p^n \simeq \mathrm{RSol}_{\mathrm{X},\mathbf{Z}/p^n}(\mathrm{M}/p^n).$$

Thus the solution functor induces the required compatible functors on reductions, and Lemma 2.3 reduces the assertion to the case modulo p.

Since we have defined RSol using proétale sheaves to handle $\mathbf{Z}_p$ coefficients well, after reducing modulo p in the target category we must use the equivalence

$$\mathrm{D}^b_{\mathrm{cons}}(\mathrm{X}_{\mathrm{pro\acute{e}t}}, \mathbf{F}_p) \simeq \mathrm{D}^b_{\mathrm{cons}}(\mathrm{X}_{\mathrm{\acute{e}t}}, \mathbf{F}_p).$$

Let $\nu : \mathrm{X}_{\mathrm{pro\acute{e}t}} \to \mathrm{X}_{\mathrm{\acute{e}t}}$ be the natural morphism of sites. Since X is smooth, it is locally Noetherian, and for each n pullback along ν identifies $\mathrm{D}^b_{\mathrm{ctf}}(\mathrm{X}_{\mathrm{\acute{e}t}}, \mathbf{Z}/p^n)$ with $\mathrm{D}^b_{\mathrm{ctf}}(\mathrm{X}_{\mathrm{pro\acute{e}t}}, \mathbf{Z}/p^n)$ by the torsion comparison of [BS15, Corollary 5.1.6 and Proposition 5.2.6], applied on quasi-compact opens. Combining this with [HRS23, Theorem 1.6(1)] gives the desired result.

Modulo p, [EK04b, Proposition 15.4.3] identifies finitely generated unit $\mathrm{D}_{\mathrm{F},\mathscr{X}}$-complexes with the usual finitely generated unit $\mathcal{O}_{\mathrm{F},\mathrm{X}}$-complexes, and [EK04b, §16.1.5] identifies their solution functors. Theorem 11.4.2 of [EK04b] therefore proves the required mod-p equivalence, while Theorem 11.5.4 gives the compatibility of its standard heart with the middle-perverse heart. This proves the equivalence.

Finally we discuss the compatibility with t-structures. Assume X has pure dimension d, let M lie in the source category's heart, and put $\mathrm{K} = \mathrm{RSol}_{\mathrm{X}}(\mathrm{M})[d]$. If M is p-torsionfree, then the derived reduction M/p lies in the mod-p heart, so K/p is perverse. The two reduction criteria in Definition 2.4 give $\mathrm{K} \in {}^{p}\mathrm{D}^{\leq 0} \subset {}^{p^+}\mathrm{D}^{\leq 0}$ and $\mathrm{K} \in {}^{p^+}\mathrm{D}^{\geq 0}$. Hence K lies in the p^+-perverse heart. If $p\mathrm{M} = 0$, one obtains

$$\mathrm{RSol}_{\mathrm{X},\mathbf{Z}_p}(\mathrm{M})[d] \simeq \mathrm{RSol}_{\mathrm{X},\mathbf{F}_p}(\mathrm{M})[d][-1],$$

which is a torsion perverse object shifted by $[-1]$, hence is p^+-perverse. The p-power torsion in a finitely generated unit module is locally bounded, so dévissage and the exact sequence

$$0 \to \mathrm{M}[p^\infty] \to \mathrm{M} \to \mathrm{M}/[p^\infty] \to 0$$

prove the claim for every M in the heart.

Conversely, the p^+-perverse heart is generated under extensions by torsionfree p-perverse sheaves and by $\mathrm{L}[-1]$ for locally bounded torsion p-perverse objects L. For the first kind, derived reduction modulo p is perverse; its inverse image under the stable equivalence has reduction in the mod-p source heart, so derived Nakayama places it in the source heart and makes it p-torsion-free. The formula $\mathrm{RSol}_{\mathrm{X},\mathbf{Z}_p}(\mathrm{M})[d] \simeq \mathrm{RSol}_{\mathrm{X},\mathbf{F}_p}(\mathrm{M})[d][-1]$ and dévissage lift the second kind. Thus $\mathrm{RSol}_{\mathrm{X}}[d]$ identifies the two hearts. □

We also record a variant for perfect rings, which we will use (in a mild way – we really only use existence of the functor and full faithfulness).

Theorem 2.6. Let R be a perfect $\mathbf{F}_p$-algebra. Then there is an equivalence

$$\mathrm{RSol} : \mathrm{D}^b_{\mathrm{fgu}}(\mathrm{W}(\mathrm{R}))^{\mathrm{op}} \simeq \mathrm{D}^b_{\mathrm{cons}}(\operatorname{Spec}\mathrm{R}, \mathbf{Z}_p)$$

given by $\mathrm{RHom}_{\mathrm{D}^{\varphi}(\mathrm{W}(\mathrm{R}))}(\mathrm{M}_{\text{proét}}, \mathrm{W}(\mathrm{R})_{\text{proét}}) \in \mathrm{D}((\operatorname{Spec}\mathrm{R})_{\text{proét}}, \mathbf{Z}_p)$. The equivalence is moreover symmetric monoidal.

Proof. The argument is similar to the previous case. Both sides are stable subcategories of their ambient derived categories closed under retracts, hence are idempotent complete and $\mathsf{Perf}(\mathbf{Z}_p)$-linear; the dual action of $\mathsf{Perf}(\mathbf{Z}_p)$ on the source makes RSol linear. Their reductions (in the sense of the setting of Lemma 2.3) are the perfect unit categories over $\mathrm{W}_n(\mathrm{R})$ and the constructible complexes whose stalks have finite Tor-dimension over $\mathbf{Z}/p^n$.

Applying lemma 2.3 now reduces the claim to [BL19, Theorem 11.4.4], together with [BL19, Theorems 12.5.4 and 12.6.1].

For the final claim, one may essentially copy the argument in [BL19, Theorem 8.4.1]. The formula for RSol gives a lax symmetric monoidal structure

$$\mathrm{RSol}(\mathrm{M}) \otimes^{\mathrm{L}}_{\mathbf{Z}_p} \mathrm{RSol}(\mathrm{N}) \longrightarrow \mathrm{RSol}(\mathrm{M} \otimes^{\mathrm{L}}_{\mathrm{W}(\mathrm{R})} \mathrm{N}).$$

Both sides are derived p-complete, so derived Nakayama reduces the assertion that this map is an equivalence to the case modulo p, where it may be checked on geometric stalks. Over an algebraically closed perfect field κ, use the dual or contravariant form of Katz's theorem described in the remark following Theorem 2.1. The fact that this is symmetric monoidal can be checked in the same way as [BL19, Theorem 8.4.1]: the category is equivalent to $\mathsf{Perf}(\mathbf{F}_p)$, and reducing to the heart (which is fine since it thickly generates) one can decompose objects as direct summands of the unit where the claim is obvious. □

3. Coherent Laurent crystals

In this section, we prove several important results about coherent Laurent crystals on the framed log affines of Definition 3.2. These will be used to understand the local structure of finitely generated unit crystals in terms of (φ, Γ)-modules.

Assumption 3.1. Let $\mathrm{X}/\mathcal{O}_K$ be smooth and qcqs, let D be a horizontal normal crossings divisor, and let $\mathcal{M} = \mathcal{M}_{\mathrm{D}}$ be the log structure induced by D.

Fix a compatible system $(\zeta_{p^m})_{m\geq 1}$ extending ζ_p, and use it to identify $\mathbf{Z}_p(1)$ with $\mathbf{Z}_p$.

In this section we assume p is odd and work over $\mathrm{W}(k)[\zeta_p]$ (mostly for expository reasons). At the end of this section in Corollary 3.19, we explain how to make modifications to obtain the result over $\mathrm{W}(k)$ (and also with more roots of unity).

Definition 3.2. A framing of $(\operatorname{Spf}\mathrm{R}, \mathcal{M})$ consists of a p-completely étale $\mathrm{W}(k)$-algebra R_0 over $\mathrm{W}(k)\langle \mathrm{T}_1, \ldots, \mathrm{T}_r, \mathrm{T}_{r+1}^{\pm}, \ldots, \mathrm{T}_n^{\pm}\rangle$ and an identification $\mathrm{R} \simeq \mathrm{R}_0\widehat{\otimes}_{\mathrm{W}(k)}\mathrm{W}(k)[\zeta_p]$ under which $\mathcal{M}$ is the pullback of the log structure with chart $\mathbf{N}^r$ whose generators map to $\mathrm{T}_1, \ldots, \mathrm{T}_r$. We denote the resulting log formal scheme $\operatorname{Spf}\mathrm{W}(k)[\zeta_p]\langle \mathrm{T}_1, \ldots, \mathrm{T}_r, \mathrm{T}_{r+1}^{\pm}, \ldots, \mathrm{T}_n^{\pm}\rangle$ by $\mathbb{T}_r^n$, the induced strict morphism to it by $\square$, and call $(\operatorname{Spf}\mathrm{R}, \mathcal{M})$ a framed log affine.
There is also a variant over $\mathcal{O}_K$ for general $K/\mathrm{W}(k)[1/p]$, obtained by replacing $\mathrm{W}(k)[\zeta_p]$ with $\mathcal{O}_K$ and requiring $\mathrm{R} \simeq \mathrm{R}_0\widehat{\otimes}_{\mathrm{W}(k)}\mathcal{O}_K$.

Remark 3.3. For a framed log affine, we use n and r as the corresponding integers in Definition 3.2. We call the étale lifts of $\mathrm{T}_1, \ldots, \mathrm{T}_r$ "boundary coordinates".

Once we have later shown descent results, the following lemma will frequently be used to reduce to the situation of a framed log affine.

Lemma 3.4. Let $(\mathrm{X}, \mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$. Then $(\mathrm{X}, \mathcal{M})$ admits a finite strict-étale cover by framed log affines over $\mathcal{O}_K$.

Proof. By the normal crossings hypothesis, a strict-étale cover separates the local branches of D and makes them a simple normal crossings divisor. Fix $x \in \mathrm{X}$. Let $\mathrm{D}_1, \ldots, \mathrm{D}_r$ be the branches of D through x, and pick a small affine Zariski neighborhood $\operatorname{Spf}\mathrm{R}$ around x so that no other component of D meets the chosen neighborhood. Since D is simple normal crossings relative to $\operatorname{Spf}\mathcal{O}_K$ and $\mathrm{X}/\operatorname{Spf}\mathcal{O}_K$ is smooth, there are functions $t_1, \ldots, t_r, s_{r+1}, \ldots, s_n \in \mathrm{R}$ such that $\mathrm{D}_i = \mathrm{V}(t_i)$ and

$$\mathrm{d}t_1, \ldots, \mathrm{d}t_r, \mathrm{d}s_{r+1}, \ldots, \mathrm{d}s_n$$

is a basis of $\widehat{\Omega}^1_{\mathrm{R}/\mathcal{O}_K}$. Choose constants $c_j \in \mathcal{O}_K$ such that $u_j = s_j + c_j$ has nonzero image in $\kappa(x)$; thus after Zariski localizing once more, each u_j is a unit. The functions t_i and u_j define a morphism

$$\square : \operatorname{Spf}\mathrm{R} \to \mathbb{T}_r^n.$$

It is induced by the ring map sending T_i to t_i for $i \leq r$ and T_j to u_j for $j > r$ (the latter are invertible, which is why we localized to make u_j units).

The pullback of $\mathrm{V}(\mathrm{T}_1 \cdots \mathrm{T}_r)$ is $\mathrm{V}(t_1 \cdots t_r) = \mathrm{D} \cap \operatorname{Spf}\mathrm{R}$, hence $\square$ is strict as a morphism of log formal schemes. The Jacobian criterion shows that the underlying morphism is p-completely étale.

Let π be a uniformizer of $\mathcal{O}_K$ with Eisenstein polynomial $\mathrm{E}(u) \in \mathrm{W}(k)[u]$ of degree e. Since $\mathcal{O}_K/p \simeq k[\pi]/(\pi^e)$, invariance of étale algebras under nilpotent thickenings descends R/p uniquely to an étale algebra $\overline{\mathrm{R}}_0$ over the reduction modulo (p,π) of the target coordinate ring. Its unique p-completely étale lift R_0 over the corresponding $\mathrm{W}(k)$-algebra satisfies $\mathrm{R} \simeq \mathrm{R}_0\widehat{\otimes}_{\mathrm{W}(k)}\mathcal{O}_K$, compatibly with the framing. $\square$

Now let $(\operatorname{Spf}\mathrm{R},\mathcal{M})$ be a framed log affine with framing $\square$. Let R_0 be the algebra in the framing, and put $\mathrm{A}^+ = \mathrm{R}_0^\square[\![q-1]\!] := \mathrm{R}_0[\![q-1]\!]$ and $\mathrm{A} = \mathrm{R}_0^\square(\!(q-1)\!)_p^\wedge := (\mathrm{A}^+[1/[p]_q])_p^\wedge$. The identification $\mathrm{A}^+/([p]_q) \simeq \mathrm{R}$ sends q to ζ_p. The log q-de Rham prism $(\mathrm{A}^+,[p]_q,\mathbf{N}^r)^a$ (in the sense of [Kos22]) uses this to witness it as an object in $(\operatorname{Spf}\mathrm{R},\mathcal{M})_{\Delta}$, and its log chart sends e_i to T_i for $i \le r$. We write $\overline{\mathrm{R}} = \mathrm{R}_0/p$, $\overline{\mathrm{A}}^+ = \mathrm{A}^+/p$, and $\overline{\mathrm{A}} = \mathrm{A}/p$.

Put $\Gamma = (1+p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$. For $\gamma = (a,b_1,\dots,b_n) \in \Gamma$, regard b_i as an element of $\mathbf{Z}_p$ through the identification fixed above. The formulas $q \mapsto q^a$ and $\mathrm{T}_i \mapsto q^{pb_i}\mathrm{T}_i$ for $i \le n$ reduce to the identity modulo $[p]_q$. They therefore lift uniquely along the framing to log prism automorphisms over $(\operatorname{Spf}\mathrm{R},\mathcal{M})$; see Lemma A.15. Thus the crystal property gives an action of Γ on every evaluation in $\mathrm{D}_\blacksquare(\mathrm{A})$ for a solid Laurent crystal as defined in Definition A.10. We use the symbol $\square$ to indicate that the Γ action on the prism depends on the framing.

Our first goal is to prove that evaluation on the log q-de Rham prism induces a t-exact conservative functor

$$\rho_{\mathrm{A}}^* : \mathrm{D}_\blacksquare((\operatorname{Spf}\mathrm{R},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge) \longrightarrow \mathrm{D}_\blacksquare(\mathrm{A})^{h\Gamma},$$

which is fully faithful when the evaluated object is p-completely nuclear (p-complete with derived mod p reduction nuclear).

We also establish a result about roots. This will be crucial in giving a workable local description of finitely generated unit crystals. Its proof uses the full faithfulness of Proposition 3.13 for coherent Laurent crystals.

Theorem 3.5. Let $(\operatorname{Spf}\mathrm{R},\mathcal{M})$ be a framed log affine and put $\overline{\mathrm{A}} = \mathrm{A}/p$. Let M be a finite $\overline{\mathrm{A}}$-module equipped with a continuous semilinear Γ-action as above. Suppose there is a Γ-equivariant injection $\mathrm{M} \hookrightarrow \varphi^*\mathrm{M}$. Then there is a coherent Laurent crystal $\mathcal{E}$ killed by p, with evaluation M, and an injection $\mathcal{E} \hookrightarrow \varphi^*\mathcal{E}$ whose evaluation is the given map.

3.1. **Laurent crystals and Γ-modules.** Fix a framed log affine $(\operatorname{Spf}\mathrm{R},\mathcal{M})$ over $\operatorname{Spf}\mathrm{W}(k)[\zeta_p]$. We let $\mathrm{A}^+ = \mathrm{R}_0^\square[\![q-1]\!]$ denote the q-de Rham prism attached to the framing $\square$. In this subsection we will define a functor

$$\rho_{\mathrm{A}}^* : \mathrm{D}_\blacksquare((\operatorname{Spf}\mathrm{R},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge) \longrightarrow \mathrm{D}_\blacksquare(\mathrm{A})^{h\Gamma}$$

and check that it is fully faithful under a nuclearity hypothesis on the source category.

While the argument is purely site-theoretic in nature, it is worth pointing out that its organization is best understood from the stacky viewpoint used in §6, where we construct an analytic stack $(\operatorname{Spf}\mathrm{R},\mathcal{M})^{\mathcal{L},\blacksquare}$ whose solid quasicoherent sheaves recover $\mathrm{D}_\blacksquare((\operatorname{Spf}\mathrm{R},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge)$.

Remark 3.6. This argument is a site-theoretic translation of a more geometric argument the author was unable to carry out, namely that using the Laurent stacks of §6 one might attempt to prove that pullback along

$$\rho_{\mathrm{A}} : [\mathrm{AnSpec}(\mathrm{A})/\Gamma] \to (\mathrm{Spf}\,\mathrm{R}, \mathcal{M})^{\mathcal{L},\blacksquare}$$

is fully faithful on $\mathrm{D}_\blacksquare$; here we use the analytic ring structure set after Lemma 3.7. If this map was a proper morphism of analytic stacks one could use the projection formula to reduce to showing $\mathcal{O}_{(\mathrm{Spf}\,\mathrm{R},\mathcal{M})^{\mathcal{L},\blacksquare}} \to \mathrm{R}\rho_{\mathrm{A},*}\mathcal{O}_{[\mathrm{Spa}(\mathrm{A})/\Gamma]}$ is an isomorphism. Let $\mathrm{A}^{(1)} = \mathrm{A}^{+,(1)}[1/[p]_q]^\wedge_p$ where $\mathrm{A}^{+,(1)}$ is the self-coproduct of A^+ in $(\mathrm{X}, \mathcal{M})_{\Delta}$. If one had base change for the square

$$\begin{tikzcd} \mathrm{D}_\blacksquare([\mathrm{AnSpec}(\mathrm{A})/\Gamma]) \arrow[r, "p_0^*"] \arrow[d, "\rho_{\mathrm{A},*}"] & \mathrm{D}_\blacksquare([\mathrm{AnSpec}\,\mathrm{A}^{(1)}/\Gamma]) \arrow[d, "\rho'_{\mathrm{A},*}"] \\ \mathrm{D}_\blacksquare((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})^{\mathcal{L},\blacksquare}) \arrow[r, "\mathrm{ev}^*"] & \mathrm{D}_\blacksquare(\mathrm{AnSpec}\,\mathrm{A}) \end{tikzcd}$$

where the bottom map is induced by the usual cover associated to a log prism of $(\mathrm{X}, \mathcal{M})^{\Delta}$, this would reduce to checking the same pushforward claim for the right vertical map $\rho'_{\mathrm{A},*}$. We do this in Lemma 3.11; Lemma 3.12 is essentially giving the base change result under nuclearity. Unfortunately it is not clear how to prove either the properness or base change claims, if they are even true (note that to some extent we used nuclearity in the proof).

Despite it being unclear whether a stack-theoretic approach is viable, we are able to give an argument which takes roughly the same form as the previous remark. The main technical content remains the cohomology calculation in Lemma 3.11, for which we will need a very explicit description of $\mathrm{A}^{(1)}$ and how it sits inside $\mathrm{C}(\Gamma, \mathrm{A})$. The following lemma computes $\mathrm{A}^{+,(1)}$ as a log prismatic envelope.

Lemma 3.7. Let $\mathrm{A}^+ = \mathrm{R}_0^\square[\![q-1]\!]$, and let $(\mathrm{A}^+)^{(1)}$ be the ring of the self-coproduct of $(\mathrm{A}^+, [p]_q, \mathcal{M})^a$ in $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}$. Write q_0, q_1 for the two copies of q, put $u_0 = q_1/q_0$, and let $u_i = \mathrm{T}_{i,1}/\mathrm{T}_{i,0}$ for $1 \le i \le n$. Then

$$(\mathrm{A}^+)^{(1)} \simeq \mathrm{R}_0^\square[\![q_0 - 1]\!][u_0^{\pm 1}, u_1^{\pm 1}, \dots, u_n^{\pm 1}]\left\{\frac{u_0 - 1}{[p]_{q_0}}, \dots, \frac{u_n - 1}{[p]_{q_0}}\right\}^\wedge_{\delta,(p,[p]_{q_0})}.$$

Here $\delta(u_j) = 0$ for $0 \le j \le n$.

Proof. This follows by verifying universal properties. For simplicity write $(\mathrm{A}^+)^{(1)}$ for $\mathrm{R}_0^\square[\![q_0 - 1]\!][u_0^{\pm 1}, u_1^{\pm 1}, \dots, u_n^{\pm 1}]\left\{\frac{u_0 - 1}{[p]_{q_0}}, \dots, \frac{u_n - 1}{[p]_{q_0}}\right\}^\wedge_{\delta,(p,[p]_{q_0})}$ with its natural log prism structure (with prismatic ideal $[p]_{q_0}$ and log structure pulled back from $\mathrm{Spf}\,\mathrm{A}^+$); we will verify that it has the required universal property.

Let us recall what this universal property should be for the self-coproduct of the log q-de Rham prism. We ask, for a test log prism $(\mathrm{B}, [p]_{q_0}, \mathcal{N})^a \in (\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}$ that in the diagram

$$\begin{tikzcd}
(\mathrm{B}, [p]_{q_0}, \mathcal{N})^a \arrow[rrd, bend left, "f_0"] \arrow[rd, dashed, "g"] \arrow[rdd, bend right, "f_1"'] & & \\
& (\mathrm{A}^{(1)}, [p]_{q_0}, \mathcal{M}^{(1)})^a \arrow[r, "p_0"] \arrow[d, "p_1"] & (\operatorname{Spf} \mathrm{A}_0^+, ([p]_{q_0}), \mathcal{M}_0)^a \arrow[d] \\
& (\operatorname{Spf} \mathrm{A}_1^+, ([p]_{q_1}), \mathcal{M}_1)^a \arrow[r] & *
\end{tikzcd}$$

there is a unique dashed arrow g (we can assume the ideal is $[p]_{q_0}$ by rigidity). Here the diagram is as sheaves in the log prismatic site, where the prisms are regarded this way via the Yoneda embedding.

First, we can replace the log structure on B with a common strict log structure to test such universal properties. One can use the same method as [DLMS24, Corollary 2.2 and proof of Lemma 2.10].

A map g into the self-coproduct (when it is represented by a log prism) then amounts to the data of strict maps of prisms f_0 and f_1. After fixing $f_0 : \mathrm{A}_0^+ \to \mathrm{B}$, the choices of f_1 are

$$\left\{(v_j)_{j=0}^n \in (\mathrm{B}^\times)^{n+1} : \delta(v_j) = 0 \text{ and } v_j \equiv 1 \pmod{[p]_{q_0}}\right\}.$$

By the universal property of log prismatic envelopes [Kos22, Propositions 3.6 and 3.7], after fixing f_0 the maps g are classified by the same data, through $g(u_j) = v_j$ on the underlying rings. Indeed, the divided generators map to $(v_j - 1)/[p]_{q_0}$, while the second chart maps e_i to (e_i, v_i) for $i \leq r$. Conversely, these assignments give the unique map g, and its composite with p_1 is f_1 by the unique étale lift along the framing. Hence the two universal properties agree. $\square$

Let $\mathrm{A} = \mathrm{R}_0^\square(\!(q-1)\!)_p^\wedge$ as usual. Our next task will be to understand $\mathrm{A}^{(1)} := \mathrm{A}^{+,(1)}[1/[p]_q]^\wedge$ as a subring of $\mathrm{C}(\Gamma, \mathrm{A})$. The topologies here are quite important as we want to speak of continuous semilinear Γ actions, even for coherent Laurent crystals, so we explain some of the content of Appendix A.3 here. We must also work with analytic rings as defined by Clausen and Scholze in [CS23], as the full faithfulness result we prove is for nuclear solid modules; see also [RC24] and [Kip26b] for more background.

We note both A and $\mathrm{C}(\Gamma, \mathrm{A})$ are equipped with natural topological ring (indeed analytic ring) structures. This is clear once we explain the structure we place on $\mathrm{B}[1/d]_p^\wedge$ where $(\mathrm{B}, (d))$ is a prism. One gives this the inverse limit analytic ring structure $\lim_n (\mathrm{B}/p^n)[1/d]$ where individually $(\mathrm{B}/p^n)[1/d]$ are given their Tate ring structures. This equips $\mathrm{B}[1/d]_p^\wedge$ with an analytic ring structure for any prism; using this we get such a structure on $\mathrm{B}^{(n)}$ (as they come from prisms). We give $\mathrm{C}(\Gamma, -)$ its natural induced analytic ring structure (as a topological ring one uses the compact-open topology).

Once this is defined, as in Definition A.10 one sets $\mathrm{D}_\blacksquare(\mathrm{B}[1/d]_p^\wedge) := \mathsf{Mod}_{\mathrm{B}[1/d]_p^\wedge}(\mathrm{D}_\blacksquare(\mathrm{Spa}(\mathrm{B}, \mathrm{B})))$ where we use Spa for the analytic ring given there.

Definition 3.8. For $\gamma = (a, b_1, \dots, b_n) \in \Gamma$, put $x_0(\gamma) = (a-1)/p$ and $x_i(\gamma) = b_i$ for $1 \le i \le n$. Set $\eta_{0,m}(x_0) = q^{m+p\binom{m}{2}}(q-1)^m\binom{x_0}{m}_{q^p}$ and $\eta_{i,m}(x_i) = q^{p\binom{m}{2}}(q-1)^m\binom{x_i}{m}_{q^p}$ for $i > 0$. Write η_α for the products of these functions in the $(n+1)$ variables and set

$$\mathrm{C}_q(\Gamma, \mathrm{A}) = \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \mathrm{A}^+\eta_\alpha\right)^\wedge_{(p,q-1)} [1/[p]_q]^\wedge_p.$$

We define $\mathrm{C}_q(\Gamma^s, \mathrm{A})$ similarly, over indices $\alpha \in \mathbf{N}^{s(n+1)}$.

There is also an alternative formula which will be of use. Pullback along (id, γ) then sends $u_i \in \mathrm{A}^{+,(1)}$ to $(q^p)^{x_i(\gamma)}$ for $0 \le i \le n$. Set

$$[m]_{q^p}!\,\gamma_{m,q^p}(x) = \prod_{j=0}^{m-1}((q^p)^x - q^{pj}),$$

where $[m]_{q^p}! = \prod_{j=1}^m [j]_{q^p}$. Then $\eta_{0,m}(x_0) = q^m[p]_q^{-m}\gamma_{m,q^p}(x_0)$ and $\eta_{i,m}(x_i) = [p]_q^{-m}\gamma_{m,q^p}(x_i)$ for $i > 0$.

The formula for η_α in Definition 3.8 shows that these functions are uniformly bounded. Consequently, an expansion whose coefficients tend to zero $(p, q-1)$-adically converges uniformly, and evaluation defines a continuous map $\mathrm{C}_q(\Gamma^s, \mathrm{A}) \to \mathrm{C}(\Gamma^s, \mathrm{A})$, where the target carries the compact-open topology. Here we give $\mathrm{C}_q(\Gamma^s, \mathrm{A})$ the natural topology induced by the completion in Definition 3.8, followed by Laurentization and p-adic completion. Lemma 3.9 shows that this map is injective, but its topology on the source is not the subspace topology.

For $s \ge 0$, let $(\mathrm{A}^+)^{(s)}$ be the degree-s term of the Čech nerve of the log q-de Rham prism in $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}$, and put $\mathrm{A}^{(s)} = ((\mathrm{A}^+)^{(s)}[1/[p]_q])^\wedge_p$.

We will show evaluation on the log q-de Rham prism induces a functor

$$\rho_{\mathrm{A}}^* : \mathrm{D}_\blacksquare((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p) \longrightarrow \mathrm{D}_\blacksquare(\mathrm{A})^{h\Gamma}.$$

The crystal property does give an abstract Γ action, but does not obviously ensure continuity and higher coherences which is why we make the following more delicate construction. Lemma A.15 writes the source as the limit

$$\mathrm{D}_\blacksquare((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p) \simeq \lim\Big(\ \mathrm{D}_\blacksquare(\mathrm{A}) \rightrightarrows \mathrm{D}_\blacksquare(\mathrm{A}^{(1)}) \substack{\rightarrow\\\rightarrow\\\rightarrow} \mathrm{D}_\blacksquare(\mathrm{A}^{(2)}) \substack{\rightarrow\\\rightarrow\\\rightarrow\\\rightarrow} \cdots\ \Big)$$

The target consists of solid A-modules with continuous semilinear Γ-action. Modulo p^n, the rings in the preceding diagram and in the continuous bar construction carry their natural analytic structures $\mathrm{Spa}(\mathrm{B}, \mathrm{B})$. The Γ-action on $\mathrm{A}^+ = \mathrm{R}_0^\square[\![q-1]\!]$, after p-completely inverting $[p]_q$ induces maps

$$\theta_s : \mathrm{A}^{(s)} \longrightarrow \mathrm{C}(\Gamma^s, \mathrm{A}).$$

A priori these land only in all functions, but we will show they land in continuous functions. This then induces ρ_{A}^* as

$$\mathrm{D}_\blacksquare((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p) \xrightarrow{\widetilde{\rho_{\mathrm{A}}^*}} \lim_{[s]\in\Delta} \mathrm{D}_\blacksquare(\mathrm{C}(\Gamma^s, \mathrm{A})) \longrightarrow \mathrm{D}_\blacksquare(\mathrm{A})^{h\Gamma}.$$

We point out that because A is induced from a topological ring that is an inverse limit of Banach rings, one can write $\mathrm{C}(\Gamma^s, \mathrm{A}) = \lim_n \mathrm{C}(\Gamma^s, \mathrm{A}/p^n)$. We also note that in the situation the quotient topology on A/p^n and the Tate topology given by $(\mathrm{A}^+/p^n)[1/(q-1)]$ agree, so there is no ambiguity (indeed this holds for any prism). Finally we also note that $\mathrm{C}(\Gamma^s, \mathrm{B})[1/d]^\wedge_p \simeq \mathrm{C}(\Gamma^s, \mathrm{B}[1/d]^\wedge_p)$ for any prism $(\mathrm{B}, (d))$ because Γ^s is profinite (combined with the previous fact about inverse limits). Thus we freely exchange these in what follows.

These maps θ_s are in fact injective as ordinary ring maps, and we show the previously defined $\mathrm{C}_q(\Gamma^s, \mathrm{A})$ are their images. In what follows we give $\mathrm{A}^{(s)}$ the analytic ring structure induced from the log prism $\mathrm{A}^{+,(s)}$ in the logarithmic Čech nerve, using $\lim_n((\mathrm{A}^{+(s)}/p^n)[1/d], (\mathrm{A}^{+(s)}/p^n))$ with their Huber pair analytic ring structures (as in the convention set in the discussion after Lemma 3.7). For $\mathrm{C}_q(\Gamma^s, \mathrm{A})$ we make a similar definition, using the completed direct sum presentation as we did for its topological ring structure.

Lemma 3.9. The degree-s map θ_s identifies $\mathrm{A}^{(s)}$ with the subring $\mathrm{C}_q(\Gamma^s, \mathrm{A}) \subset \mathrm{C}(\Gamma^s, \mathrm{A})$:

$$\mathrm{A}^{(s)} \xrightarrow{\sim} \mathrm{C}_q(\Gamma^s, \mathrm{A}).$$

This is further an isomorphism of analytic rings.

The map $\mathrm{C}_q(\Gamma^s, \mathrm{A}) \to \mathrm{C}(\Gamma^s, \mathrm{A})$ is continuous as topological rings where the target is equipped with the compact-open topology; the same assertion holds when regarded as solid modules.

Proof. As above, we use $\mathrm{A} = \mathrm{R}_0^\square(\!(q-1)\!)^\wedge_p$. We first treat $s = 1$. We actually will show a stronger claim, which is that there is a natural isomorphism of adic rings

$$\theta_1 : \mathrm{A}^{+,(1)} \simeq \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \mathrm{A}^+ \eta_\alpha \right)^\wedge_{(p,q-1)}.$$

After p-completely inverting $[p]_q$ or equivalently $q-1$, this isomorphism of ordinary adic rings implies the desired isomorphism of analytic rings (as in this setting passing from topological rings to analytic rings is fully faithful).

We do this by using the explicit log prismatic envelope presentation of $\mathrm{A}^{+,(1)}$. Recall $(\mathrm{A}^+)^{(1)}$ from Lemma 3.7 is the $(p, q-1)$-adic (equivalently $(p, [p]_q)$) completion of

$$\mathrm{A}^+[u_0^{\pm 1}, \ldots, u_n^{\pm 1}] \left\{ \frac{u_0 - 1}{[p]_{q_0}}, \ldots, \frac{u_n - 1}{[p]_{q_0}} \right\}_\delta.$$

We used $u_0 = q_1/q_0$ and $u_i = \mathrm{T}_{i,1}/\mathrm{T}_{i,0}$ as shorthand. The map θ_1 can be made explicit integrally. For $g \in \Gamma$, using the coordinates $g = (x_0, \ldots, x_n)$ of Definition 3.8 write the evaluation of $\theta_1(a) \in \mathrm{Map}(\Gamma, \mathrm{A}^+)$[3] for $a \in \mathrm{A}^+$ at g as the pullback map determined by $u_i \mapsto (q^p)^{x_i(g)}$ for $0 \leq i \leq n$.

Using [GLSQ23, Theorem 4.4], the cyclotomic factor $\mathrm{A}^+[u_0^{\pm 1}] \left\{ \frac{u_0 - 1}{[p]_{q_0}} \right\}^\wedge_\delta$ is, up to a change of variables, the $(p, q-1)$-completion of the free A^+-module on their Δ-divided powers $\omega_{0,\Delta}^{\{n\}}$ (see [GLSQ23, Definition 1.7]) as a $(p, q-1)$-adic ring. Indeed from the discussion preceding

[3] We haven't yet checked the output is a continuous function.

Theorem 4.4. in *loc. cit.* they identify $\widehat{\bigoplus}_{n\geq 0}\mathrm{A}^+\omega_{0,\Delta}^{\{n\}}$ with the prismatic envelope of the δ-pair $(\mathrm{A}^+[\![\xi]\!],(\xi))$, which checking universal properties is indeed our ring (one has $\xi\mapsto q_0(u_0-1)$).

Now we see where θ_1 sends this basis. Remark [GLSQ23, Remark 1.8(5)] provides a "blowing up" map Bl, yielding

$$\mathrm{A}^+\langle\xi\rangle_{q_0^p}\xrightarrow{\mathrm{Bl}}\widehat{\bigoplus}_{n\geq 0}\mathrm{A}^+\omega_{0,\Delta}^{\{n\}}\simeq\mathrm{A}^+[u_0^{\pm 1}]\left\{\frac{u_0-1}{[p]_{q_0}}\right\}_\delta^\wedge\xrightarrow{\theta_1}\mathrm{Map}(\Gamma,\mathrm{A}^+)$$

such that the q-divided power $\xi^{[m]_{q_0^p}}$ is sent to $[p]_{q_0}^m\omega_{0,\Delta}^{\{m\}}$ under Bl. This is helpful, because we know where $\xi^{[m]_{q_0^p}}$ is sent to under the composite map $\theta_1\circ\mathrm{Bl}$: ξ is sent by θ_1 to

$$q((q^p)^{x_0}-1)\in\mathrm{Map}(\Gamma,\mathrm{A}^+).$$

Thus we may deduce

$$[p]_{q_0}^m\theta_1(\omega_{0,\Delta}^{\{m\}})=q^m\gamma_{m,q^p}(x_0),$$

so in particular using the alternative formula after Definition 3.8 we see $\theta_1(\omega_{0,\Delta}^{\{m\}})=\eta_{0,m}$ using $[p]_{q_0}$-torsionfreeness.

Next, we handle the remaining factors corresponding to u_i for $i>0$. The argument is more or less the same. Using [GLSQ22, Definition 1.6 and Corollary 5.6], the corresponding factor $\mathrm{A}^+[u_i^{\pm 1}]\left\{\frac{u_i-1}{[p]_{q_0}}\right\}_\delta^\wedge$ is the completion of the free A^+-module on its negative-level divided powers $\omega_{i,\Delta}^{\{m\}}$. The analogous map in [GLSQ22, Proposition 1.7] let us deduce the formula

$$[p]_q^m\theta_1(\omega_{i,\Delta}^{\{m\}})=\gamma_{m,q^p}(x_i).$$

Similarly we find $\theta_1(\omega_{i,\Delta}^{\{m\}})=\eta_{i,m}$. Thus θ_1 induces a surjection onto

$$\left(\bigoplus_{\alpha\in\mathbf{N}^{n+1}}\mathrm{A}^+\eta_\alpha\right)^\wedge_{(p,q-1)}.$$

It is injective: evaluation at nonnegative integral values of each x_i is triangular, with $\eta_{i,m}(j)=0$ for $m>j$ and $\eta_{i,m}(m)$ a unit times $(q-1)^m$. Only finitely many terms contribute at each such value, and A^+ is $(q-1)$-torsion-free. Hence the integral map is, as desired, an isomorphism onto $(\bigoplus_\alpha\mathrm{A}^+\eta_\alpha)^\wedge_{(p,q-1)}$; the isomorphism is of $(p,q-1)$-adic rings, so the claim about analytic ring structures also follows. One sees θ_1 lands in $\mathrm{C}(\Gamma,\mathrm{A})$ by the discussion after Definition 3.8.

For general s, the degree-s Čech term is an iterated tensor product $\mathrm{A}^{+,(s)}=(\mathrm{A}^{+(1)})^{\widehat{\otimes}^s_{\mathrm{A}^+}}$ (one alternates using the two maps $p_0,p_1:\mathrm{A}^+\to\mathrm{A}^{+,(1)}$ in the tensor product). The identification at $s=1$ is compatible with both projections. Using this one may check

$$(\mathrm{A}^+)^{(s)}\simeq\left(\bigoplus_{\alpha\in\mathbf{N}^{s(n+1)}}\mathrm{A}^+\eta_\alpha\right)^\wedge_{(p,q-1)},$$

so by the same argument the general claim follows. $\square$

The functor $\widetilde{\rho}_{\mathrm{A}}^{*}$ takes values in $\lim_{[s]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{C}(\Gamma^{s},\mathrm{A}))$, whereas ρ_{A}^{*} takes values in $\mathrm{D}_{\blacksquare}(\mathrm{A})^{h\Gamma}$. The next lemma identifies these two targets on objects whose underlying A-module is p-completely nuclear.

We first fix notation. Let G be a profinite group and let B be a derived p-complete analytic ring with continuous G-action. Write $\mathrm{D}_{\blacksquare}(\mathrm{B})^{h\mathrm{G}} := \mathsf{Rep}_{\mathrm{B}^{t}_{\blacksquare}}(\mathrm{G})$ for the category of modules over the twisted solid group algebra $\mathrm{B}^{t}_{\blacksquare}[\mathrm{G}]$; equivalently, this is the category of solid B-modules with continuous semilinear G-action. Here $\mathbf{Z}_{\blacksquare}[\mathrm{G}]$ denotes the solid group algebra. We set

$$\mathrm{R}\Gamma_{\mathrm{solid}}(\mathrm{G},\mathrm{M}) := \mathrm{RHom}_{\mathbf{Z}_{\blacksquare}[\mathrm{G}]}(\mathbf{Z}_{\blacksquare},\mathrm{M}) \simeq \mathrm{RHom}_{\mathrm{B}^{t}_{\blacksquare}[\mathrm{G}]}(\mathrm{B},\mathrm{M}).$$

For $\mathrm{M} \in \mathrm{D}_{\blacksquare}(\mathrm{B})$, we write

$$\mathrm{C}(\mathrm{G},\mathrm{M}) := \underline{\mathrm{RHom}}_{\mathrm{B}}(\mathrm{B}[\underline{\mathrm{G}}],\mathrm{M}).$$

Recall that $\mathrm{D}_{\blacksquare}(\mathrm{B})^{\mathrm{nuc}}$ consists of the objects M for which, for every compact $\mathrm{P} \in \mathrm{D}_{\blacksquare}(\mathrm{B})^{\omega}$, the canonical map

$$\mathrm{P}^{\vee} \otimes^{\mathrm{L},\blacksquare}_{\mathrm{B}} \mathrm{M} \longrightarrow \underline{\mathrm{RHom}}_{\mathrm{B}}(\mathrm{P},\mathrm{M})$$

is an equivalence. For a discrete analytic ring this condition is automatic; for example, $\mathrm{D}_{\blacksquare}(\mathbf{Z}_{\mathrm{disc}})^{\mathrm{nuc}} = \mathrm{D}(\mathbf{Z})$.

Lemma 3.10. Let B be a derived p-complete solid analytic ring with a continuous action of a compact p-adic Lie group G. Give $\mathrm{C}(\mathrm{G}^{s},\mathrm{B})$ its natural analytic ring structure. The natural functor

$$\lim_{[s]\in\Delta} \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{C}(\mathrm{G}^{s},\mathrm{B})) \longrightarrow \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{B})^{h\mathrm{G}}$$

restricts to an equivalence from objects whose evaluation at [0] is p-completely nuclear[a] to representations whose underlying B-module is p-completely nuclear. Here $\widehat{\mathrm{D}}_{\blacksquare}$ denotes the derived category of solid derived p-complete modules. If $\mathrm{M} \in \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{B})$ is the underlying solid B-module under this equivalence,

$$\varprojlim_{[s]\in\Delta} \mathrm{C}(\mathrm{G}^{s},\mathrm{B}) \otimes^{\mathrm{L},\blacksquare}_{\mathrm{B}} \mathrm{M} \simeq \varprojlim_{[s]\in\Delta} \mathrm{C}(\mathrm{G}^{s},\mathrm{M}) \simeq \mathrm{R}\Gamma_{\mathrm{solid}}(\mathrm{G},\mathrm{M}).$$

[a]By this we mean it satisfies the nuclearity property for derived p-complete objects; equivalently derived p-complete and nuclear mod p.

Proof. The Barr-Beck-Lurie theorem, as in [Mik26, Remark 1.49, Corollary 1.50] (which is a similar argument), identifies the source and target with coalgebras in $\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{B})$ for the following comonads. The first is induced by the endofunctor

$$\mathbb{T} : \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{B}) \xrightarrow{\mathrm{C}(\mathrm{G},\mathrm{B})\otimes^{\mathrm{L},\blacksquare}_{\mathrm{B}} -} \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{C}(\mathrm{G},\mathrm{B})) \longrightarrow \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{B}).$$

Here the first morphism uses the map $\mathrm{B} \to \mathrm{C}(\mathrm{G},\mathrm{B})$ sending $b \in \mathrm{B}$ to the constant function (valued at b). The second morphism is pushforward along the action map $\mathrm{act} : \mathrm{B} \to \mathrm{C}(\mathrm{G},\mathrm{B})$ (i.e. b is sent to the function sending g to $g(b)$). The comonad structure is the obvious one (the comultiplication is induced by the multiplication on G)

The second is induced by the endofunctor $\mathbb{S} : \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{B}) \to \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{B})$ given by

$$\mathbb{S}(\mathrm{M}) = \mathrm{C}(\mathrm{G},\mathrm{M}),$$

where $(bf)(g) = g(b)f(g)$ on $\mathbb{S}(\mathrm{M})$. Thus

$$\varprojlim_{[s]\in\Delta} \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{C}(\mathrm{G}^s, \mathrm{B})) \simeq \mathrm{CoAlg}_{\mathbb{T}}(\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{B})).$$

Similarly, $\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{B})^{h\mathrm{G}} \simeq \mathrm{CoAlg}_{\mathbb{S}}(\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{B}))$.

The map $f \otimes m \mapsto (g \mapsto f(g)m)$ defines a natural transformation $\mathbb{T} \to \mathbb{S}$ compatible with the counits and comultiplications, hence a map of comonads.

Suppose now that M is p-completely nuclear. For the compact object $\mathrm{P} = \mathrm{B}[\underline{\mathrm{G}}]$, nuclearity and the identifications $\mathrm{P}^\vee \simeq \mathrm{C}(\mathrm{G}, \mathrm{B})$ and $\underline{\mathrm{RHom}}_{\mathrm{B}}(\mathrm{P}, \mathrm{M}) \simeq \mathrm{C}(\mathrm{G}, \mathrm{M})$ give

$$\mathbb{T}(\mathrm{M}) = \mathrm{C}(\mathrm{G}, \mathrm{B}) \otimes_{\mathrm{B}}^{\mathrm{L},\blacksquare} \mathrm{M} \simeq \mathrm{P}^\vee \otimes_{\mathrm{B}}^{\mathrm{L},\blacksquare} \mathrm{M} \simeq \underline{\mathrm{RHom}}_{\mathrm{B}}(\mathrm{P}, \mathrm{M}) \simeq \mathrm{C}(\mathrm{G}, \mathrm{M}) = \mathbb{S}(\mathrm{M}).$$

This calculation shows both comonads restrict to endofunctors of the full subcategory of p-completely nuclear modules $\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{B})^{\mathrm{nuc}}$ (since $\mathrm{C}(\mathrm{G}, \mathrm{B})$ is nuclear and the subcategory of nuclear objects is closed under tensor products); moreover it also shows $\mathbb{S}$ and $\mathbb{T}$ become isomorphic comonads. It follows that on p-completely nuclear objects we can describe the source and target as coalgebras over isomorphic comonads, thus the claim follows.

The cohomological comparison follows by taking RHom with the unit. □

To compute solid group cohomology, we use the resolution of [Bos23, Proposition B.3 and Lemma B.4]. If $\mathrm{G} \simeq \mathbf{Z}_p^r$, let Λ_{G} be the solid algebra associated with $\mathbf{Z}_p[\![\mathrm{G}]\!]$. For topological generators $\gamma_1, \ldots, \gamma_r$, there is a finite projective resolution in solid Λ_{G}-modules

$$0 \longrightarrow \Lambda_{\mathrm{G}}^{\binom{r}{r}} \longrightarrow \cdots \longrightarrow \Lambda_{\mathrm{G}}^{\binom{r}{1}} \xrightarrow{(\gamma_1 - 1, \ldots, \gamma_r - 1)} \Lambda_{\mathrm{G}} \longrightarrow (\mathbf{Z}_p)_{\blacksquare} \longrightarrow 0. \tag{1}$$

The omitted arrows are the Koszul differentials. Thus solid group cohomology of a complex of solid $\mathbf{Z}_p$-modules with continuous G-action is computed by the Koszul complex of the operators $\gamma_i - 1$. The topology of Definition 3.8 ensures that the translations and homotopies below are morphisms of solid modules.

Lemma 3.11. Let $\mathrm{M} \in \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{A})$ be p-completely nuclear, and give its underlying solid $\mathbf{Z}_p$-module the trivial Γ-action. Right translation defines a continuous Γ-action on $\mathrm{C}_q(\Gamma, \mathrm{A}) \otimes_{\mathrm{A},p_0}^{\mathrm{L},\blacksquare} \mathrm{M}$ by $\gamma(f \otimes m)(g) = f(g\gamma) \otimes m$. Then $m \mapsto 1 \otimes m$ induces an equivalence

$$\mathrm{M} \xrightarrow{\sim} \mathrm{R}\Gamma_{\mathrm{solid}}\big(\Gamma, \mathrm{C}_q(\Gamma, \mathrm{A}) \otimes_{\mathrm{A},p_0}^{\mathrm{L},\blacksquare} \mathrm{M}\big).$$

Proof. Recall from Definition 3.8 that we write $g = (x_0, \ldots, x_n) \in \Gamma$ in the coordinates $x_0(g) = (a-1)/p$ and $x_i(g) = b_i$ for $g = (a, b_1, \ldots, b_n)$. By Lemma 3.9 and the compatibility of $\theta_\bullet$ with the Čech nerve maps $p_0, p_1 : \mathrm{A} \to \mathrm{A}^{(1)}$, right translation preserves $\mathrm{C}_q(\Gamma, \mathrm{A})$ and is continuous, while $f(g) \mapsto g^{-1}(f(g))$ restricts to a continuous (p_1 to p_0)-linear map of $\mathrm{C}_q(\Gamma, \mathrm{A})$. Under θ_1, interchanging the two factors of the Čech self-product sends $f(g)$ to $g(f(g^{-1}))$. Applying $f(g) \mapsto g^{-1}(f(g))$ after this factor interchange gives the p_0-linear involution $f(g) \mapsto f(g^{-1})$. Since it fixes constant functions, it induces an isomorphism of solid Γ-modules from the action in the statement to the left translation action $\gamma(f \otimes m)(g) = f(\gamma^{-1}g) \otimes m$. We compute using the latter action.

For $1 \leq i \leq n$, choose the generator γ_i of the ith factor of $\mathbf{Z}_p(1)^n$ whose ith coordinate is -1. Left translation by γ_i^{-1} sends x_i to $x_i + 1$, and Definition 3.8 gives

$$(\gamma_i - 1)\eta_{i,m+1} = (q-1)(q^p)^{x_i}\eta_{i,m}.$$

The factor $(q-1)(q^p)^{x_i}$ is a unit: $(q^p)^{x_i}$ corresponds under θ_1 to u_i, while $q-1$ is a unit since $[p]_q$ is invertible and $[p]_q \equiv (q-1)^{p-1} \pmod{p}$. Let h_i be the composite

$$\mathrm{C}_q(\Gamma, \mathrm{A}) \xrightarrow{\cdot((q-1)(q^p)^{x_i})^{-1}} \mathrm{C}_q(\Gamma, \mathrm{A}) \xrightarrow{\eta_{i,m} \mapsto \eta_{i,m+1}} \mathrm{C}_q(\Gamma, \mathrm{A}).$$

Writing the basis of Definition 3.8 as $\eta_\alpha = \prod_{j=0}^n \eta_{j,\alpha_j}$, the second arrow sends η_α to $\eta_{i,\alpha_i+1}\prod_{j\neq i}\eta_{j,\alpha_j}$. Both arrows are continuous and p_0-linear, and the recurrence gives $(\gamma_i - 1)h_i = \mathrm{id}$ and $h_i(\gamma_i-1)(f) = f - f|_{x_i=0}$. Let $\mathrm{C}_q(1+p\mathbf{Z}_p, \mathrm{A}) \subset \mathrm{C}_q(\Gamma, \mathrm{A})$ be the closed subring of functions independent of $x_1, \ldots, x_n$; by Definition 3.8,

$$\mathrm{C}_q(1+p\mathbf{Z}_p, \mathrm{A}) = \left(\bigoplus_{m\geq 0} \mathrm{A}^+\eta_{0,m}\right)^{\wedge}_{(p,q-1)} [1/[p]_q]^{\wedge}_p.$$

By (1), solid $\mathbf{Z}_p(1)^n$-cohomology is the iterated fiber of $\gamma_1 - 1, \ldots, \gamma_n - 1$. On $\mathrm{C}_q(\Gamma, \mathrm{A}) \otimes^{\mathrm{L},\blacksquare}_{\mathrm{A},p_0} \mathrm{M}$, the identities for $h_i \otimes^{\mathrm{L},\blacksquare} \mathrm{id}_{\mathrm{M}}$ give a retraction of the two term complex for the fiber of $\gamma_i - 1$ onto the submodule obtained by setting $x_i = 0$. Applying these retractions successively for $i = 1, \ldots, n$ gives

$$\mathrm{C}_q(1+p\mathbf{Z}_p, \mathrm{A}) \otimes^{\mathrm{L},\blacksquare}_{\mathrm{A},p_0} \mathrm{M} \xrightarrow{\sim} \mathrm{R}\Gamma_{\mathrm{solid}}\big(\mathbf{Z}_p(1)^n, \mathrm{C}_q(\Gamma, \mathrm{A}) \otimes^{\mathrm{L},\blacksquare}_{\mathrm{A},p_0} \mathrm{M}\big).$$

This equivalence is induced by the $(1+p\mathbf{Z}_p)$-equivariant inclusion of the functions independent of $x_1, \ldots, x_n$.

Under the identification in the proof of Lemma 3.9, [GLSQ23, Proposition 5.14] gives a continuously split p_0-linear exact sequence

$$0 \longrightarrow \mathrm{A}^+ \xrightarrow{p_0} \left(\bigoplus_{m\geq 0} \mathrm{A}^+\eta_{0,m}\right)^{\wedge}_{(p,q-1)} \xrightarrow{\mathrm{L}_\Delta} \left(\bigoplus_{m\geq 0} \mathrm{A}^+\eta_{0,m}\right)^{\wedge}_{(p,q-1)} \longrightarrow 0.$$

This is a split sequence of solid A^+-modules and remains so after inverting $[p]_q$ and p-adically completing. Let R_{p+1} denote right translation by $p+1$ on $\mathrm{C}_q(1+p\mathbf{Z}_p, \mathrm{A})$, so $(\mathrm{R}_{p+1}f)(g) = f(g(p+1))$ for $g \in 1+p\mathbf{Z}_p$. By [GLSQ23, Remarks 5.11(2)(a)],

$$\mathrm{R}_{p+1} - 1 = p_1(q^2 - q)\mathrm{L}_\Delta.$$

The element $p_1(q^2-q)$ is a unit because $p_1([p]_q)$ is a unit and $p_1([p]_q) \equiv p_1(q-1)^{p-1} \pmod{p}$. The split sequence therefore gives a deformation retraction

$$\Big[\mathrm{C}_q(1+p\mathbf{Z}_p, \mathrm{A}) \xrightarrow{\mathrm{R}_{p+1}-1} \mathrm{C}_q(1+p\mathbf{Z}_p, \mathrm{A})\Big] \simeq \mathrm{A}.$$

The map $f(g) \mapsto f(g^{-1})$ conjugates R_{p+1} to $f(g) \mapsto f((p+1)^{-1}g)$, the action of the topological generator $p+1$ of $1+p\mathbf{Z}_p$. Applying (1) and tensoring this with M gives

$$\mathrm{M} \xrightarrow{\sim} \mathrm{R}\Gamma_{\mathrm{solid}}\big(1+p\mathbf{Z}_p, \mathrm{C}_q(1+p\mathbf{Z}_p, \mathrm{A}) \otimes^{\mathrm{L},\blacksquare}_{\mathrm{A},p_0} \mathrm{M}\big).$$

Since $\mathbf{Z}_p(1)^n$ is normal in $\Gamma = (1 + p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$, solid homotopy fixed points may be taken successively:

$$\mathrm{R}\Gamma_{\mathrm{solid}}(\Gamma, -) \simeq \mathrm{R}\Gamma_{\mathrm{solid}}\big(1 + p\mathbf{Z}_p, \mathrm{R}\Gamma_{\mathrm{solid}}(\mathbf{Z}_p(1)^n, -)\big).$$

Combining the $\mathbf{Z}_p(1)^n$ and $(1 + p\mathbf{Z}_p)$-cohomology computations with this equivalence proves the claim. □

We next explain some facts about the maps in (Laurentized) Čech nerve of $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$. Let $p_0, p_1 : \mathrm{A} \to \mathrm{A}^{(1)}$ be the two projections to its first term. Under the identification $\mathrm{A}^{(1)} \simeq \mathrm{C}_q(\Gamma, \mathrm{A})$ of Lemma 3.9, the action on the second vertex is given by precomposition with right multiplication:

$$(\gamma f)(g) = f(g\gamma).$$

Indeed, $p_0(a)(g) = a$ and $p_1(a)(g) = g(a)$, so p_0 is invariant and $\gamma p_1(a) = p_1(\gamma a)$. For a semilinear Γ-action on M, the induced actions are

$$\gamma \cdot (f \otimes m) = \begin{cases} (\gamma f) \otimes \gamma(m) & \text{on } p_1^*\mathrm{M}, \\ (\gamma f) \otimes m & \text{on } p_0^*\mathrm{M}. \end{cases}$$

In what follows we consider the colimit-preserving functor between presentable categories

$$\widetilde{\rho}_{\mathrm{A}}^* : \widehat{\mathrm{D}}_{\blacksquare}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}) \to \varprojlim_{[s]\in\Delta} \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{C}(\Gamma^s, \mathrm{A})),$$

so it has a right adjoint $\widetilde{\rho}_{\mathrm{A},*}$ by the adjoint functor theorem.

Lemma 3.12. Let M be the evaluation of $\mathcal{E} \in \widehat{\mathrm{D}}_{\blacksquare}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$ on the log q-de Rham prism attached to $\mathrm{A}^+ = \mathrm{R}_0^{\square}[\![q-1]\!]$. If M is p-completely nuclear, then there is a natural equivalence

$$\big(\widetilde{\rho}_{\mathrm{A},*}\widetilde{\rho}_{\mathrm{A}}^*\mathcal{E}\big)(\mathrm{A}) \simeq \mathrm{R}\Gamma_{\mathrm{solid}}\big(\Gamma, \mathrm{C}_q(\Gamma, \mathrm{A}) \otimes_{\mathrm{A},p_0}^{\mathrm{L},\blacksquare} \mathrm{M}\big).$$

Here $\gamma \cdot (f \otimes m) = (\gamma f) \otimes m$, where $(\gamma f)(g) = f(g\gamma)$. Under this equivalence, the evaluation of the adjunction unit is the map of Lemma 3.11.

Proof. The target of $\widetilde{\rho}_{\mathrm{A}}^*$ is $\mathrm{CoAlg}_{\mathbb{T}}(\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{A}))$, where $\mathbb{T}$ is the comonad of Lemma 3.10. Lemma A.15 identifies the source with coalgebras in $\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{A})$ over the comonad

$$\mathrm{M} \mapsto (p_0)_*(p_1)^*\mathrm{M}.$$

Denote this comonad by $\mathbb{L}$ (for Laurent crystals). The comparison defining $\widetilde{\rho}_{\mathrm{A}}^*$ is induced by a morphism $\mathbb{L} \to \mathbb{T}$. On p-completely nuclear objects, Lemma 3.10 identifies $\mathbb{T}$ with $\mathbb{S}$.

For one of these comonads, let $\mathrm{For}_{-} : \mathrm{CoAlg}_{-}(\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{A})) \to \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{A})$ be the forgetful functor (i.e. evaluation) and let Cofr_{-} be its right adjoint. Corestriction along $\mathbb{L} \to \mathbb{T}$ is the identity on underlying objects. Hence, for $\mathcal{F} \in \mathrm{CoAlg}_{\mathbb{L}}(\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{A}))$ and $\mathrm{P} \in \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{A})$,

$$\begin{aligned} \mathrm{Map}_{\mathrm{CoAlg}_{\mathbb{L}}}\big(\mathcal{F}, \widetilde{\rho}_{\mathrm{A},*}\,\mathrm{Cofr}_{\mathbb{T}}(\mathrm{P})\big) &\simeq \mathrm{Map}_{\mathrm{CoAlg}_{\mathbb{T}}}\big(\widetilde{\rho}_{\mathrm{A}}^*\mathcal{F}, \mathrm{Cofr}_{\mathbb{T}}(\mathrm{P})\big) \\ &\simeq \mathrm{Map}_{\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{A})}\big(\mathrm{For}_{\mathbb{L}}\,\mathcal{F}, \mathrm{P}\big) \\ &\simeq \mathrm{Map}_{\mathrm{CoAlg}_{\mathbb{L}}}\big(\mathcal{F}, \mathrm{Cofr}_{\mathbb{L}}(\mathrm{P})\big). \end{aligned}$$

Thus $\widetilde{\rho}_{\mathrm{A},*}\operatorname{Cofr}_{\mathbb{T}}(\mathrm{P}) \simeq \operatorname{Cofr}_{\mathbb{L}}(\mathrm{P})$. The canonical cofree resolution of the $\mathbb{T}$-coalgebra $\widetilde{\rho}^*_{\mathrm{A}}\mathcal{E}$ is

$$\widetilde{\rho}^*_{\mathrm{A}}\mathcal{E} \simeq \operatorname{Tot}\Big(\operatorname{Cofr}_{\mathbb{T}}(\mathrm{M}) \Longrightarrow \operatorname{Cofr}_{\mathbb{T}}(\mathbb{T}\mathrm{M}) \Rrightarrow \operatorname{Cofr}_{\mathbb{T}}(\mathbb{T}^2\mathrm{M}) \cdots \Big).$$

Indeed, the augmentation is induced by the $\mathbb{T}$-coaction on M, and the counit of $\mathbb{T}$ splits it after applying $\operatorname{For}_{\mathbb{T}}$. Since $\widetilde{\rho}_{\mathrm{A},*}$ preserves limits and $\widetilde{\rho}_{\mathrm{A},*}\operatorname{Cofr}_{\mathbb{T}}(\mathrm{P}) \simeq \operatorname{Cofr}_{\mathbb{L}}(\mathrm{P})$, we obtain

$$\widetilde{\rho}_{\mathrm{A},*}\widetilde{\rho}^*_{\mathrm{A}}\mathcal{E} \simeq \operatorname{Tot}\Big(\operatorname{Cofr}_{\mathbb{L}}(\mathrm{M}) \Longrightarrow \operatorname{Cofr}_{\mathbb{L}}(\mathbb{T}\mathrm{M}) \Rrightarrow \operatorname{Cofr}_{\mathbb{L}}(\mathbb{T}^2\mathrm{M}) \cdots \Big). \tag{2}$$

The presentation in Lemma 3.7 and [AM24, Lemma 2.18(i)] show that $\mathrm{A}^{(1)}$ is p-completely nuclear over A through either p_0 or p_1. Using p-complete nuclearity and Lemma 3.10, we obtain identifications

$$\mathbb{L}\mathbb{T}^s(\mathrm{M}) \xrightarrow{\sim} \mathrm{C}\big(\Gamma^s, \mathbb{L}(\mathrm{M})\big).$$

natural in Γ and M. This then upgrades to

$$\operatorname{Cofr}_{\mathbb{L}}(\mathbb{T}^s\mathrm{M}) \xrightarrow{\sim} \mathrm{C}\big(\Gamma^s, \operatorname{Cofr}_{\mathbb{L}}(\mathrm{M})\big),$$

as $\mathbb{L}$-coalgebras, so under p-complete nuclearity

$$\widetilde{\rho}_{\mathrm{A},*}\widetilde{\rho}^*_{\mathrm{A}}\mathcal{E} \simeq \operatorname{Tot}\Big(\operatorname{Cofr}_{\mathbb{L}}(\mathrm{M}) \Longrightarrow \mathrm{C}\big(\Gamma^1, \operatorname{Cofr}_{\mathbb{L}}(\mathrm{M})\big) \Rrightarrow \mathrm{C}\big(\Gamma^2, \operatorname{Cofr}_{\mathbb{L}}(\mathrm{M})\big) \cdots \Big).$$

Now termwise applying $\operatorname{For}_{\mathbb{L}}$ would give $\mathrm{C}\big(\Gamma^s, \mathbb{L}(\mathrm{M})\big)$, but this operation does not necessarily commute with totalizations. In this case there is no issue because solid Γ-cohomology is computed by a finite resolution.

The resolution (1), applied to $\mathbf{Z}_p(1)^n$ and to $1+p\mathbf{Z}_p$ with generator $(p+1)^{-1}$, gives finite resolutions computing continuous invariants in the coalgebra category. The solid Hochschild-Serre equivalence used in Lemma 3.11 gives

$$\mathrm{R}\Gamma_{\text{solid}}\big(\Gamma, \operatorname{Cofr}_{\mathbb{L}}(\mathrm{M})\big) \simeq \mathrm{R}\Gamma_{\text{solid}}\big(1+p\mathbf{Z}_p, \mathrm{R}\Gamma_{\text{solid}}(\mathbf{Z}_p(1)^n, \operatorname{Cofr}_{\mathbb{L}}(\mathrm{M}))\big).$$

Since the differentials in these resolutions respect the $\mathbb{L}$-coactions, this is a finite iterated fiber of $\mathbb{L}$-coalgebras. The comonad $\mathbb{L} = (p_0)_*(p_1)^*$ is exact, so $\operatorname{For}_{\mathbb{L}}$ preserves this finite limit. Since $\operatorname{For}_{\mathbb{L}}\operatorname{Cofr}_{\mathbb{L}}(\mathrm{M}) = \mathbb{L}(\mathrm{M})$ we obtain

$$\big(\widetilde{\rho}_{\mathrm{A},*}\widetilde{\rho}^*_{\mathrm{A}}\mathcal{E}\big)(\mathrm{A}) \simeq \mathrm{R}\Gamma_{\text{solid}}\big(\Gamma, \mathbb{L}(\mathrm{M})\big).$$

The crystal equivalence

$$\alpha_{\mathcal{E}} : p_1^*\mathrm{M} \xrightarrow{\sim} p_0^*\mathrm{M}$$

identifies $\mathbb{L}(\mathrm{M})$ with $\mathrm{A}^{(1)} \otimes^{\mathrm{L},\blacksquare}_{\mathrm{A},p_0} \mathrm{M}$, with the right translation action, and carries the degree-zero component of the adjunction unit to $m \mapsto 1 \otimes m$. Lemma 3.9 therefore identifies the resulting equivalence and adjunction unit with

$$\mathrm{M} \longrightarrow \mathrm{R}\Gamma_{\text{solid}}(\Gamma, \mathrm{A}^{(1)} \otimes^{\mathrm{L},\blacksquare}_{\mathrm{A},p_0} \mathrm{M}) \simeq \mathrm{R}\Gamma_{\text{solid}}\big(\Gamma, \mathrm{C}_q(\Gamma, \mathrm{A}) \otimes^{\mathrm{L},\blacksquare}_{\mathrm{A},p_0} \mathrm{M}\big),$$

where the first arrow is induced by $m \mapsto 1 \otimes m$. This proves the claim. □

Proposition 3.13. Evaluation on the log q-de Rham prism attached to $\mathrm{A}^+ = \mathrm{R}^{\square}_0[\![q-1]\!]$ induces a t-exact conservative functor

$$\rho^*_{\mathrm{A}} : \mathrm{D}_{\blacksquare}((\operatorname{Spf}\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p) \longrightarrow \mathrm{D}_{\blacksquare}(\mathrm{A})^{h\Gamma}.$$

It is fully faithful on objects whose evaluation on this prism is p-completely nuclear.

Proof. Lemma A.15 identifies the source as

$$\mathrm{D}_{\blacksquare}((\operatorname{Spf}\mathrm{R},\mathcal{M})_{\mathbb{\Delta}},\mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]_p^{\wedge}) \simeq \lim_{[s]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{A}^{(s)})$$

over the (Laurentized) Čech nerve terms from Lemma 3.7. Proposition 4.13 of [Kos22] makes the maps before Laurent localization $(p,[p]_q)$-completely flat. After inverting $[p]_q$ and p-adically completing, each map $\mathrm{A}\to\mathrm{A}^{(m)}$ is p-completely flat. Thus $\mathrm{A}^{(m)}/p$ is flat over A/p and $\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{A}^{(m)},\mathrm{A}/p)=0$. Since A is Noetherian, [Bha20, Lemma 5.15] makes $\mathrm{A}^{(m)}$ flat over A. Hence termwise truncation defines the t-structure, and this t-structure is detected by the conservative projection

$$\varprojlim_{[m]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{A}^{(m)}) \longrightarrow \mathrm{D}_{\blacksquare}(\mathrm{A}^{(0)}) = \mathrm{D}_{\blacksquare}(\mathrm{A}).$$

After forgetting the Γ-action, ρ_{A}^* is this projection, while the target t-structure is detected by the forgetful functor. Hence ρ_{A}^* is t-exact and conservative.

Consider the adjunction

$$\widetilde{\rho}_{\mathrm{A}}^* : \widehat{\mathrm{D}}_{\blacksquare}((\operatorname{Spf}\mathrm{R},\mathcal{M})_{\mathbb{\Delta}},\mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]_p^{\wedge}) \rightleftarrows \varprojlim_{[s]\in\Delta} \widehat{\mathrm{D}}_{\blacksquare}(\mathrm{C}(\Gamma^s,\mathrm{A})) : \widetilde{\rho}_{\mathrm{A},*}.$$

A left adjoint is fully faithful on any full subcategory on which its adjunction unit is an equivalence.

Let $\mathcal{E}$ have p-completely nuclear evaluation M on the log q-de Rham prism $\mathrm{R}_0^{\square}[\![q-1]\!]$. Lemmas 3.12 and 3.11 show that evaluation on this prism of the adjunction unit is an equivalence. Degree-zero evaluation is conservative by Lemma A.15, so the adjunction unit is an equivalence. Thus $\widetilde{\rho}_{\mathrm{A}}^*$ is fully faithful on this subcategory. If $\mathcal{E}$ and $\mathcal{F}$ have p-completely nuclear evaluations on this prism, Lemma 3.10 gives

$$\mathrm{RHom}\big(\widetilde{\rho}_{\mathrm{A}}^*\mathcal{E},\widetilde{\rho}_{\mathrm{A}}^*\mathcal{F}\big) \simeq \mathrm{RHom}\big(\rho_{\mathrm{A}}^*\mathcal{E},\rho_{\mathrm{A}}^*\mathcal{F}\big).$$

Hence ρ_{A}^* is fully faithful there. □

3.2. **Roots in Laurent crystals.** We now turn our attention to proving Theorem 3.5. As we now impose a perfectness condition, condensed mathematics is not relevant in this subsection. In Appendix A we show in Corollary A.17 that perfect Laurent crystals have a canonical t-structure in the normal crossings setting, allowing us to define $\mathsf{Coh}((\operatorname{Spf}\mathrm{R},\mathcal{M})_{\mathbb{\Delta}},\mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]_p^{\wedge})$.

For a framed log affine, put $\mathrm{A}=\mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$ as before. We use A to mean this throughout this subsection, and use $\overline{\mathrm{A}} := \mathrm{A}/p$. One can understand the category of perfect A-complexes with continuous Γ action as the totalization

$$\lim\left(\mathsf{Perf}(\mathrm{A}) \overset{p_0}{\underset{p_1}{\rightrightarrows}} \mathsf{Perf}(\mathrm{C}(\Gamma,\mathrm{A})) \;\substack{\longrightarrow\\[-2pt]\longrightarrow\\[-2pt]\longrightarrow}\; \mathsf{Perf}(\mathrm{C}(\Gamma^2,\mathrm{A})) \;\substack{\longrightarrow\\[-2pt]\longrightarrow\\[-2pt]\longrightarrow\\[-2pt]\longrightarrow}\; \cdots\right) \tag{3}$$

where A has the topology specified after Lemma 3.7.

- This is compatible with the previous very general description $\lim_{[s]\in\Delta}\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{C}(\Gamma^s,\mathrm{A}))$ by taking dualizable objects, which will only depend on the underlying ring (i.e. evaluation on $*$ is an equivalence with ordinary perfect complexes). These are automatically p-completely nuclear, so by Lemma 3.10 there is no subtlety about the relation to modules with semilinear Γ-action.

- Upon restricting to Coh the description is also compatible with classical theory, where one equips finite modules with their weak topology as in [SV11, Lemmas 8.2 and 8.22] (which discusses the case of a point). By taking the topological module with the weak topology and passing to the condensed module, one obtains the canonical solid structure on the corresponding static dualizable object of $\mathrm{D}_{\blacksquare}(\mathrm{A})$.[4] We retain the solid tensor product to record the weak topology on a finite module; it agrees with the ordinary tensor product in this setting.

For coherent objects, to write down the data it is enough to keep the terms through $\mathrm{C}(\Gamma^2, \overline{\mathrm{A}})$ in (3). The descent data can naturally be encoded as a comodule over the Hopf algebroid $(\overline{\mathrm{A}}, \mathrm{C}(\Gamma, \overline{\mathrm{A}}))$. Explicitly, an object of the three-term totalization is a finite $\overline{\mathrm{A}}$-module M together with an isomorphism

$$\alpha : \mathrm{C}(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{p_1,\overline{\mathrm{A}}} \mathrm{M} \xrightarrow{\sim} \mathrm{C}(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{p_0,\overline{\mathrm{A}}} \mathrm{M}$$

satisfying the identity and cocycle conditions. Writing $p_0(a)(g) = a$ and $p_1(a)(g) = g(a)$, the rule $\delta(m) = \alpha(1 \otimes m)$ identifies α with a p_1-semilinear coaction $\delta : \mathrm{M} \to \mathrm{C}(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{p_0,\overline{\mathrm{A}}} \mathrm{M}$. The identity and cocycle conditions become the counit and coassociativity axioms, respectively, giving the equivalence with finite comodules over the Hopf algebroid.

Lemmas A.15 and A.11 give the same dictionary for Laurent crystals. Put $\overline{\mathrm{A}}^{(m)} = \mathrm{A}^{(m)}/p$. In the coaction formulation, coherent Laurent crystals are described by replacing $\mathrm{C}(\Gamma, \overline{\mathrm{A}})$ and $\mathrm{C}(\Gamma^2, \overline{\mathrm{A}})$ with $\overline{\mathrm{A}}^{(1)}$ and $\overline{\mathrm{A}}^{(2)}$. For the following lemma, we note there is a natural identification $\mathrm{C}(\Gamma^s, \mathrm{N}) \simeq \mathrm{C}(\Gamma^s, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N}$ for a finite $\overline{\mathrm{A}}$-module N equipped with its weak topology.

Lemma 3.14. Let $\overline{\mathrm{A}} = \mathrm{A}/p$, let $s = 1, 2$, and let $p_0 : \overline{\mathrm{A}} \to \overline{\mathrm{A}}^{(s)}$ be the zeroth Čech projection. For every finite $\overline{\mathrm{A}}$-module N, evaluation induces an injection

$$\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N} \longrightarrow \mathrm{C}(\Gamma^s, \mathrm{N}).$$

If $\mathrm{M} \subset \mathrm{N}$ is finite, then inside $\mathrm{C}(\Gamma^s, \mathrm{N})$ one has

$$\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{M} = \big(\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N}\big) \cap \mathrm{C}(\Gamma^s, \mathrm{M}).$$

Proof. By Definition 3.8 and Lemma 3.9, an element of $\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N}$ has a completed expansion

$$f = \sum_{\alpha \in \mathbf{N}^{s(n+1)}} m_\alpha \eta_\alpha.$$

In $\overline{\mathrm{A}}$ one has $[p]_q = (q-1)^{p-1}$, and $[p]_q$ is invertible; hence $q-1$ is a unit. The definition of γ_{m,q^p} gives

$$\eta_{i,m}(\ell) = \begin{cases} 0 & 0 \le \ell < m, \\ q^{p\binom{m}{2}}(q-1)^m & \ell = m \text{ and } i > 0, \\ q^{m+p\binom{m}{2}}(q-1)^m & \ell = m \text{ and } i = 0. \end{cases}$$

[4] By virtue of being dualizable, they admit a two-term $\mathrm{A}^n \to \mathrm{A}^m \to \mathrm{M}$ resolution as solid modules, where $\mathrm{A}^n \in \mathrm{D}_{\blacksquare}(\mathrm{A})$ has its natural product solid module structure. Thus the solid module structure is constructed in the same way as the weak topology.

Thus, for $\alpha, \ell \in \mathbf{N}^{s(n+1)}$, one has $\eta_\alpha(\ell) = 0$ unless $\alpha \leq \ell$ coordinatewise, while $\eta_\ell(\ell) \in \overline{\mathrm{A}}^\times$. If f evaluates to zero, then

$$0 = f(\ell) = \sum_{\alpha \leq \ell} m_\alpha \eta_\alpha(\ell).$$

The sum is finite. Induction on $|\ell|$ therefore gives $m_\ell = 0$ for every ℓ, proving injectivity.

The map $p_0 : \mathrm{A} \to \mathrm{A}^{(s)}$ is p-completely flat: before localizing p-completely at $[p]_q$ this follows from [Kos22, Proposition 4.13], and the property is preserved by inverting $[p]_q$ and applying p-adic completion. Since A is Noetherian, [Bha20, Lemma 5.15] makes $p_0 : \mathrm{A} \to \mathrm{A}^{(s)}$ flat. Its reduction $p_0 : \overline{\mathrm{A}} \to \overline{\mathrm{A}}^{(s)}$ is then flat. Consider the commutative square

$$\begin{tikzcd}
\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}}, p_0} \mathrm{N} \arrow[r] \arrow[d] & \mathrm{C}(\Gamma^s, \mathrm{N}) \arrow[d] \\
\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}}, p_0} (\mathrm{N}/\mathrm{M}) \arrow[r] & \mathrm{C}(\Gamma^s, \mathrm{N}/\mathrm{M}).
\end{tikzcd}$$

The lower horizontal map is injective by the first part. Flatness identifies the kernel of the left vertical map with $\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}}, p_0} \mathrm{M}$. If an element of the upper left term maps into $\mathrm{C}(\Gamma^s, \mathrm{M})$, its image in the lower right term vanishes. It therefore vanishes in the lower left term and belongs to $\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}}, p_0} \mathrm{M}$ (using injectivity of the lower horizontal map). The reverse inclusion follows from naturality, proving

$$\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}}, p_0} \mathrm{M} = \left(\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}}, p_0} \mathrm{N}\right) \cap \mathrm{C}(\Gamma^s, \mathrm{M}).$$

□

Via the coordinates $x = (x_0, \ldots, x_n) : \Gamma \to \mathbf{Z}_p^{n+1}$ of Definition 3.8[5], a continuous function $f : \Gamma \to \mathrm{M}$ with values in a complete separated module has a unique *Mahler expansion* $f(x) = \sum_\alpha a_\alpha \binom{x}{\alpha}$. Its Mahler coefficient of degree α is $a_\alpha = (\Delta_0^{\alpha_0} \cdots \Delta_n^{\alpha_n} f)(0)$, where $\Delta_i f(x) = f(x + e_i) - f(x)$ and e_i is the ith standard basis vector of $\mathbf{Z}_p^{n+1}$. We will use this in the following to give an alternative characterization of C_q using Mahler expansions, at least modulo p.

Lemma 3.15. Let M be a finite $\overline{\mathrm{A}}$-module with continuous semilinear Γ-action. For $k \gg 0$, the coaction of $\varphi^{*k}\mathrm{M}$ factors through

$$\mathrm{C}_q(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{\overline{\mathrm{A}}, p_0} \varphi^{*k}\mathrm{M} \subset \mathrm{C}(\Gamma, \varphi^{*k}\mathrm{M}).$$

The same holds with Γ^2 as needed for the cocycle condition.

Proof. Put $\overline{\mathrm{A}}^+ = \mathrm{A}^+/p$, and choose a finite $\overline{\mathrm{A}}^+$-submodule $\mathrm{M}^+ \subset \mathrm{M}$ such that $\mathrm{M}^+[1/(q-1)] = \mathrm{M}$. Picking a finite set of generators S for M^+, its orbit $\Gamma\mathrm{S}$ must be contained $(q-1)^{-N}\mathrm{M}^+$ by compactness of Γ and continuity of the action. Putting $\mathrm{M}^{+'} = \overline{\mathrm{A}}^+ \cdot \Gamma\mathrm{S} \subset (q-1)^{-N}\mathrm{M}^+$, since $\overline{\mathrm{A}}^+$ is Noetherian this is a finite Γ-stable module so we may assume without loss of

[5]The map x is only a homeomorphism, and does not respect the group structures.

generality that M^+ is Γ-stable. We equip M with a complete separated valuation $\nu_{q-1} : m \mapsto \sup\{r \in \mathbf{Z} : m \in (q-1)^r \mathrm{M}^+\}$. The action is isometric since Γ preserves M^+ and sends $q-1$ to a unit multiple of itself; we will need this structure to apply results in [BR25] later in the proof.

We first establish a practical criterion for $f \in \mathrm{C}(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{M} \simeq \mathrm{C}(\Gamma, \mathrm{M})$ to belong to $\mathrm{C}_q(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{M}$.

Lemma 3.16. Let $f \in \mathrm{C}(\Gamma, \mathrm{M})$. Consider its Mahler expansion

$$f(x) = \sum_{\alpha \in \mathbf{N}^{n+1}} a_\alpha \binom{x}{\alpha},$$

where $\binom{x}{\alpha} = \prod_i \binom{x_i}{\alpha_i}$. Suppose there are constants $\sigma > 1$ and C for which

$$a_\alpha \in (q-1)^{\lceil \sigma |\alpha| - \mathrm{C} \rceil} \mathrm{M}^+$$

for every multi-index α. Then $f \in \mathrm{C}_q(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{M}$.

Proof. We will prove the stronger assertion

$$\left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \overline{\mathrm{A}}^+ \eta_\alpha \right)^{\wedge}_{(q-1)} = \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \overline{\mathrm{A}}^+ (q-1)^{|\alpha|} \binom{x}{\alpha} \right)^{\wedge}_{(q-1)}.$$

In one variable x, set $\tilde{\eta}_m = q^{p\binom{m}{2}} \binom{x}{m}_{q^p}$, a renormalized version of $\eta_{i,m}$ from Definition 3.8 (by a factor of $(q-1)^m$). For $m \geq 1$, the q-Pascal recurrence

$$\tilde{\eta}_m(x+1) - \tilde{\eta}_m(x) = (q^{pm} - 1)\tilde{\eta}_m(x) + q^{p(m-1)} \tilde{\eta}_{m-1}(x)$$

shows that the coefficient of $\binom{x}{j}$ in $\tilde{\eta}_m$ vanishes for $j < m$, is a unit for $j = m$, and belongs to $(q^p - 1)^{j-m} \overline{\mathrm{A}}^+$ for $j > m$. Here we use that the finite difference operator $\Delta(f) = f(x+1) - f(x)$ extracts the coefficient of $\binom{x}{j}$ in its Mahler expansion as $\Delta^j f(0)$. The first claim is simply because $\tilde{\eta}_m(j) = 0$ when $j < m$, the second because we obtain $q^{p\binom{m}{2}}$ at $j = m$, and the third follows by induction using the q-Pascal recurrence.

In $\overline{\mathrm{A}}^+$ one has $q^p - 1 = (q-1)^p$, and hence for all $0 \leq i \leq n$ (rescaling by the unit q^m at $i = 0$) one has

$$\eta_{i,m} = u_{i,m} (q-1)^m \binom{x_i}{m} + \sum_{j > m} c_{i,m,j} (q-1)^j \binom{x_i}{j}.$$

Here $u_{i,m} \in (\overline{\mathrm{A}}^+)^\times$, and $c_{i,m,j} \in (q-1)^{(p-1)(j-m)} \overline{\mathrm{A}}^+$. This shows the $(q-1)^{|\alpha|}\binom{x}{\alpha}$ and the η_α bases are related by an operator $\mathrm{D}(1 + \mathrm{N})$ where D is diagonal and invertible and

$$\mathrm{N}^r \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \overline{\mathrm{A}}^+ (q-1)^{|\alpha|} \binom{x}{\alpha} \right)^{\wedge}_{(q-1)} \subset (q-1)^{(p-1)r} \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \overline{\mathrm{A}}^+ (q-1)^{|\alpha|} \binom{x}{\alpha} \right)^{\wedge}_{(q-1)},$$

hence N is topologically nilpotent. Thus we deduce the claimed isomorphism. Choosing a finite free surjection $\overline{\mathrm{A}}^{+\oplus n} \twoheadrightarrow \mathrm{M}^+$ and inverting $q-1$ proves the criterion for finite coefficients. ☐

Equipped with this criterion, our goal is now to show that a high Frobenius pullback automatically satisfies the criterion. For $j \geq 0$, put $\Gamma_j = \{g^{p^j} : g \in \Gamma\}$. Then Γ is uniform and

$$\Gamma_j = (1 + p^{j+1}\mathbf{Z}_p) \ltimes p^j\mathbf{Z}_p(1)^n.$$

Moreover, $gg'^{-1} \in \Gamma_j$ if and only if $x_i(g) - x_i(g') \in p^j\mathbf{Z}_p$ for every i. Thus $x = (x_0, \dots, x_n) : \Gamma \to \mathbf{Z}_p^{n+1}$ is a coordinate in the sense of [BR25, Proposition 1.1.1].

Continuity on a finite set of generators, together with the formulas for the action of Γ on A, gives $\ell \geq 0$ and $d > 0$ such that

$$(g-1)(q-1)^r\mathrm{M}^+ \subset (q-1)^{r+d}\mathrm{M}^+$$

for $g \in \Gamma_\ell$ and $r \in \mathbf{Z}$. If $g \in \Gamma_{\ell+j}$, write $g = g_0^{p^j}$ with $g_0 \in \Gamma_\ell$. Since $g - 1 = (g_0 - 1)^{p^j}$ in characteristic p, we obtain

$$(g-1)(q-1)^r\mathrm{M}^+ \subset (q-1)^{r+dp^j}\mathrm{M}^+.$$

It follows from [BR25, Lemmas 1.2.8 and 1.2.10] that every orbit map is super-Hölder with $e = 1$ and $\lambda = \log_p(d) - \ell$ in the sense of [BR25, Definition 1.2.1]. Applying [BR25, Theorem 4.3.4] in the coordinate x, its Mahler coefficients satisfy

$$a_\alpha \in (q-1)^{\lceil c\max_i \alpha_i - \mathrm{C}\rceil}\mathrm{M}^+ \subset (q-1)^{\lceil c|\alpha|/(n+1) - \mathrm{C}\rceil}\mathrm{M}^+.$$

Here $c > 0$ and C are independent of α.

Since $\varphi^k(q-1) = (q-1)^{p^k}$ modulo p, the orbit map of $1 \otimes m$, for $m \in \mathrm{M}^+$, has Mahler coefficients in

$$(q-1)^{\lceil p^k c|\alpha|/(n+1) - p^k\mathrm{C}\rceil}(\overline{\mathrm{A}}^+ \otimes_{\varphi^k, \overline{\mathrm{A}}^+} \mathrm{M}^+).$$

Choose k so that $p^k c/(2(n+1)) > 1$. The criterion in Lemma 3.16 now shows that the coaction on $1 \otimes m$ belongs to $\mathrm{C}_q(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{\overline{\mathrm{A}}, p_0} \varphi^{*k}\mathrm{M}$. Since $\mathrm{M}^+[1/(q-1)] = \mathrm{M}$, the elements $1 \otimes m$ with $m \in \mathrm{M}^+$ generate $\varphi^{*k}\mathrm{M}$ over $\overline{\mathrm{A}}$. The coaction is p_1-linear, and $p_1(\overline{\mathrm{A}}) \subset \mathrm{C}_q(\Gamma, \overline{\mathrm{A}})$ by Lemma 3.9, so this proves the assertion over Γ.

The same argument in the $2(n+1)$ product coordinates proves the assertion over Γ^2. $\square$

Proposition 3.17. Let M be a finite $\overline{\mathrm{A}}$-module with continuous semilinear Γ-action. For $k \gg 0$, the module $\varphi^{*k}\mathrm{M}$ arises from a coherent Laurent crystal killed by p.

Proof. Choose k large enough for Lemma 3.15 in degrees one and two, and put $\mathrm{N} = \varphi^{*k}\mathrm{M}$. Lemmas 3.9 and 3.14 give injective vertical maps in the commutative diagram

$$\begin{tikzcd}
\mathrm{N} \arrow[r, shift left] \arrow[r, shift right] \arrow[d, equal] & \mathrm{C}(\Gamma, \mathrm{N}) \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] & \mathrm{C}(\Gamma^2, \mathrm{N}) \\
\mathrm{N} \arrow[r, shift left] \arrow[r, shift right] & \overline{\mathrm{A}}^{(1)} \otimes^{\blacksquare}_{\overline{\mathrm{A}}, p_0} \mathrm{N} \arrow[u, hook] \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] & \overline{\mathrm{A}}^{(2)} \otimes^{\blacksquare}_{\overline{\mathrm{A}}, p_0} \mathrm{N} \arrow[u, hook]
\end{tikzcd}$$

The coaction $m \mapsto (g \mapsto g(m))$ factors uniquely through the lower degree one term by Lemma 3.15; use this factorization for the corresponding lower coface. Compatibility of the evaluation maps with the Čech maps makes the diagram commute. The counit and coassociativity identities hold in the upper row, hence in the lower row by injectivity in degrees one and two. Thus the lower row defines a descent datum on N as a coherent Laurent crystal. □

Lemma 3.18. Let $\mathrm{M} \subset \mathrm{N}$ be finite $\overline{\mathrm{A}}$-modules with continuous semilinear Γ-action. If N is the evaluation of a coherent Laurent crystal killed by p and M is Γ-stable, then the crystal coaction restricts to M.

Proof. One can identify the crystal coaction with

$$\mathrm{N} \longrightarrow \overline{\mathrm{A}}^{(1)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N}.$$

For $x \in \mathrm{M}$, its image in $\overline{\mathrm{A}}^{(1)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} (\mathrm{N}/\mathrm{M})$ evaluates to zero because M is Γ-stable. Lemma 3.14 makes this image zero, so the coaction restricts to M. The same argument after exchanging the two factors restricts the inverse coaction. For $s = 1, 2$, flatness of $p_0 : \overline{\mathrm{A}} \to \overline{\mathrm{A}}^{(s)}$ makes

$$\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{M} \longrightarrow \overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N}$$

injective. The restricted coactions are therefore inverse, and the cocycle identity follows from that for N. The counit identity follows after composing with $\mathrm{M} \hookrightarrow \mathrm{N}$. Since $\overline{\mathrm{A}}$ is Noetherian regular and M is finite, the resulting subcrystal is coherent. □

Proof of Theorem 3.5. Let $\beta : \mathrm{M} \hookrightarrow \varphi^*\mathrm{M}$ be the given map. Since $\overline{\mathrm{A}}$ is regular, Frobenius is flat by Kunz's theorem. Iterating β therefore embeds both M and $\varphi^*\mathrm{M}$ as Γ-stable submodules of $\mathrm{N} = \varphi^{*k}\mathrm{M}$ for every $k \geq 1$.

Choose k large enough for Proposition 3.17, and let $\mathcal{H}$ be the resulting coherent crystal with evaluation N. Lemma 3.18 gives coherent subcrystals $\mathcal{E}, \mathcal{E}_1 \subset \mathcal{H}$ with evaluations M and $\varphi^*\mathrm{M}$. These evaluations are perfect, hence p-completely nuclear, over the regular ring $\overline{\mathrm{A}}$. Proposition 3.13 lifts β uniquely to a morphism

$$\widetilde{\beta} : \mathcal{E} \longrightarrow \mathcal{E}_1.$$

Its kernel has zero evaluation, so t-exactness and conservativity make $\widetilde{\beta}$ injective. Frobenius compatibility shows that $\mathcal{E}_1$ and $\varphi^*\mathcal{E}$ both evaluate to $\varphi^*\mathrm{M}$ with the same Γ-action. Full faithfulness gives the unique equivalence between them evaluating to the identity of $\varphi^*\mathrm{M}$. Under this equivalence, $\widetilde{\beta}$ is the required map $\mathcal{E} \hookrightarrow \varphi^*\mathcal{E}$. □

As promised we now investigate other bases, which is a straightforward modification.

Corollary 3.19. Both Proposition 3.13 and Theorem 3.5 hold over $\mathrm{W}(k)[\zeta_{p^\ell}]$ for $\ell \geq 0$, by changing to $\Gamma = \mathbf{Z}_p^\times \ltimes \mathbf{Z}_p(1)^n$ when $\ell = 0$ and $(1 + p^\ell \mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$ for $\ell > 0$.

Proof. For $\ell > 0$ the modification is very easy, and can be done by simply setting the prismatic ideal in the q-de Rham prism to be $\varphi^{\ell-1}([p]_q)$. The toric factor acts by $\mathrm{T}_i \mapsto q^{p^\ell b_i}\mathrm{T}_i$, so this action is trivial modulo $\varphi^{\ell-1}([p]_q)$. For $\ell = 0$ and p odd, over $\mathrm{W}(k)$ one can show all sth terms $\mathrm{A}^{(s)}$ of the (Laurentized) Čech nerve of the log q-de Rham over $(\mathrm{Spf}\,\mathrm{R}_0, \mathcal{M})$ are μ_{p-1}^s-torsors for the Čech nerve over $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$. Note this claim is not true before p-completely inverting $[p]_q$.

Applying the Čech descent argument of Theorem A.14 gives

$$\mathrm{D}_{\blacksquare}((\operatorname{Spf}\mathrm{R}_0,\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)\simeq \mathrm{D}_{\blacksquare}((\operatorname{Spf}\mathrm{R},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)^{h\mu_{p-1}},$$

and the claims follow by descent. □

4. Finitely generated unit Laurent crystals

4.1. **Basic constructions.** Following [EK04b] and the more general version in [BL19] (which explains how to remove finiteness hypotheses), we first generalize the notion of finitely generated unit modules.

Definition 4.1. Let $(\mathrm{A}^+,\mathrm{I})$ be a prism with $\mathrm{I}=(d)$. Equip $\mathrm{A}=\mathrm{A}^+[1/d]^{\wedge}_p$ with the analytic ring structure in Definition A.10, and we use $\mathrm{D}_{\blacksquare}(\mathrm{A})$ as defined there. Note A has a natural Frobenius φ induced by the Frobenius on A^+ (since $\varphi(d)\equiv d^p \pmod p$). Write $\mathrm{A}\langle \mathrm{F}\rangle$ for

$$\widehat{\bigoplus}_{i\geq 0}\mathrm{A}\mathrm{F}^i,$$

where $\mathrm{F}x=\varphi(x)\mathrm{F}$. It is naturally a noncommutative algebra object of $\mathrm{D}_{\blacksquare}(\mathrm{A})$. We consider $\mathrm{M}\in\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{A}\langle\mathrm{F}\rangle):=\mathsf{Mod}_{\mathrm{A}\langle\mathrm{F}\rangle}\widehat{\mathrm{D}}_{\blacksquare}(\operatorname{Spa}(\mathrm{A}^+,\mathrm{A}^+))$, or derived p-complete solid modules over $\mathrm{A}\langle\mathrm{F}\rangle$.
We say M is *unit* if

$$\varphi_{\mathrm{A}}^*\mathrm{M}:=\mathrm{A}\otimes^{\mathrm{L},\blacksquare}_{\mathrm{A},\varphi_{\mathrm{A}}}\mathrm{M}\longrightarrow\mathrm{M}$$

given by $a\otimes m\mapsto a\mathrm{F}_{\mathrm{M}}(m)$ is an equivalence. A unit object is *finitely generated unit* if it is perfect over $\mathrm{A}\langle\mathrm{F}\rangle$; these objects form the full subcategory $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A})$. Here, we mean perfect in the sense that it lies in the thick subcategory generated by $\mathrm{A}\langle\mathrm{F}\rangle$ as we work over a noncommutative base.

In fact one can show the category on a fixed prism is insensitive to the solid structure, but we will still remember it so that our crystals live in an ambient category which is a stack.

Lemma 4.2. Let $(\mathrm{A}^+,(d))$ be an orientable bounded prism, and let $\mathrm{A}=\mathrm{A}^+[1/d]^{\wedge}_p$. Then

$$\mathrm{Thick}_{\mathrm{A}\langle\mathrm{F}\rangle}\widehat{\mathrm{D}}(\mathrm{A}\langle\mathrm{F}\rangle)\simeq\mathrm{Thick}_{\mathrm{A}\langle\mathrm{F}\rangle}\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{A}\langle\mathrm{F}\rangle)$$

via $\mathrm{M}\mapsto\mathrm{M}(*)$. The solid modules in the target thick subcategory are all p-completely nuclear.

Proof. The comparison is exact and carries $\mathrm{A}\langle\mathrm{F}\rangle$ to the corresponding solid module. We will show the natural functor induces an equivalence on their mapping spectra. It is then fully faithful on the thick closure of the generator, and its essential image is a thick subcategory containing the target generator.

We can compute

$$\mathrm{Map}_{\mathsf{Mod}_{\mathrm{A}\langle\mathrm{F}\rangle}(\widehat{\mathrm{D}}_{\blacksquare}(\operatorname{Spa}(\mathrm{A}^+,\mathrm{A}^+)))}\big(\mathrm{A}\langle\mathrm{F}\rangle,\mathrm{A}\langle\mathrm{F}\rangle\big)\simeq\mathrm{Map}_{\widehat{\mathrm{D}}_{\blacksquare}(\operatorname{Spa}(\mathrm{A}^+,\mathrm{A}^+))}\big(\mathrm{A}^+,\mathrm{A}\langle\mathrm{F}\rangle\big)\simeq\mathrm{A}\langle\mathrm{F}\rangle.$$

The first equivalence uses the free-forgetful adjunction. The second equivalence can be checked modulo p by derived Nakayama. It then follows from compactness of A^+/p: modulo p localization at d is a filtered colimit and $(\mathrm{A}\langle\mathrm{F}\rangle)/p=\bigoplus_{i\geq 0}(\mathrm{A}/p)\mathrm{F}^i$.

The unit A^+ is nuclear by [AM24, Lemma 2.18(i)]. Since A is the filtered colimit of multiplication by d (in derived p-complete solid modules), it is p-completely nuclear. The p-completely nuclear objects are closed under colimits, so the underlying module of $\mathrm{A}\langle \mathrm{F}\rangle = \widehat{\bigoplus}_{i\geq 0}\mathrm{AF}^i$ is nuclear. The forgetful functor from Frobenius modules preserves finite colimits and retracts, so the same holds throughout the thick subcategory generated by the free $\mathrm{A}\langle F\rangle$-module. ☐

Definition 4.3. Let $(\mathrm{X},\mathcal{M})$ be a bounded fs log p-adic formal scheme. A crystal

$$\mathcal{E}\in \mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})^{\varphi=1}$$

is finitely generated unit if it is t-bounded and, for every object $(\mathrm{A},\mathrm{I},\mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^a$ of $(\mathrm{X},\mathcal{M})_{\Delta}$, its evaluation is perfect in

$$\mathrm{Mod}_{\mathrm{A}[1/\mathrm{I}]^{\wedge}_{p}\langle \mathrm{F}\rangle}\big(\mathrm{D}_{\blacksquare}(\mathrm{A}[1/\mathrm{I}]^{\wedge}_{p})\big).$$

We write $\mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ for the full subcategory of finitely generated unit crystals, and $\mathsf{Mod}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ for its intersection with the heart of $\mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$.

Remark 4.4. Let B be a p-adically complete associative ring in which p is central, and let M be a derived p-complete left B-module. Then

$$\mathrm{M}\in \mathsf{Perf}(\mathrm{B}) \iff (\mathrm{B}/p)\otimes^{\mathrm{L}}_{\mathrm{B}}\mathrm{M}\in \mathsf{Perf}(\mathrm{B}/p).$$

Indeed, apply [Sta26, Lemma 15.99.4, Tag 09AW] to the tower $\{\mathrm{M}/p^n\}_{n\geq 1}$; its proof applies unchanged when (p) is central. The same holds in the solid setting; the unit Frobenius condition can also be checked modulo p by derived Nakayama.

By Remark 4.12, it suffices to test Definition 4.3 on the log q-de Rham prisms associated to framed log affines.

Our first main goals will be to give an explicit local description of this category, as well as deduce strict-étale descent. We note that while $\mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})^{\varphi=1}$ is a strict-étale stack (by Corollary A.16), it is not immediate that the finitely generated unit subcategory is also a strict-étale stack. While ordinary dualizable objects can be detected termwise in a limit, because we work over $\mathrm{A}\langle \mathrm{F}\rangle$ we use the thick subcategory generated by the unit (which satisfies no such property). Instead, descent of the fgu condition will use its characterization by coherent roots, which we now study.

Definition 4.5. Let A be as in 4.1, and assume that A is Noetherian regular. If M is a unit Frobenius module over A, a *root* of M is a coherent A-submodule $\mathrm{M}_0\subset \mathrm{M}$ together with an injective map $\beta:\mathrm{M}_0\to \varphi_{\mathrm{A}}^{*}\mathrm{M}_0$ whose iterates identify

$$\mathrm{M}\simeq \mathrm{colim}_n\,\varphi_{\mathrm{A}}^{*n}\mathrm{M}_0.$$

It is *minimal* if it has no proper subroot. A distinguished minimal root is a minimal root contained in every other root of M as a submodule of the same ambient unit module.

One can make the analogous definitions for any unit object in the heart of $\mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_{p})$. When referring to a root of a module we mean the former, and when referring to a root of a crystal we mean the latter.

The essential idea of the proof of strict-étale descent is now the following: Proposition 4.7 identifies the existence of distinguished minimal roots (as Laurent crystals) with the finitely generated unit condition. Distinguished minimal roots are unique as subcrystals and therefore glue on intersections, once we know strict-étale descent for the ambient category $\mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_{p})^{\varphi=1}$. This reduces the descent problem to descent of coherent Laurent crystals, to which the main results of [Mat22] apply. There are of course many technical details to dispense of, but the core argument is very simple.

Let $(\operatorname{Spf}\mathrm{R},\mathcal{M})$ be a framed log affine in the sense of Definition 3.2 and put $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)^{\wedge}_{p}$, the Laurent version of its logarithmic q-de Rham prism. The following theorem identifies finitely generated unit crystals with (φ,Γ)-modules and constructs distinguished minimal roots. A key aspect is that we may ignore solid modules locally and work with continuous semilinear Γ-actions on topological modules. By $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A})^{h\Gamma}$, we mean

$$\lim_{[s]\in\Delta} \mathrm{Thick}_{\mathrm{C}(\Gamma^s,\mathrm{A}\langle\mathrm{F}\rangle)}\widehat{\mathrm{D}}(\mathrm{C}(\Gamma^s,\mathrm{A}\langle\mathrm{F}\rangle))$$

with no solid structures involved. Here $\mathrm{A}\langle\mathrm{F}\rangle$ has its natural topology (induced from A and the completed direct sum). Note $\mathrm{C}(\Gamma^s,\mathrm{A})\widehat{\otimes}_{\mathrm{A}}\mathrm{A}\langle\mathrm{F}\rangle \simeq \mathrm{C}(\Gamma^s,\mathrm{A}\langle\mathrm{F}\rangle)$ as topological rings.

The condition of lying in these thick subcategories is stable under the pullbacks $\mathrm{C}(\Gamma^s,\mathrm{A})\otimes^{\mathrm{L}}_{\mathrm{A}} -$ by nuclearity of $\mathrm{A}\langle\mathrm{F}\rangle$ (passing to the associated solid condensed module recovers the solid tensor product). On the heart one may take a more classical approach, endowing a finitely generated unit module with a weak topology induced by a surjection from $\mathrm{A}\langle\mathrm{F}\rangle^{\oplus i}$ (see the discussion at the beginning of §3.2); this is essentially what we are doing here, just in a more derived way.

Theorem 4.6. Let $(\operatorname{Spf}\mathrm{R},\mathcal{M})$ be a framed log affine in the sense of Definition 3.2 and put $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)^{\wedge}_{p}$. Evaluation on the log q-de Rham prism $(\mathrm{R}_0^{\square}[\![q-1]\!], [p]_q, \mathcal{M})^a$ induces an equivalence

$$\rho^*_{\mathrm{A}} : \mathrm{D}_{\mathrm{fgu}}((\operatorname{Spf}\mathrm{R},\mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_{p}) \xrightarrow{\sim} \mathrm{D}_{\mathrm{fgu}}(\mathrm{A})^{h\Gamma}.$$

Proof. Lemma 4.2 shows that the evaluation of every finitely generated unit crystal is p-completely nuclear. Proposition 3.13 therefore gives derived full faithfulness, where the target is solid finitely generated unit modules. Lemma 4.2 identifies the target with ordinary finitely generated unit complexes with a continuous Γ-action, as defined before the theorem as $\lim_{[s]\in\Delta}\mathrm{Thick}_{\mathrm{C}(\Gamma^s,\mathrm{A}\langle\mathrm{F}\rangle)}\widehat{\mathrm{D}}(\mathrm{C}(\Gamma^s,\mathrm{A}\langle\mathrm{F}\rangle))$. Strictly speaking it applies directly to only the first term $s=0$, but an identical argument covers all terms.

We first prove essential surjectivity on the heart modulo p. Put $\overline{\mathrm{A}} = \mathrm{A}/p$, and let M be a finitely generated unit $\overline{\mathrm{A}}$-module with continuous semilinear Γ-action. Since $\overline{\mathrm{A}}$ is regular and F-finite, [Bli08, Corollary 2.26] gives a distinguished minimal root

$$\beta : \mathrm{M}_{\min} \hookrightarrow \varphi^*\mathrm{M}_{\min}.$$

Every Γ-translate is again a root. Moreover distinguished containment (i.e. its defining property) applied to a translate and its inverse gives $\gamma \mathrm{M}_{\min} = \mathrm{M}_{\min}$, so β is Γ-equivariant. Theorem 3.5 lifts it to a coherent Laurent crystal $\mathcal{E}_0$. Its unitalization $\mathcal{E}$ satisfies $\rho_{\mathrm{A}}^* \mathcal{E} \simeq \mathrm{M}$. By the t-exactness and conservativity of Proposition 3.13, $\mathcal{E}$ lies in the heart.

The existence of a root $\mathcal{E}_0$ so that $\mathcal{E} \simeq \operatorname{colim}_k \varphi^{*k}\mathcal{E}_0$ shows that $\mathcal{E}$ is finitely generated unit, as this presentation applied to evaluation on any log prism induces a two-term resolution showing it lies in the thick subcategory of Definition 4.1.

Since evaluation is compatible with shifts and fibers, [Hau24, Lemma 5.4.3] upgrades essential surjectivity on the heart modulo p to essential surjectivity on the derived categories. Thus evaluation is an equivalence modulo p. Lemma 2.3 now gives the claim. $\square$

By applying Theorem 3.5 we can prove a root criterion for the finitely generated unit condition.

Proposition 4.7. Let $(\operatorname{Spf} \mathrm{R}, \mathcal{M})$ be a framed log affine. A crystal $\mathcal{E} \in \mathrm{D}_{\blacksquare}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)^{\varphi=1}$ is finitely generated unit if and only if it is t-bounded and each crystal $\mathrm{H}^i(\mathcal{E}_{p=0})$ admits a distinguished minimal root in $\mathsf{Coh}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)_{p=0}$.

Proof. Put $\mathrm{A} = \mathrm{R}_0^{\square}((q-1))^{\wedge}_p$ and $\overline{\mathrm{A}} = \mathrm{A}/p$. Suppose first that the condition on roots holds. By Remark 4.4, it is enough to work modulo p. A coherent root is a perfect Laurent crystal, so its evaluation on every log prism is perfect over the coefficient ring. On a log prism $(\mathrm{B}, \mathrm{J}, \mathcal{N})^a$, the two-term induced resolution of [EK04b, Proposition 5.3.3] makes its unitalization perfect over $(\mathrm{B}[1/\mathrm{J}]^{\wedge}_p/p)[\mathrm{F}]$. Thus every cohomology module of the evaluation of $\mathcal{E}_{p=0}$ on the log q-de Rham prism is perfect over $\overline{\mathrm{A}}[\mathrm{F}]$. The unit condition is detected on cohomology by [EK04b, 5.4.1]. Since $\mathcal{E}$ is t-bounded and perfect objects are closed under finite extensions, its evaluation is finitely generated unit. Theorem 4.6 and Proposition 3.13 show that $\mathcal{E}_{p=0}$ is finitely generated unit.

Conversely, Proposition 3.13 makes evaluation on the mod-p log q-de Rham prism t-exact and conservative. After forgetting the solid structure, Lemma 4.2 makes the evaluated complex perfect over $\overline{\mathrm{A}}[\mathrm{F}]$. Proposition 11.3.9 of [BL19] shows that the evaluation M of $\mathrm{H}^i(\mathcal{E}_{p=0})$ is an ordinary finitely generated unit module. As in the proof of Theorem 4.6, its distinguished minimal root $\mathrm{M}_{\min}$ is Γ-stable and its root map is Γ-equivariant. Theorem 3.5 lifts this map to a coherent Laurent crystal $\mathcal{E}_{\min}$. Proposition 3.13 identifies its unitalization with $\mathrm{H}^i(\mathcal{E}_{p=0})$, since both evaluate to M. Transporting the canonical inclusion into the unitalization gives

$$\mathcal{E}_{\min} \hookrightarrow \mathrm{H}^i(\mathcal{E}_{p=0}).$$

It is injective by the t-exactness and conservativity of Proposition 3.13, and its iterates unitalize to the ambient crystal. Thus it is a root.

Let $\mathcal{E}_0'$ be any other coherent root, with evaluation M_0'. Distinguished containment gives a Γ-equivariant inclusion $\mathrm{M}_{\min} \hookrightarrow \mathrm{M}_0'$. Proposition 3.13 lifts it uniquely to $\mathcal{E}_{\min} \to \mathcal{E}_0'$ and makes the lift injective by t-exact conservativity. The two composites to $\varphi^*\mathcal{E}_0'$ evaluate to the same map, so the same proposition makes the inclusion compatible with the root maps. Hence $\mathcal{E}_{\min}$ is contained in every root. Any subroot evaluates to a subroot of $\mathrm{M}_{\min}$, so minimality and conservativity in Proposition 3.13 make it equal to $\mathcal{E}_{\min}$. Thus $\mathcal{E}_{\min}$ is the distinguished minimal root as a crystal. $\square$

4.2. **Strict-étale descent.** Proposition 4.7 reduces strict-étale descent to descent of coherent Laurent crystals on $(\mathrm{X}, \mathcal{M})_{\mathbb{A}}$. The following lemmas give the required compatibility of minimal roots with completion and étale base change.

Lemma 4.8. Let A be a regular local F-finite $\mathbf{F}_p$-algebra, let $\widehat{\mathrm{A}}$ be its completion, and let $\mathrm{N}_0 \subset \mathrm{N}$ be a root of a finitely generated unit A[F]-module. Then N_0 is minimal if and only if $\widehat{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{N}_0 \subset \widehat{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{N}$ is minimal.

Proof. The forward implication is [Bli04, Corollary 2.11]. Conversely, assume the completed root is minimal and let $\mathrm{N}_1 \subset \mathrm{N}_0$ be a subroot. Its completion is a subroot, hence $\widehat{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{N}_1 = \widehat{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{N}_0$.

Since $\mathrm{A} \to \widehat{\mathrm{A}}$ is faithfully flat, it follows that $\mathrm{N}_1 = \mathrm{N}_0$. Thus N_0 has no proper subroot, hence is minimal. $\square$

Lemma 4.9. Let R be a $(q-1)$-adically complete Noetherian $k[\![q-1]\!]$-algebra, topologically of finite type and formally smooth over $k[\![q-1]\!]$, and let S be another such algebra. Suppose $\mathrm{R}(\!(q-1)\!) \to \mathrm{S}(\!(q-1)\!)$ induces an étale morphism of $k(\!(q-1)\!)$-affinoid spaces. Then scalar extension sends minimal roots of finitely generated unit modules to minimal roots.

Proof. Put $\mathrm{A} = \mathrm{R}(\!(q-1)\!)$ and $\mathrm{B} = \mathrm{S}(\!(q-1)\!)$, and let $\mathrm{N}_0 \subset \mathrm{N}$ be a minimal root over A, with root map $\beta : \mathrm{N}_0 \to \varphi_{\mathrm{A}}^* \mathrm{N}_0$. Both rings are regular F-finite, so the hypotheses in [Bli08] for the results we will use are satisfied. Flatness of $\mathrm{A} \to \mathrm{B}$ and naturality of Frobenius pullback show that $\mathrm{B} \otimes_{\mathrm{A}} \mathrm{N}_0$ is a root of $\mathrm{B} \otimes_{\mathrm{A}} \mathrm{N}$, so we only concern ourselves with minimality. More generally, flat pullbacks send roots to roots.

Following [Bli08, §2.4], let $\sigma_\beta^{-1}(\mathrm{N}_0) \subset \mathrm{N}_0$ be the smallest submodule N_0' such that $\beta(\mathrm{N}_0) \subset \varphi_{\mathrm{A}}^* \mathrm{N}_0'$. By [Bli08, Lemma 2.17 and Proposition 2.18] this construction commutes with localization, and a root is minimal precisely when $\sigma_\beta^{-1}(\mathrm{N}_0) = \mathrm{N}_0$. It follows that a root over A is minimal if and only if its localization at every maximal ideal is minimal, and similarly over B. Together with Lemma 4.8, this shows that minimality may be checked on the *completed* local rings at the classical points of the rigid spaces.

Fix a maximal ideal $\mathfrak{m} \subset \mathrm{B}$, let $x \in \mathrm{Spa}(\mathrm{B})$ be the corresponding rigid point, and put $\mathfrak{n} = \mathfrak{m} \cap \mathrm{A}$. By [KL15, Definition 8.2.16] and the affinoid basis following [KL15, Definition 8.2.19], the morphism factors near x into affinoid open immersions and a finite étale morphism. Since affinoid open immersions induce isomorphisms on completed local rings by [BGR84, 7.3.2/3], the map

$$\widehat{\mathrm{A}_{\mathfrak{n}}} \longrightarrow \widehat{\mathrm{B}_{\mathfrak{m}}}$$

is finite étale. Both rings are again regular and F-finite. It therefore remains to show that finite étale scalar extension preserves minimal roots.

Let $\mathrm{A} \to \mathrm{B}$ now be a finite étale map between regular F-finite rings. Put $\mathrm{D}_{\mathrm{A}}^{(1)} = \mathrm{End}_{\mathrm{A}}(\varphi_{\mathrm{A},*}\mathrm{A})$, and define $\mathrm{D}_{\mathrm{B}}^{(1)}$ similarly. The relative Frobenius is an isomorphism by [Sta26, Lemma 41.14.3, Tag 0EBS], and F-finiteness together with Kunz's theorem gives

$$\mathrm{B} \otimes_{\mathrm{A}} \mathrm{D}_{\mathrm{A}}^{(1)} \simeq \mathrm{D}_{\mathrm{B}}^{(1)}.$$

Blickle's description in [Bli08, §2.4] then gives, for the base change β_{B} of a root map β,

$$\sigma_{\beta_{\mathrm{B}}}^{-1}(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{N}_0) = \mathrm{B} \otimes_{\mathrm{A}} \sigma_{\beta}^{-1}(\mathrm{N}_0),$$

where faithful flatness of φ_{B} descends the equality after Frobenius pullback. Thus finite étale scalar extension preserves minimal roots. Applying this to the completed local rings above and using the completed-local criterion proves the claim. □

Corollary 4.10. Let $f : (\mathrm{Spf}\,\mathrm{S}, \mathcal{M}_{\mathrm{Spf}\,\mathrm{S}}) \to (\mathrm{Spf}\,\mathrm{R}, \mathcal{M}_{\mathrm{Spf}\,\mathrm{R}})$ be a strict-étale map of framed log affines compatible with the framings. If $\mathcal{E}$ is a finitely generated unit crystal killed by p with distinguished minimal root $\mathcal{E}_{\min}$, then $f^*\mathcal{E}$ is finitely generated unit and $f^*\mathcal{E}_{\min}$ is its distinguished minimal root.

Proof. The pullback is t-exact because f is strict-étale. Applying Lemma 4.9 to the evaluations on the logarithmic q-de Rham prisms for the compatible framings shows that $f^*\mathcal{E}_{\min}$ is minimal. By [Bli08, Corollary 2.26], the minimal root is unique and contained in every root, so it is distinguished. The claim for crystals now follows from Theorem 4.6 and the argument of Proposition 4.7. □

We can now deduce strict-étale descent.

Theorem 4.11. Let $(\mathrm{X}, \mathcal{M})$ satisfy Assumption 3.1 over $\mathrm{W}(k)[\zeta_p]$. Then the assignment

$$(\mathrm{X}, \mathcal{M}) \mapsto \mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$$

is a stack for the strict-étale topology on such $(\mathrm{X}, \mathcal{M})$.

Proof. By Corollary A.16, $\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})^{\varphi=1}$ satisfies strict-étale descent. The pullback is t-exact, and the cover may be taken finite, so boundedness is local. Let $\{\rho_i : \mathrm{U}_i \to (\mathrm{X}, \mathcal{M})\}_{i \in \mathrm{I}}$ be a strict-étale cover such that every $\rho_i^*\mathcal{E}$ is finitely generated unit.

Using Remark 4.4 and Proposition 4.7 we may reduce the claim to showing

$$\left\{\rho_i^*\mathcal{E} \in \mathsf{Mod}_{\mathrm{fgu}}((\mathrm{U}_i)_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})_{p=0}\right\} \implies \left\{\mathcal{E} \in \mathsf{Mod}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})_{p=0}\right\}$$

for $\mathcal{E} \in \mathsf{QCoh}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})^{\varphi=1}$ killed by p, or that the condition can be detected locally on a strict-étale cover. Note that the converse is obvious, because any test log prism for the U_i is also one for $(\mathrm{X}, \mathcal{M})$, so the finitely generated unit condition on the right side already includes the conditions imposed on the left hand side.

By Lemma 3.4, after a finite strict-étale refinement we may assume without loss of generality that every $\mathrm{U}_i = \mathrm{Spf}\,\mathrm{R}_i$ is a framed log affine. Put $\mathrm{U}_{ij} = \mathrm{U}_i \times_{(\mathrm{X}, \mathcal{M})} \mathrm{U}_j$ with projections p_i, p_j; it admits the structure of a framing through either U_i or U_j.

Proposition 4.7 gives distinguished minimal roots $\mathcal{R}_i \subset \rho_i^*\mathcal{E}$. Corollary 4.10, applied with each framing, makes $p_i^*\mathcal{R}_i$ and $p_j^*\mathcal{R}_j$ distinguished minimal roots on any open affine in $\mathcal{E}|_{\mathrm{U}_{ij}}$.

Since U_{ij} is quasicompact by virtue of X being quasiseparated, we may cover U_{ij} by finitely many such framed log affines.

Their intrinsic uniqueness in Proposition 4.7 gives

$$p_i^*\mathcal{R}_i = p_j^*\mathcal{R}_j \subset \mathcal{E}|_{\mathrm{U}_{ij}}.$$

Indeed by the remark after Corollary A.17 one has descent for coherent Laurent crystals $\mathsf{Coh}^\varphi((\mathrm{X},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge)$ with a lax Frobenius. Thus we can check agreement locally on the chosen finite open cover of U_{ij} by framed log affines, which also shows agreement of the root maps. These equalities also satisfy the cocycle condition on triple overlaps.

Strict-étale descent for $\mathsf{Coh}^\varphi((\mathrm{X},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge)_{p=0}$ and $\mathrm{D}^\varphi_\blacksquare((\mathrm{X},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge)_{p=0}^\heartsuit$ produces a global coherent root. Restricting any subroot or any other root to the cover shows that the descended root is again distinguished minimal. Injectivity and unitalization are local, so the unitalization of the descended coherent root is $\mathcal{E}$. On a log prism $(\mathrm{A},\mathrm{I},\mathrm{M})^a$, its two-term induced resolution is perfect over $(\mathrm{A}[1/\mathrm{I}]_p^\wedge/p)[\mathrm{F}]$. Hence $\mathcal{E}$ is finitely generated unit. ☐

Remark 4.12. Combined with Theorem 4.6, this shows that the finitely generated unit condition can equivalently be tested upon evaluation by a covering family of log q-de Rham prisms as claimed in the introduction.

Let us record an important fact we showed in the proof of the theorem.

Corollary 4.13. Let $(\mathrm{X},\mathcal{M})$ satisfy Assumption 3.1 over $\mathrm{W}(k)[\zeta_p]$. A crystal $\mathcal{E} \in \mathrm{D}_\blacksquare((\mathrm{X},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge)^{\varphi=1}$ is finitely generated unit if and only if it is t-bounded and, for every ℓ, the unit Laurent crystal $\mathrm{H}^\ell(\mathcal{E} \otimes^{\mathrm{L},\blacksquare} \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]/p)$ admits a global coherent root.

Proof. The forward implication is the construction in the proof of Theorem 4.11, applied to the cohomology sheaves of $\mathcal{E} \otimes^{\mathrm{L},\blacksquare} \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]/p$ using Proposition 4.7 locally. The converse follows from the first paragraph of the proof of Proposition 4.7 and Theorem 4.11. ☐

Remark 4.14. This is the Laurent analogue of the root criterion for finitely generated unit modules on a k-scheme X. By [EK04b, Theorem 6.1.3], a finitely generated unit $\mathcal{O}_{\mathrm{F},\mathrm{X}}$-module admits a coherent root on every quasicompact open; as X is quasicompact, the root is global. Such a root need not carry a crystal structure like in the previous corollary, but it turns out there always exists a root carrying such a structure. Cartier descent gives a fully faithful functor

$$\mathsf{QCoh}(\mathrm{X}^{(1)}) \to \mathsf{QCoh}((\mathrm{X}/k)_{\mathrm{cris}})$$

with essential image the crystals with vanishing p-curvature. If $\mathrm{Fr} : \mathrm{X} \to \mathrm{X}^{(1)}$ is relative Frobenius, it sends $\mathscr{F}$ to $(\mathrm{Fr}^*\mathscr{F}, \nabla_{\mathrm{can}})$, where $(\mathrm{Fr}^*\mathscr{F})^{\nabla_{\mathrm{can}}=0} \simeq \mathscr{F}$.
Write the absolute Frobenius as $\mathrm{F}_\mathrm{X} = \pi \circ \mathrm{Fr}$, with $\pi : \mathrm{X}^{(1)} \to \mathrm{X}$. Given a coherent root $\beta : \mathrm{M}_0 \to \mathrm{F}_\mathrm{X}^*\mathrm{M}_0$, replace it by

$$\mathrm{F}_\mathrm{X}^*\beta : \mathrm{F}_\mathrm{X}^*\mathrm{M}_0 \longrightarrow \mathrm{F}_\mathrm{X}^{*2}\mathrm{M}_0.$$

Since $\mathrm{F_X}$ is finite flat, this map is injective and has the same unitalization. Moreover, $\mathrm{F_X^*M_0} = \mathrm{Fr}^*\pi^*\mathrm{M_0}$ and $\mathrm{F_X^*}\beta = \mathrm{Fr}^*(\pi^*\beta)$, so the map is horizontal for the canonical Cartier connections. It is therefore a coherent crystalline root. Uniqueness in Cartier descent identifies its unitalization with the crystal structure on M (similarly) induced by being an $\mathcal{O}_\mathrm{X}$-module with unit Frobenius.

Corollary 4.15. Let $f : (\mathrm{Y}, \mathcal{N}) \to (\mathrm{X}, \mathcal{M})$ be a morphism between pairs satisfying Assumption 3.1 over $\mathrm{W}(k)[\zeta_p]$. Then $\mathrm{L}f^*$ preserves finitely generated unit crystals.

Proof. Let $\mathcal{E}$ be a finitely generated unit crystal on $(\mathrm{X}, \mathcal{M})_{\Delta}$, and let $(\mathrm{B}, \mathrm{I}, \mathrm{M_{Spf\,B}})^a$ be a log prism over $(\mathrm{Y}, \mathcal{N})$. Composition with f makes it a log prism over $(\mathrm{X}, \mathcal{M})$, and pullback of crystals gives a Frobenius-equivariant equivalence

$$(\mathrm{L}f^*\mathcal{E})(\mathrm{B}, \mathrm{I}, \mathrm{M_{Spf\,B}}) \simeq \mathcal{E}(\mathrm{B}, \mathrm{I}, \mathrm{M_{Spf\,B}}).$$

Thus every evaluation of $\mathrm{L}f^*\mathcal{E}$ is perfect over $\mathrm{B}[1/\mathrm{I}]_p^\wedge\langle\mathrm{F}\rangle$ by Definition 4.3. ☐

Theorem 4.11 and Lemma 3.4 permit strict-étale localization to framed log affines, where Theorem 4.6 describes finitely generated unit crystals explicitly.

4.3. **The Kummer-étale realization and constructibility.** We construct a functor

$$\mathrm{T_{\acute{e}t}} : \mathrm{D_{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge)^{\mathrm{op}} \to \mathrm{D}((\mathrm{X}, \mathcal{M})_{\eta,\mathrm{qpk\acute{e}t}}, \mathbf{Z}_p).$$

Its essential image consists of p-Kummer complexes constructible for the logarithmic stratification induced by D. We first note that we can use several equivalent formalisms to access the target category under our assumptions.

Lemma 4.16. Let $(\mathrm{X}, \mathcal{M})$ be a locally Noetherian fs log p-adic formal scheme. Then the natural comparison functors induce equivalences of topoi

$$(\mathrm{X}, \mathcal{M})_{\eta,\mathrm{prok\acute{e}t}} \simeq (\mathrm{X}, \mathcal{M})_{\eta,\mathrm{qpk\acute{e}t}} \simeq (\mathrm{X}, \mathcal{M})^{\diamond}_{\eta,\mathrm{qpk\acute{e}t}}.$$

Proof. Put $(\mathrm{Y}, \mathcal{M}_\mathrm{Y}) = (\mathrm{X}, \mathcal{M})_\eta$. By [DLLZ23, Propositions 5.3.11 and 5.3.12], the log affinoid perfectoid objects form a basis B of $\mathrm{Y_{prok\acute{e}t}}$ stable under fiber products. For $\mathrm{U} \in \mathrm{B}$, let $\underline{\mathrm{U}}$ be its associated affinoid perfectoid space. Its characteristic monoid is divisible and saturated by [DLLZ23, Definition 5.3.1 and Remark 5.3.3].

Lemma 5.3.8 of [DLLZ23], together with [Sch26, Lemma 15.6 and the proof of Proposition 14.8], gives functorial equivalences

$$(\mathrm{Y_{prok\acute{e}t}})_{/\mathrm{U}} \simeq \underline{\mathrm{U}}_{\mathrm{pro\acute{e}t}} \simeq \underline{\mathrm{U}}^{\diamond}_{\mathrm{qpro\acute{e}t}}.$$

The last term is the corresponding localization of the diamond quasi-pro-Kummer-étale topos by [KY25, Corollary 7.23]; the same basis computes the adic quasi-pro-Kummer-étale localization.

Saturated fiber products commute with passage to log diamonds by the proof of [KY25, Proposition 7.22], and quasi-pro-Kummer-étale morphisms are stable under saturated base change by [KY25, Proposition 7.16(2)]. Thus these equivalences identify fiber products and covering families on the three bases. The comparison lemma for sites proves the claim. ☐

Thus it suffices to define $\mathrm{T}_{\text{ét}}$ on log perfectoid test objects.

Definition 4.17. Let $(\mathrm{X},\mathcal{M})$ be a bounded fs log formal scheme, and let $\mathcal{E} \in \mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_{p})$. For a saturated log perfectoid space $(\mathrm{Spa}(\mathrm{S},\mathrm{S}^{+}),\mathcal{M}_{\mathrm{S}}) \to (\mathrm{X},\mathcal{M})^{\diamond}_{\eta}$, let $\mathbb{\Delta}_{\mathrm{S}^{+}}$ be the perfect saturated log prism with underlying prism $(\mathrm{W}(\mathrm{S}^{+,\flat}), \ker\theta)$ and log structure induced by the tilted log structure on $\mathrm{S}^{\flat}$. Set

$$\mathrm{T}_{\text{ét}}(\mathcal{E})(\mathrm{Spa}(\mathrm{S},\mathrm{S}^{+}),\mathcal{M}_{\mathrm{S}}) := \mathrm{RHom}_{\mathrm{D}^{\varphi}_{\blacksquare}(\mathrm{W}(\mathrm{S}^{\flat}))}(\mathcal{E}(\mathbb{\Delta}_{\mathrm{S}^{+}}), \mathrm{W}(\mathrm{S}^{\flat})) \in \widehat{\mathrm{D}}(\mathbf{Z}_{p}).$$

These assignments define a presheaf on the saturated log affinoid perfectoid basis. We denote its right Kan extension to the qpkét site by $\mathrm{T}_{\text{ét}}(\mathcal{E})$.

Here $\mathrm{D}_{\blacksquare}(\mathrm{W}(\mathrm{S}^{\flat}))$ is the inverse limit over n of the categories of $\mathrm{W}(\mathrm{S}^{\flat})/p^{n}$-modules in the ordinary solid category of the analytic ring $\mathrm{Spa}(\mathbb{A}_{\mathrm{inf}}(\mathrm{S}^{+})/p^{n}, \mathbb{A}_{\mathrm{inf}}(\mathrm{S}^{+})/p^{n})$. We remark that the perfect (log) prism we use has underlying ring

$$\mathrm{W}(\mathrm{S}^{+,\flat}) = \mathbb{A}_{\mathrm{inf}}(\mathrm{S}^{+}),$$

whereas its Laurent variant is

$$\mathrm{W}(\mathrm{S}^{\flat}) = \mathbb{A}_{\mathrm{inf}}(\mathrm{S}^{+})[1/\xi]^{\wedge}_{p},$$

with $(\xi) = \ker(\theta)$ the prismatic ideal in $\mathbb{A}_{\mathrm{inf}}(\mathrm{S}^{+})$ (one can see these are equal using that they are both strict p-rings). Thus $\mathcal{E}(\mathbb{\Delta}_{\mathrm{S}^{+}})$ in Definition 4.17 denotes the Laurent evaluation, which is naturally a module over $\mathrm{W}(\mathrm{S}^{\flat})$.

Lemma 4.18. Let $(\mathrm{X},\mathcal{M})$ be a bounded fs log formal scheme. Definition 4.17 yields a well-defined functor

$$\mathrm{T}_{\text{ét}} : \mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_{p})^{\mathrm{op}} \to \mathrm{D}((\mathrm{X},\mathcal{M})^{\diamond}_{\eta,\mathrm{qpk\acute{e}t}}, \mathbf{Z}_{p}).$$

Proof. It is enough to check hyperdescent on the saturated log affinoid perfectoid basis of $(\mathrm{X},\mathcal{M})^{\diamond}_{\eta,\mathrm{qpk\acute{e}t}}$. After refining by saturated log affinoid perfectoids with divisible characteristic monoid, the quasi-pro-Kummer-étale and quasi-pro-étale sites agree by [KY25, Corollary 7.23]. Thus let $(\mathrm{Spa}(\mathrm{S},\mathrm{S}^{+}),\mathcal{M}_{\mathrm{S}})$ be such a test object and let $(\mathrm{Spa}(\mathrm{S}_{\bullet},\mathrm{S}^{+}_{\bullet}),\mathcal{M}_{\mathrm{S}_{\bullet}}) \to (\mathrm{Spa}(\mathrm{S},\mathrm{S}^{+}),\mathcal{M}_{\mathrm{S}})$ be a quasi-pro-étale hypercover by objects of the same kind.

Put $\mathrm{A} = \mathrm{W}(\mathrm{S}^{\flat})$ and $\mathrm{A}_{m} = \mathrm{W}(\mathrm{S}^{\flat}_{m})$. The v-acyclicity of the tilted structure sheaf on affinoid perfectoid spaces [Sch26, Theorem 1.2] identifies the derived reductions modulo p of A and $\mathrm{R}\varprojlim_{m} \mathrm{A}_{m}$ (which agree with their ordinary mod p reductions). Both are derived p-complete, so derived Nakayama gives

$$\mathrm{A} \simeq \mathrm{R}\varprojlim_{[m]\in\Delta} \mathrm{A}_{m}$$

in $\mathrm{D}^{\varphi}_{\blacksquare}(\mathrm{A})$, where Frobenius compatibility follows from functoriality of Witt vectors.

Put $\mathrm{M} = \mathcal{E}(\mathbb{\Delta}_{\mathrm{S}^{+}})$ and $\mathrm{M}_{m} = \mathcal{E}(\mathbb{\Delta}_{\mathrm{S}^{+}_{m}})$. The crystal property shows M_{m} is simply a base change of M. Using $\mathrm{A} \simeq \mathrm{R}\varprojlim_{[m]\in\Delta} \mathrm{A}_{m}$, we obtain

$$\mathrm{R}\varprojlim_{[m]\in\Delta} \mathrm{RHom}_{\mathrm{D}^{\varphi}_{\blacksquare}(\mathrm{A}_{m})}(\mathrm{M}_{m},\mathrm{A}_{m}) \simeq \mathrm{RHom}_{\mathrm{D}^{\varphi}_{\blacksquare}(\mathrm{A})}(\mathrm{M},\mathrm{A}).$$

Thus $\mathrm{T}_{\text{ét}}(\mathcal{E})$ is a hypercomplete sheaf. $\square$

It is easy to see that for every morphism $f : (\mathrm{Y}, \mathcal{N}) \to (\mathrm{X}, \mathcal{M})$ between pairs satisfying Assumption 3.1, the construction is compatible with pullback:

$$\mathrm{T}_{\text{ét},\mathrm{Y}}(\mathrm{L}f^{*}\mathcal{E}) \simeq \mathrm{L}f_{\eta}^{*}\mathrm{T}_{\text{ét},\mathrm{X}}(\mathcal{E}).$$

The functor $\mathrm{T}_{\text{ét}}$ actually lands in the full subcategory of p-Kummer sheaves, which we now define.

Definition 4.19. Let $(\mathrm{X}, \mathcal{M})^{\diamond}$ be an fs log diamond. A complex $\mathrm{K} \in \mathrm{D}((\mathrm{X}, \mathcal{M})^{\diamond}_{\text{qpkét}}, \mathbf{Z}_p)$ is *p-Kummer* if, for every saturated log perfectoid $(\mathrm{Y}, \mathcal{M}_{\mathrm{Y}}) \to (\mathrm{X}, \mathcal{M})^{\diamond}$ admitting charts quasi-pro-étale locally and such that $\mathcal{M}_{\mathrm{Y}}/\mathcal{M}_{\mathrm{Y}}^{\times}$ is p-divisible, its pullback lies in the essential image of

$$\mathrm{D}(\mathrm{Y}_{\text{proét}}, \mathbf{Z}_p) \longrightarrow \mathrm{D}((\mathrm{Y}, \mathcal{M}_{\mathrm{Y}})_{\text{qpkét}}, \mathbf{Z}_p).$$

For a derived lisse complex, this agrees with [IKY26, Lemma 1 and Definition 2].

We next prove that $\mathrm{T}_{\text{ét}}(\mathcal{E})$ is constructible for the stratification induced by D. We first need the following descent result.

Lemma 4.20. On pairs satisfying Assumption 3.1 over $\mathcal{O}_K$, the full subcategories of p-Kummer sheaves in

$$\mathrm{D}^{(b)}_{\text{cons},\mathrm{D}}((\mathrm{X}, \mathcal{M})_{\eta}, \mathbf{Z}_p)$$

form a stack for the strict-étale topology.

Proof. Let $g : (\mathrm{U}, \mathcal{N}) \to (\mathrm{X}, \mathcal{M})$ be a strict-étale cover, and let $\mathrm{U}^{(\bullet)}$ be its Čech nerve. Passage to the log adic generic fiber preserves fiber products and carries g to a strict-étale cover. Descent for quasi-pro-Kummer-étale sheaves gives

$$\mathrm{D}^{(b)}((\mathrm{X}, \mathcal{M})_{\eta,\text{qpkét}}, \mathbf{Z}_p) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}^{(b)}((\mathrm{U}^{(m)}, \mathcal{N}^{(m)})_{\eta,\text{qpkét}}, \mathbf{Z}_p).$$

Here local boundedness descends because it is strict-étale local. By strictness, if $\mathrm{X}_{\eta}^{(a)}$ and $\mathrm{U}_{\eta}^{(a)}$ denote the loci where $\mathrm{rk}_{\mathbf{Z}}\,\overline{\mathcal{M}}_{\bar{x}}^{\text{gp}} = a$ (respectively for $\mathcal{N}$), then

$$\mathrm{U}_{\eta}^{(a)} = \mathrm{U}_{\eta} \times_{\mathrm{X}_{\eta}} \mathrm{X}_{\eta}^{(a)}.$$

The logarithmic strata are therefore compatible, reducing descent for $\mathrm{D}^{(b)}_{\text{cons},\mathrm{D}}((\mathrm{X}, \mathcal{M})_{\eta}, \mathbf{Z}_p)$ to descent for lisse sheaves.

It remains to check the claim still holds when we restrict to p-Kummer sheaves. By [IKY26, Proposition 5], a derived local system is p-Kummer if and only if prime-to-p logarithmic inertia acts trivially on its cohomology at every geometric log point. However strictness canonically identifies the logarithmic inertia groups, so the same reduction to lisse sheaves shows the claim. $\square$

By Lemmas 3.4 and 4.20, constructibility for the logarithmic stratification induced by D may be checked on a framed log affine $\operatorname{Spf}\mathrm{R}$. Let $\mathrm{R}_0^{\square}[\![q-1]\!]_{\text{perf}}$ be the completed perfection of the framed log q-de Rham prism. If $\operatorname{Spf} \mathrm{R}_m$ is the p-completed pullback of the framing along the Kummer map $\mathrm{T}_i \mapsto \mathrm{T}_i^{p^m}$ over $\mathrm{W}(k)[\zeta_{p^m}]$, then

$$(\mathrm{R}_0^\square)_\infty := \mathrm{R}_0^\square[\![q-1]\!]_{\mathrm{perf}}/([p]_q) = \left(\varinjlim_m \mathrm{R}_m\right)_p^\wedge .$$

Its log structure $\mathcal{M}_\infty$ is induced by the saturated chart $\mathbf{N}[1/p]^r \to (\mathrm{R}_0^\square)_\infty$ sending e_i/p^m to T_i^{1/p^m} for $1 \le i \le r$. Write $\mathrm{X}_{\infty,\eta} = \operatorname{Spa}((\mathrm{R}_0^\square)_\infty[1/p], (\mathrm{R}_0^\square)_\infty)$. We use ϖ for the tilted pseudouniformizer, and write $(\mathrm{R}_0^\square)_{\infty,\eta}^\flat = ((\mathrm{R}_0^\square)_\infty)^\flat[1/\varpi]$.

For $m \ge 1$, the Γ-action extends to this tower, for $1 \le i \le n$, by

$$\gamma(\mathrm{T}_i^{1/p^m}) = q^{pb_i/p^m}\mathrm{T}_i^{1/p^m} = \zeta_{p^m}^{b_i}\mathrm{T}_i^{1/p^m}.$$

Lemma 4.21. Let $(\operatorname{Spf} \mathrm{R}, \mathcal{M})$ be a framed log affine and let $\mathrm{K} \in \mathrm{D}^{(b)}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\eta,\mathrm{qpk\acute{e}t}}, \mathbf{Z}_p)$. For $\Lambda \subseteq \{1, \dots, r\}$, put

$$\mathrm{X}_{\infty,\eta,\Lambda} = \mathrm{X}_{\infty,\eta} \times_{\mathrm{X}_\eta} \mathrm{X}_{\eta,\Lambda},$$

and let $\epsilon_\infty : (\mathrm{X}_{\infty,\eta}, \mathcal{M}_\infty)_{\mathrm{qpk\acute{e}t}} \to (\mathrm{X}_{\infty,\eta})_{\mathrm{pro\acute{e}t}}$ be the natural morphism of sites. The following are equivalent:

(1) K is constructible for the stratification induced by D and is p-Kummer;
(2) its pullback to $\mathrm{X}_{\infty,\eta}$ is $\epsilon_\infty^*\mathcal{L}_\infty$ for some $\mathcal{L}_\infty \in \mathrm{D}^{(b)}((\mathrm{X}_{\infty,\eta})_{\mathrm{pro\acute{e}t}}, \mathbf{Z}_p)$ which is constructible for the finite stratification $\mathrm{X}_{\infty,\eta} = \bigsqcup_\Lambda \mathrm{X}_{\infty,\eta,\Lambda}$.

Proof. We first note that ϵ_∞^* is fully faithful on bounded complexes constructible for the pullback stratification on $\mathrm{X}_{\infty,\eta}$. If S is a stratum and $\mathbb{L}$ is derived lisse on S, then the projection formula and [KY25, Lemma 6.9], applied on the saturated log affinoid perfectoid basis, give

$$\mathrm{R}\epsilon_{\infty,S,*}\epsilon_{\infty,S}^*(\mathbb{L}/p^n) \simeq (\mathbb{L}/p^n) \otimes_{\mathbf{Z}/p^n}^{\mathrm{L}} \mathrm{R}\epsilon_{\infty,S,*}\mathbf{Z}/p^n \simeq \mathbb{L}/p^n.$$

One can use recollement to generalize to constructible sheaves, and then derived p-completeness gives the assertion.

Suppose that (1) holds. The derived form of [IKY26, Lemma 1], applied on each stratum, shows that the pullback of K lies stratumwise in the image of ϵ_∞^*. Full faithfulness lifts the extension maps in the finite recollement filtration, giving a constructible $\mathbb{L}_\infty$ and an equivalence

$$\epsilon_\infty^*\mathbb{L}_\infty \simeq \pi^*\mathrm{K}.$$

Thus (2) holds. Here $\pi : \mathrm{X}_{\infty,\eta} \to \mathrm{X}_\eta$ is the natural map.

Conversely, suppose that (2) holds. Derived lissity descends along π on each stratum, so K is constructible. The logarithmic inertia at $\bar{x}_\infty$ acts trivially on the cohomology of $\epsilon_\infty^*\mathbb{L}_\infty$. At a geometric point $\bar{x}$ of $\mathrm{X}_{\eta,\Lambda}$ and a lift $\bar{x}_\infty$ to $\mathrm{X}_{\infty,\eta}$, the map on characteristic groups is $\mathbf{Z}^{|\Lambda|} \to \mathbf{Z}[1/p]^{|\Lambda|}$. The cyclotomic tower and the roots of the remaining coordinates are strict pro-étale, so

$$\mathrm{I}_{\bar{x}_\infty}^{\log} \simeq \operatorname{Hom}\left(\mathbf{Z}[1/p]^{|\Lambda|}, \widehat{\mathbf{Z}}(1)\right) \simeq \mathrm{I}_{\bar{x}}^{\log,(p')}.$$

Thus the prime-to-p logarithmic inertia at $\bar{x}$ acts trivially on the cohomology of K. By [IKY26, Proposition 5], applied to the cohomology local systems, K is p-Kummer. $\square$

Let $\mathrm{T}_{\text{ét}}^{\text{perf}}$ denote the scheme-theoretic solution functor on $\mathrm{Spec}(\mathrm{R}_0^{\square})^{\flat}_{\infty,\eta}$ of Theorem 2.6. For a finitely generated unit module M over $\mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$, we use the convention

$$\mathrm{T}_{\text{ét}}^{\text{perf}}(\mathrm{M}) := \mathrm{T}_{\text{ét}}^{\text{perf}}\left(\mathrm{M} \otimes^{\mathrm{L},\blacksquare}_{\mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}} \mathrm{W}((\mathrm{R}_0^{\square})^{\flat}_{\infty,\eta})\right),$$

For $\mathrm{M} \in \mathrm{D}_{\text{fgu}}(\mathrm{W}((\mathrm{R}_0^{\square})^{\flat}_{\infty,\eta}))$, its value after base change to $\mathrm{Spec}\,\mathrm{S}^{\flat}$ is

$$\mathrm{RHom}_{\widehat{\mathrm{D}}^{\varphi}(\mathrm{W}(\mathrm{S}^{\flat}))}\left(\left(\mathrm{M} \otimes^{\mathrm{L}}_{\mathrm{W}((\mathrm{R}_0^{\square})^{\flat}_{\infty,\eta})} \mathrm{W}(\mathrm{S}^{\flat})\right)^{\wedge}_{p}, \mathrm{W}(\mathrm{S}^{\flat})\right) \in \mathrm{D}(\mathbf{Z}_p).$$

The following lemma controls the Fitting ideals on each stratum. Using the minimal root characterization of being a finitely generated unit module, this will be crucial in sufficiently constraining $\mathsf{Mod}_{\text{fgu}}(\mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge})^{h\Gamma}$ enough to show the essential image of $\mathrm{T}_{\text{ét}}$ is constructible.

Lemma 4.22. Let $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ be a framed log affine with $\overline{\mathrm{R}} = \mathrm{R}_0/p$. If $\mathfrak{a} \subset \overline{\mathrm{R}}^{\square}(\!(q-1)\!)$ is stable under $\mathbf{Z}_p(1)^n \subset \Gamma$, then, on each connected component of

$$\mathrm{Spa}\,\overline{\mathrm{R}}^{\square}(\!(q-1)\!) \setminus \mathrm{V}(\mathrm{T}_1 \cdots \mathrm{T}_r)$$

the image of $\mathfrak{a}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]$ is either zero or the unit ideal.

Proof. Put $\mathrm{X} = \mathrm{Spa}\,\overline{\mathrm{R}}^{\square}(\!(q-1)\!)$ and $\mathrm{X}^{\circ} = \mathrm{X} \setminus \mathrm{V}(\mathrm{T}_1 \cdots \mathrm{T}_r)$. The framing gives an étale map

$$\mathrm{X} \longrightarrow \mathrm{Spa}\,k(\!(q-1)\!)\langle \mathrm{T}_1, \ldots, \mathrm{T}_r, \mathrm{T}_{r+1}^{\pm 1}, \ldots, \mathrm{T}_n^{\pm 1}\rangle.$$

The target is reduced, hence so is X, and every T_i is a unit on X°. The ith copy of $\mathbf{Z}_p(1)$ acts by $\mathrm{T}_i \mapsto q^{pb_i}\mathrm{T}_i$ and fixes the other coordinates.

Put $\mathcal{I} = \mathfrak{a}\mathcal{O}_{\mathrm{X}^{\circ}}$. Let x be a classical point at which $\mathcal{I}_x$ is proper. After replacing $k(\!(q-1)\!)$ by a finite extension containing the residue field of x, choose a rational lift $\widetilde{x}$ of x. The rigid analytic inverse function theorem identifies a neighborhood of $\widetilde{x}$ with a polydisk. We note vanishing may be checked after passage to this extension. Taking the elements of $\mathbf{Z}_p(1)^n \leq \Gamma$ with coordinates $b_i = p^{m_i - 1}$ for integers m_i, stability gives zeros of every $f \in \mathfrak{a}$ at the points with coordinates $q^{p^{m_i}}\mathrm{T}_i(\widetilde{x})$ for large $m_i \geq 1$ lying in this polydisk. Since

$$q^{p^m} - 1 = (q-1)^{p^m} \longrightarrow 0.$$

these points form a product of convergent sequences.

Successive applications of the single variable rigid analytic identity theorem, as in [Pen25, Lemma 3.8], shows that a finite set of generators of $\mathfrak{a}$ vanishes on a neighborhood of $\widetilde{x}$. Faithful flatness gives $\mathcal{I}_x = 0$; if $\mathcal{I}_x$ is not proper, then $\mathcal{I}_x = \mathcal{O}_{\mathrm{X}^{\circ},x}$.

The closed analytic locus

$$\mathrm{Supp}(\mathcal{I}) \cap \mathrm{Supp}(\mathcal{O}_{\mathrm{X}^{\circ}}/\mathcal{I})$$

has no classical points. If it were nonempty, a point in it would admit an affinoid neighborhood in X°. Its intersection with this neighborhood would be a nonempty closed analytic locus and hence would contain a classical point by the affinoid Nullstellensatz [BGR84, 5.2.6/3 and 7.1.3/1], a contradiction. The zero and unit loci of $\mathcal{I}$ are therefore complementary open and closed subsets of X°. On each connected component, $\mathcal{I}$ is zero or the unit ideal. $\square$

Lemma 4.23. Let $\mathrm{A} = \overline{\mathrm{R}}^{\square}(\!(q-1)\!)$ for a framed log affine, and let $\mathrm{J} = (\mathrm{T}_i \mid i \in \Lambda)$ for some $\Lambda \subset \{1, \dots, r\}$. If $\mathrm{M} \in \mathrm{D}_{\mathrm{fgu}}(\mathrm{A})^{h\Gamma}$, then

$$(\mathrm{A}/\mathrm{J}) \otimes^{\mathrm{L}}_{\mathrm{A}} \mathrm{M}$$

has bounded finitely generated unit cohomology over A/J and carries the induced Γ-action.

Proof. Frobenius preserves J, and extension of scalars gives

$$(\mathrm{A}/\mathrm{J})[\mathrm{F}] \otimes^{\mathrm{L}}_{\mathrm{A}[\mathrm{F}]} \mathrm{M} \simeq (\mathrm{A}/\mathrm{J}) \otimes^{\mathrm{L}}_{\mathrm{A}} \mathrm{M}.$$

The resulting complex is unit because Frobenius pullback commutes with extension of scalars. It is perfect over (A/J)[F], so its underlying A/J-complex is bounded. Since A/J is regular, Noetherian, and F-finite, [BL19, Proposition 11.3.9] makes its cohomology modules finitely generated unit. Finally, J is Γ-stable and Γ commutes with Frobenius, so the quotient map is Γ-equivariant. □

Proposition 4.24. Let $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M}_{\mathrm{Spf}\,\mathrm{R}})$ be a framed log affine with framing □, and let $\mathcal{E}$ be an object of $\mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M}_{\mathrm{Spf}\,\mathrm{R}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)$. Then $\mathrm{T}^{\mathrm{perf}}_{\text{ét}}(\mathcal{E}(\mathrm{R}^{\square}_0(\!(q-1)\!)^{\wedge}_p))$ is constructible for the stratification of $\mathrm{Spec}(\mathrm{R}^{\square}_0)^{\flat}_{\infty,\eta}$ cut out by the divisor $\mathrm{D}_{\infty} = \bigcup_i \mathrm{V}(\mathrm{T}^{\flat}_i)$.

Proof. The evaluated crystal is perfect over $\mathrm{W}((\mathrm{R}^{\square}_0)^{\flat}_{\infty,\eta})\langle \mathrm{F} \rangle$. Consequently,

$$\mathrm{T}^{\mathrm{perf}}_{\text{ét}}(\mathcal{E}(\mathrm{R}^{\square}_0(\!(q-1)\!)^{\wedge}_p)) \otimes^{\mathrm{L}}_{\mathbf{Z}_p} \mathbf{F}_p \simeq \mathrm{T}^{\mathrm{perf}}_{\text{ét}}((\mathcal{E}/p)(\mathrm{R}^{\square}_0(\!(q-1)\!)^{\wedge}_p)).$$

Constructibility is detected modulo p, so assume that $\mathcal{E}$ is p-torsion and write $\mathrm{A} = \overline{\mathrm{R}}^{\square}_0(\!(q-1)\!)$. Evaluation identifies each cohomology module with a finitely generated unit Γ-module M. Let $\beta : \mathrm{M}_0 \to \varphi^*\mathrm{M}_0$ be its distinguished minimal coherent root. It is Γ-equivariant by the distinguished property.

Each $\mathrm{Fitt}_i(\mathrm{M}_0)$ is $\mathbf{Z}_p(1)^n$-stable. On a connected component of the open stratum $\mathrm{Spec}\,\overline{\mathrm{R}}^{\square}_0(\!(q-1)\!)[1/\mathrm{T}_1, \dots, 1/\mathrm{T}_r]$, let ρ be the least index for which the restriction of $\mathrm{Fitt}_{\rho}(\mathrm{M}_0)$ is the unit ideal (we see $\mathrm{Fitt}_{-1} = 0$). After restriction to this component, Lemma 4.22 gives $\mathrm{Fitt}_{\rho-1}(\mathrm{M}_0) = 0$ and $\mathrm{Fitt}_{\rho}(\mathrm{M}_0) = \mathcal{O}$. The Fitting ideal criterion [Sta26, Lemma 15.8.8, Tag 07ZD] makes M_0 locally free of rank ρ there. We note $\varphi^*\mathrm{M}_0$ has the same rank.

Thus β is an injection between vector bundles of the same rank. Moreover β is Γ-equivariant, so by the same argument $\mathrm{coker}\,\beta$ must again be a vector bundle. This is forced to be rank zero, and we conclude that β is an isomorphism: in particular restricted to the open stratum $\mathrm{M} = \mathrm{M}_0$. It follows $\mathrm{T}^{\mathrm{perf}}_{\text{ét}}(\mathcal{E}(\mathrm{R}^{\square}_0(\!(q-1)\!)^{\wedge}_p))$ is lisse on the open stratum by (the dualized version of) Theorem 2.1. One can make a similar argument for any locally closed stratum, by first taking an appropriate quotient of $\overline{\mathrm{R}}^{\square}_0(\!(q-1)\!)$ before inverting any of the T_i (using Lemma 4.23). □

Remark 4.25. For the trivial log structure there is only the open stratum, so the same argument gives a derived lisse complex.

Let $\nu : (\mathrm{X}^{\flat}_{\infty,\eta})_{\text{ét}} \to \mathrm{Spec}((\mathrm{R}_0^{\square})^{\flat}_{\infty,\eta})_{\text{ét}}$ be the support map of [Bha25, Notation 6.2.1], and let $\lambda : (\mathrm{X}^{\flat}_{\infty,\eta})_{\text{proét}} \to (\mathrm{X}^{\flat}_{\infty,\eta})_{\text{ét}}$ be the natural morphism. For a constructible $\mathbf{Z}_p$-complex K, set

$$\mathrm{an}^*\mathrm{K} := \mathrm{R}\varprojlim_m \lambda^*\nu^*(\mathrm{K}/p^m) \in \mathrm{D}((\mathrm{X}^{\flat}_{\infty,\eta})_{\text{proét}}, \mathbf{Z}_p).$$

For a Γ-equivariant constructible complex K, set

$$\mathrm{an}^*_{\Gamma}\mathrm{K} := \left(\epsilon^*_{\infty}((\mathrm{an}^*\mathrm{K})^{\flat})\right)^{h\Gamma} \in \mathrm{D}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\eta,\text{qpkét}}, \mathbf{Z}_p),$$

where $(-)^{\flat}$ denotes transport across the tilting equivalence and $h\Gamma$ denotes descent along $\mathrm{X}_{\infty,\eta} \to \mathrm{X}_{\eta}$.

Lemma 4.26. Let $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ be a framed log affine with framing $\square : (\mathrm{Spf}\,\mathrm{R}, \mathcal{M}) \to \mathbb{T}^n_r$. Evaluation on $\Delta_{(\mathrm{R}_0^{\square})_{\infty}}$ gives a natural commutative diagram

$$\begin{tikzcd} \mathrm{D}_{\text{fgu}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)^{\mathrm{op}} \arrow[r, "\mathrm{T}_{\text{ét}}"] \arrow[d, "\mathcal{E}\mapsto\mathcal{E}(\Delta_{(\mathrm{R}_0^{\square})_{\infty}})"'] & \mathrm{D}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\eta,\text{qpkét}}, \mathbf{Z}_p) \\ \left(\mathrm{D}_{\text{fgu}}(\mathrm{W}((\mathrm{R}_0^{\square})^{\flat}_{\infty,\eta}))^{h\Gamma}\right)^{\mathrm{op}} \arrow[r, "\mathrm{T}^{\text{perf}}_{\text{ét}}"] & \mathrm{D}^{(b)}_{\text{cons}}(\mathrm{Spec}(\mathrm{R}_0^{\square})^{\flat}_{\infty,\eta}, \mathbf{Z}_p)^{h\Gamma} \arrow[u, "\mathrm{an}^*_{\Gamma}"'] \end{tikzcd}$$

Proof. Put $\mathrm{A}_{\infty} = \mathrm{W}((\mathrm{R}_0^{\square})^{\flat}_{\infty,\eta})$, $\mathrm{A}_{\mathrm{S}} = \mathrm{W}(\mathrm{S}^{\flat})$, and $\mathrm{U} = (\mathrm{Spa}(\mathrm{S}, \mathrm{S}^+), \mathcal{M}_{\mathrm{S}})$. By Lemma 4.2, choose M_{∞} with $\mathrm{M}_{\infty}(*) \simeq \mathcal{E}(\Delta_{(\mathrm{R}_0^{\square})_{\infty}})$. Since $(\mathrm{X}_{\infty,\eta}, \mathcal{M}_{\infty}) \to (\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\eta}$ is the Γ-cover defining $h\Gamma$-descent, it suffices to compare the two paths on U before descent. The base-change formula for $\mathrm{T}^{\text{perf}}_{\text{ét}}$, the definition of an^*, tilting, and $((\mathrm{X}_{\infty,\eta}, \mathcal{M}_{\infty})_{\text{qpkét}})_{/\mathrm{U}} \simeq \mathrm{Spa}(\mathrm{S}, \mathrm{S}^+)_{\text{proét}}$ give

$$\epsilon^*_{\infty}((\mathrm{an}^*\mathrm{T}^{\text{perf}}_{\text{ét}}(\mathrm{M}_{\infty}))^{\flat})(\mathrm{U}) \simeq \mathrm{RHom}_{\widehat{\mathrm{D}}^{\varphi}(\mathrm{A}_{\mathrm{S}})}\left(\left(\mathrm{M}_{\infty} \otimes^{\mathrm{L}}_{\mathrm{A}_{\infty}} \mathrm{A}_{\mathrm{S}}\right)^{\wedge}_p, \mathrm{A}_{\mathrm{S}}\right).$$

On the other hand, Definition 4.17 gives

$$\mathrm{T}_{\text{ét}}(\mathcal{E})(\mathrm{U}) \simeq \mathrm{RHom}_{\mathrm{D}^{\varphi}_{\blacksquare}(\mathrm{A}_{\mathrm{S}})}\left(\mathcal{E}(\Delta_{\mathrm{S}^+}), \mathrm{A}_{\mathrm{S}}\right).$$

The comparison functor of Lemma 4.2 and the crystal property give

$$\left(\left(\mathrm{M}_{\infty} \otimes^{\mathrm{L}}_{\mathrm{A}_{\infty}} \mathrm{A}_{\mathrm{S}}\right)^{\wedge}_p\right)(*) \simeq \mathrm{M}_{\infty}(*) \otimes^{\mathrm{L},\blacksquare}_{\mathrm{A}_{\infty}(*)} \mathrm{A}_{\mathrm{S}}(*) \simeq \mathcal{E}(\Delta_{\mathrm{S}^+}).$$

The first equivalence holds on the generator $\mathrm{A}_{\infty}\langle \mathrm{F}\rangle$, where both sides are $\mathrm{A}_{\mathrm{S}}(*)\langle \mathrm{F}\rangle$, and hence on its thick closure. Moreover $0 \to \mathrm{A}_{\mathrm{S}}\langle \mathrm{F}\rangle \xrightarrow{\cdot(\mathrm{F}-1)} \mathrm{A}_{\mathrm{S}}\langle \mathrm{F}\rangle \to \mathrm{A}_{\mathrm{S}} \to 0$ places A_{S} in this thick closure. Lemma 4.2 now identifies the two mapping complexes.

These equivalences are natural under log prism automorphisms and hence are Γ-equivariant. It follows on the saturated log affinoid perfectoid basis that the two paths agree; taking $h\Gamma$ proves the claim. □

Remark 4.27. The factorization in Lemma 4.26 shows that Definition 4.17 is symmetric monoidal. For $\mathrm{T}^{\text{perf}}_{\text{ét}}$, the mod-$p$ tensor comparison follows from [BL19, Theorem 8.4.1] and its identification with the Emerton-Kisin solution functor in [BL19, Theorems 12.5.4

and 12.6.1]; the integral statement follows by derived p-completeness. Analytification, tilting, and descent are symmetric monoidal.

The same factorization controls the essential image of $\mathrm{T}_{\text{ét}}$.

Proposition 4.28. Every object in the essential image of $\mathrm{T}_{\text{ét}}$ on $\mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\mathrm{op}}$ is a p-Kummer sheaf in $\mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}}((\mathrm{X},\mathcal{M})_\eta,\mathbf{Z}_p)$.

Proof. The source satisfies strict-étale descent by Theorem 4.11, while Lemma 4.20 shows that being p-Kummer and constructible is strict-étale local. Lemma 3.4 therefore reduces the claim to a framed log affine.

Put $\mathscr{F}=\mathrm{T}^{\mathrm{perf}}_{\text{ét}}(\mathcal{E}(\Delta_{(\mathrm{R}_0^{\square})_\infty}))$. By Proposition 4.24, it is constructible for the stratification cut out by $\bigcup_i \mathrm{V}(\mathrm{T}_i^\flat)$. Analytification and tilting give a pro-étale complex whose restriction to every stratum cut out by $\bigcup_i \mathrm{V}(\mathrm{T}_i)$ is derived lisse. Lemma 4.26 gives

$$\pi^*\mathrm{T}_{\text{ét}}(\mathcal{E})\simeq \epsilon_\infty^*((\mathrm{an}^*\mathscr{F})^\flat).$$

Finally, Lemma 4.21 proves the claim. □

5. The main equivalence

We now prove that

$$\mathrm{T}_{\text{ét}}:\mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\mathrm{op}}\longrightarrow \mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}}((\mathrm{X},\mathcal{M})_\eta,\mathbf{Z}_p)$$

is fully faithful, with essential image the p-Kummer complexes. Theorem 4.11 and Lemmas 3.4 and 4.20 reduce the assertion to framed affines, where Theorem 4.6 describes the source by (φ,Γ)-modules.

The first step constructs crystals realizing $j_!\mathbb{L}$ for pro-étale $\mathbf{Z}_p$-local systems on $\mathrm{U}_\eta=\mathrm{X}_\eta\setminus\mathrm{D}_\eta$. It begins with $\mathcal{O}_{\Delta}(*\mathrm{D})$, whose realization is $j_!\mathbf{Z}_p$, where

$$j_!:\mathsf{Loc}_{\mathbf{Z}_p}(\mathrm{U}_\eta)\to \mathrm{D}((\mathrm{X},\mathcal{M})_{\eta,\mathrm{qpkét}},\mathbf{Z}_p).$$

Lemma 4.26 reduces full faithfulness to that of analytification, supplied by Bhatt-Lurie's prismatic Riemann-Hilbert functor [Bha25].

For essential surjectivity on the open stratum, the basic strategy is as follows. Lütkebohmert's theorem extends $\mathbb{L}$ to a Kummer-étale local system $\mathbb{L}'$ on $(\mathrm{X},\mathcal{M})_\eta$; after this a finite prime-to-p Kummer Galois cover $\pi:\mathrm{Y}_\eta\to\mathrm{X}_\eta$ kills its prime-to-p monodromy. This allows us to apply Theorem 7 of [IKY26], which produces a crystal $\mathcal{E}_{\mathrm{Y}}$ for $\pi^*\mathbb{L}'$; tensoring with $\mathcal{O}_{\Delta}(*\mathrm{D}_{\mathrm{Y}})$ realizes $j_{\mathrm{Y}_\eta,!}\pi^*_{\mathrm{U}_\eta}\mathbb{L}$, which we prove descends to a crystal realizing $j_!\mathbb{L}$.

We next prove a Γ-equivariant variant of Kashiwara's lemma, reducing to [EK04b]. This gives the induction on dimension, apart from full faithfulness for maps between open and closed strata. For essential surjectivity, the target is thickly generated by $j_!\mathbb{L}$ for inclusions j of locally closed strata, and the preceding constructions realize these generators.

5.1. **The open stratum.** We construct $\mathcal{O}_{\Delta}(*\mathrm{D})$ from the line bundle with section attached to the log structure, following the analogous log-crystalline construction of [Tsu99, §5]. Assume $(\mathrm{X}, \mathcal{M})$ satisfies Assumption 3.1 over $\mathrm{W}(k)[\zeta_p]$

For a sheaf of groups G, a right G-torsor P, and a left G-sheaf V, we write $\mathrm{P} \times^{\mathrm{G}} \mathrm{V}$ for the contracted product, with $(pg, v) \sim (p, gv)$.

Definition 5.1. Let $\mathcal{M}(-\mathrm{D})$ be the $\mathcal{O}_{\mathrm{X}}^{\times}$-torsor of local sections $m \in \mathcal{M}$ such that $\alpha_{\mathrm{X}}(m)\mathcal{O}_{\mathrm{X}} = \mathcal{O}_{\mathrm{X}}(-\mathrm{D})$. Let $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}})^a$ be a saturated log prism in $(\mathrm{X}, \mathcal{M})_{\Delta}$, with structure map

$$f : (\mathrm{Spf\,A/I}, \mathrm{M}_{\mathrm{Spf\,A/I}}) \to (\mathrm{X}, \mathcal{M}).$$

Let $i : \mathrm{Spf\,A/I} \to \mathrm{Spf\,A}$ be the closed immersion. On $(\mathrm{Spf\,A/I})_{\text{ét}}$, set $f^*\mathcal{M}(-\mathrm{D}) := f^{-1}\mathcal{M}(-\mathrm{D}) \times^{f^{-1}\mathcal{O}_{\mathrm{X}}^{\times}} \mathcal{O}_{\mathrm{Spf\,A/I}}^{\times}$. The log structure map gives $f^*\mathcal{M}(-\mathrm{D}) \to \mathrm{M}_{\mathrm{Spf\,A/I}}$. On $(\mathrm{Spf\,A})_{\text{ét}}$, set $\widetilde{\mathcal{M}}_{\mathrm{A}}(-\mathrm{D}) := \mathrm{M}_{\mathrm{Spf\,A}} \times_{i_*\mathrm{M}_{\mathrm{Spf\,A/I}}} i_*f^*\mathcal{M}(-\mathrm{D})$. Define

$$\mathcal{O}_{\Delta}(-\mathrm{D})(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}}) := \Gamma\left(\mathrm{Spf\,A}, \widetilde{\mathcal{M}}_{\mathrm{A}}(-\mathrm{D}) \times^{\mathcal{O}_{\mathrm{Spf\,A}}^{\times}} \mathcal{O}_{\mathrm{Spf\,A}}\right).$$

From this, we can construct an important finitely generated unit crystal corresponding to $j_!\mathbf{Z}_p$ under $\mathrm{T}_{\text{ét}}$.

Definition 5.2. For $n \geq 0$, set

$$\mathcal{O}_{\Delta}(n\mathrm{D}) := \underline{\mathrm{Hom}}_{\mathcal{O}_{\Delta}}\left(\mathcal{O}_{\Delta}(-\mathrm{D})^{\otimes^{\blacksquare} n}, \mathcal{O}_{\Delta}\right).$$

The map $\mathcal{O}_{\Delta}(-\mathrm{D}) \to \mathcal{O}_{\Delta}$ gives transition maps $\mathcal{O}_{\Delta}(n\mathrm{D}) \to \mathcal{O}_{\Delta}((n+1)\mathrm{D})$. We define

$$\mathcal{O}_{\Delta}(*\mathrm{D}) := \left((\mathrm{colim}_{n\geq 0}\, \mathcal{O}_{\Delta}(n\mathrm{D}))_p^{\wedge}\,[1/\mathrm{I}_{\Delta}]\right)_p^{\wedge}.$$

Strict-étale locally on a framed log affine, if D is cut out by parameters $\mathrm{T}_1, \ldots, \mathrm{T}_r$, then $\mathcal{O}_{\Delta}(-\mathrm{D})$ is trivial and its map to $\mathcal{O}_{\Delta}$ is multiplication by a lift of $\mathrm{T}_1 \cdots \mathrm{T}_r$. Hence $\mathcal{O}_{\Delta}(*\mathrm{D})$ evaluates on the corresponding log q-de Rham prism as $\mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]_p^{\wedge}$. Logarithmic Frobenius carries the canonical section to its pth power up to a unit, so Frobenius on $\mathcal{O}_{\Delta}$ extends to $\mathcal{O}_{\Delta}(*\mathrm{D})$.

Since $\mathcal{O}_{\Delta}(*\mathrm{D})$ is a sheaf of rings, we can consider $\mathsf{Perf}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1}$.

This category has the following local description. On a framed log affine, put $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$ and $\mathrm{A}^{\circ} = \mathrm{A}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]_p^{\wedge}$. If $\mathrm{A}^{(m)}$ is the degree m term of the Laurentized log q-de Rham Čech nerve, set

$$\mathrm{A}^{\circ,(m)} = \mathrm{A}^{(m)} \otimes_{\mathrm{A}}^{\mathrm{L},\blacksquare} \mathrm{A}^{\circ} \simeq \mathrm{A}^{(m)}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]_p^{\wedge}.$$

Here the T_i denote their pullbacks to $\mathrm{A}^{(m)}$; strictness makes the resulting localization independent of the projection to the degree zero term. These rings are the evaluations of $\mathcal{O}_{\Delta}(*\mathrm{D})$ on the Čech nerve. Lemma A.15 and the compatibility of module objects with limits give

$$\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D})) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{A}^{\circ,(m)}).$$

Here Perf denotes dualizable objects. Dualizability is detected termwise in a limit of symmetric monoidal categories, so this equivalence restricts to

$$\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}(*\mathrm{D}))\simeq \varprojlim_{[m]\in\Delta}\mathsf{Perf}(\mathrm{A}^{\circ,(m)}).$$

Applying $\varphi=1$, since this is an equalizer of stable ∞-categories, gives the analogous description.

Lemma 5.3. We have $\mathcal{O}_{\Delta}(*\mathrm{D})\in\mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)$. Moreover any crystal in $\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1}$ is also in $\mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)$.

Proof. This can be checked strict-étale locally by Theorem 4.11 using the log q-de Rham prism $(\mathrm{R}_0^{\square}[\![q-1]\!],[p]_q,\mathrm{M})^a$; set $\mathrm{A}=\mathrm{R}_0^{\square}(\!(q-1)\!)^{\wedge}_p$. Then we can apply the criterion in Theorem 4.6 to check the finitely generated unit condition. On a framed chart where D is defined by $\mathrm{T}_1\cdots\mathrm{T}_r=0$, the mod p coherent root is generated by $(\mathrm{T}_1\cdots\mathrm{T}_r)^{-1}$, and Frobenius carries it to the root generated by $(\mathrm{T}_1\cdots\mathrm{T}_r)^{-p}$. Its unitalization is precisely $\mathrm{A}[1/\mathrm{T}_1,\dots,1/\mathrm{T}_r]^{\wedge}_p$.

For the second assertion, Remark 4.4 and a dévissage using [Hau24, Lemma 5.4.3] reduce to a finitely generated unit Frobenius module P over $\mathrm{A}[1/\mathrm{T}_1,\dots,1/\mathrm{T}_r]/p$ whose underlying coefficient module is perfect. It is therefore coherent. Put $\mathrm{T}=\mathrm{T}_1\cdots\mathrm{T}_r$, choose a coherent A/p-submodule $\mathrm{P}_0\subset\mathrm{P}$ with $\mathrm{P}_0[1/\mathrm{T}]=\mathrm{P}$, and let $\beta:\mathrm{P}\to\varphi^*\mathrm{P}$ be the inverse linearized Frobenius. For some $\mathrm{N}\geq 0$,

$$\beta(\mathrm{P}_0)\subseteq\mathrm{T}^{-\mathrm{N}}\varphi^*\mathrm{P}_0.$$

If $(p-1)a>\mathrm{N}$, then $\mathrm{T}^{-a}\mathrm{P}_0\to\varphi^*(\mathrm{T}^{-a}\mathrm{P}_0)$ is a coherent root. Indeed, if $\Phi:\varphi^*\mathrm{P}\to\mathrm{P}$ is the linearized Frobenius and $d=(p-1)a-\mathrm{N}>0$, then

$$\Phi^n\big(\varphi^{*n}(\mathrm{T}^{-a}\mathrm{P}_0)\big)\supseteq\mathrm{T}^{-d(1+p+\cdots+p^{n-1})}\mathrm{T}^{-a}\mathrm{P}_0.$$

Its unitalization is therefore P. $\square$

Theorem 5.4. Let $(\mathrm{X},\mathcal{M})$ be as in Assumption 3.1 over $\mathrm{W}(k)[\zeta_p]$. Then $\mathrm{T}_{\text{ét}}$ fits in a commutative diagram

$$\begin{tikzcd}
\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1,\mathrm{op}} \arrow[r, "\sim"] \arrow[d, hook] & \mathrm{D}^{(b)}_{\mathrm{lisse}}((\mathrm{X}_\eta\setminus\mathrm{D}_\eta)_{\text{proét}},\mathbf{Z}_p) \arrow[d, hook, "j_!"] \\
\mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)^{\mathrm{op}} \arrow[r, "\mathrm{T}_{\text{ét}}"] & \mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}}((\mathrm{X},\mathcal{M})_\eta,\mathbf{Z}_p).
\end{tikzcd}$$

The top horizontal arrow is an equivalence.

Remark 5.5. Although [BS23, Corollary 3.7] describes the target using ordinary Laurent F-crystals, it is not directly suited to the functor $j_!$. The complement $\mathrm{X}_\eta\setminus\mathrm{D}_\eta$ need not be quasicompact; for example, $\mathbf{P}^1_{\mathbf{Q}_p}\setminus\{\infty\}=\mathbf{A}^1_{\mathbf{Q}_p}$. Moreover, a formal model of this complement need not map to the fixed smooth proper model X with generic fiber j. Raynaud models require noncanonical admissible blowups and need not remain smooth. We fix this problem by using log formal schemes, as in Assumption 3.1.

Lemma 4.26 computes $\mathrm{T}_{\text{ét}}(\mathcal{O}_{\Delta}(*\mathrm{D})) \simeq j_!\mathbf{Z}_p$, and Remark 4.27 identifies the bottom horizontal functor in the diagram of Theorem 5.4 with applying $j_!$ to the lisse sheaf on the open stratum.

Lemma 5.6. Let $(\mathrm{X}, \mathcal{M})$ be as in Assumption 3.1 over $\mathrm{W}(k)[\zeta_p]$. The category

$$\mathsf{Perf}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1,\mathrm{op}}$$

is a stack for the strict-étale topology on such $(\mathrm{X}, \mathcal{M})$.

Proof. Let $g : (\mathrm{U}, \mathcal{N}) \to (\mathrm{X}, \mathcal{M})$ be strict-étale and put $\mathrm{D}_{\mathrm{U}} = \mathrm{D} \times_{\mathrm{X}} \mathrm{U}$. There is a canonical Frobenius-equivariant equivalence

$$g^*\mathcal{O}_{\Delta,\mathrm{X}}(*\mathrm{D}) \simeq \mathcal{O}_{\Delta,\mathrm{U}}(*\mathrm{D}_{\mathrm{U}}).$$

Indeed, strictness identifies $g^*\mathcal{M}(-\mathrm{D})$ with $\mathcal{N}(-\mathrm{D}_{\mathrm{U}})$. In particular $\mathcal{O}_{\Delta}(*\mathrm{D})$ commutes with strict-étale pullback.

Let $\mathrm{U}^{(\bullet)}$ be the Čech nerve of a strict-étale cover of X. Corollary A.16 gives symmetric monoidal descent for solid logarithmic Laurent crystals, under which $\mathcal{O}_{\Delta,\mathrm{X}}(*\mathrm{D})$ corresponds to the algebras $\mathcal{O}_{\Delta,\mathrm{U}^{(m)}}(*\mathrm{D}_{\mathrm{U}^{(m)}})$. One then also obtains descent for their module categories. Dualizability is detected termwise in a limit, and imposing the unit Frobenius structure is a limit. Hence the same descent statement holds for the category in the lemma. Taking opposites preserves limits. □

Remark 5.7. Let $\pi = \zeta_p - 1$ be a uniformizer of $\mathrm{W}(k)[\zeta_p]$. If one does not invert I_{Δ} and instead takes the unitalization of $\mathcal{O}_{\Delta}(\mathrm{D}) \to \varphi^*\mathcal{O}_{\Delta}(\mathrm{D})$, one can calculate that the crystalline realization (i.e. the unitalization $\mathcal{E}$ of $\mathcal{O}_{\mathrm{cris}}(\mathrm{D}) \to \varphi^*\mathcal{O}_{\mathrm{cris}}(\mathrm{D})$) satisfies

$$\mathrm{RHom}_{\widehat{\mathrm{D}}(((\mathrm{X},\mathcal{M})_{\pi=0}/\mathrm{W}(k))_{\mathrm{cris}},\mathcal{O}_{\mathrm{cris}})}(\mathcal{E}, \mathcal{O}_{\mathrm{cris}}) \simeq \mathrm{R}\Gamma_{\mathrm{cris},c}((\mathrm{X}, \mathcal{M})_{\pi=0}/\mathrm{W}(k)).$$

Here compactly supported log crystalline cohomology is as defined in [Tsu99]. This shows the crystalline realization of the natural non-Laurentized variant is compatible with the expectation one would have from the open C_{st} conjecture.

Lemma 5.8. Let $(\operatorname{Spf} \mathrm{R}, \mathcal{M})$ be a framed log affine. Then

$$\mathsf{Perf}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1} \simeq \mathsf{Perf}(\mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]_p^{\wedge})^{\varphi=1,h\Gamma}.$$

Proof. Put $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$ and $\mathrm{A}^{\circ} = \mathrm{A}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]_p^{\wedge}$. Evaluation is symmetric monoidal and carries $\mathcal{O}_{\Delta}(*\mathrm{D})$ to A° with its Γ-action. Lemma 5.3 embeds the source into the fgu category, and Theorem 4.6 is fully faithful there. Modules over an idempotent localization form a full subcategory, so the displayed functor is fully faithful.

Conversely, let M be a perfect unit Frobenius A°-complex with Γ action. The algebraic argument in the proof of Lemma 5.3 makes its restriction to A finitely generated unit. Theorem 4.6 gives an fgu crystal $\mathcal{E}$ with evaluation M. The localization map

$$\mathcal{E} \longrightarrow \mathcal{E} \otimes \mathcal{O}_{\Delta}(*\mathrm{D})$$

is an equivalence after evaluation, hence is an equivalence by conservativity. Thus $\mathcal{E}$ is an $\mathcal{O}_{\Delta}(*\mathrm{D})$-module. The local description $\mathsf{Perf}((\operatorname{Spf}\mathrm{R},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}(*\mathrm{D}))\simeq\varprojlim_{[m]\in\Delta}\mathsf{Perf}(\mathrm{A}^{\circ,(m)})$ and the crystal property identify its degree m evaluation with

$$\mathrm{A}^{\circ,(m)}\widehat{\otimes}^{\mathrm{L}}_{\mathrm{A}^\circ}\mathrm{M},$$

which is perfect over $\mathrm{A}^{\circ,(m)}$. Since dualizability is detected termwise, $\mathcal{E}$ is perfect over $\mathcal{O}_{\Delta}(*\mathrm{D})$. Hence the fgu equivalence restricts to the displayed equivalence. ◻

We now tackle full faithfulness. Our strategy will be to show this in the perfectoid setting (using the formulation of $\mathrm{T}_{\text{ét}}$ in Lemma 4.26). However unlike Laurent F-crystals in perfect complexes, it is not possible to use only the unit Frobenius to reduce to the perfectoid setting. We explain this further in Remark 5.13.

Proposition 5.9. Let $(\operatorname{Spf}\mathrm{R},\mathcal{M})$ be a framed log affine. The base-change functor

$$\mathsf{Perf}((\operatorname{Spf}\mathrm{R},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1}\to\mathsf{Perf}(\mathbb{A}_{\inf}((\mathrm{R}_0^{\square})_\infty)[1/\xi,1/[\mathrm{T}_1^\flat],\dots,1/[\mathrm{T}_r^\flat]]^\wedge_p)^{\varphi=1,h\Gamma}$$

is fully faithful.

Proof. As usual, associated to the framed log affine we have a logarithmic q-de Rham prism, so we let $\mathrm{A}=\mathrm{R}_0^{\square}(\!(q-1)\!)^\wedge_p$. For this particular proof we only care about a localization, so put $\mathrm{A}^\circ:=\mathrm{A}[1/\mathrm{T}_1,\dots,1/\mathrm{T}_r]^\wedge_p$ and $\mathrm{A}^\circ_{\mathrm{perf}}=\mathbb{A}_{\inf}((\mathrm{R}_0^{\square})_\infty)[1/\xi,1/[\mathrm{T}_1^\flat],\dots,1/[\mathrm{T}_r^\flat]]^\wedge_p$. We can express the source as $\mathsf{Perf}(\mathrm{A}^\circ)^{\varphi=1,h\Gamma}$ using Lemma 5.8; the target is already $\mathsf{Perf}(\mathrm{A}^\circ_{\mathrm{perf}})^{\varphi=1,h\Gamma}$.

By the tensor-Hom adjunction, it suffices to show that

$$\mathrm{R}\Gamma_{\mathrm{cont}}(\Gamma,\mathrm{M})^{\varphi=1}\longrightarrow\mathrm{R}\Gamma_{\mathrm{cont}}(\Gamma,\mathrm{M}\otimes_{\mathrm{A}^\circ}\mathrm{A}^\circ_{\mathrm{perf}})^{\varphi=1}\tag{4}$$

is an isomorphism for $\mathrm{M}\in\mathsf{Perf}(\mathrm{A}^\circ)^{\varphi=1,h\Gamma}$. We can also use the ordinary tensor product $\mathrm{M}\otimes_{\mathrm{A}^\circ}\mathrm{A}^\circ_{\mathrm{perf}}$ in the target because the relevant map $\mathrm{A}^\circ\to\mathrm{A}^\circ_{\mathrm{perf}}$ is faithfully flat. Indeed, Kunz's theorem makes $\mathrm{A}^+/p\to\mathrm{A}^+_{\mathrm{perf}}/p$ faithfully flat, so $\mathrm{A}^+\to\mathrm{A}^+_{\mathrm{perf}}$ is p-completely faithfully flat. The map $\mathrm{A}^\circ\to\mathrm{A}^\circ_{\mathrm{perf}}$ is its p-completed base change after inverting $[p]_q,\mathrm{T}_1,\dots,\mathrm{T}_r$, so it is p-completely faithfully flat; since A° is Noetherian, it is faithfully flat.

The source and target also both have standard t-structures obtained by restricting the t-structure on $\mathrm{D}_\blacksquare(\mathrm{A})$ and $\mathrm{D}_\blacksquare(\mathrm{A}_{\mathrm{perf}})$. To check these are well-defined one must check truncations remain perfect, which can be done modulo p by derived p-completeness. In this setting the t-structure exists even before applying $(-)^{h\Gamma}$, which is implicit in the proof [BS23, Proposition 3.4]: in particular they show that the top cohomology of a perfect unit Frobenius module over an $\mathbf{F}_p$-algebra is a vector bundle. By derived Nakayama we can work modulo p. Using a Postnikov dévissage ([Hau24, Lemma 5.4.3]) we reduce to testing the claim for p-torsion M in the heart, which must lie in $\mathsf{Vect}(\mathrm{A}^\circ/p)^{\varphi=1,h\Gamma}$.

We now prove (4) for M in $\mathsf{Vect}(\mathrm{A}^\circ/p)^{\varphi=1,h\Gamma}$. The faithfully flat map $\mathrm{A}^\circ/p\to\mathrm{A}^\circ_{\mathrm{perf}}/p$ is injective, with cokernel $(\mathrm{A}^\circ_{\mathrm{perf}}/\mathrm{A}^\circ)/p$. Since M is finite projective over A°/p, tensoring this sequence with M is exact. It is therefore enough to prove $\mathrm{R}\Gamma_{\mathrm{cont}}(\Gamma,\mathrm{M}\otimes_{\mathrm{A}^\circ}(\mathrm{A}^\circ_{\mathrm{perf}}/\mathrm{A}^\circ))^{\varphi=1}=0$.

Choose $m\gg 0$ and let $(\mathrm{R}_0^{\square})_m$ be obtained from the framing by adjoining p^mth roots of the coordinates T_i. Define $\varphi^{-m}\mathrm{A}^\circ=(\mathrm{R}_0^{\square})_m(\!(q^{1/p^m}-1)\!)[1/\mathrm{T}_1,\dots,1/\mathrm{T}_r]$ and $\varphi^{-m}\mathrm{A}^{\circ,\mathrm{cyc}}=$

$(\mathrm{R}_0^\square)_m(\!(q^{1/p^\infty}-1)\!)^\wedge[1/\mathrm{T}_1,\ldots,1/\mathrm{T}_r]$, viewed as intermediate subrings

$$\mathrm{A}^\circ \subset \varphi^{-m}\mathrm{A}^\circ \subset \varphi^{-m}\mathrm{A}^{\circ,\mathrm{cyc}} \subset \mathrm{A}^\circ_{\mathrm{perf}}.$$

Since φ and Γ commute, the unit Frobenius of M gives a Γ-equivariant equivalence

$$\mathrm{R}\Gamma_{\mathrm{cont}}(\Gamma,\mathrm{M})^{\varphi=1} \simeq \mathrm{R}\Gamma_{\mathrm{cont}}\big(\Gamma,\mathrm{M}\otimes_{\mathrm{A}^\circ}\varphi^{-m}\mathrm{A}^\circ\big)^{\varphi=1}.$$

For $m \gg 0$, the exact triangles associated to $\varphi^{-m}\mathrm{A}^\circ/\mathrm{A}^\circ \longrightarrow \mathrm{A}^\circ_{\mathrm{perf}}/\mathrm{A}^\circ \longrightarrow \mathrm{A}^\circ_{\mathrm{perf}}/\varphi^{-m}\mathrm{A}^\circ$ and to $\varphi^{-m}\mathrm{A}^{\circ,\mathrm{cyc}}/\varphi^{-m}\mathrm{A}^\circ \longrightarrow \mathrm{A}^\circ_{\mathrm{perf}}/\varphi^{-m}\mathrm{A}^\circ \longrightarrow \mathrm{A}^\circ_{\mathrm{perf}}/\varphi^{-m}\mathrm{A}^{\circ,\mathrm{cyc}}$ show

$$\mathrm{R}\Gamma_{\mathrm{cont}}(\Gamma,\mathrm{M}\otimes_{\mathrm{A}^\circ}(\mathrm{A}^\circ_{\mathrm{perf}}/\mathrm{A}^\circ))^{\varphi=1}=0$$

by applying Lemmas 5.11 and 5.12. The claim then follows. □

We now provide the needed technical lemmas. We retain the notation of Proposition 5.9, and use the t-structure constructed in the proof.

Put $\mathrm{T}=\mathrm{T}_1\cdots\mathrm{T}_r$. Through the identification in Theorem 4.6 (using a resolution in $\mathrm{D}_\blacksquare(\mathrm{A})^{h\Gamma}$ by $\mathrm{A}\langle\mathrm{F}\rangle$ induced by a root), the mod-p coefficient ring

$$\overline{\mathrm{A}}^\circ := \mathrm{A}^\circ/p = \bigcup_{i\geq 0}\mathrm{T}^{-i}\overline{\mathrm{R}}^\square(\!(q-1)\!)$$

is equipped with the inductive limit topology, where $\mathrm{T}^{-i}\overline{\mathrm{R}}^\square[\![q-1]\!]$ is an open lattice in $\mathrm{T}^{-i}\overline{\mathrm{R}}^\square(\!(q-1)\!)$ with its $(q-1)$-adic topology. This is a strict inductive limit of Banach modules with closed transition maps, and every compact subset is contained in $\mathrm{T}^{-i}\overline{\mathrm{R}}^\square(\!(q-1)\!)$ for some i. It is worth noting that this topology agrees with the weak topology induced on a finitely generated unit module, by using the resolution by $\mathrm{A}\langle\mathrm{F}\rangle$ induced by the root and showing this weak topology, modulo p, agrees with the natural topology on the colimit presentation by the root (which is what we use here).

Choose a finite $\overline{\mathrm{R}}^\square[\![q-1]\!]$-submodule $\mathrm{M}_0\subset\mathrm{M}$ with $\mathrm{M}_0[1/(q-1),1/\mathrm{T}_1,\ldots,1/\mathrm{T}_r]=\mathrm{M}$. The orbit under the compact group Γ of a finite generating set of M_0 is contained in $(q-1)^{-c}\mathrm{T}^{-i}\mathrm{M}_0$ for some $c,i\geq 0$. Its $\overline{\mathrm{R}}^\square[\![q-1]\!]$-span is therefore a finite Γ-stable lattice $\mathrm{M}^+\subset\mathrm{M}$ with $\mathrm{M}^+[1/(q-1),1/\mathrm{T}_1,\ldots,1/\mathrm{T}_r]=\mathrm{M}$.

Let $\mathrm{H}=(1+p^m\mathbf{Z}_p)\ltimes(p^{m-1}\mathbf{Z}_p(1))^n\subset\Gamma$. Choose a topological generator γ_0 of $1+p^m\mathbf{Z}_p$ and generators $\gamma_1,\ldots,\gamma_n$ of $(p^{m-1}\mathbf{Z}_p(1))^n$ such that $\gamma_i(\mathrm{T}_i)=q^{p^m}\mathrm{T}_i$. For $N\geq 0$, put $\mathrm{M}^{(N)}=\mathrm{T}^{-N}\mathrm{M}^+[1/(q-1)]$ and $\mathrm{Fil}^a\,\mathrm{M}^{(N)}=\mathrm{T}^{-N}(q-1)^a\mathrm{M}^+$. Then $\mathrm{M}=\varinjlim_N\mathrm{M}^{(N)}$ with its strict inductive limit topology. Increasing m if necessary, H is normal and uniform, and

$$(\gamma_i-1)(\mathrm{Fil}^a\,\mathrm{M}^{(N)})\subset\mathrm{Fil}^{a+1}\,\mathrm{M}^{(N)} \text{ for } 0\leq i\leq n.$$

We use the following elementary lemma, which is similar to [KL16, Lemmas 5.6.3, 5.6.4]. A strict LF filtered module is a strict inductive limit $\mathrm{M}=\varinjlim_N\mathrm{M}^{(N)}$ of complete filtered modules with closed filtration-preserving transition maps. Write $\widehat{\bigoplus}_j\mathrm{M}_j$ for the completion of $\bigoplus_j\mathrm{M}_j$ with its termwise filtration, and for compatible strict LF presentations put

$$\widehat{\bigoplus_j}^{\mathrm{LF}}\mathrm{M}_j := \varinjlim_N\widehat{\bigoplus}_j\mathrm{M}_j^{(N)}.$$

A continuous cochain model will mean one which computes continuous cohomology and commutes with these completed direct sums and strict inductive limits. In the applications it is a finite completed Koszul complex or the mapping fiber of an endomorphism of one. A complex is *strictly contractible* if it admits a continuous contracting homotopy. Such a homotopy is *uniformly bounded* if it preserves every LF stage and, for some $c \geq 0$ independent of all indices, satisfies

$$h^i(\mathrm{Fil}^a \mathrm{C}^i) \subseteq \mathrm{Fil}^{a-c} \mathrm{C}^{i-1}.$$

Lemma 5.10. Let H be a compact p-adic Lie group and let

$$\mathrm{B}_0 \subseteq \mathrm{B}_1 \subseteq \cdots \subseteq \mathrm{B}$$

be topological rings with continuous H-actions, where B is the completion of $\bigcup_j \mathrm{B}_j$. Suppose these rings have compatible strict LF filtered structures, the inclusions are strict, and there are continuous H-equivariant B_j-linear strict filtered retractions $\psi_j : \mathrm{B}_{j+1} \longrightarrow \mathrm{B}_j$ preserving every LF stage. Assume that the natural map is an isomorphism

$$\widehat{\bigoplus_{j\geq 0}}^{\mathrm{LF}} \ker(\psi_j) \xrightarrow{\sim} \mathrm{B}/\mathrm{B}_0.$$

Fix a continuous cochain model for H. Let $\mathrm{M} \in \mathsf{Vect}(\mathrm{B}_0)$ have continuous semilinear H-action, and write $\mathrm{M}_j = \mathrm{B}_j \widehat{\otimes}_{\mathrm{B}_0} \mathrm{M}$ and $\mathrm{M}_\mathrm{B} = \mathrm{B} \widehat{\otimes}_{\mathrm{B}_0} \mathrm{M}$. Give

$$\mathrm{M}_j \widehat{\otimes}_{\mathrm{B}_j} \ker(\psi_j)$$

the strict LF structures induced by the ring data, and suppose their stages are H-stable. If their continuous cochain complexes are strictly contractible by compatible uniformly bounded homotopies, then $\mathrm{R\Gamma}_{\mathrm{cont}}(\mathrm{H}, \mathrm{M}_\mathrm{B}/\mathrm{M}) = 0$.

Proof. Finite projectivity of M carries the ring decomposition to a strict LF decomposition

$$\mathrm{M}_\mathrm{B}/\mathrm{M} \simeq \widehat{\bigoplus_{j\geq 0}}^{\mathrm{LF}} \left(\mathrm{M}_j \widehat{\otimes}_{\mathrm{B}_j} \ker(\psi_j)\right).$$

The cochain model carries this to the corresponding completed direct sum of cochain complexes. The uniform bounds extend the contracting homotopies to this completed direct sum and then to its strict inductive limit. ☐

Lemma 5.11. Let M be p-torsion and in the heart of $\mathsf{Perf}(\mathrm{A}^\circ)^{\varphi=1,h\Gamma}$. For $m \gg 0$,

$$\mathrm{R\Gamma}_{\mathrm{cont}}\big(\Gamma, \mathrm{M} \otimes_{\mathrm{A}^\circ} (\mathrm{A}^\circ_{\mathrm{perf}}/\varphi^{-m}\mathrm{A}^{\circ,\mathrm{cyc}})\big) = 0.$$

Proof. By the discussion in Proposition 5.9 (around t-structures) we see M is a vector bundle. In Lemma 5.10 take

$$\mathrm{B}_j = \varphi^{-(m+j)}(\mathrm{A}^{\circ,\mathrm{cyc}})/p$$

and $\mathrm{B} = \mathrm{A}^\circ_{\mathrm{perf}}/p$, and take the coefficient module to be $\mathrm{B}_0 \widehat{\otimes}_{\mathrm{A}^\circ/p} \mathrm{M}$. We have the decomposition

$$\mathrm{B}_{j+1} = \bigoplus_{\alpha \in \{0,\ldots,p-1\}^n} \mathrm{B}_j \mathrm{T}_1^{a_1/p^{m+j+1}} \cdots \mathrm{T}_n^{a_n/p^{m+j+1}}.$$

Bounding the poles by T^{-N} and using the $(q-1)$-adic filtration gives compatible strict LF filtered structures. Projection $\psi_j : \mathrm{B}_{j+1} \to \mathrm{B}_j$ onto $\alpha = \vec{0}$ is a Γ-equivariant strict filtered retraction preserving every LF stage. The resulting map

$$\widehat{\bigoplus}^{\mathrm{LF}}_{j\geq 0} \ker(\psi_j) \xrightarrow{\sim} (\mathrm{A}^{\circ}_{\mathrm{perf}}/\varphi^{-m}\mathrm{A}^{\circ,\mathrm{cyc}})/p$$

is an isomorphism of strict LF filtered modules.

We apply Lemma 5.10 to the mod p reductions of these decompositions and for the closed normal subgroup $\mathrm{H} = \overline{\langle \gamma_1, \ldots, \gamma_n \rangle}$. For $\alpha \neq 0$, let i be the least index with $a_i \neq 0$. On the corresponding monomial summand write the action of γ_i as $q^{a_i/p^{j+1}}\mathrm{U}_{i,j}$, where $\mathrm{U}_{i,j}$ acts on the coefficient factor. The choice of m and the action on $\mathrm{T}_i^{1/p^{m+j}}$ give

$$\mathrm{ord}_{q-1}((\mathrm{U}_{i,j}-1)y) \geq \mathrm{ord}_{q-1}(y) + p^{-j}$$

and $\mathrm{ord}_{q-1}(q^{a_i/p^{j+1}}-1) = p^{-j-1}$. Hence, on this summand,

$$(\gamma_i - 1)^{-1} = \sum_{\ell\geq 0} \left(-\frac{q^{a_i/p^{j+1}}}{q^{a_i/p^{j+1}}-1}(\mathrm{U}_{i,j}-1) \right)^{\ell} (q^{a_i/p^{j+1}}-1)^{-1}.$$

The operator in parentheses raises $(q-1)$-adic order by at least $(p-1)p^{-j-1}$, so the series converges. For every N, the inverse preserves the submodule with poles bounded by T^{-N} and lowers the integer filtration by at most one. One can use this to give the needed contracting homotopies on the full Koszul complex. Lemma 5.10 gives the desired vanishing for H; Hochschild-Serre gives it for Γ. ☐

Lemma 5.12. Let M be p-torsion and in the heart of $\mathsf{Perf}(\mathrm{A}^{\circ})^{\varphi=1,h\Gamma}$. For $m \gg 0$,

$$\mathrm{R}\Gamma_{\mathrm{cont}}\big(\Gamma, \mathrm{M} \otimes_{\mathrm{A}^{\circ}} (\varphi^{-m}\mathrm{A}^{\circ,\mathrm{cyc}}/\varphi^{-m}\mathrm{A}^{\circ})\big) = 0.$$

Proof. For the same reasons we know M is a vector bundle over A°/p. For Lemma 5.10, take $\mathrm{B} = \varphi^{-m}\mathrm{A}^{\circ,\mathrm{cyc}}/p$, $\mathrm{B}_j = (\overline{\mathrm{R}}^{\square}_0)_m(\!(q^{1/p^{m+j}}-1)\!)[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]$, and coefficient module $\mathrm{B}_0 \widehat{\otimes}_{\mathrm{A}^{\circ}/p}\mathrm{M}$. There are strict decompositions

$$\mathrm{B}_{j+1} = \bigoplus_{a=0}^{p-1} \mathrm{B}_j q^{a/p^{m+j+1}}.$$

Bounding the poles by T^{-N} and using the $(q-1)$-adic filtration gives compatible strict LF filtered structures. The projection $\psi_j : \mathrm{B}_{j+1} \to \mathrm{B}_j$ onto $a = 0$ is a Γ-equivariant strict filtered retraction preserving every LF stage, and

$$\widehat{\bigoplus}^{\mathrm{LF}}_{j\geq 0} \ker(\psi_j) \xrightarrow{\sim} (\varphi^{-m}\mathrm{A}^{\circ,\mathrm{cyc}}/\varphi^{-m}\mathrm{A}^{\circ})/p$$

is an isomorphism of strict LF filtered modules.

For the group in Lemma 5.10, take

$$\mathrm{H} = (1 + p^m\mathbf{Z}_p) \ltimes (p^{m-1}\mathbf{Z}_p(1))^n$$

and also consider the normal subgroup $\mathrm{N} = (p^{m-1}\mathbf{Z}_p(1))^n$. Choose a generator γ_0 of the first factor in H and write $\gamma_0 = 1 + p^m u$ with $u \in \mathbf{Z}_p^\times$. Since N is normal in H, continuous H-cohomology is computed by

$$\mathrm{C}^\bullet_{\mathrm{cont}}(\mathrm{H}, -) = \mathrm{fib}\left(\gamma_0 - 1 : \mathrm{Kos}^\bullet_{\mathrm{N}}(-) \longrightarrow \mathrm{Kos}^\bullet_{\mathrm{N}}(-)\right),$$

where γ_0 acts on the completed Koszul complex by conjugation.

The point is that, on every nonzero summand, the scalar part of $\gamma_0 - 1$ has smaller $(q-1)$-adic order than the remaining chain action. Indeed, on the summand indexed by $a \neq 0$, write the chain action as $q^{au/p^{j+1}}\mathrm{U}_{0,j}$. Then

$$\mathrm{ord}_{q-1}((\mathrm{U}_{0,j} - 1)y) \geq \mathrm{ord}_{q-1}(y) + p^{-j} - p^{-(m+j)}$$

and $\mathrm{ord}_{q-1}(q^{au/p^{j+1}} - 1) = p^{-j-1}$. The first estimate also includes the action on the Koszul generators: on the generator corresponding to γ_i, the conjugation correction is

$$\frac{\gamma_i^{1+p^m u} - 1}{\gamma_i - 1} \equiv 1 \pmod{\gamma_i - 1},$$

which follows from $1 + p^m u \equiv 1 \pmod p$. Hence

$$(\gamma_0 - 1)^{-1} = \sum_{\ell \geq 0} \left(-\frac{q^{au/p^{j+1}}}{q^{au/p^{j+1}} - 1}(\mathrm{U}_{0,j} - 1)\right)^{\ell} (q^{au/p^{j+1}} - 1)^{-1}.$$

The operator in parentheses raises $(q-1)$-adic order by at least

$$p^{-j} - p^{-(m+j)} - p^{-j-1} = p^{-j}(1 - p^{-m} - p^{-1}) > 0,$$

so the series converges. For every N, the inverse preserves the submodule with poles bounded by T^{-N} and lowers the integer filtration by at most one. Thus $\gamma_0 - 1$ is an invertible chain map on each summand of $\mathrm{Kos}^\bullet_{\mathrm{N}}$, so the displayed fiber is strictly contractible with the required uniform bound. Lemma 5.10 gives the vanishing for H, and Hochschild-Serre for the open normal subgroup $\mathrm{H} \subset \Gamma$ gives the assertion. □

Remark 5.13. It is tempting to try to argue the previous result would only need to use φ, as this is the case for ordinary (φ, Γ) modules using invariance of étale φ-modules under completed perfections. Consider the map

$$\mathrm{A} = k((q-1))\langle \mathrm{T} \rangle [1/\mathrm{T}] \to \mathrm{B} = k((q^{1/p^\infty} - 1))^\wedge \langle (\mathrm{T}^\flat)^{1/p^\infty} \rangle [1/\mathrm{T}^\flat],$$

with $\mathrm{T} \mapsto \mathrm{T}^\flat$. If full faithfulness followed just from invariance under this completed perfection, then $\mathsf{Perf}(\mathrm{A})^{\varphi=1} \to \mathsf{Perf}(\mathrm{B})^{\varphi=1}$ would be fully faithful. By Katz's theorem this would force $\mathrm{H}^1_{\text{ét}}(\mathrm{A}, \mathbf{F}_p) \to \mathrm{H}^1_{\text{ét}}(\mathrm{B}, \mathbf{F}_p)$ to be an isomorphism. One can check that the Artin-Schreier class on B defined by

$$f = \sum_{n \geq 2} (q-1)^{p^n} (\mathrm{T}^\flat)^{-\left(\frac{1}{p} + \frac{1}{p^n}\right)}$$

is not pulled back from A, so the étale cohomology groups are not isomorphic via the natural base change map. The issue is passage to B mixes completed and decompleted perfections.

Proposition 5.14. The top horizontal functor in Theorem 5.4 is fully faithful.

Proof. Lemma 5.6 and strict-étale descent for $\mathrm{D}^{(b)}_{\mathrm{lisse}}((\mathrm{X}_\eta \setminus \mathrm{D}_\eta)_{\mathrm{pro\acute{e}t}}, \mathbf{Z}_p)$, together with Lemma 3.4, reduce the claim to a framed log affine. Lemma 5.8 identifies the source with $\mathsf{Perf}(\mathrm{R}_0^\square(\!(q-1)\!)^\wedge_p[1/\mathrm{T}_1,\dots,1/\mathrm{T}_r]^\wedge_p)^{\varphi=1,h\Gamma}$, and Proposition 5.9 makes base change to $\mathbb{A}_{\mathrm{inf}}((\mathrm{R}_0^\square)_\infty)[1/\xi, 1/[\mathrm{T}_1^\flat],\dots,1/[\mathrm{T}_r^\flat]]^\wedge_p$ fully faithful. Lemma 4.26 then identifies $\mathrm{T}_{\acute{e}t}$ with tilting, analytification, and continuous descent along the perfectoid pro-Kummer Γ-torsor. Tilting and descent are equivalences.

Thus we must investigate full faithfulness of analytification. Put

$$\mathrm{U} = \mathrm{Spec}(\mathrm{R}_0^\square)^\flat_{\infty,\eta}[1/\mathrm{T}_1^\flat,\dots,1/\mathrm{T}_r^\flat].$$

For mod-p objects $\mathbb{L},\mathbb{L}' \in \mathrm{D}^{(b)}_{\mathrm{lisse}}(\mathrm{U}_{\acute{e}t}, \mathbf{F}_p)$, the monodromy of each $\mathrm{H}^i\underline{\mathrm{RHom}}(\mathbb{L},\mathbb{L}')$ is contained in a finite group $\mathrm{GL}_{m_i}(\mathbf{F}_p)$. Boundedness gives one finite étale cover $\mathrm{U}' \to \mathrm{U}$ on which $\mathrm{H}^i\underline{\mathrm{RHom}}(\mathbb{L},\mathbb{L}') \simeq \mathbf{F}_p^{m_i}$ for every i. On U', [Bha25, Remark 6.2.7] gives

$$\mathrm{R}\Gamma(\mathrm{U}', \mathbf{F}_p) \xrightarrow{\sim} \mathrm{R}\Gamma(\mathrm{U}'^{\,\mathrm{an}}, \mathbf{F}_p).$$

Postnikov dévissage and finite étale descent give full faithfulness modulo p; derived Nakayama gives the assertion. □

We next prove essential surjectivity after passing, on a framed strict-étale chart, to a finite prime-to-p Kummer cover. Proposition 5.15 descends the object constructed on the cover, using the full faithfulness already proved.

Proposition 5.15. Let $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ be a framed log affine with boundary coordinates $\mathrm{T}_1,\dots,\mathrm{T}_r$, and let N be prime to p. Put

$$\mathrm{R}_\mathrm{N} = (\mathrm{R}[\mathrm{S}_1,\dots,\mathrm{S}_r]/(\mathrm{S}_1^\mathrm{N} - \mathrm{T}_1,\dots,\mathrm{S}_r^\mathrm{N} - \mathrm{T}_r))^\wedge_p,$$

and let $(\mathrm{Spf}\,\mathrm{R}_\mathrm{N}, \mathcal{M}_\mathrm{N})$ with its induced log structure. Then we have

$$\big(\mathsf{Perf}((\mathrm{Spf}\,\mathrm{R}_\mathrm{N}, \mathcal{M}_\mathrm{N})_{\mathbbl{\Delta}}, \mathcal{O}_{\mathbbl{\Delta}}(*\mathrm{D}_\mathrm{N}))^{\varphi=1,\mathrm{op}}\big)^{h\mu_\mathrm{N}^r} \simeq \mathsf{Perf}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\mathbbl{\Delta}}, \mathcal{O}_{\mathbbl{\Delta}}(*\mathrm{D}))^{\varphi=1,\mathrm{op}}.$$

Here $(-)^{h\mu_\mathrm{N}^r}$ denotes the totalization of the μ_N^r-action groupoid; after localizing along D_N, this is the Čech nerve of a finite étale torsor. The equivalence is compatible with finite étale descent for the étale realization.

Proof. Let $f_\mathrm{N} : (\mathrm{Spf}\,\mathrm{R}_\mathrm{N}, \mathcal{M}_\mathrm{N}) \to (\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ be the Kummer map. Since $f_\mathrm{N}^*\mathrm{D} = \mathrm{N}\mathrm{D}_\mathrm{N}$, there is a canonical equivalence

$$f_\mathrm{N}^*\mathcal{O}_{\mathbbl{\Delta}}(*\mathrm{D}) \simeq \mathcal{O}_{\mathbbl{\Delta}}(*(\mathrm{N}\mathrm{D}_\mathrm{N})) \simeq \mathcal{O}_{\mathbbl{\Delta}}(*\mathrm{D}_\mathrm{N}).$$

Thus pullback induces a functor on the corresponding module categories

$$\mathsf{Perf}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\mathbbl{\Delta}}, \mathcal{O}_{\mathbbl{\Delta}}(*\mathrm{D}))^{\varphi=1,\mathrm{op}} \to \mathsf{Perf}((\mathrm{Spf}\,\mathrm{R}_\mathrm{N}, \mathcal{M}_\mathrm{N})_{\mathbbl{\Delta}}, \mathcal{O}_{\mathbbl{\Delta}}(*\mathrm{D}_\mathrm{N}))^{\varphi=1,\mathrm{op}}$$

we will use to induce the desired equivalence.

Base change of the framing along $\mathrm{T}_i \mapsto \mathrm{S}_i^\mathrm{N}$ gives R_N a framing with boundary coordinates $\mathrm{S}_1,\dots,\mathrm{S}_r$, so it also has an associated q-de Rham prism. On the degree zero log q-de Rham term, the pullback is base change along

$$\mathrm{A} := \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]_p^{\wedge} \longrightarrow \mathrm{B} := \mathrm{A}[\mathrm{S}_1, \ldots, \mathrm{S}_r]/(\mathrm{S}_1^{\mathrm{N}} - \mathrm{T}_1, \ldots, \mathrm{S}_r^{\mathrm{N}} - \mathrm{T}_r).$$

Because $\mathrm{N} \in \mathbf{Z}_p^{\times}$ and the T_i are units in A, this is a finite étale μ_{N}^r-torsor.

After rescaling the first r coordinates by N^{-1}, the two actions of Γ make $\mathrm{A} \to \mathrm{B}$ a Γ-equivariant μ_{N}^r-torsor. It is also Frobenius-equivariant by functoriality of the q-de Rham prisms. Lemma 5.8 identifies pullback of localized perfect crystals with extension of scalars from A to B. The proposition therefore reduces to finite étale descent in (φ, Γ)-equivariant perfect complexes:

$$\mathsf{Perf}(\mathrm{A})^{\varphi=1, h\Gamma} \simeq \left(\mathsf{Perf}(\mathrm{B})^{\varphi=1, h\Gamma}\right)^{h\mu_{\mathrm{N}}^r}.$$

Taking opposites proves the assertion.

Pullback compatibility is clear. □

Remark 5.16. On a framed log affine, $[(\mathrm{X}_{\mathrm{N}}, \mathcal{M}_{\mathrm{N}})/\mu_{\mathrm{N}}^r]$ is the Nth root stack of $(\mathrm{X}, \mathcal{M})$. Thus the proposition is the local form of finite étale descent from this root stack; the local form suffices because the equivalence can be checked strict-étale locally.

Corollary 5.17. The top horizontal functor in Theorem 5.4 is an equivalence.

Proof. By Proposition 5.14, it remains to prove essential surjectivity. Let $\mathbb{K} \in \mathrm{D}^{(b)}_{\mathrm{lisse}}((\mathrm{X}_\eta \setminus \mathrm{D}_\eta)_{\mathrm{pro\acute{e}t}}, \mathbf{Z}_p)$. Since X_η is quasicompact and smooth, it has finitely many connected components, each irreducible; their nonempty intersections with $\mathrm{X}_\eta \setminus \mathrm{D}_\eta$ remain irreducible. The amplitude of a derived-lisse complex is locally constant, so $\mathbb{K}$ is globally bounded. Bounded objects are generated under shifts and fibers by their hearts [Hau24, Lemma 5.4.3], so it suffices to treat $\mathbb{L} \in \mathrm{D}^{(b)}_{\mathrm{lisse}}((\mathrm{X}_\eta \setminus \mathrm{D}_\eta)_{\mathrm{pro\acute{e}t}}, \mathbf{Z}_p)^{\heartsuit}$.

This assertion is strict-étale local on X. Indeed, if a local system $\mathbb{L}$ is realized locally by objects $\mathcal{E}_i$, then full faithfulness gives unique identifications of the $\mathcal{E}_i$ on pairwise overlaps inducing the identity on $\mathbb{L}$. The cocycle condition holds after applying the fully faithful realization functor, and Lemma 5.6 glues the $\mathcal{E}_i$. Thus we may assume that $\mathrm{X} = \mathrm{Spf}\,\mathrm{R}$ is framed.

Let $\mathbb{L}$ be such a local system. By [DLLZ23, Corollary 6.3.4], whose proof uses the rigid Abhyankar lemma, $\mathbb{L}^{\mathrm{ext}} := j_{\mathrm{k\acute{e}t},*}\mathbb{L}$ is a Kummer-étale $\mathbf{Z}_p$-local system on the log generic fiber. At every log geometric point, the congruence kernel of $\mathrm{Aut}_{\mathbf{Z}_p}(\mathbb{L}^{\mathrm{ext}}_{\bar{x}}) \to \mathrm{Aut}_{\mathbf{F}_p}(\mathbb{L}^{\mathrm{ext}}_{\bar{x}}/p)$ is pro-p, as follows from its filtration by the congruence kernels modulo p^n. Its intersection with the image of the prime-to-p Kummer inertia is therefore trivial, so this image is finite and injects into $\mathrm{Aut}_{\mathbf{F}_p}(\mathbb{L}^{\mathrm{ext}}_{\bar{x}}/p)$.

Working componentwise, these images have exponents dividing the prime-to-p exponent of some $\mathrm{GL}_m(\mathbf{F}_p)$. Choose N prime to p divisible by these exponents, and let $f_{\mathrm{N}} : (\mathrm{X}_{\mathrm{N}}, \mathcal{M}_{\mathrm{N}}) \to (\mathrm{X}, \mathcal{M})$ be the cover of Proposition 5.15. Then $f_{\mathrm{N}}^*\mathbb{L}^{\mathrm{ext}}$ is p-Kummer.

By the (dualized) Koshikawa-Yao correspondence [IKY26, Theorem 7]

$$\mathsf{Perf}((\mathrm{X}_{\mathrm{N}}, \mathcal{M}_{\mathrm{N}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\varphi=1,\mathrm{op}} \simeq \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{lisse}}((\mathrm{X}_{\mathrm{N}}, \mathcal{M}_{\mathrm{N}})_\eta, \mathbf{Z}_p),$$

this p-Kummer local system is realized by a Laurent prismatic F-crystal $\mathcal{E}$ on $(\mathrm{X_N}, \mathcal{M}_\mathrm{N})_{\Delta}$, so $\mathrm{T_{\acute{e}t}}(\mathcal{E}) \simeq f_\mathrm{N}^* \mathbb{L}^{\mathrm{ext}}$. Let $j_\mathrm{N} : \mathrm{X}_{\mathrm{N},\eta} \setminus \mathrm{D}_{\mathrm{N},\eta} \hookrightarrow (\mathrm{X_N}, \mathcal{M}_\mathrm{N})_\eta$ denote the open-stratum inclusion. Since $\mathrm{T_{\acute{e}t}}(\mathcal{O}_{\Delta}(*\mathrm{D_N})) \simeq j_{\mathrm{N},!}\mathbf{Z}_p$ by the compatibility of the vertical arrows before Theorem 5.4, the symmetric monoidality of $\mathrm{T_{\acute{e}t}}$ gives

$$\mathrm{T_{\acute{e}t}}(\mathcal{E} \otimes \mathcal{O}_{\Delta}(*\mathrm{D_N})) \simeq j_{\mathrm{N},!} f_\mathrm{N}^* \mathbb{L}.$$

The sheaf $j_{\mathrm{N},!} f_\mathrm{N}^* \mathbb{L}$ has its canonical descent datum along the finite étale torsor on the open stratum. The Čech nerve of the localized map in Proposition 5.15 is finite étale in every degree. Proposition 5.14 therefore lifts the descent datum uniquely to $\mathcal{E} \otimes \mathcal{O}_{\Delta}(*\mathrm{D_N})$, and full faithfulness gives the cocycle condition. Proposition 5.15 now descends this object to $\mathsf{Perf}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1,\mathrm{op}}$. Compatibility with realization shows that it realizes $j_! \mathbb{L}$. $\square$

5.2. **Proof of the main theorem.** We will show that

$$\mathrm{T_{\acute{e}t}} : \mathrm{D_{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge)^{\mathrm{op}} \longrightarrow \mathrm{D}^{(b)}_{\mathrm{cons,D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$$

is an equivalence onto the full subcategory of p-Kummer sheaves $\mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons,D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$. Strict-étale descent on the source and target, established in Theorem 4.11 and Lemma 4.20, permits us to work on framed log affines for the proof.

The remainder of the proof beyond the already established equivalence for the open stratum is an induction on dimension using a variant of Kashiwara's lemma in Lemma 5.18 for closed strata, together with full faithfulness for maps between open and closed strata. The following toy model summarizes the argument.

A toy model. Consider the disk $\mathbb{D}_{\mathbf{Q}_p}$ punctured at the origin; we use the coordinate T so $\mathbb{D} = \mathrm{Spa}\, \mathbf{Q}_p\langle \mathrm{T} \rangle$, equipped with the log structure $\mathcal{M}_{\mathrm{V(T)}}$. We use the usual formal model $(\mathrm{Spf}\, \mathbf{Z}_p\langle \mathrm{T} \rangle, \mathcal{M}_{\mathrm{V(T)}})$.

Let $j : \mathbb{D}^\times \hookrightarrow \mathbb{D}$ and $i : \mathrm{V(T)} \hookrightarrow \mathbb{D}$ denote the two strata. If $\mathrm{A} = \mathbf{Z}_p\langle \mathrm{T} \rangle (\!(q-1)\!)_p^\wedge$ and $\Gamma = (1 + p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)$, the setup of the argument is summarized by the diagram

$$\begin{tikzcd}
\left\{\begin{matrix}\text{open stratum unit}\\ \text{Laurent crystals}\end{matrix}\right\}^{\mathrm{op}} \arrow[r, hook, "-\otimes^{\mathrm{L},\blacksquare}\mathcal{O}_{\Delta}(*\mathrm{V(T)})"] \arrow[d, "\mathrm{T_{\acute{e}t}}"] & \left\{\begin{matrix}\text{fgu Laurent crystals}\\ \text{on } (\mathrm{Spf}\, \mathbf{Z}_p\langle \mathrm{T}\rangle, \mathcal{M}_{\mathrm{V(T)}})_{\Delta}\end{matrix}\right\}^{\mathrm{op}} \arrow[d, "\mathrm{T_{\acute{e}t}}"] & \left(\mathrm{D_{fgu}}(\mathrm{A}/(\mathrm{T}))^{h\Gamma}\right)^{\mathrm{op}} \arrow[l, hook', dashed, "(i_+(-))[-1]"'] \arrow[d, "\mathrm{T_{\acute{e}t}}"] \\
\left\{\begin{matrix}p\text{-Kummer local systems}\\ \text{on } \mathbb{D}^\times\end{matrix}\right\} \arrow[r, hook, "j_!"] & \left\{\begin{matrix}p\text{-Kummer constructible sheaves}\\ \text{on } (\mathbb{D}, \mathcal{M}_{\mathrm{V(T)}})\end{matrix}\right\} & \left\{\begin{matrix}p\text{-Kummer local systems}\\ \text{on } (\mathrm{V(T)}, i^*\mathcal{M}_{\mathrm{V(T)}})\end{matrix}\right\}. \arrow[l, hook', "i_*"']
\end{tikzcd}$$

The left vertical arrow is an equivalence by Theorem 5.4, and tensoring with $\mathcal{O}_{\Delta}(*\mathrm{V(T)})$ identifies with $j_!$ under $\mathrm{T_{\acute{e}t}}$. The factor of Γ corresponding to T acts trivially on $\mathrm{A}/(\mathrm{T})$, but its action on objects records the normal Kummer inertia on the logarithmic point. The dashed arrow is supplied by the local Kashiwara equivalence below: since $i^! = \mathrm{L}i^*[-1]$ and $\mathrm{T_{\acute{e}t}}$ is contravariant, $(i_+(-))[-1]$ identifies with i_* under realization.

With Kashiwara's lemma what we have so far (generalized to a hollow log point to deal with the origin) is enough for essential surjectivity, because étale sheaves of the form $j_!\mathbb{L}$ and $i_*\mathbb{L}'$ generate $\mathrm{D}^{(b)}_{\mathrm{cons,V(T)}}((\mathbb{D}, \mathcal{M}_{\mathrm{V(T)}}), \mathbf{Z}_p)$ under shifts and fibers; the same holds after imposing the p-Kummer condition. The functor $\mathrm{T_{\acute{e}t}}$ lands in the full subcategory of p-Kummer constructible sheaves along the stratification $(\mathbb{D}^\times, \mathrm{V}(\mathrm{T}))$ of the log adic generic fiber and is compatible with shifts and fibers. One need only note that $\mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathbf{Z}_p\langle \mathrm{T}\rangle, \mathcal{M})_{\mathbb{A}}, \mathcal{O}_{\mathbb{A}}[1/\mathrm{I}_{\mathbb{A}}]^\wedge_p)$ is closed under shifts and fibers to conclude the argument.

For full faithfulness, it is not quite enough. The strategy will be to place any $\mathcal{E} \in \mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathbf{Z}_p\langle \mathrm{T}\rangle, \mathcal{M})_{\mathbb{A}}, \mathcal{O}_{\mathbb{A}}[1/\mathrm{I}_{\mathbb{A}}]^\wedge_p)$ into a fiber sequence with an "open" and "closed" part. We will use the corresponding stratum on the adic generic fiber as a subscript to indicate the support of a crystal. When comparing $\mathrm{RHom}(\mathcal{E}, \mathcal{E}')$ with $\mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathcal{E}'), \mathrm{T}_{\text{ét}}(\mathcal{E}))$ to deduce the comparison one must ultimately show full faithfulness in the form

$$\mathrm{RHom}(\mathcal{E}_{\mathbb{D}^\times}, \mathcal{E}'_{\mathrm{V}(\mathrm{T})}) \simeq \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathcal{E}'_{\mathrm{V}(\mathrm{T})}), \mathrm{T}_{\text{ét}}(\mathcal{E}_{\mathbb{D}^\times})).$$

Here the corresponding étale realizations are supported on the logarithmic point at the origin and on the punctured disk $\mathbb{D}^\times$. The trick is similar to what was already done in the proof of essential surjectivity of Theorem 5.4. After passing to a prime-to-p Kummer cover $\mathbb{D}_{\mathrm{N}}$, put $\mathbb{L} = \mathrm{T}_{\text{ét}}(\mathcal{E}_{\mathbb{D}_{\mathrm{N}}^\times})$ and $i_*\mathbb{L}' = \mathrm{T}_{\text{ét}}(\mathcal{E}'_{\mathrm{V}(\mathrm{T})})$. We may assume that $\mathbb{L}$ extends across the origin to the p-Kummer derived local system $\mathrm{R}j_*\mathbb{L}$. Thus there is $\mathcal{E}_{\mathbb{D}_{\mathrm{N}}} \in \mathsf{Perf}((\mathrm{Spf}\,\mathbf{Z}_p\langle \mathrm{T}^{1/\mathrm{N}}\rangle, \mathcal{M})_{\mathbb{A}}, \mathcal{O}_{\mathbb{A}}[1/\mathrm{I}_{\mathbb{A}}]^\wedge_p)^{\varphi=1}$ such that $\mathrm{T}_{\text{ét}}(\mathcal{E}_{\mathbb{D}_{\mathrm{N}}}) \simeq \mathrm{R}j_*\mathbb{L}$ and $\mathcal{E}_{\mathbb{D}_{\mathrm{N}}^\times} \simeq \mathcal{E}_{\mathbb{D}_{\mathrm{N}}}(*\mathrm{V}(\mathrm{T}))$. On the Kummer-étale side, one has

$$\mathrm{RHom}(i_*\mathbb{L}', \mathrm{R}j_*\mathbb{L}) = 0.$$

Indeed, the adjunction between j^* and $\mathrm{R}j_*$ identifies this with $\mathrm{RHom}(j^*i_*\mathbb{L}', \mathbb{L}) = 0$. Applying $\mathrm{RHom}(i_*\mathbb{L}', -)$ to the triangle

$$j_!\mathbb{L} \longrightarrow \mathrm{R}j_*\mathbb{L} \longrightarrow i_*i^*\mathrm{R}j_*\mathbb{L}$$

and using the adjunction between i^* and i_* gives $\mathrm{RHom}(i_*\mathbb{L}', j_!\mathbb{L}) \simeq \mathrm{RHom}(\mathbb{L}', i^*\mathrm{R}j_*\mathbb{L})[-1]$. This now consists of sheaves supported at the origin $\mathrm{V}(\mathrm{T}) \subset \mathbb{D}_{\mathrm{N}}$, so producing an analogous isomorphism on the crystal side would reduce the needed result down one dimension to a point where the claim is known (we will use an induction on dimension). For this to be plausible one needs $\mathrm{R}j_*\mathbb{L}$ to be p-Kummer, so there is a corresponding crystal; this is the point of passing to the cover. Using exactly this strategy, one can show

$$\mathrm{RHom}(\mathcal{E}_{\mathbb{D}_{\mathrm{N}}^\times}, \mathcal{E}'_{\mathrm{V}(\mathrm{T})}) \simeq \mathrm{RHom}(\mathrm{fib}(\mathcal{E}_{\mathbb{D}_{\mathrm{N}}} \to \mathcal{E}_{\mathbb{D}_{\mathrm{N}}}(*\mathrm{V}(\mathrm{T}))), \mathcal{E}'_{\mathrm{V}(\mathrm{T})})[-1].$$

The remaining closed term is computed over the quotient by T, so it is indeed a dimension lower. Thus one can reduce to a hollow point where the claim can be checked directly.

Notation for this section. Fix a framed log affine $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ with framing $\square$ as in Definition 3.2, and put $\mathrm{A} = \mathrm{R}_0^\square(\!(q-1)\!)^\wedge_p$. For $\Lambda \subseteq \{1, \ldots, r\}$, put

$$\mathrm{J}_\Lambda = (\mathrm{T}_i \mid i \in \Lambda).$$

For $\mathrm{Z}_\Lambda = \mathrm{V}(\mathrm{J}_\Lambda)$, put $\mathrm{D}|_{\mathrm{Z}_\Lambda} = \bigcup_{i \notin \Lambda}(\mathrm{D}_i \cap \mathrm{Z}_\Lambda)$ and consider $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_\Lambda)^{h\Gamma}$. The group $\Gamma_\Lambda = \prod_{i\in\Lambda} \mathbf{Z}_p(1)_i$ fixes $\mathrm{A}/\mathrm{J}_\Lambda$. Give Z_Λ the pullback log structure, which now has a hollow component. Since we continue using the full group Γ, we see $\mathrm{T}_{\text{ét}}$ factors as

$$\mathrm{T}_{\text{ét}} : \left(\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_\Lambda)^{h\Gamma}\right)^{\mathrm{op}} \longrightarrow \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons},\mathrm{D}|_{\mathrm{Z}_\Lambda}}((\mathrm{Z}_\Lambda, \mathcal{M}|_{\mathrm{Z}_\Lambda})_\eta, \mathbf{Z}_p)$$

followed by extension by zero. One can see this using Lemma 4.26. The target is constructible for the remaining coordinate stratification.

We now prove a local version of Kashiwara's lemma. We write $\mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\mathbb{A}}, \mathcal{O}_{\mathbb{A}}[1/\mathrm{I}_{\mathbb{A}}]^\wedge_p)_{\mathrm{Z}}$ for the full subcategory of finitely generated unit Laurent crystals supported on Z. Let us explain the general globalized notion of this when $\mathrm{D} = \bigcup_i \mathrm{D}_i$ is a simple normal crossings divisor with irreducible components D_i. For any nonempty intersection $\mathrm{Z} = \mathrm{D}_1 \cap \cdots \cap \mathrm{D}_c$ of distinct irreducible components, define

$$\mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})_{\mathrm{Z}} := \bigcap_{a=1}^{c} \ker\left(- \otimes^{\mathrm{L},\blacksquare} \mathcal{O}_{\Delta}(*\mathrm{D}_a)\right).$$

Similarly one may make a definition for connected components. This covers all Z which are smooth closed log strata of the stratification induced by D; connected components are necessarily of pure codimension. On a compatible framing, Theorem 4.6 identifies this supported ambient category with the source of the following lemma.

Lemma 5.18 (Kashiwara's lemma). Let $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ be a framed log affine, put $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$ and $\mathrm{J} = \mathrm{J}_\Lambda = (\mathrm{T}_\lambda \mid \lambda \in \Lambda)$, and let $i : \mathrm{Z} = \mathrm{V}(\mathrm{J}) \hookrightarrow \mathrm{Spf}\,\mathrm{R}$ so that under Theorem 4.6 one has

$$\mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})_{\mathrm{Z}} \simeq \mathrm{D}_{\mathrm{fgu}}(\mathrm{A})_{\mathrm{J}}^{h\Gamma},$$

the analogous category of finitely generated unit modules supported on Z.
Then one has an equivalence

$$i^! := \left((\mathrm{A}/\mathrm{J}) \otimes_{\mathrm{A}}^{\mathrm{L},\blacksquare} -\right)[-|\Lambda|] : \mathrm{D}_{\mathrm{fgu}}(\mathrm{A})_{\mathrm{J}}^{h\Gamma} \xrightarrow{\sim} \mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J})^{h\Gamma}$$

We write i_+ for the inverse equivalence.

Proof. Put $\overline{\mathrm{A}} = \mathrm{A}/p$ and $\overline{\mathrm{J}} = \mathrm{J}\overline{\mathrm{A}}$. The categories and functor above satisfy the hypotheses of Lemma 2.3; the support condition is detected modulo p by derived Nakayama. It is therefore enough to prove that

$$i^! := \left((\overline{\mathrm{A}}/\overline{\mathrm{J}}) \otimes_{\overline{\mathrm{A}}}^{\mathrm{L},\blacksquare} -\right)[-|\Lambda|] : \mathrm{D}_{\mathrm{fgu}}(\overline{\mathrm{A}})_{\overline{\mathrm{J}}}^{h\Gamma} \longrightarrow \mathrm{D}_{\mathrm{fgu}}(\overline{\mathrm{A}}/\overline{\mathrm{J}})^{h\Gamma}$$

is an equivalence. The functor is well-defined: one can use Lemma 4.23.

In this case, the functor is an equivalence before we apply $(-)^{h\Gamma}$. One may generalize [EK04b, Corollary 5.11.3] to the present regular F-finite situation (indeed the proof still applies, noting we have access to important ingredients like roots by [Bli08]). Thus one obtains an adjoint equivalence

$$i_+ : \mathrm{D}_{\mathrm{fgu}}(\overline{\mathrm{A}}/\overline{\mathrm{J}}) \rightleftarrows \mathrm{D}_{\mathrm{fgu}}(\overline{\mathrm{A}})_{\overline{\mathrm{J}}} : i^!.$$

This survives after applying a bar construction $(-)^{h\Gamma}$ to both sides, because evaluation on the degree zero term is conservative. Thus one may test $i^! i_+ \simeq \mathrm{id}$ and $i_+ i^! \simeq \mathrm{id}$ after evaluation on the degree zero term, at which point it reduces to the ordinary Kashiwara lemma for finitely generated unit modules. ☐

For $i : \mathrm{Z}_\Lambda \hookrightarrow \mathrm{Spf}\,\mathrm{R}$ and $\mathrm{M} \in \mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_\Lambda)^{h\Gamma}$, the equivalence above in Lemma 5.18 gives

$$\mathrm{T}_{\text{ét}}(i_+\mathrm{M}) \simeq i_* \mathrm{T}_{\text{ét}}(\mathrm{M})[-|\Lambda|].$$

It is also easy to give a formula for $i_+ i^!(-)$ even when the input is not supported on Z. On the same chart, we also remark that local cohomology along Z is computed by

$$\mathrm{R}\Gamma_{\mathrm{Z}}(\mathrm{M}) \simeq \mathrm{M} \otimes_{\mathrm{A}}^{\mathrm{L},\blacksquare} \bigotimes_{i\in\Lambda}^{\mathrm{L},\blacksquare} \left[\mathrm{A} \longrightarrow (\mathrm{A}[1/\mathrm{T}_i])_p^\wedge\right].$$

Completed base change to A/J kills every localized term, so $i^!\mathrm{R}\Gamma_{\mathrm{Z}}(\mathrm{M}) \simeq i^!\mathrm{M}$. The complex $\mathrm{R}\Gamma_{\mathrm{Z}}(\mathrm{M})$ is canonically Γ-equivariant, since each $\gamma \in \Gamma$ carries every T_i to a unit multiple of itself. Checking $\mathrm{R}\Gamma_{\mathrm{Z}}(\mathrm{M})$ is finitely generated unit is straightforward, so Kashiwara's lemma therefore gives

$$\mathrm{R}\Gamma_{\mathrm{Z}}(\mathrm{M}) \simeq i_+ i^!\mathrm{M}.$$

In the toy model at the start of the subsection, the following is the Laurent crystal analogue of $\mathrm{RHom}(i_*\mathbb{L}', \mathrm{R}j_*\mathbb{L}) = 0$ (any Kummer-étale local system in the second argument can be written this way using [DLLZ23]).

Lemma 5.19. Let $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ be a framed log affine, put $\mathrm{A} = \mathrm{R}_0^\square(\!(q-1)\!)_p^\wedge$, and fix one boundary component $\mathrm{Z} = \mathrm{D}_i = \mathrm{V}(\mathrm{T}_i)$. If

$$\mathrm{E} \in \mathsf{Perf}(\mathrm{A})^{\varphi=1,h\Gamma}$$

and $\mathrm{Q} \in \mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/(\mathrm{T}_i))^{h\Gamma}$, then

$$\mathrm{RHom}(\mathrm{E}, i_+\mathrm{Q}) = 0.$$

Proof. Put $\mathrm{T} = \mathrm{T}_i$ and $\overline{\mathrm{A}} = \mathrm{A}/p$. Since Kashiwara's equivalence commutes with reduction modulo p, derived Nakayama reduces the assertion modulo p. A Postnikov dévissage (see [Hau24]) then reduces to Q concentrated in degree zero.

We use the distinguished minimal root of Q to give an explicit formula for $i_+\mathrm{Q}$. Consider

$$\mathcal{L} = \mathrm{T}^{-1}\overline{\mathrm{A}}/\overline{\mathrm{A}} \simeq (\mathrm{T}/\mathrm{T}^2)^\vee.$$

The inclusion $\mathrm{T}^{-1}\overline{\mathrm{A}}/\overline{\mathrm{A}} \to \mathrm{T}^{-p}\overline{\mathrm{A}}/\overline{\mathrm{A}}$ defines an injective map $\mathcal{L} \to \varphi^*\mathcal{L}$; this is the root for $i_+(\overline{\mathrm{A}}/(\mathrm{T}))$. It is a distinguished minimal root, so it is Γ-equivariant.

Let Q_0 be the distinguished minimal root of Q, which is Γ-stable by its distinguished property, and put

$$\mathrm{K} = \mathrm{colim}_{n\geq 0}\, \varphi^{*n}\big(\mathcal{L} \otimes_{\overline{\mathrm{A}}/(\mathrm{T})}^{\blacksquare} \mathrm{Q}_0\big).$$

This is a finitely generated unit Γ-module supported on V(T), and

$$i^!\mathrm{K} \simeq \Big((\overline{\mathrm{A}}/(\mathrm{T})) \otimes_{\overline{\mathrm{A}}}^{\mathrm{L},\blacksquare} \mathrm{K}\Big)[-1] \simeq \mathrm{Q}.$$

Lemma 5.18 therefore identifies K with $i_+\mathrm{Q}$.

There is a natural exhaustive "pole order" filtration on $i_+\mathrm{Q}$, with $\mathrm{Fil}_0 = 0$ and

$$\mathrm{Fil}_m = \mathrm{colim}_{p^n \geq m}\, \mathrm{T}^{p^n - m}\varphi^{*n}(\mathcal{L} \otimes^{\blacksquare} \mathrm{Q}_0).$$

Put $\mathrm{C} = \underline{\mathrm{RHom}}_{\overline{\mathrm{A}}/(\mathrm{T})}((\overline{\mathrm{A}}/(\mathrm{T})) \otimes^{\mathrm{L},\blacksquare}_{\overline{\mathrm{A}}} \mathrm{E}, \mathrm{Q})$. Using the unit Frobenius isomorphisms of Q, its graded pieces give isomorphisms of underlying complexes

$$\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E}, \mathrm{gr}_m(i_+\mathrm{Q})) \simeq \mathrm{C} =: \mathrm{C}_m.$$

We equip C_m with the Γ-action induced by the left side; relative to the natural action on the right side, it is twisted by the character through which Γ acts on T^{-m}.

The complex C is a finitely generated unit (φ, Γ)-module, and each C_m has the same underlying module with a twisted Γ-action. Every $\mathrm{H}^i(\mathrm{C})$ is $\mathrm{F}_{\overline{\mathrm{A}}/(\mathrm{T})}$-finite in the convention of [Hoc07], using the inverse of its unit isomorphism. By [Hoc07, Theorem 5.1], $\mathrm{End}_{\mathrm{F}_{\overline{\mathrm{A}}/(\mathrm{T})}}(\mathrm{H}^j(\mathrm{C}))$ is finite. The subgroup $\mathbf{Z}_p(1)_i \subset \Gamma$ corresponding to T fixes $\overline{\mathrm{A}}/(\mathrm{T})$ and commutes with Frobenius, so its action on each $\mathrm{H}^j(\mathrm{C})$ factors through the finite group of invertible Frobenius-linear endomorphisms. This subgroup is normal: every $\mathbf{Z}_p(1)_j$ commutes with it, while conjugation by $a \in 1 + p\mathbf{Z}_p$ sends $b \in \mathbf{Z}_p(1)_i$ to ab. Since C is bounded, $\mathrm{H} = p^\ell \mathbf{Z}_p(1)_i$ acts trivially on all its cohomology modules for some ℓ. Let γ generate H, with $\gamma(\mathrm{T}) = q^{p^{\ell+1}}\mathrm{T}$.

Once we restrict to this $\mathrm{H} \subset \mathbf{Z}_p(1)_i$, only the twisted part of the action remains. On each C_m, the action of γ on $\mathrm{H}^j(\mathrm{C}_m)$ is therefore multiplication by $q^{-mp^{\ell+1}}$. Thus $\gamma - 1$ acts by the unit $q^{-mp^{\ell+1}} - 1$ modulo p, since $q - 1$ is invertible. For fixed m the filtration on $\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E}, \mathrm{Fil}_m(i_+\mathrm{Q}))$ is finite, so $\gamma - 1$ is an automorphism on this complex.

The filtration on $i_+\mathrm{Q}$ is exhaustive. Since E is perfect, $\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E}, -) \simeq \mathrm{E}^\vee \otimes^{\mathrm{L},\blacksquare}_{\overline{\mathrm{A}}} -$ preserves filtered colimits. Hence $\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E}, i_+\mathrm{Q}) \simeq \mathrm{colim}_m \, \underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E}, \mathrm{Fil}_m(i_+\mathrm{Q}))$, and $\gamma - 1$ is an equivalence on this complex. Since $\mathrm{H} \simeq \mathbf{Z}_p$, its standard two-term resolution to compute continuous cohomology then gives

$$\mathrm{R\Gamma}_{\mathrm{cont}}(\mathrm{H}, \underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E}, i_+\mathrm{Q})) \simeq [\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E}, i_+\mathrm{Q}) \xrightarrow{\gamma-1} \underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E}, i_+\mathrm{Q})] = 0.$$

Applying Hochschild-Serre gives

$$\mathrm{RHom}(\mathrm{E}, i_+\mathrm{Q}) \simeq \mathrm{R\Gamma}_{\mathrm{cont}}\big(\Gamma/\mathrm{H}, \mathrm{R\Gamma}_{\mathrm{cont}}(\mathrm{H}, \underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E}, i_+\mathrm{Q}))\big)^{\varphi=1} = 0.$$

Thus the claim follows. □

We now introduce an assumption which will be our induction step for the main theorem.

Assumption 5.20. Let $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ be a framed log affine. Assume that the restriction of $\mathrm{T}_{\text{ét}}$ to $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\{i\}})^{h\Gamma}$ is an equivalence onto the full subcategory of $\mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons},\mathrm{D}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_\eta, \mathbf{Z}_p)$ consisting of sheaves supported on $\mathrm{V}(\mathrm{T}_i)$ for all i.

As in the toy model, what we will try to do is reduce the general claim to a claim which is one dimension lower, which is where the assumption appears.

Lemma 5.21. Suppose Assumption 5.20 holds.
If $\mathcal{E}, \mathcal{E}' \in \mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p)$ satisfy

$$\mathcal{E}(*\mathrm{D}) = \mathcal{E}'(*\mathrm{D}) = 0,$$

then the natural map

$$\mathrm{RHom}(\mathcal{E},\mathcal{E}') \longrightarrow \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathcal{E}'),\mathrm{T}_{\text{ét}}(\mathcal{E}))$$

is an equivalence.
Moreover, if $\mathcal{E}_0 \in \mathsf{Perf}((\mathrm{Spf}\,\mathrm{R},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)^{\varphi=1}$ and $\mathrm{Q} \in \mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)$ satisfies $\mathrm{Q}(*\mathrm{D}) = 0$, then $\mathrm{RHom}(\mathcal{E}_0,\mathrm{Q}) = 0$.

Proof. Write $\mathrm{D} = \bigcup_{i=1}^r \mathrm{D}_i$, where $\mathrm{D}_i = \mathrm{V}(\mathrm{T}_i)$. Kashiwara's equivalence identifies the category supported on D_i with $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\{i\}})^{h\Gamma}$, and its compatibility with realization identifies the supported comparison with the restriction of $\mathrm{T}_{\text{ét}}$ to this category.

The first statement does not immediately follow from Lemma 5.18, because D need not be smooth. However it is not too far from the statement: we will use localization triangles to break D into smooth components by peeling off pieces supported on D_i one at a time.

Let $\mathcal{E}(*\mathrm{D}) = 0$. Put $\mathcal{E}_0 = \mathcal{E}$ and $\mathcal{E}_i = \mathcal{E}_{i-1}(*\mathrm{D}_i)$. For each i,

$$\mathrm{R}\Gamma_{\mathrm{D}_i}(\mathcal{E}_{i-1}) \longrightarrow \mathcal{E}_{i-1} \longrightarrow \mathcal{E}_i$$

is a localization triangle. Indeed if $\mathcal{E}$ is bounded finitely generated unit, then so are $\mathcal{E}(*\mathrm{D}_i)$ and $\mathrm{R}\Gamma_{\mathrm{D}_i}(\mathcal{E})$ by the calculation following Lemma 5.18 (both also inherit continuous Γ actions commuting with Frobenius). Symmetric monoidality and the identification of $\mathrm{T}_{\text{ét}}(\mathcal{O}_{\Delta}(*\mathrm{D}_i))$ with extension by zero from the complement of D_i show $\mathrm{T}_{\text{ét}}$ produces the corresponding Kummer-étale localization triangle. Moreover,

$$\mathcal{E}_r \simeq \mathcal{E} \otimes^{\mathrm{L},\blacksquare} \mathcal{O}_{\Delta}(*\mathrm{D}_1) \otimes^{\mathrm{L},\blacksquare} \cdots \otimes^{\mathrm{L},\blacksquare} \mathcal{O}_{\Delta}(*\mathrm{D}_r) \simeq \mathcal{E}(*\mathrm{D}) = 0.$$

Each $\mathrm{R}\Gamma_{\mathrm{D}_i}(\mathcal{E}_{i-1})$ is supported on D_i. For fixed $\mathcal{E}'$, the comparison is a natural transformation of exact functors

$$\mathrm{RHom}(-,\mathcal{E}') \longrightarrow \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathcal{E}'),\mathrm{T}_{\text{ét}}(-)).$$

Applying it to the preceding triangles for $\mathcal{E}$ and working downward from $\mathcal{E}_r = 0$ reduces the first assertion to the case where $\mathcal{E}$ is supported on some D_i. We can run a similar reduction for $\mathcal{E}'$. The localization triangle for $\mathcal{E}'$ is

$$\mathrm{R}\Gamma_{\mathrm{D}_i}(\mathcal{E}') \longrightarrow \mathcal{E}' \longrightarrow \mathcal{E}'(*\mathrm{D}_i).$$

Localization gives $\mathrm{RHom}(\mathcal{E},\mathcal{E}'(*\mathrm{D}_i)) = 0$. One sees $\mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathcal{E}'(*\mathrm{D}_i)),\mathrm{T}_{\text{ét}}(\mathcal{E})) = 0$, so the comparison is therefore reduced to

$$\mathrm{RHom}(\mathcal{E},\mathrm{R}\Gamma_{\mathrm{D}_i}(\mathcal{E}')) \longrightarrow \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{R}\Gamma_{\mathrm{D}_i}(\mathcal{E}')),\mathrm{T}_{\text{ét}}(\mathcal{E})).$$

Both crystals are now supported on D_i. Kashiwara's equivalence and full faithfulness of $\mathrm{T}_{\text{ét}}$ on $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\{i\}})^{h\Gamma}$ show that this map is an equivalence.

The localization dévissage above shows that every Q with $\mathrm{Q}(*\mathrm{D}) = 0$ lies in the thick subcategory generated by bounded finitely generated unit crystals supported on the smooth divisors D_i. By Lemma 5.18, each such crystal is of the form $i_+\mathrm{Q}'$ for $\mathrm{Q}' \in \mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\{i\}})^{h\Gamma}$; hence Lemma 5.19 and exactness of $\mathrm{RHom}(\mathcal{E}_0,-)$ give the last assertion.

The same proof applies in $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\Lambda})^{h\Gamma}$, using the remaining boundary coordinates. Its induction hypothesis is full faithfulness of $\mathrm{T}_{\text{ét}}$ on $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\Lambda\cup\{i\}})^{h\Gamma}$ for $i \notin \Lambda$. ∎

We next treat maps between objects supported on the open stratum and its complement.

Lemma 5.22. Under Assumption 5.20, let
$$\mathrm{P} \in \mathsf{Perf}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1}$$
and let $\mathrm{Q} \in \mathrm{D}_{\mathrm{fgu}}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ satisfy $\mathrm{Q}(*\mathrm{D}) = 0$. Then the natural map
$$\mathrm{RHom}(\mathrm{P}, \mathrm{Q}) \longrightarrow \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathrm{T}_{\text{ét}}(\mathrm{P}))$$
is an equivalence.

Proof. Let $j : \mathrm{U}_{\eta} \hookrightarrow (\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\eta}$ be the complement of D. Theorem 5.4 identifies $\mathrm{T}_{\text{ét}}(\mathrm{P})$ with $j_{!}\mathbb{L}$ for a p-Kummer derived local system $\mathbb{L}$ on the open stratum.

By [DLLZ23, Corollary 6.3.4], the cohomology sheaves of $\mathbb{L}$ extend across the boundary as Kummer-étale local systems. The vanishing in [DLLZ23, Theorem 4.6.1], applied modulo p^n and passed to the inverse limit, makes the spectral sequence
$$\mathrm{E}_2^{a,b} = \mathrm{R}^a j_* \mathrm{H}^b(\mathbb{L}) \Longrightarrow \mathrm{H}^{a+b}(\mathrm{R}j_*\mathbb{L})$$
degenerate. Thus $\mathrm{R}j_*\mathbb{L}$ is derived lisse, and, as in the essential surjectivity component of Theorem 5.4, the prime-to-p boundary inertia of each cohomology sheaf has finite image. A Postnikov dévissage as in [Hau24, Lemma 5.4.3] and boundedness give a common exponent for these finite images. Hence a single prime-to-p root cover X_{N} along the boundary makes $\mathrm{R}j_*\mathbb{L}$ p-Kummer.

On the log q-de Rham chart $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)^{\wedge}_{p}$, for the Kummer cover the corresponding ring is
$$\mathrm{A}_{\mathrm{N}} = \mathrm{A}[\mathrm{S}_1, \ldots, \mathrm{S}_r]/(\mathrm{S}_1^{\mathrm{N}} - \mathrm{T}_1, \ldots, \mathrm{S}_r^{\mathrm{N}} - \mathrm{T}_r).$$
The algebra A_{N} is finite free of rank N^r, including along D. Its normalized trace $\tau = \mathrm{N}^{-r}\mathrm{Tr}_{\mathrm{A}_{\mathrm{N}}/\mathrm{A}}$ satisfies $\tau(\sum_{0 \leq a_i < \mathrm{N}} x_{\alpha} \prod_i \mathrm{S}_i^{a_i}) = x_{\vec{0}}$. Since $(p, \mathrm{N}) = 1$, multiplication by p permutes the nonzero residue classes modulo N, so τ is Frobenius-equivariant. It is also Γ-equivariant and splits $\mathrm{A} \to \mathrm{A}_{\mathrm{N}}$.

The Kummer cover X_{N} we chose is a torsor under the finite étale group scheme μ_{N}^r. After a finite strict-étale refinement obtained from an unramified extension containing μ_{N} (i.e. enlarging k), this group scheme becomes constant, and averaging defines a normalized corestriction on Kummer-étale sheaves. The construction descends from the refinement. There both normalized maps are projection onto the invariant summand, so pullback compatibility of $\mathrm{T}_{\text{ét}}$ identifies the trace with corestriction. Applying these two normalized traces gives

$$\begin{CD}
\mathrm{RHom}(\mathrm{P}, \mathrm{Q}) @>>> \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathrm{T}_{\text{ét}}(\mathrm{P})) \\
@V{f_{\mathrm{N}}^*}VV @VV{f_{\mathrm{N}}^*}V \\
\mathrm{RHom}(f_{\mathrm{N}}^*\mathrm{P}, f_{\mathrm{N}}^*\mathrm{Q}) @>>> \mathrm{RHom}(\mathrm{T}_{\text{ét}}(f_{\mathrm{N}}^*\mathrm{Q}), \mathrm{T}_{\text{ét}}(f_{\mathrm{N}}^*\mathrm{P})).
\end{CD}$$

They give compatible retractions of the vertical maps. Hence the top arrow is a retract of the bottom arrow, so it suffices to prove that the bottom arrow is an equivalence.

On the root cover, $\mathrm{R}j_*\mathbb{L}$ is p-Kummer. The Koshikawa-Yao correspondence [IKY26, Theorem 7] therefore gives a perfect unit Laurent crystal $\mathcal{E}_0$ realizing it. Since $\mathcal{E}_0(*\mathrm{D}_\mathrm{N})$ and $f_\mathrm{N}^*\mathrm{P}$ both realize $j_!f_\mathrm{N}^*\mathbb{L}$, open-stratum full faithfulness gives

$$f_\mathrm{N}^*\mathrm{P} \simeq \mathcal{E}_0(*\mathrm{D}_\mathrm{N})$$

with $\mathcal{E}_0 \in \mathsf{Perf}((\mathrm{X}_\mathrm{N}, \mathcal{M}_\mathrm{N})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge)^{\varphi=1}$. We now work on the root cover and omit the pullbacks and subscripts. Put $\mathcal{E}_\mathrm{Z} = \mathrm{R}\Gamma_\mathrm{D}(\mathcal{E}_0)$. The localization triangle is

$$\mathcal{E}_\mathrm{Z} \longrightarrow \mathcal{E}_0 \longrightarrow \mathrm{P}.$$

The localization and coherent-root arguments in the proofs of [EK04b, Propositions 5.11.5 and 6.8.1] show that $\mathcal{E}_\mathrm{Z}$ is bounded finitely generated unit and satisfies $\mathcal{E}_\mathrm{Z}(*\mathrm{D}) = 0$. The last assertion of Lemma 5.21 gives $\mathrm{RHom}(\mathcal{E}_0, \mathrm{Q}) = 0$.

By construction, $\mathrm{T}_{\text{ét}}(\mathcal{E}_0) \simeq \mathrm{R}j_*\mathbb{L}$. Since $\mathrm{Q}(*\mathrm{D}) = 0$, symmetric monoidality and the calculation preceding Theorem 5.4 give $j^*\mathrm{T}_{\text{ét}}(\mathrm{Q}) = 0$. The adjunction between j^* and $\mathrm{R}j_*$ therefore gives

$$\mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathrm{T}_{\text{ét}}(\mathcal{E}_0)) \simeq \mathrm{RHom}(j^*\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathbb{L}) = 0.$$

Applying $\mathrm{RHom}(-, \mathrm{Q})$ to the localization triangle and $\mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), -)$ to its realization gives a map of fiber sequences

$$\begin{tikzcd}
\mathrm{RHom}(\mathrm{P}, \mathrm{Q}) \arrow[r] \arrow[d] & \mathrm{RHom}(\mathcal{E}_0, \mathrm{Q}) \arrow[r] \arrow[d] & \mathrm{RHom}(\mathcal{E}_\mathrm{Z}, \mathrm{Q}) \arrow[d] \\
\mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathrm{T}_{\text{ét}}(\mathrm{P})) \arrow[r] & \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathrm{T}_{\text{ét}}(\mathcal{E}_0)) \arrow[r] & \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathrm{T}_{\text{ét}}(\mathcal{E}_\mathrm{Z})).
\end{tikzcd}$$

The two middle terms vanish by the preceding two paragraphs, while the right vertical map is an equivalence by the first assertion of Lemma 5.21. The left vertical map is therefore an equivalence. This proves the comparison on the root cover; the trace retractions constructed above then prove it downstairs.

The same argument applies in $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_\Lambda)^{h\Gamma}$. One takes the root cover only in the remaining boundary coordinates. The normalized trace is Γ-equivariant, including for Γ_Λ. After forgetting the Γ_Λ-action, apply the perfect unit correspondence to $\mathrm{R}j_*\mathbb{L}$; full faithfulness lifts the continuous Γ_Λ-action and its cocycle uniquely. Thus the mixed comparison holds for $\mathrm{T}_{\text{ét}}$ whenever the boundary-supported comparison holds in $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_\Lambda)^{h\Gamma}$. □

Proposition 5.23. Let $(\operatorname{Spf}\mathrm{R}, \mathcal{M})$ be a framed log affine, and suppose Assumption 5.20 holds. Then the étale realization

$$\mathrm{T}_{\text{ét}} : \mathrm{D}_{\mathrm{fgu}}((\operatorname{Spf}\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge)^{\mathrm{op}} \longrightarrow \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons,D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$$

is an equivalence.

Proof. The hypothesis gives us full faithfulness on full subcategories of $\mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge)$ that have the form

$$\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_\Lambda)^{h\Gamma}$$

with $|\Lambda| = 1$ under Theorem 4.6 (here as usual $\mathrm{A} = \mathrm{R}_0^\square(\!(q-1)\!)_p^\wedge$ with its Γ action).

The desired claim is for $\Lambda = \emptyset$, so we will show full faithfulness $|\Lambda| = 1$ implies the claim for $\Lambda = \emptyset$. We reduce full faithfulness to maps between two objects supported on D, between two objects supported on the open stratum, and from an object supported on the open stratum to an object supported on D. These cases are Lemma 5.21, Theorem 5.4, and Lemma 5.22, respectively. The middle case does not use the induction hypothesis, but the others do.

Let $\mathcal{E}, \mathcal{E}' \in \mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$. Put $\mathcal{E}_{\mathrm{Z}} = \mathrm{R}\Gamma_{\mathrm{D}}(\mathcal{E})$ and $\mathcal{E}_{\mathrm{U}} = \mathcal{E}(*\mathrm{D})$, and use primes for the variants applied to $\mathcal{E}'$. It is straightforward to check that $\mathcal{E}_{\mathrm{Z}}$ (see the discussion after Lemma 5.18 which identifies this with $i_+ i^!$) and $\mathcal{E}_{\mathrm{U}}$ remain finitely generated unit, and similarly for $\mathcal{E}'_{\mathrm{Z}}$ and $\mathcal{E}'_{\mathrm{U}}$. We will later check the open terms are actually in $\mathsf{Perf}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1}$.

The associated localization triangles give a square of fiber sequences

$$\begin{tikzcd}
\mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{Z}}) \arrow[r] \arrow[d] & \mathrm{RHom}(\mathcal{E}, \mathcal{E}'_{\mathrm{Z}}) \arrow[r] \arrow[d] & \mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}'_{\mathrm{Z}}) \arrow[d] \\
\mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}') \arrow[r] \arrow[d] & \mathrm{RHom}(\mathcal{E}, \mathcal{E}') \arrow[r] \arrow[d] & \mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}') \arrow[d] \\
\mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{U}}) \arrow[r] & \mathrm{RHom}(\mathcal{E}, \mathcal{E}'_{\mathrm{U}}) \arrow[r] & \mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}'_{\mathrm{U}}).
\end{tikzcd}$$

In fact every row and column is a fiber sequence. Since $\mathcal{E}_{\mathrm{Z}}(*\mathrm{D}) = 0$ and $\mathcal{E}'_{\mathrm{U}} \simeq \mathcal{E}'_{\mathrm{U}}(*\mathrm{D})$, the lower right term $\mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}'_{\mathrm{U}})$ vanishes. After applying $\mathrm{T}_{\text{ét}}$, let $j : \mathrm{U}_{\eta} \hookrightarrow (\mathrm{X}, \mathcal{M})_{\eta}$ denote the complement of D. The adjunction between $j_!$ and j^* identifies the analogous term with

$$\mathrm{RHom}(j^* \mathrm{T}_{\text{ét}}(\mathcal{E}'_{\mathrm{U}}), j^* \mathrm{T}_{\text{ét}}(\mathcal{E}_{\mathrm{Z}})) = 0,$$

since $\mathrm{T}_{\text{ét}}(\mathcal{E}_{\mathrm{Z}})$ is supported on the complement of U. The same vanishing gives $\mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}') \simeq \mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}'_{\mathrm{Z}})$ and $\mathrm{RHom}(\mathcal{E}, \mathcal{E}'_{\mathrm{U}}) \simeq \mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{U}})$. Taking the total fiber of the bottom right square by columns gives $\mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{Z}})$; by rows, it is the fiber of

$$\mathrm{RHom}(\mathcal{E}, \mathcal{E}') \to \mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}'_{\mathrm{Z}}) \oplus \mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{U}})$$

and hence

$$\mathrm{RHom}(\mathcal{E}, \mathcal{E}') \longrightarrow \mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}'_{\mathrm{Z}}) \oplus \mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{U}}) \longrightarrow \mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{Z}}[1]).$$

after rotating the fiber sequence. Exactness of the étale realization gives the corresponding fiber sequence after applying $\mathrm{T}_{\text{ét}}$, and naturality gives a map between them. To deduce that the map induced by $\mathrm{T}_{\text{ét}}$ on the first term is an equivalence, since it is a fiber sequence we need only check the induced maps on the second and third terms are isomorphisms.

We verify that the open terms satisfy the hypotheses of Theorem 5.4 and Lemma 5.22. Put $\mathrm{A}^{\circ} = \mathrm{A}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]^{\wedge}_{p}$. Theorem 4.6 identifies $\mathcal{E}_{\mathrm{U}}$ with an object $\mathrm{M} \in \mathrm{D}_{\mathrm{fgu}}(\mathrm{A})^{h\Gamma}$ satisfying $\mathrm{M} \simeq \mathrm{A}^{\circ} \otimes^{\mathrm{L},\blacksquare}_{\mathrm{A}} \mathrm{M}$. Modulo p, the Fitting ideal argument in the proof of Proposition 4.24 makes the minimal roots of the cohomology modules of M vector bundles over A°/p and their root maps isomorphisms. Hence M/p is perfect over A°/p, and therefore M is perfect over A°. Lemma 5.8 then gives

$$\mathcal{E}_{\mathrm{U}} \in \mathsf{Perf}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1}.$$

The same argument of course applies to $\mathcal{E}'_{\mathrm{U}}$.

Lemma 5.21 treats $\mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}'_{\mathrm{Z}})$ using the induction hypothesis. Theorem 5.4 treats $\mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{U}})$. Lemma 5.22 treats $\mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{Z}}[1])$ again under the induction hypothesis. Hence the comparison for $\mathcal{E}$ and $\mathcal{E}'$ is an equivalence.

Finally, we address essential surjectivity. Since the source is idempotent complete and $\mathrm{T}_{\text{ét}}$ is exact and fully faithful, its essential image is thick. The target is thickly generated by extensions by zero from the open stratum and by its full subcategories supported on the divisors $\mathrm{V}(\mathrm{T}_i)$. The former lie in the essential image by Theorem 5.4, and the latter do by Assumption 5.20 and Lemma 5.18. □

Corollary 5.24. Let $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ be a framed log affine. Then

$$\mathrm{T}_{\text{ét}} : \mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})^{\mathrm{op}} \longrightarrow \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons,D}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\eta}, \mathbf{Z}_p)$$

is an equivalence.

Proof. We run a descending induction on $|\Lambda|$ for the categories $\left(\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\Lambda})^{h\Gamma}\right)^{\mathrm{op}}$, in the notation notation fixed at the start of this subsection. We claim these are equivalent to the full subcategories of $\mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons,D}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\eta}, \mathbf{Z}_p)$ supported on $(\bigcap_{i\in\Lambda} \mathrm{D}_i)_{\eta}$ (with its pullback log structure). Equivalently, they should be equivalent to p-Kummer constructible sheaves on $(\bigcap_{i\in\Lambda} \mathrm{D}_i)_{\eta}$ with its induced stratification by restricting the divisor.

In the base case $|\Lambda| = r$, the equivalence can be checked directly. Here this is because the corresponding stratum on the log adic generic fiber is an ordinary framed affine $\mathrm{Z} = \bigcap_{i=1}^{r} \mathrm{D}_i$ over a rank $n-r$ torus with a rank r hollow log structure. That is we get an induced framing

$$\square : \mathrm{Z} \to \mathrm{Spf}\,\mathrm{W}(k)[\zeta_p]\langle \mathrm{T}^{\pm}_{r+1}, \ldots, \mathrm{T}^{\pm}_{n}\rangle,$$

where the map is strict-étale for the rank r hollow log structure on the torus. The torus has a natural perfectoid cover $\mathrm{Spf}\,\mathrm{W}(k)^{\mathrm{cyc}}\langle \mathrm{T}^{\pm 1/p^{\infty}}_{r+1}, \ldots, \mathrm{T}^{\pm 1/p^{\infty}}_{n}\rangle$ with hollow log structure $\mathbf{N}[1/p]^r \to 0$, so using $\square$ we obtain an associated perfectoid cover $(\mathrm{R}^{\square}_0)_{\infty,\mathrm{Z}}[1/p]$ again with the chart $\mathbf{N}[1/p]^r \to 0$; we let $(\mathrm{A}/\mathrm{J}_{\Lambda})_{\mathrm{perf}}$ denote $\mathbb{A}_{\mathrm{inf}}((\mathrm{R}^{\square}_0)_{\infty,\mathrm{Z}}[1/p]^{\flat})$. Katz's theorem, tilting, and pro-Kummer-étale descent for the same full group Γ give

$$\left(\mathsf{Perf}((\mathrm{A}/\mathrm{J}_{\Lambda})_{\mathrm{perf}})^{\varphi=1,h\Gamma}\right)^{\mathrm{op}} \xrightarrow{\sim} \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{lisse}}\left((\mathrm{Z}, \mathbf{N}^r)_{\eta}, \mathbf{Z}_p\right) \xhookrightarrow{i_*} \mathrm{D}^{(b)}_{\mathrm{cons,D}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\eta}, \mathbf{Z}_p).$$

In this case the finitely generated unit condition is equivalent to being perfect with unit Frobenius by the Fitting ideal argument of Proposition 4.24. Moreover,

$$\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\Lambda})^{h\Gamma} \simeq \mathsf{Perf}((\mathrm{A}/\mathrm{J}_{\Lambda})_{\mathrm{perf}})^{\varphi=1,h\Gamma}$$

by [BS23, Proposition 3.6] and Lemma 2.3; thus the desired claim follows. We note Lemma 4.26 identifies construction with $\mathrm{T}_{\text{ét}}$.

The proof of Proposition 5.23 applied to $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\Lambda})^{h\Gamma}$ supplies the induction step, deducing the claim for Λ from the claims for $\Lambda \cup \{i\}$ for all $i \notin \Lambda$. Strictly speaking, as stated the proposition would only apply with the group Γ/Γ_{Λ}, where $\Gamma_{\Lambda} = \prod_{i\in\Lambda} \mathbf{Z}_p(1)_i \simeq \mathbf{Z}_p(1)^{|\Lambda|}$. We actually need to deduce the equivalence

$$\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\Lambda})^{h\Gamma,\mathrm{op}} \simeq \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons,D}|\mathrm{Z}_{\Lambda}}((\mathrm{Z}_{\Lambda}, \mathcal{M}|_{\mathrm{Z}_{\Lambda}})_{\eta}, \mathbf{Z}_p)$$

for the induction step; here the target has a hollow component to its log structure, and is therefore not covered by Assumption 3.1. Fortunately, the proof still applies with the full group Γ. The only nontrivial modified input is a variant of the open stratum comparison (Theorem 5.4) where a rank $|\Lambda|$ hollow log structure is allowed on the open stratum: after inverting T_j for $j \notin \Lambda$, the base case argument applies with the rank $|\Lambda|$ hollow log structure, and the deperfection argument of Theorem 5.4 is unchanged. □

The main theorem immediately follows.

Theorem 5.25. Let $(\mathrm{X}, \mathcal{M})$ satisfy Assumption 3.1 over $\mathrm{W}(k)[\zeta_p]$. Let $\mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons,D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$ denote the full subcategory of p-Kummer constructible sheaves in $\mathrm{D}^{(b)}_{\mathrm{cons,D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$. Then

$$\mathrm{T}_{\text{ét}} : \mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\triangle}, \mathcal{O}_{\triangle}[1/\mathrm{I}_{\triangle}]^{\wedge}_p)^{\mathrm{op}} \xrightarrow{\sim} \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons,D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$$

is a symmetric monoidal equivalence.

Proof. By Lemma 3.4 and strict-étale descent (for the source Theorem 4.11 and for the target Lemma 4.20), we may first work on a framed log affine $(\mathrm{Spf}\, \mathrm{R}, \mathcal{M})$. The claim is then Corollary 5.24.

The fact that the equivalence is symmetric monoidal can also be tested locally, where via Lemma 4.26 it reduces to symmetric monoidality in Theorem 2.6. □

We must also understand perverse t-exactness. We use a definition of the perverse t-structure on the Kummer-étale side of the correspondence which is similar to [BH22].

Definition 5.26. For each connected logarithmic stratum $j_{\mathrm{U}} : \mathrm{U}_\eta \hookrightarrow (\mathrm{X}, \mathcal{M})_\eta$, put $d_{\mathrm{U}} = \dim(\mathrm{U}_\eta)$. The p-perverse t-structure of [Gab04] on the full subcategory of p-Kummer sheaves in $\mathrm{D}^{(b)}_{\mathrm{cons,D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$ is defined by

$$^p\mathrm{D}^{\leq 0} = \{\mathrm{K} \mid j_{\mathrm{U}}^*\mathrm{K} \in \mathrm{D}^{\leq -d_{\mathrm{U}}}(\mathrm{U}_\eta, \mathbf{Z}_p) \text{ for every } \mathrm{U}_\eta\},$$
$$^p\mathrm{D}^{\geq 0} = \{\mathrm{K} \mid j_{\mathrm{U}}^!\mathrm{K} \in \mathrm{D}^{\geq -d_{\mathrm{U}}}(\mathrm{U}_\eta, \mathbf{Z}_p) \text{ for every } \mathrm{U}_\eta\}.$$

Write $^p\mathrm{H}^i$ for its cohomology functors. The p^+-perverse t-structure is

$$^{p^+}\mathrm{D}^{\leq 0} = \{\mathrm{K} \in {}^p\mathrm{D}^{\leq 1} \mid {}^p\mathrm{H}^1(\mathrm{K}) \text{ is locally bounded } p\text{-power torsion}\},$$
$$^{p^+}\mathrm{D}^{\geq 0} = \{\mathrm{K} \mid \mathrm{K}/p \in {}^p\mathrm{D}^{\geq 0}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{F}_p)\}.$$

One may check these t-structures restrict to the p-Kummer subcategory. The argument of [BH22, Construction 4.3 and Proposition 4.6], applied to the defining bounds after j_{U}^* and $j_{\mathrm{U}}^!$, gives the following reduction criteria. For every $n \geq 1$, derived reduction modulo p^n is right t-exact for p-perversity and left t-exact for p^+-perversity. Moreover, $\mathrm{K} \in {}^p\mathrm{D}^{\leq 0}$ if and only if $\mathrm{K}/p \in {}^p\mathrm{D}^{\leq 0}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{F}_p)$, while $\mathrm{K} \in {}^{p^+}\mathrm{D}^{\geq 0}$ if and only if $\mathrm{K}/p \in {}^p\mathrm{D}^{\geq 0}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{F}_p)$.

Using the equivalence with the essential image, the main content of the following proposition can be reduced to the analogous perverse t-exactness of the Emerton-Kisin correspondence.

Proposition 5.27. Assume that $(\mathrm{X},\mathcal{M})_\eta$ is pure of dimension d_{X}. The equivalence of Theorem 5.25 with its essential image, shifted by $[d_{\mathrm{X}}]$, is t-exact for the standard t-structure on the source and the p^+-perverse t-structure of Definition 5.26 on the target.

Proof. The source t-structure is strict-étale local by Corollary A.17. On the essential image, Lemma 4.20 and strict-étale base change for j_{U}^* and $j_{\mathrm{U}}^!$ make the conditions of Definition 5.26 strict-étale local. We may therefore work on a framed chart and, by the reduction criteria in Definition 5.26, first work modulo p.

Put $\overline{\mathrm{A}} = \overline{\mathrm{R}}^{\square}(\!(q-1)\!)$. The Emerton-Kisin solution functor gives an equivalence

$$\mathrm{RSol}_{\overline{\mathrm{A}}} : (\mathrm{D}_{\mathrm{fgu}}(\overline{\mathrm{A}})^{h\Gamma})^{\mathrm{op}} \xrightarrow{\sim} \mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}}(\operatorname{Spec}\overline{\mathrm{A}}, \mathbf{F}_p)^{h\Gamma}.$$

In this generality we can use [BL19, Theorem 11.4.4], as when working with the cobar construction formulation of $(-)^{h\Gamma}$ it is convenient to have a correspondence which works for an arbitrary $\mathbf{F}_p$ algebra. One can also combine [Bau23, Corollary 5.1.15, Proposition 5.2.6, and Theorem 5.2.7] and [BB11, Theorem 5.9 and Proposition 5.12] to deduce an equivalence (induced by the same functor) just for the perverse heart, and construct $h\Gamma$ classically: as previously discussed one can equip finitely generated unit modules with a weak topology and consider continuous Γ actions on both sides (one need only make sure they correspond under the equivalence, which is checkable on the unit).

Theorems 4.6 and 5.25 therefore define an equivalence

$$\alpha : \mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}}(\operatorname{Spec}\overline{\mathrm{A}}, \mathbf{F}_p)^{h\Gamma} \xrightarrow{\sim} \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons},\mathrm{D}}((\mathrm{X},\mathcal{M})_\eta, \mathbf{F}_p)$$

by $\alpha\circ\mathrm{RSol}_{\overline{\mathrm{A}}} = \mathrm{T}_{\text{ét},\mathbf{F}_p}$. On a coordinate stratum, the corresponding equivalence is induced from $\mathrm{T}_{\text{ét}}$ on $\mathrm{D}_{\mathrm{fgu}}(\overline{\mathrm{A}}/(\mathrm{T}_i \mid i\in\Lambda))^{h\Gamma}$ and retains the full Γ-action; when no boundary coordinates remain, perverse t-exactness is unchanged by the normal equivariantization.

A double induction, first on dimension and then on the number of irreducible components of D, shows that α preserves the perverse t-structures. Choose one component D_1, and let $j : \mathrm{U} = \mathrm{X}\setminus\mathrm{D}_1 \hookrightarrow \mathrm{X}$ and $i : \mathrm{D}_1 \hookrightarrow \mathrm{X}$. In the following diagram, α_{D_1} is induced from $\mathrm{T}_{\text{ét}}$ on $\mathrm{D}_{\mathrm{fgu}}(\overline{\mathrm{A}}/(\mathrm{T}_1))^{h\Gamma}$ after the Emerton-Kisin equivalence. The compatibilities established thus far yield recollement diagrams

$$\begin{tikzcd}
\mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}|_{\mathrm{D}_1}}(\operatorname{Spec}(\overline{\mathrm{A}}/(\mathrm{T}_1)), \mathbf{F}_p)^{h\Gamma} \arrow[r, "\alpha_{\mathrm{D}_1}"] \arrow[d, "i_*"] & \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons},\mathrm{D}|_{\mathrm{D}_1}}((\mathrm{D}_1, \mathcal{M}|_{\mathrm{D}_1})_\eta, \mathbf{F}_p) \arrow[d, "i_*"] \\
\mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}}(\operatorname{Spec}\overline{\mathrm{A}}, \mathbf{F}_p)^{h\Gamma} \arrow[r, "\alpha_{\mathrm{X}}"] \arrow[u, bend left, "i^*"] \arrow[u, bend right, "i^!"'] \arrow[d, "j^*"] & \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons},\mathrm{D}}((\mathrm{X},\mathcal{M})_\eta, \mathbf{F}_p) \arrow[u, bend left, "i^*"] \arrow[u, bend right, "i^!"'] \arrow[d, "j^*"] \\
\mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}\setminus\mathrm{D}_1}(\operatorname{Spec}\overline{\mathrm{A}}[1/\mathrm{T}_1], \mathbf{F}_p)^{h\Gamma} \arrow[r, "\alpha_{\mathrm{U}}"] \arrow[u, bend left, "j_!"] \arrow[u, bend right, "j_*"'] & \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons},\mathrm{D}\setminus\mathrm{D}_1}((\mathrm{U}, \mathcal{M}|_{\mathrm{U}})_\eta, \mathbf{F}_p). \arrow[u, bend left, "j_!"] \arrow[u, bend right, "j_*"']
\end{tikzcd}$$

These are exchanged by α; here we define $\mathrm{D}|_{\mathrm{D}_1} := \bigcup_{i>1}\mathrm{D}_1\cap\mathrm{D}_i$. When there are no irreducible components of D, i.e. the base case where we only have lisse sheaves, the claim is clear; this covers the dimension zero or point case as well. The induction hypothesis gives perverse t-exactness for α_{U} as well as α_{D_1} so we get the claim for α_{X}. Note we have only actually verified compatibility of α with i_* and $j_!$, but the others follow from uniqueness of left and right adjoints.

Perverse t-exactness of the Emerton-Kisin correspondence now shows that $\mathrm{T}_{\text{ét},\mathbf{F}_p}[d_{\mathrm{X}}]$ is perverse t-exact. If $p\mathcal{E}=0$, one obtains

$$\mathrm{T}_{\text{ét},\mathbf{Z}_p}(\mathcal{E})[d_{\mathrm{X}}] \simeq \mathrm{T}_{\text{ét},\mathbf{F}_p}(\mathcal{E})[d_{\mathrm{X}}][-1].$$

The reduction criteria in Definition 5.26, derived Nakayama, and the torsion/torsion-free dévissage from the proof of Theorem 2.5 therefore lift the assertion to p^+-perverse t-exactness over $\mathbf{Z}_p$. □

It is also possible to extract ordinary perverse sheaves from the correspondence. We warn that extracting ordinary constructible sheaves in the derived sense cannot work without altering the crystal side of the correspondence, as these are not a full subcategory of the derived category of Kummer-étale sheaves (simply because Kummer-étale cohomology and étale cohomology differ).

Remark 5.28. Following [KL16, Theorem 7.5.6], call $\mathcal{E} \in \mathsf{Mod}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)$ *effective* if, on every framed strict-étale chart, its associated (φ,Γ)-module M satisfies the following condition for every boundary coordinate T_i: the ith $\mathbf{Z}_p(1)$-factor acts trivially on every cohomology module of

$$(\mathrm{A}/(\mathrm{T}_i)) \otimes^{\mathrm{L},\blacksquare}_{\mathrm{A}} \mathrm{M}.$$

This condition is strict-étale local and independent of the framing. Under Theorem 5.25, it is equivalent to trivial Kummer monodromy along each local branch. If X_η is pure of dimension d, Theorem 5.25 and Proposition 5.27 therefore give an equivalence

$$\mathrm{T}_{\text{ét}}[d] : \mathsf{Mod}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)^{\mathrm{eff},\mathrm{op}} \xrightarrow{\sim} {}^{p^+}\mathrm{D}^{\heartsuit}_{\mathrm{cons},\mathrm{D}}(\mathrm{X}_{\eta,\text{proét}}, \mathbf{Z}_p).$$

In the derived setting, it seems plausible that one could take the "effective part" of an RHom to obtain the derived category of constructible sheaves along D.

We remark that trivial modifications of the previous arguments allow one to deduce the theorem over a slightly more general base. In particular our restriction to $\mathrm{W}(k)[\zeta_p]$ up until this point is mainly for expository reasons to avoid tracking additional variables and cases throughout.

Corollary 5.29. Theorem 5.25 and all preceding results stated over $\mathrm{W}(k)[\zeta_p]$ admit corresponding variants over $\mathrm{W}(k)[\zeta_{p^\ell}]$ for every $\ell \geq 0$.

Proof. Corollary 3.19 is the only nontrivial change needed in the argument. With this, one can deduce Theorem 4.11 verbatim (as it uses nothing about Γ). The following arguments are insensitive to the change in prismatic ideal or Γ, except for Proposition 5.9; however there the reduction is to vanishing of a cohomology group, so changing Γ from $(1+p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$ to $\mu_{p-1} \times (1+p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$ does not change the result by using Hochschild-Serre. □

Remark 5.30. The author expects some variant of the theory of (φ, τ)-modules could allow $\mathcal{O}_K$ as a base. It seemed plausible there could be an integral variant of $\mathbf{A}_K$ which is a prism (indeed the entire theory should work locally if one moves to $\mathbf{A}_K$; the obstruction is entirely in globalization), but unfortunately it seems such a prism cannot exist in general. See [Wat26, Appendix 2, Proposition 23] for more details.

The following explicit (φ, Γ)-module gives a constructible sheaf with nontrivial gluing between the two strata.

Example 5.31. Let $(\mathrm{X}, \mathcal{M}) = (\mathrm{Spf}\, \mathbf{Z}_p[\zeta_p]\langle \mathrm{T}\rangle, \mathbf{N})$ with log structure induced by $\mathrm{D} = \{\mathrm{T} = 0\}$, and let $j : \mathrm{U}_\eta = \mathrm{X}_\eta \setminus \mathrm{D}_\eta \hookrightarrow (\mathrm{X}_\eta, \mathbf{N})$. Put $\overline{\mathrm{R}} = \mathbf{F}_p\langle \mathrm{T}\rangle$ and $\mathrm{A} = \overline{\mathrm{R}}^{\square}((q-1))$. For $\gamma \in \Gamma$, define

$$\mathrm{H}_\gamma = -\sum_{n\geq 0} \varphi^n(\gamma(\mathrm{T}) - \mathrm{T}).$$

If $\gamma = (a, b)$, then $\gamma(\mathrm{T}) - \mathrm{T} = (q^{pb} - 1)\mathrm{T}$, whose $(q-1)$-adic valuation is at least p. Frobenius multiplies this valuation by p, so the series converges. Since the actions of Γ and φ on A commute, one has $\mathrm{H}_{\gamma\delta} = \mathrm{H}_\gamma + \gamma(\mathrm{H}_\delta)$ and $\mathrm{H}_\gamma - \varphi(\mathrm{H}_\gamma) = \mathrm{T} - \gamma(\mathrm{T})$.
Set

$$\mathrm{M} = \mathrm{A}[1/\mathrm{T}]e_1 \oplus \mathrm{A}e_2.$$

Define $\varphi(e_1) = e_1$ and $\varphi(e_2) = e_2 + \mathrm{T}e_1$, and define $\gamma(e_1) = e_1$ and $\gamma(e_2) = e_2 + \mathrm{H}_\gamma e_1$. The preceding identities give a continuous semilinear Γ-action (for M equipped with its weak topology) commuting with Frobenius, so we obtain a finitely generated unit (φ, Γ)-module. There is an exact sequence of (φ, Γ)-modules

$$0 \longrightarrow \mathrm{A}[1/\mathrm{T}]e_1 \longrightarrow \mathrm{M} \longrightarrow \mathrm{A}\overline{e}_2 \longrightarrow 0.$$

It remains nonsplit after inverting T: a Frobenius-equivariant splitting would give $g \in \mathrm{A}[1/\mathrm{T}]$ satisfying

$$g - \varphi(g) = \mathrm{T}.$$

The $(q-1)$-adic valuation forces $v_{q-1}(g) = 0$. Its constant coefficient would then satisfy $a(\mathrm{T}) - a(\mathrm{T}^p) = \mathrm{T}$ in $\mathbf{F}_p[\mathrm{T}^{\pm 1}]$, which recursively forces the coefficients of T^{p^n} to equal one for every n (which is not possible).
Let $\mathscr{F} = \mathrm{T}_{\text{ét}}(\mathrm{M})$. Contravariance and the previous exact sequence give

$$0 \longrightarrow \mathbf{F}_{p,(\mathrm{X}_\eta,\mathbf{N})} \longrightarrow \mathscr{F} \longrightarrow j_!\mathbf{F}_{p,\mathrm{U}_\eta} \longrightarrow 0.$$

Its restriction to U_η is still nonsplit, so in particular this gives a nontrivial example (i.e. not a direct sum of trivial examples).

6. Pushforward compatibility

This section proves site-theoretic pushforward compatibility after establishing Poincaré duality for perfect logarithmic Laurent crystals.

6.1. **Poincaré duality.** There are several results proving Poincaré duality for ordinary prismatic crystals in the literature. For example, see [GR24]. However, because we will need such a duality for Laurent prismatic crystals, these methods are unavailable to us: ultimately their approach is to use the Hodge-Tate comparison to reduce to a known case of Poincaré duality, but for Laurent crystals one cannot use such a comparison.

The category $\mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\mathbb{\Delta}},\mathcal{O}_{\mathbb{\Delta}})$ is large enough that the six-functor results needed for Poincaré duality imply their Laurent analogues. For ordinary formal schemes, this is carried out in [Kip26b, Kip26a] by constructing an analytic stack $\mathrm{X}^{\mathbb{\Delta},\blacksquare}$. We construct the logarithmic analogue and compare it with our site-theoretic category. This proves Poincaré duality for strict log smooth proper maps with coefficients in $\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\mathbb{\Delta}},\mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]_p^{\wedge})$.

Let $(\mathrm{X},\mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$ for a finite extension $K/\mathbf{Q}_p$. As explained in more detail in Definition A.1, the ordinary log prismatic stack is defined by the fiber product

$$\begin{tikzcd} (\mathrm{X},\mathcal{M})^{\mathbb{\Delta}} \arrow[r] \arrow[d] & \mathrm{X}^{\mathbb{\Delta}} \arrow[d] \\ \mathbf{Z}_p^{\mathbb{\Delta}} \times \mathrm{Log} \arrow[r] & \mathrm{Log}^{\mathbb{\Delta}} \end{tikzcd}$$

where Log is the formal p-completion of Olsson's log stack [Ols03]. This is a relative prismatic stack over Log, viewed as a "δ-stack", so the constructions of [Kip26a] apply by base change.

In [Kip26a, Construction 2.2.7], the éid topology is defined (one takes the union of the étale topology and integral descendable maps). By [Kip26a, Theorem 2.3.3(A) and §3.6.1], one has a colimit and finite limit preserving functor

$$(-)^{\blacksquare} : (\mathbf{Z}_p^{\mathbb{\Delta}})_{\text{éid}} \longrightarrow \mathrm{AnStack}_{/\mathbf{Z}_p^{\mathbb{\Delta},\blacksquare}}.$$

Construction 3.7.1 and Definition 3.7.2 of [Kip26a] construct $\mathrm{X}^{\mathbb{\Delta},\blacksquare}$ as this functor $(-)^{\blacksquare}$ applied to (the éid sheafification of) $\mathrm{X}^{\mathbb{\Delta}}$. We may then define

$$\begin{tikzcd} (\mathrm{X},\mathcal{M})^{\mathbb{\Delta},\blacksquare} \arrow[r] \arrow[d] & \mathrm{X}^{\mathbb{\Delta},\blacksquare} \arrow[d] \\ (\mathbf{Z}_p^{\mathbb{\Delta}} \times \mathrm{Log})^{\blacksquare} \arrow[r] & \mathrm{Log}^{\mathbb{\Delta},\blacksquare} \end{tikzcd}$$

where we regard the formal stacks on the bottom as sheafified $(\mathbf{Z}_p^{\mathbb{\Delta}})_{\text{éid}}$. Noting $(-)^{\blacksquare}$ has compatibility with finite limits shows one may also construct $(\mathrm{X},\mathcal{M})^{\mathbb{\Delta},\blacksquare}$ by simply applying $(-)^{\blacksquare}$ to the already existing formal stack over $\mathbf{Z}_p^{\mathbb{\Delta}}$.

The logarithmic Hodge-Tate substack is the fiber product $(\mathrm{X},\mathcal{M})^{\mathbb{\Delta},\blacksquare} \times_{\mathbf{Z}_p^{\mathbb{\Delta},\blacksquare}} \mathbf{Z}_p^{\mathrm{HT},\blacksquare}$, where $\mathbf{Z}_p^{\mathrm{HT},\blacksquare}$ is as in [Kip26a]. We write $(\mathrm{X},\mathcal{M})^{\mathcal{L},\blacksquare}$ for the corresponding Laurent variant, namely the *open* complement of the logarithmic Hodge-Tate substack in $(\mathrm{X},\mathcal{M})^{\mathbb{\Delta},\blacksquare}$ as analytic stacks.

Since this stack is obtained from $\mathrm{X}^{\mathbb{\Delta},\blacksquare}$ by base change, the relevant results of [Kip26a] apply to it. See [Kip26a, §1.1] for cohomological smoothness and weak cohomological properness in six-functor formalisms.

Lemma 6.1. Let $f : \mathrm{X} \to \mathrm{Y}$ be a morphism of analytic stacks, and suppose it is cohomologically smooth and weakly cohomologically proper. Then given a square

$$\begin{tikzcd} \mathrm{X} \times_{\mathrm{Y}} \mathrm{Z} \arrow[r, "\iota_{\mathrm{X}}"] \arrow[d, "f_{\mathrm{Z}}"] & \mathrm{X} \arrow[d, "f"] \\ \mathrm{Z} \arrow[r, "\iota_{\mathrm{Y}}"] & \mathrm{Y} \end{tikzcd}$$

the map $f_{\mathrm{Z}} : \mathrm{X} \times_{\mathrm{Y}} \mathrm{Z} \to \mathrm{Z}$ is cohomologically smooth and weakly cohomologically proper. Moreover, $f_{\mathrm{Z}}^{!}\mathcal{O}_{\mathrm{Z}} \simeq \iota_{\mathrm{X}}^{*} f^{!}\mathcal{O}_{\mathrm{Y}}$.

Proof. The property of being weakly cohomologically proper is stable under arbitrary base change by [Kip26a, Remark 1.1.8], and cohomological smoothness is defined after arbitrary base change, so f_{Z} is again cohomologically smooth and weakly cohomologically proper. Put $\omega_f = f^{!}\mathcal{O}_{\mathrm{Y}}$. For a cohomologically smooth map, ω_f is invertible, the natural map $\omega_f \otimes f^{*}(-) \to f^{!}(-)$ is an equivalence, and ω_f commutes with arbitrary base change; compare [Kip26b, Definition 1.2.1.1]. Hence the displayed square gives $\omega_{f_{\mathrm{Z}}} \simeq \iota_{\mathrm{X}}^{*}\omega_f$. Applying this to the unit object gives $f_{\mathrm{Z}}^{!}\mathcal{O}_{\mathrm{Z}} \simeq \omega_{f_{\mathrm{Z}}} \simeq \iota_{\mathrm{X}}^{*} f^{!}\mathcal{O}_{\mathrm{Y}}$. □

Lemma 6.2. Let $f : \mathrm{X} \to \mathrm{Y}$ be a cohomologically smooth and weakly cohomologically proper morphism of analytic stacks, and put $\omega_f = f^{!}\mathcal{O}_{\mathrm{Y}}$. Suppose further that

$$\mathrm{R}f_{!} \simeq \mathrm{R}f_{*}$$

by a natural equivalence.
Then $\mathrm{R}f_{*}$ preserves dualizable objects, and for every dualizable object $\mathcal{E} \in \mathrm{D}_{\blacksquare}(\mathrm{X})$ there is a natural equivalence

$$(\mathrm{R}f_{*}\mathcal{E})^{\vee} \simeq \mathrm{R}f_{*}(\mathcal{E}^{\vee} \otimes \omega_f).$$

Proof. For every $\mathcal{K} \in \mathrm{D}_{\blacksquare}(\mathrm{Y})$, adjunction, the formula $f^{!}(-) \simeq \omega_f \otimes f^{*}(-)$, and the projection formula (using the chosen equivalence $\mathrm{R}f_{!} \simeq \mathrm{R}f_{*}$) give

$$\underline{\mathrm{RHom}}_{\mathrm{Y}}(\mathrm{R}f_{*}\mathcal{E}, \mathcal{K}) \simeq \mathrm{R}f_{*}\underline{\mathrm{RHom}}_{\mathrm{X}}(\mathcal{E}, f^{!}\mathcal{K}) \simeq \mathrm{R}f_{*}(\mathcal{E}^{\vee} \otimes \omega_f) \otimes \mathcal{K}.$$

Thus $\mathrm{R}f_{*}\mathcal{E}$ is dualizable with dual $\mathrm{R}f_{*}(\mathcal{E}^{\vee} \otimes \omega_f)$. □

Lemma 6.3. Let $f : (\mathrm{X}, \mathcal{M}_{\mathrm{X}}) \to (\mathrm{Y}, \mathcal{M}_{\mathrm{Y}})$ be a strict log smooth proper morphism of bounded fs log p-adic formal schemes of pure relative dimension d. Then the induced maps on $(\mathrm{X}, \mathcal{M}_{\mathrm{X}})^{\mathbb{A},\blacksquare}$ and $(\mathrm{X}, \mathcal{M}_{\mathrm{X}})^{\mathcal{L},\blacksquare}$ are cohomologically smooth and weakly cohomologically proper, and on both stacks one has

$$f^{!}\mathcal{O} \simeq \mathcal{O}\{d\}[2d].$$

Moreover there is a canonical natural equivalence $\mathrm{R}f_{!} \simeq \mathrm{R}f_{*}$.

Proof. The underlying map of formal schemes is smooth and proper. Theorem 4.1.3(C),(F) and Example 4.3.2 of [Kip26a] therefore show that the induced map

$$\mathrm{X}^{\mathbb{A},\blacksquare} \to \mathrm{Y}^{\mathbb{A},\blacksquare}$$

is cohomologically smooth and weakly cohomologically proper.[6] We show in addition that there is a canonical equivalence $\mathrm{R}f_{!} \simeq \mathrm{R}f_{*}$.

[6] We note [Kip26a, Lemma 3.5.6] is false for a general connective perfect complex. In its applications to smooth morphisms, the relevant cotangent complex is a vector bundle, in which case there is no issue.

It suffices to verify the hypothesis of [Mik25, Lemma 1.50] for the diagonal Δ_f. Since f is smooth and separated, this is a regular closed immersion. Work on the semiperfectoid atlas of Construction 3.7.1 of [Kip26a]. After a Zariski refinement, its pullback is induced by a derived quotient $\mathrm{S}_0 = \mathrm{R}_0/(g_1, \dots, g_d)$. Choose a perfectoid ring R surjecting onto R_0, lift the g_i to $f_i \in \mathrm{R}$, and put $\mathrm{S} = \mathrm{R}/(f_1, \dots, f_d)$, again as a derived quotient. Then

$$\mathrm{S}_0 \simeq \mathrm{R}_0 \otimes^{\mathrm{L}}_{\mathrm{R}} \mathrm{S}.$$

Since solid prismatization preserves finite limits, the map over the original chart is the base change of

$$\mathrm{Spa}(\mathbb{\Delta}_{\mathrm{S}}) \longrightarrow \mathrm{Spa}(\mathbb{\Delta}_{\mathrm{R}}).$$

It is therefore enough to show that this map belongs to P. The quotient calculation in the proof of [Kip26a, Theorem 3.6.2(c) and Proposition 4.1.5] applies to this derived quotient. Indeed, $L_{\mathrm{S}/\mathrm{R}}[-1] \simeq \mathrm{S}^{\oplus d}$, and the animated divided-power-envelope description shows that $\pi_0(\mathbb{\Delta}_{\mathrm{S}}/(p, \mathrm{I}))$ is generated over $\pi_0(\mathrm{S}/p)$ by divided power classes. Each such class has pth power zero, so

$$\pi_0(\mathrm{S}/p) \longrightarrow \pi_0(\mathbb{\Delta}_{\mathrm{S}}/(p, \mathrm{I}))$$

is integral. Since $\pi_0(\mathrm{S}/p)$ is a quotient of R/p, the composite from R/p is integral as well. As (p, I) is an ideal of definition, [Kip26a, Example 2.1.13] shows that the analytic map above is proper in the sense of [Kip26a, §2.1.10], equivalently it belongs to the class P of [Mik25, Definition 1.5(2)]. Membership in P is preserved by base change, while the semiperfectoid atlases are universal covers for descent of $\mathrm{D}_{\blacksquare}(-)^*$ by [Kip26a, Corollary 3.6.3(c) and Construction 3.7.1]. The same argument after arbitrary base change therefore shows that Δ_f is universally $\mathrm{D}_{\blacksquare}(-)^*$-locally on its target represented by morphisms in P.

Let π_2 denote the second projection from $\mathrm{X}^{\mathbb{\Delta},\blacksquare} \times_{\mathrm{Y}^{\mathbb{\Delta},\blacksquare}} \mathrm{X}^{\mathbb{\Delta},\blacksquare}$. Weak cohomological properness and [Mik25, Corollary 1.28] give a codualizing object

$$\delta_f \simeq \mathrm{R}\pi_{2,*}\mathrm{R}\Delta_{f,!}\mathcal{O}$$

and a natural equivalence

$$\mathrm{R}f_!(\delta_f \otimes -) \xrightarrow{\sim} \mathrm{R}f_*.$$

The preceding local description of Δ_f and [Mik25, Lemma 1.50] give a natural equivalence $\mathrm{R}\Delta_{f,!} \simeq \mathrm{R}\Delta_{f,*}$. It follows canonically that

$$\delta_f \simeq \mathrm{R}\pi_{2,*}\mathrm{R}\Delta_{f,!}\mathcal{O} \simeq \mathrm{R}\pi_{2,*}\mathrm{R}\Delta_{f,*}\mathcal{O} \simeq \mathcal{O},$$

where the last equivalence uses $\pi_2 \circ \Delta_f = \mathrm{id}$. Substituting this identification in the preceding equivalence gives $\mathrm{R}f_! \simeq \mathrm{R}f_*$. The equivalence $\mathrm{R}\Delta_{f,!} \simeq \mathrm{R}\Delta_{f,*}$ in Lemma 1.50 is constructed by descent from the comparisons for morphisms in P, and [Mik25, Remark 1.49] identifies these local comparisons with the corresponding codualizing trivializations. It is therefore independent of all choices. The local comparisons and the transformation of [Mik25, Corollary 1.28] are preserved by arbitrary base change, while [Mik25, Corollary 1.31(2)] identifies the base change of δ_f; hence the resulting equivalence $\mathrm{R}f_! \simeq \mathrm{R}f_*$ is canonical and compatible with arbitrary base change.

Theorem 4.1.3(F) of [Kip26a] identifies the dualizing object with the d-fold Tate object. Kipp's homological shift $[n]$ is our cohomological shift $[-n]$, so his formula $\mathcal{O}\{d\}[-2d]$ becomes $\mathcal{O}\{d\}[2d]$. Strictness gives the cartesian identification

$$(\mathrm{X}, \mathcal{M}_{\mathrm{X}})^{\mathbb{\Delta},\blacksquare} \simeq \mathrm{X}^{\mathbb{\Delta},\blacksquare} \times_{\mathrm{Y}^{\mathbb{\Delta},\blacksquare}} (\mathrm{Y}, \mathcal{M}_{\mathrm{Y}})^{\mathbb{\Delta},\blacksquare}.$$

The logarithmic prismatic assertion therefore follows from Lemma 6.1 and base-change compatibility of $\mathrm{R}f_! \simeq \mathrm{R}f_*$. The Laurent open complement is compatible with base change by its fiber-product definition, so the same argument proves the Laurent assertion. ☐

The following proposition bridges the gap between the stack and the site-theoretic approach we have taken thus far.

Proposition 6.4. For every log formal scheme $(\mathrm{X}, \mathcal{M})$ satisfying Assumption 3.1, there is a natural equivalence

$$\mathrm{an}_{(\mathrm{X},\mathcal{M})} : \mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}) \xrightarrow{\sim} \mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}).$$

Proof. For every orientable object $(\mathrm{B}, (d), \mathcal{N})^{a}$ of the strict log prismatic site, Lemma A.4 and solidification give

$$\mathrm{Spa}(\mathrm{B}, \mathrm{B}) \longrightarrow (\mathrm{X}, \mathcal{M})^{\Delta,\blacksquare}.$$

The Hodge-Tate pullback is cut out by d, so Definition A.10 gives

$$\mathrm{D}_{\blacksquare}\big(\mathrm{Spa}(\mathrm{B}, \mathrm{B}) \times_{(\mathrm{X},\mathcal{M})^{\Delta,\blacksquare}} (\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}\big) \simeq \mathrm{D}_{\blacksquare}(\mathrm{B}[1/d]^{\wedge}_{p}).$$

Indeed, the analytic localization on the left is computed by the pairs $((\mathrm{B}/^{\mathrm{L}}p^{n})[1/d], \mathrm{B}/^{\mathrm{L}}p^{n})$. Taking the limit of the compatible pullback functors defines

$$\mathrm{res}_{(\mathrm{X},\mathcal{M})} : \mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}) \longrightarrow \mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}).$$

Both sides satisfy strict-étale descent. For the target this is Corollary A.16. For the source, for $(\mathrm{X}, \mathcal{M})^{\Delta,\blacksquare}$ by strictness this reduces to [Kip26a, Theorem 2.3.3(A),(B) and Construction 3.7.1] which give étale descent for $\mathrm{X}^{\Delta,\blacksquare}$ (i.e. étale covers are sent to effective epimorphisms of analytic stacks). The same holds for the Laurent stack $(\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}$, using the analogous result for the Hodge-Tate stack from [Kip26a]. Descent for $\mathrm{D}_{\blacksquare}$ then proves the claim. Lemma 3.4 therefore reduces us to a framed log affine; choose the logarithmic q-de Rham Breuil-Kisin prism $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}})^{a}$ attached to the framing, and let $\mathrm{A}^{(m)}$ be its Čech nerve in the strict log prismatic site. By rigidity of prisms, we also write I for its extension to every term of this nerve.

Proposition A.20 and Laurent base change give

$$\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{A}^{(m)}[1/\mathrm{I}]^{\wedge}_{p}).$$

Theorem A.14 identifies this limit with $\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$. Both equivalences are induced by the augmentations, and their composite is $\mathrm{res}_{(\mathrm{X},\mathcal{M})}$. Thus restriction is an equivalence, with inverse $\mathrm{an}_{(\mathrm{X},\mathcal{M})}$. ☐

Lemma 6.5. For every $(\mathrm{X}, \mathcal{M})$ satisfying Assumption 3.1, there is a functorial symmetric monoidal equivalence

$$\mathrm{an}_{(\mathrm{X},\mathcal{M})} : \mathsf{Perf}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}) \xrightarrow{\sim} \mathsf{Perf}((\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}).$$

It is compatible with pullback and $\underline{\mathrm{RHom}}$. If $(\mathrm{Y},\mathcal{N})$ also satisfies Assumption 3.1, $f:(\mathrm{X},\mathcal{M})\to(\mathrm{Y},\mathcal{N})$ is strict log smooth and proper, and $f_{\mathcal{L}}$ is the induced map of Laurent analytic stacks, then the site-theoretic pushforward preserves perfect objects and

$$\mathrm{an}_{(\mathrm{Y},\mathcal{N})}\,\mathrm{R}f_{\Delta,*}\simeq\mathrm{R}f_{\mathcal{L},*}\,\mathrm{an}_{(\mathrm{X},\mathcal{M})}\,.$$

Proof. All functors in the construction of Proposition 6.4 are symmetric monoidal. Thus its equivalence restricts to dualizable objects and is compatible with $\underline{\mathrm{RHom}}$. Its compatibility with pullback gives

$$\begin{array}{ccc}\mathrm{D}_{\blacksquare}((\mathrm{Y},\mathcal{N})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}) & \xrightarrow{\mathrm{an}_{(\mathrm{Y},\mathcal{N})}} & \mathrm{D}_{\blacksquare}((\mathrm{Y},\mathcal{N})^{\mathcal{L},\blacksquare})\\ \big\downarrow{\scriptstyle \mathrm{L}f^{*}} & & \big\downarrow{\scriptstyle \mathrm{L}f^{*}_{\mathcal{L}}}\\ \mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}) & \xrightarrow{\mathrm{an}_{(\mathrm{X},\mathcal{M})}} & \mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})^{\mathcal{L},\blacksquare}).\end{array}$$

Taking right adjoints gives the asserted comparison of pushforwards on the ambient categories. Lemma 6.3 gives $\mathrm{R}f_{\mathcal{L},!}\simeq\mathrm{R}f_{\mathcal{L},*}$, so Lemma 6.2 shows that $\mathrm{R}f_{\mathcal{L},*}$ preserves perfect objects. The comparison and the equivalence $\mathrm{an}_{(\mathrm{Y},\mathcal{N})}$ give the same assertion for $\mathrm{R}f_{\Delta,*}$. ☐

Corollary 6.6. Let $f:(\mathrm{X},\mathcal{M})\to(\mathrm{Y},\mathcal{N})$ be as in Lemma 6.3, and suppose both source and target satisfy Assumption 3.1. Transport $f^{!}$ on perfect objects from the map $(\mathrm{X},\mathcal{M})^{\mathcal{L},\blacksquare}\to(\mathrm{Y},\mathcal{N})^{\mathcal{L},\blacksquare}$ through Lemma 6.5. Then $f^{!}\mathcal{G}\simeq f^{*}\mathcal{G}\{d\}[2d]$ for $\mathcal{G}\in\mathsf{Perf}((\mathrm{Y},\mathcal{N})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$, the site-theoretic pushforward preserves perfect complexes, and

$$\mathrm{R}f_{*}\mathcal{E}\simeq\underline{\mathrm{RHom}}_{\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}}\big(\mathrm{R}f_{*}(\mathcal{E}^{\vee}\{d\}),\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}[-2d]\big)$$

for every $\mathcal{E}\in\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$.

Proof. Transporting Lemma 6.3 through Lemma 6.5 gives $f^{!}\mathcal{G}\simeq f^{*}\mathcal{G}\{d\}[2d]$. Lemma 6.2, together with the pushforward comparison in Lemma 6.5, gives

$$(\mathrm{R}f_{*}\mathcal{E})^{\vee}\simeq\mathrm{R}f_{*}(\mathcal{E}^{\vee}\{d\})[2d].$$

Dualizing, this equivalence gives the claim. ☐

We also note that the Poincaré pairing is compatible with Frobenius. The identification $f^{!}\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}\simeq\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}\{d\}[2d]$ is Frobenius equivariant since it is upgraded to an identification of F-gauges in [Kip26a, Theorem 4.1.3(F)].

6.2. **Pushforward compatibility.** Let $f:(\mathrm{X},\mathcal{M}_{\mathrm{X}})\to(\mathrm{Y},\mathcal{M}_{\mathrm{Y}})$ be a strict log smooth proper morphism of pure relative dimension d between log p-adic formal schemes whose adic generic fibers have pure dimensions d_{X} and d_{Y}. We assume $(\mathrm{X},\mathrm{M}_{\mathrm{X}})$ and $(\mathrm{Y},\mathrm{M}_{\mathrm{Y}})$ satisfy Assumption 3.1 over one of the bases $\mathrm{Spf}\,\mathrm{W}(k)[\zeta_{p^{\ell}}]$ covered in Corollary 5.29; closed strata are equipped with their pullback log structures.

Let $d = d_{\mathrm{X}} - d_{\mathrm{Y}}$. In such a setup, define

$$\mathrm{R}f_{+}(-) := \mathrm{R}f_{*}(-)\{d\}[d].$$

We will show that $\mathrm{R}f_{+}$ induces a functor

$$\mathrm{R}f_{+} : \mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M}_{\mathrm{X}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_{p}^{\wedge}) \to \mathrm{D}_{\mathrm{fgu}}((\mathrm{Y}, \mathcal{M}_{\mathrm{Y}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_{p}^{\wedge}).$$

We prove preservation first for perfect crystals, then for the open and closed strata, and conclude by dévissage.

We first show a variant of [GR24, Lemma 8.8] that works over any ring. We allow connective algebras because we will need to handle derived mod p reductions of arbitrary prisms, which may allow p-torsion.

Lemma 6.7. Let B be a connective algebra over $\mathbf{F}_p$ with an endomorphism inducing Frobenius on $\pi_0(\mathrm{B})$. If $\mathrm{M}, \mathrm{C} \in \mathsf{Perf}(\mathrm{B})$ and

$$\mathrm{M} \simeq \varphi^{*}\mathrm{M} \oplus \mathrm{C},$$

then $\mathrm{C} = 0$.

Proof. Let $\mathfrak{m} \subset \pi_0(\mathrm{B})$ be maximal and put $k = \kappa(\mathfrak{m})$. We have

$$\varphi^{-1}(\mathfrak{m}) = \{x : x^p \in \mathfrak{m}\} = \mathfrak{m}.$$

Therefore base change gives

$$(\varphi^{*}\mathrm{M}) \otimes^{\mathrm{L}}_{\mathrm{B}} k \simeq \varphi_k^{*}(\mathrm{M} \otimes^{\mathrm{L}}_{\mathrm{B}} k).$$

Pullback along the Frobenius of k preserves the dimension of every cohomology group. The direct sum decomposition therefore gives $\mathrm{C} \otimes^{\mathrm{L}}_{\mathrm{B}} k = 0$. However, taking all fibers at $\kappa(\mathfrak{m})$ is conservative on perfect complexes over a connective ring, so $\mathrm{C} = 0$. ☐

We now treat perfect Laurent F-crystals.

Lemma 6.8. Let $f : (\mathrm{X}, \mathrm{M}_{\mathrm{X}}) \to (\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ be strict log smooth proper. If

$$\mathcal{E} \in \mathsf{Perf}((\mathrm{X}, \mathrm{M}_{\mathrm{X}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_{p}^{\wedge})^{\varphi=1},$$

then $\mathrm{R}f_{*}\mathcal{E}$ is a perfect unit Laurent crystal.

Proof. The claim is strict-étale local on Y. Indeed, Lemma 6.5 transports strict-étale base change for $\mathrm{R}f_{\mathcal{L},*}$ to the site, while Lemma 6.3 identifies it with base change for $\mathrm{R}f_{\mathcal{L},!}$. Lemma 3.4 therefore reduces us to the case where $(\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ is a framed log affine and the map has pure relative dimension d on the adic generic fiber. Corollary 6.6 makes $\mathrm{R}f_{*}\mathcal{E}$ perfect, as well as $\mathrm{R}f_{+}\mathcal{E}$. The unit Frobenius of $\mathcal{E}$ and Frobenius base change give a lax Frobenius map

$$\alpha : \varphi^{*}\mathrm{R}f_{+}\mathcal{E} \longrightarrow \mathrm{R}f_{+}\mathcal{E}.$$

It remains to prove that α is an equivalence.

For $\ell = 0$, Corollary 3.19 allows us to make the following check after passage to $\mathrm{W}(k)[\zeta_p]$, so assume $\ell > 0$. Evaluate on the framed log q-de Rham prism $(\mathrm{R}_0^{\square}[\![q-1]\!], \varphi^{\ell-1}([p]_q), \mathrm{M}_{\mathrm{Y}})$ for

$(\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ and set $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$. Let M be the evaluation of the perfect Laurent crystal $\mathrm{R}f_+\mathcal{E}$, viewed as an A-module. Poincaré duality gives a perfect dual $\mathrm{M}^{\vee}$ and a perfect pairing

$$\mathrm{M} \otimes_{\mathrm{A}}^{\mathrm{L},\blacksquare} \mathrm{M}^{\vee} \longrightarrow \mathrm{A}.$$

The Frobenius structures on $\mathcal{E}$ and its Poincaré dual induce maps $\alpha : \varphi^*\mathrm{M} \to \mathrm{M}$ and $\beta : \varphi^*\mathrm{M}^{\vee} \to \mathrm{M}^{\vee}$. The Frobenius-equivariant cup product and trace from Poincaré duality give a commutative diagram of perfect pairings

$$\begin{tikzcd}
\varphi^*\mathrm{M} \otimes_{\mathrm{A}}^{\mathrm{L},\blacksquare} \varphi^*\mathrm{M}^{\vee} \arrow[r] \arrow[d, "\alpha\otimes\beta"] & \varphi^*\mathrm{A} \arrow[d, "\gamma"] \\
\mathrm{M} \otimes_{\mathrm{A}}^{\mathrm{L},\blacksquare} \mathrm{M}^{\vee} \arrow[r] & \mathrm{A}.
\end{tikzcd}$$

Here γ is the Frobenius map on the target induced by the twist $\{d\}$ in the trace map, and β is the Frobenius map on the twisted Poincaré dual. Since the prismatic generator $\varphi^{\ell-1}([p]_q)$ is invertible in A, the map γ is an isomorphism.[7]

Lemma A.11 identifies this diagram with a diagram of ordinary perfect A-complexes. By [GR24, Proposition 8.6], α is a split injection with perfect cofiber, so

$$\mathrm{M} \simeq \varphi^*\mathrm{M} \oplus \mathrm{C}$$

for a perfect A-complex C. After reduction modulo p, Lemma A.11 identifies the terms with ordinary perfect complexes. Lemma 6.7 gives $\mathrm{C}/p = 0$, and derived Nakayama gives $\mathrm{C} = 0$.

The map α is therefore an equivalence after evaluation on the framed log q-de Rham prism. Conservativity of evaluation on this log prism completes the argument. □

We now extend the preceding argument to the hollow log structures obtained by restricting to closed strata. This is the only hollow case needed below.

Lemma 6.9. Let $(\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ satisfy Assumption 3.1 over $\mathrm{W}(k)[\zeta_{p^\ell}]$, and let $i : (\mathrm{Z}, \mathrm{M}_{\mathrm{Z}}) \hookrightarrow (\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ be the inclusion of a smooth closed stratum, with $\mathrm{M}_{\mathrm{Z}} = i^*\mathrm{M}_{\mathrm{Y}}$. There is a natural symmetric monoidal equivalence

$$\mathrm{an}_{(\mathrm{Z},\mathrm{M}_{\mathrm{Z}})} : \mathrm{D}_{\blacksquare}((\mathrm{Z}, \mathrm{M}_{\mathrm{Z}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}) \xrightarrow{\sim} \mathrm{D}_{\blacksquare}((\mathrm{Z}, \mathrm{M}_{\mathrm{Z}})^{\mathcal{L},\blacksquare}).$$

It is compatible with pullback and $\underline{\mathrm{RHom}}$. The pushforward comparison and the conclusions of Corollary 6.6 hold for strict log smooth proper morphisms obtained by base change to such strata.

Proof. The pointwise construction in the proof of Proposition 6.4 gives a canonical symmetric monoidal restriction functor

$$\mathrm{res}_{(\mathrm{Z},\mathrm{M}_{\mathrm{Z}})} : \mathrm{D}_{\blacksquare}((\mathrm{Z}, \mathrm{M}_{\mathrm{Z}})^{\mathcal{L},\blacksquare}) \longrightarrow \mathrm{D}_{\blacksquare}((\mathrm{Z}, \mathrm{M}_{\mathrm{Z}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}).$$

[7] This is special to mixed characteristic. For the Emerton-Kisin correspondence done on the crystalline site, one *cannot* use the ordinary site-theoretic pushforward: the unit condition is not preserved. This is clear if one considers pushforward of $\mathcal{O}_{\mathrm{cris}}$ from $\mathbf{P}^1_k \to \operatorname{Spec} k$ and observes the resulting F-isocrystal is no longer slope zero.

Both sides satisfy strict-étale descent for the pullback of a strict-étale cover of Y. On the site side this follows from the unique completely étale lifting property for strict log prisms [Kos22, Lemma 4.12] and descent for $\mathrm{D}_\blacksquare$; on the stack side it follows from étale descent for ordinary prismatization and the Hodge–Tate stack. We may therefore assume that Y is affine and choose a logarithmic Breuil–Kisin atlas (A, I) for it. Let $\mathrm{A}^{(\bullet)}$ denote its log prismatic Čech nerve.

Since i is a regular strict immersion, [AKN26, Example 9.3 and Theorem 9.6], [BL22b, Corollary 7.18], and [Kos22, Propositions 3.6 and 3.7] identify its relative strict log prismatic envelope over A with a bounded fine strict log prism C representing

$$\operatorname{Spf} \mathrm{C} = \operatorname{Spf} \mathrm{A} \times_{(\mathrm{Y},\mathrm{M}_\mathrm{Y})^{\mathbb{\Delta}}} (\mathrm{Z}, \mathrm{M}_\mathrm{Z})^{\mathbb{\Delta}}$$

over $(\mathrm{Z}, \mathrm{M}_\mathrm{Z})$. The maps $\mathrm{A} \to \mathrm{A}^{(m)}$ are completely flat, so the corresponding base changes of i remain regular. Their relative envelopes represent the Čech terms $\mathrm{C}^{(m)}$, and

$$\operatorname{Spf} \mathrm{C}^{(m)} \simeq \operatorname{Spf} \mathrm{A}^{(m)} \times_{(\mathrm{Y},\mathrm{M}_\mathrm{Y})^{\mathbb{\Delta}}} (\mathrm{Z}, \mathrm{M}_\mathrm{Z})^{\mathbb{\Delta}}.$$

Proposition A.20, left exactness of éid sheafification, and finite limit preservation of $(-)^\blacksquare$ show that $\operatorname{Spa}(\mathrm{C}, \mathrm{C}) \to (\mathrm{Z}, \mathrm{M}_\mathrm{Z})^{\mathbb{\Delta},\blacksquare}$ is an effective epimorphism with Čech nerve $\operatorname{Spa}(\mathrm{C}^{(\bullet)}, \mathrm{C}^{(\bullet)})$. After Laurent base change, analytic descent gives

$$\mathrm{D}_\blacksquare((\mathrm{Z}, \mathrm{M}_\mathrm{Z})^{\mathcal{L},\blacksquare}) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}_\blacksquare(\mathrm{C}^{(m)}[1/\mathrm{I}]^\wedge_p).$$

Let $(\mathrm{B}, (d), \mathcal{N})^a$ be an orientable strict log prism over $(\mathrm{Z}, \mathrm{M}_\mathrm{Z})$. Since i is strict, it is also a strict log prism over $(\mathrm{Y}, \mathrm{M}_\mathrm{Y})$, and associativity of fiber products gives

$$\operatorname{Spf} \mathrm{B} \times_{(\mathrm{Z},\mathrm{M}_\mathrm{Z})^{\mathbb{\Delta}}} \operatorname{Spf} \mathrm{C}^{(m)} \simeq \operatorname{Spf} \mathrm{B} \times_{(\mathrm{Y},\mathrm{M}_\mathrm{Y})^{\mathbb{\Delta}}} \operatorname{Spf} \mathrm{A}^{(m)}.$$

For $m = 0$, Lemma A.5 identifies the right side with $\operatorname{Spf}(\mathrm{A} \coprod_* \mathrm{B})$, and the displayed simplicial object is the Čech nerve of $\mathrm{B} \to \mathrm{A} \coprod_* \mathrm{B}$. Lemma A.13 and the proof of Theorem A.14 therefore give

$$\mathrm{D}_\blacksquare((\mathrm{Z}, \mathrm{M}_\mathrm{Z})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^\wedge_p) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}_\blacksquare(\mathrm{C}^{(m)}[1/\mathrm{I}]^\wedge_p).$$

Under these two descriptions, $\mathrm{res}_{(\mathrm{Z},\mathrm{M}_\mathrm{Z})}$ is the identity on the common limit. Its inverse is the required equivalence $\mathrm{an}_{(\mathrm{Z},\mathrm{M}_\mathrm{Z})}$, which is symmetric monoidal and compatible with pullback and $\underline{\mathrm{RHom}}$.

If $f : (\mathrm{X}, \mathrm{M}_\mathrm{X}) \to (\mathrm{Y}, \mathrm{M}_\mathrm{Y})$ is strict log smooth proper, put $\mathrm{Z}_\mathrm{X} = \mathrm{X} \times_\mathrm{Y} \mathrm{Z}$. Strictness gives

$$(\mathrm{Z}_\mathrm{X}, \mathrm{M}_{\mathrm{Z}_\mathrm{X}})^{\mathcal{L},\blacksquare} \simeq (\mathrm{X}, \mathrm{M}_\mathrm{X})^{\mathcal{L},\blacksquare} \times_{(\mathrm{Y},\mathrm{M}_\mathrm{Y})^{\mathcal{L},\blacksquare}} (\mathrm{Z}, \mathrm{M}_\mathrm{Z})^{\mathcal{L},\blacksquare}.$$

Compatibility with pullback and passage to right adjoints give the pushforward comparison. Lemma 6.3 and Lemma 6.2 then give preservation of perfect objects and the Frobenius compatible Poincaré duality formula of Corollary 6.6. □

Lemma 6.10. In the notation of Lemma 6.9, let $f_\mathrm{Z} : (\mathrm{Z}_\mathrm{X}, \mathrm{M}_{\mathrm{Z}_\mathrm{X}}) \to (\mathrm{Z}, \mathrm{M}_\mathrm{Z})$ be the base change of a strict log smooth proper morphism. Then $\mathrm{R}f_{\mathrm{Z},*}$ preserves perfect unit Laurent crystals.

Proof. Fix $\mathcal{E} \in \mathsf{Perf}((\mathrm{Z_X}, \mathrm{M_{Z_X}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\varphi=1}$. Lemma 6.9 and Lemma 6.2 show that $\mathrm{R}f_{\mathrm{Z},*}$ preserves perfect objects and supply the Frobenius compatible Poincaré pairing. Let α be the lax Frobenius on $\mathrm{R}f_{\mathrm{Z},+}\mathcal{E}$. On an orientable prism of the strict site, let M be the evaluation of $\mathrm{R}f_{\mathrm{Z},+}\mathcal{E}$. The formal argument of [GR24, Proposition 8.6] gives a decomposition

$$\mathrm{M} \simeq \varphi^*\mathrm{M} \oplus \mathrm{C}$$

with C perfect. Its target map is Frobenius on the Breuil-Kisin twisted coefficient and is therefore invertible after Laurentization.

Reduce this decomposition modulo p. Lemma A.11 identifies its terms with perfect complexes over the resulting connective $\mathbf{F}_p$-algebra, and Lemma 6.7 gives $\mathrm{C}/p = 0$. Since C is derived p-complete, derived Nakayama gives $\mathrm{C} = 0$. Thus α is an equivalence on every orientable prism, and hence on the strict site. □

We can now prove the claim for open strata.

Proposition 6.11. Let $f : (\mathrm{X}, \mathrm{M_X}) \to (\mathrm{Y}, \mathrm{M_Y})$ be strict log smooth proper, with source and target satisfying Assumption 3.1 over $\mathrm{Spf}\, \mathrm{W}(k)[\zeta_{p^\ell}]$. Let $\mathrm{D_X}$ and $\mathrm{D_Y}$ be their associated normal crossings divisors. Then $\mathrm{R}f_+$ induces a functor

$$\mathrm{R}f_+ : \mathsf{Perf}((\mathrm{X}, \mathrm{M_X})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D_X}))^{\varphi=1} \to \mathsf{Perf}((\mathrm{Y}, \mathrm{M_Y})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D_Y}))^{\varphi=1}.$$

The same holds for smooth closed strata with their pullback log structures and divisors.

Proof. On a framed logarithmic q-de Rham evaluation of either X or Y, let A be the evaluation of $\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}$, let $\mathrm{T}_1, \dots, \mathrm{T}_r$ cut out the boundary, and put

$$\mathrm{A}^\circ = \mathrm{A}[1/\mathrm{T}_1, \dots, 1/\mathrm{T}_r]_p^{\wedge}.$$

This is the evaluation of $\mathcal{O}_{\Delta}(*\mathrm{D})$, and $\mathrm{A}^\circ \otimes_{\mathrm{A}}^{\mathrm{L},\blacksquare} \mathrm{A}^\circ \simeq \mathrm{A}^\circ$. Thus one may check locally both $\mathcal{O}_{\Delta}(*\mathrm{D_X})$ and $\mathcal{O}_{\Delta}(*\mathrm{D_Y})$ are idempotent $\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}$-algebras. Strictness gives

$$\mathrm{L}f^*\mathcal{O}_{\Delta}(*\mathrm{D_Y}) \simeq \mathcal{O}_{\Delta}(*\mathrm{D_X}),$$

so the map on the localized Laurent categories is obtained from the original map by base change along this idempotent algebra. We may test the claim locally on Y, so assume it is a framed log affine and let A° denote the evaluation of $\mathcal{O}_{\Delta}(*\mathrm{D_Y})$ on this prism.

Put $\mathcal{L}_{\mathrm{Y}} = \mathcal{O}_{\Delta}(*\mathrm{D_Y})$ and $\mathcal{L}_{\mathrm{X}} = \mathcal{O}_{\Delta}(*\mathrm{D_X})$. If $\mathcal{E}$ is $\mathcal{L}_{\mathrm{X}}$-local, the projection formula gives

$$\mathcal{L}_{\mathrm{Y}} \otimes^{\mathrm{L},\blacksquare} \mathrm{R}f_*\mathcal{E} \simeq \mathrm{R}f_*(\mathcal{L}_{\mathrm{X}} \otimes^{\mathrm{L},\blacksquare} \mathcal{E}) \simeq \mathrm{R}f_*\mathcal{E}.$$

The same calculation applies to $\mathrm{R}f_!$, so the canonical equivalence $\mathrm{R}f_! \simeq \mathrm{R}f_*$ and the Poincaré duality argument restrict to the module categories over these idempotent algebras. It follows that $\mathrm{R}f_+$ produces a crystal in $\mathsf{Perf}((\mathrm{Y}, \mathrm{M_Y})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D_Y}))$. To check the unit Frobenius one may pass to the associated perfect A°-module and repeat the argument of Lemma 6.8 using the localized Poincaré duality pairing.

For the second assertion the same proof applies using Lemma 6.9 and localizing only at the remaining boundary components. □

We next extend proper base change to bounded finitely generated unit objects.

Lemma 6.12. Let $u : (\mathrm{Z}, \mathrm{M}_\mathrm{Z}) \to (\mathrm{Y}, \mathrm{M}_\mathrm{Y})$ be either strict-étale or the inclusion of a smooth closed stratum equipped with the pullback log structure. Form the cartesian square

$$\begin{tikzcd} \mathrm{X}_\mathrm{Z} \arrow[r, "\widetilde{u}"] \arrow[d, "f_\mathrm{Z}"] & \mathrm{X} \arrow[d, "f"] \\ \mathrm{Z} \arrow[r, "u"] & \mathrm{Y}. \end{tikzcd}$$

For every bounded finitely generated unit Laurent crystal $\mathcal{E}$ on X, the base change morphism

$$\mathrm{L}u^*\mathrm{R}f_*\mathcal{E} \longrightarrow \mathrm{R}f_{\mathrm{Z},*}\mathrm{L}\widetilde{u}^*\mathcal{E}$$

is an equivalence.

Proof. Strictness gives a cartesian square of Laurent analytic stacks. In the strict-étale case, naturality of Proposition 6.4 identifies the analytification of the base change morphism with

$$\mathrm{L}u_\mathcal{L}^*\mathrm{R}f_{\mathcal{L},*}\,\mathrm{an}_{(\mathrm{X},\mathrm{M}_\mathrm{X})}\,\mathcal{E} \longrightarrow \mathrm{R}f_{\mathrm{Z},\mathcal{L},*}\mathrm{L}\widetilde{u}_\mathcal{L}^*\,\mathrm{an}_{(\mathrm{X},\mathrm{M}_\mathrm{X})}\,\mathcal{E}.$$

In the closed stratum case, the same identification follows from Lemma 6.9. Lemma 6.3 identifies both direct images with the corresponding exceptional direct images, compatibly with base change. This is the base change equivalence for $\mathrm{R}f_!$ and is Frobenius compatible. □

Thus we deduce pushforward compatibility on closed strata.

Lemma 6.13. Let $i : (\mathrm{Z}, \mathrm{M}_\mathrm{Z}) \hookrightarrow (\mathrm{Y}, \mathrm{M}_\mathrm{Y})$ be a smooth closed stratum of pure codimension c, with the pullback log structure, and let D_Z be the union of the remaining boundary components. There is a fully faithful functor

$$i_+ : \mathsf{Perf}((\mathrm{Z}, \mathrm{M}_\mathrm{Z})_\triangle, \mathcal{O}_\triangle[1/\mathrm{I}_\triangle]_p^\wedge)^{\varphi=1} \longrightarrow \mathrm{D}_{\mathrm{fgu}}((\mathrm{Y}, \mathrm{M}_\mathrm{Y})_\triangle, \mathcal{O}_\triangle[1/\mathrm{I}_\triangle]_p^\wedge)$$

characterized by

$$\mathrm{T}_{\text{ét},\mathrm{Y}}(i_+\mathcal{E}) \simeq i_*\mathrm{T}_{\text{ét},\mathrm{Z}}(\mathcal{E})[-c].$$

The same construction gives a fully faithful functor

$$i_+ : \mathsf{Perf}((\mathrm{Z}, \mathrm{M}_\mathrm{Z})_\triangle, \mathcal{O}_\triangle(*\mathrm{D}_\mathrm{Z}))^{\varphi=1} \longrightarrow \mathrm{D}_{\mathrm{fgu}}((\mathrm{Y}, \mathrm{M}_\mathrm{Y})_\triangle, \mathcal{O}_\triangle[1/\mathrm{I}_\triangle]_p^\wedge).$$

Strict étale locally both functors agree with the transfer functor of Lemma 5.18; in particular $i^! = \mathrm{L}i^*[-c]$ is inverse to i_+ on their essential images.

Proof. The dualized lisse correspondence of [IKY26, Theorem 7] applies to the hollow pair $(\mathrm{Z}, \mathrm{M}_\mathrm{Z})$. Since i is strict, extension by zero carries p-Kummer lisse complexes on $(\mathrm{Z}, \mathrm{M}_\mathrm{Z})_\eta$ to p-Kummer constructible complexes on $(\mathrm{Y}, \mathrm{M}_\mathrm{Y})_\eta$. For $\mathcal{E}$ in the first source category, define $i_+\mathcal{E}$ by

$$i_+\mathcal{E} = \mathrm{T}_{\text{ét},\mathrm{Y}}^{-1}\big(i_*\mathrm{T}_{\text{ét},\mathrm{Z}}(\mathcal{E})[-c]\big).$$

Put $\mathrm{U}_{\mathrm{Z},\eta} = \mathrm{Z}_\eta \setminus \mathrm{D}_{\mathrm{Z},\eta}$ and let $j : \mathrm{U}_{\mathrm{Z},\eta} \hookrightarrow (\mathrm{Z}, \mathrm{M}_\mathrm{Z})_\eta$. The open-stratum argument of Theorem 5.4, in the hollow form used in the induction step of Corollary 5.24, gives an equivalence

$$\mathsf{Perf}((\mathrm{Z}, \mathrm{M}_\mathrm{Z})_\triangle, \mathcal{O}_\triangle(*\mathrm{D}_\mathrm{Z}))^{\varphi=1,\mathrm{op}} \xrightarrow{\sim} \mathrm{D}_{\mathrm{lisse}}^{(b),\mathrm{pKum}}(\mathrm{U}_{\mathrm{Z},\eta}, \mathbf{Z}_p).$$

If $\mathbb{L}_{\mathcal{E}}$ denotes the image of $\mathcal{E}$ under this equivalence, define

$$i_+\mathcal{E} = \mathrm{T}_{\text{ét},\mathrm{Y}}^{-1}\big(i_*j_!\mathbb{L}_{\mathcal{E}}[-c]\big).$$

Both functors are fully faithful because i_* and $j_!$ are fully faithful.

To compare these constructions with Lemma 5.18, work on a framed neighborhood, relabel the boundary coordinates so that $\mathrm{Z} = \mathrm{V}(\mathrm{T}_1, \dots, \mathrm{T}_c)$, and put

$$\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}.$$

Writing $\mathrm{J} = (\mathrm{T}_1, \dots, \mathrm{T}_c)$, the quotient comparisons in the proof of Corollary 5.24 identify the two source categories with

$$\mathsf{Perf}(\mathrm{A}/\mathrm{J})^{\varphi=1,h\Gamma}$$

and

$$\mathsf{Perf}\left(((\mathrm{A}/\mathrm{J})[1/\mathrm{T}_{c+1}, \dots, 1/\mathrm{T}_r])_p^{\wedge}\right)^{\varphi=1,h\Gamma},$$

viewed as the corresponding full subcategories of $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J})^{h\Gamma}$. The transfer equivalence

$$i_+ : \mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J})^{h\Gamma} \xrightarrow{\ \sim\ } \mathrm{D}_{\mathrm{fgu}}(\mathrm{A})_{\mathrm{J}}^{h\Gamma}$$

of Lemma 5.18 restricts to these subcategories. For the first one, the realization compatibility following that lemma gives

$$\mathrm{T}_{\text{ét}}(i_+\mathrm{M}) \simeq i_*\mathrm{T}_{\text{ét}}(\mathrm{M})[-c].$$

For the second one, the same compatibility and the open-stratum calculation give

$$\mathrm{T}_{\text{ét}}(i_+\mathrm{M}) \simeq i_*j_!\mathrm{T}_{\text{ét}}(\mathrm{M})[-c].$$

Full faithfulness of Theorem 5.25 therefore identifies the local transfer functors with the two constructions above. Since $i^! = \mathrm{L}i^*[-c]$ is inverse to the local transfer equivalence, it is inverse to both functors on their essential images. □

Corollary 6.14. Let $i : (\mathrm{Z}, \mathrm{M}_{\mathrm{Z}}) \hookrightarrow (\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ be a smooth closed stratum, and let $i_{\mathrm{X}} : (\mathrm{Z}_{\mathrm{X}}, \mathrm{M}_{\mathrm{Z}_{\mathrm{X}}}) \hookrightarrow (\mathrm{X}, \mathrm{M}_{\mathrm{X}})$ be its pullback along a strict log smooth proper morphism f. There is a commutative diagram

$$\begin{tikzcd}
\mathsf{Perf}((\mathrm{Z}_{\mathrm{X}}, \mathrm{M}_{\mathrm{Z}_{\mathrm{X}}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\varphi=1} \arrow[r, "i_{\mathrm{X},+}"] \arrow[d, "\mathrm{R}f_{\mathrm{Z},+}"] & \mathrm{D}^{\varphi}((\mathrm{X}, \mathrm{M}_{\mathrm{X}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}) \arrow[d, "\mathrm{R}f_+"] \\
\mathsf{Perf}((\mathrm{Z}, \mathrm{M}_{\mathrm{Z}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\varphi=1} \arrow[r, "i_+"] & \mathrm{D}^{\varphi}((\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}).
\end{tikzcd}$$

Proof. Work strict étale locally on a framed target $\mathrm{Y} = \operatorname{Spf}\mathrm{R}$, write $\mathrm{Z} = \mathrm{V}(\mathrm{T}_1, \dots, \mathrm{T}_c)$, and put $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$. Fix $\mathcal{E} \in \mathsf{Perf}((\mathrm{Z}_{\mathrm{X}}, \mathrm{M}_{\mathrm{Z}_{\mathrm{X}}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\varphi=1}$. Lemma 6.10 shows that $\mathrm{R}f_{\mathrm{Z},*}\mathcal{E}$ is perfect unit.

Write η and ε for the unit and counit of the adjunction $\mathrm{L}f_{\mathrm{Z}}^* \leftrightarrows \mathrm{R}f_{\mathrm{Z},*}$. Pullback functoriality gives $\mathrm{L}f^*i_+ \simeq i_{\mathrm{X},+}\mathrm{L}f_{\mathrm{Z}}^*$. Lemma 6.13 sends $\varepsilon_{\mathcal{E}}$ to a unique map

$$\mathrm{L}f^*i_+\mathrm{R}f_{\mathrm{Z},*}\mathcal{E} \longrightarrow i_{\mathrm{X},+}\mathcal{E}.$$

Adjunction gives a Frobenius equivariant map

$$\theta : i_+\mathrm{R}f_{\mathrm{Z},*}\mathcal{E} \longrightarrow \mathrm{R}f_*i_{\mathrm{X},+}\mathcal{E}.$$

It remains to show that θ is an equivalence.

Lemma 6.12 identifies $(\mathrm{L}i^*\theta)[-c]$ with

$$\mathrm{R}f_{\mathrm{Z},*}\mathcal{E} \xrightarrow{\eta} \mathrm{R}f_{\mathrm{Z},*}\mathrm{L}f_{\mathrm{Z}}^*(\mathrm{R}f_{\mathrm{Z},*}\mathcal{E}) \xrightarrow{\mathrm{R}f_{\mathrm{Z},*}(\varepsilon)} \mathrm{R}f_{\mathrm{Z},*}\mathcal{E}.$$

This is the identity, so $\mathrm{L}i^*\mathrm{cofib}(\theta) = 0$.

The source of θ is supported on Z. For $1 \leq a \leq c$, strictness gives $\mathrm{L}f_{\mathcal{L}}^*\mathcal{O}_{\Delta}(*\mathrm{D}_a) \simeq \mathcal{O}_{\Delta}(*f^{-1}\mathrm{D}_a)$, so the projection formula and $\mathrm{R}f_{\mathcal{L},!} \simeq \mathrm{R}f_{\mathcal{L},*}$ give

$$\mathcal{O}_{\Delta}(*\mathrm{D}_a) \otimes^{\mathrm{L},\blacksquare} \mathrm{R}f_{\mathcal{L},*}i_{\mathrm{X},+}\mathcal{E} \simeq \mathrm{R}f_{\mathcal{L},!}\big(\mathrm{L}f_{\mathcal{L}}^*\mathcal{O}_{\Delta}(*\mathrm{D}_a) \otimes^{\mathrm{L},\blacksquare} i_{\mathrm{X},+}\mathcal{E}\big) \simeq 0. \tag{5}$$

Evaluate on the log q-de Rham prism and put

$$\mathrm{K}_j := \rho_{\mathrm{A}}^*(\mathrm{cofib}(\theta)) \otimes_{\mathrm{A}}^{\mathrm{L},\blacksquare} \mathrm{A}/(\mathrm{T}_1, \ldots, \mathrm{T}_j).$$

Each K_j is derived p-complete: the pushforward commutes with inverse limits and, by the projection formula, with reduction modulo p^n. The quotient pullback calculation gives $\mathrm{K}_c = 0$. The regular sequence $\mathrm{T}_1, \ldots, \mathrm{T}_c$ gives triangles $\mathrm{K}_{j-1} \xrightarrow{\mathrm{T}_j} \mathrm{K}_{j-1} \to \mathrm{K}_j$. If $\mathrm{K}_j = 0$, then T_j acts invertibly on K_{j-1}, so derived p-completeness identifies K_{j-1} with its p-completed localization at T_j, which vanishes by (5). Thus downward induction gives $\mathrm{K}_0 = 0$, and conservativity of evaluation shows that θ is an equivalence. Since i_+ commutes with twists and shifts, the same identity holds for $\mathrm{R}f_+$. □

Theorem 6.15. Let $f : (\mathrm{X}, \mathrm{M}_{\mathrm{X}}) \to (\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ be strict log smooth proper, where the source and target satisfy Assumption 3.1 over $\mathrm{Spf}\, \mathrm{W}(k)[\zeta_{p^\ell}]$. Then $\mathrm{R}f_+$ induces a functor

$$\mathrm{R}f_+ : \mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathrm{M}_{\mathrm{X}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}) \to \mathrm{D}_{\mathrm{fgu}}((\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}).$$

Proof. The assertion is strict-étale local on Y. The objects whose images under $\mathrm{R}f_+$ are finitely generated unit form a thick subcategory of the source. The proof of Theorem 5.25 shows that the source is thickly generated by shifts of $i_{\mathrm{X},+}\mathcal{E}$, where $i_{\mathrm{X}} : \mathrm{Z}_{\mathrm{X}} \hookrightarrow \mathrm{X}$ is the pullback of the closure $i : \mathrm{Z} \hookrightarrow \mathrm{Y}$ of a logarithmic stratum and

$$\mathcal{E} \in \mathsf{Perf}((\mathrm{Z}_{\mathrm{X}}, \mathrm{M}_{\mathrm{Z}_{\mathrm{X}}})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D}_{\mathrm{Z}_{\mathrm{X}}}))^{\varphi=1}.$$

After a further strict-étale localization, the finite étale sheaf $\pi_0(\mathrm{X}/\mathrm{Y})$ is constant, so its open and closed components may be treated separately. Strictness identifies each logarithmic stratum of X with a connected component of the inverse image of a stratum of Y, and smooth flatness identifies its closure with the corresponding component of the pullback of the closure. Proposition 6.11 shows that $\mathrm{R}f_{\mathrm{Z},+}\mathcal{E}$ is perfect unit over $\mathcal{O}_{\Delta}(*\mathrm{D}_{\mathrm{Z}})$. The same argument in the localized module categories gives

$$\mathrm{R}f_+i_{\mathrm{X},+}\mathcal{E} \simeq i_+\mathrm{R}f_{\mathrm{Z},+}\mathcal{E}.$$

The right side is finitely generated unit by Lemma 6.13. Thus every generator belongs to the indicated thick subcategory. □

Theorem 6.16. Let $f : (\mathrm{X}, \mathrm{M}_{\mathrm{X}}) \to (\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ be a strict log smooth proper morphism of pure relative dimension d where both satisfy Assumption 3.1 over $\mathrm{Spf}\, \mathrm{W}(k)[\zeta_{p^\ell}]$ whose log adic generic fibers are pure of dimensions d_{X} and d_{Y} respectively. Let $\mathrm{D_X}$ and $\mathrm{D_Y}$ be horizontal normal crossings divisors inducing $\mathrm{M_X}$ and $\mathrm{M_Y}$. There is a commutative diagram

$$\begin{tikzcd}
\mathrm{D_{fgu}}((\mathrm{X},\mathrm{M_X})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\mathrm{op}} \arrow[r, "\mathrm{T}_{\text{ét},\mathrm{X}}[d_{\mathrm{X}}]"] \arrow[d, "\mathrm{R}f_*\{d\}[d]"] & \mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D_X}}((\mathrm{X},\mathrm{M_X})_\eta,\mathbf{Z}_p) \arrow[d, "\mathrm{R}f_{\eta,*}"] \\
\mathrm{D_{fgu}}((\mathrm{Y},\mathrm{M_Y})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\mathrm{op}} \arrow[r, "\mathrm{T}_{\text{ét},\mathrm{Y}}[d_{\mathrm{Y}}]"] & \mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D_Y}}((\mathrm{Y},\mathrm{M_Y})_\eta,\mathbf{Z}_p).
\end{tikzcd}$$

Proof. Theorem 6.15 and Corollary 4.15 allow us to restrict the relevant functors to the finitely generated unit categories. For $\mathcal{E}$ on X and $\mathcal{G}$ on Y, transporting Lemma 6.3 through Proposition 6.4 and using the projection formula give

$$\begin{aligned}\mathrm{RHom}(\mathrm{R}f_+\mathcal{E}[d],\mathcal{G}) &\simeq \mathrm{RHom}(\mathrm{R}f_*(\mathcal{E}\{d\}[2d]),\mathcal{G})\\ &\simeq \mathrm{RHom}(\mathcal{E}\{d\}[2d], f^!\mathcal{G}) \simeq \mathrm{RHom}(\mathcal{E},\mathrm{L}f^*\mathcal{G}).\end{aligned}$$

Thus $\mathrm{R}f_+[d]$ is left adjoint to $\mathrm{L}f^*$. The functor $\mathrm{R}f_{\eta,*}$ preserves p-Kummer constructible complexes, so the adjunction $\mathrm{L}f_\eta^* \leftrightarrows \mathrm{R}f_{\eta,*}$ restricts to these categories. Indeed, proper smooth pushforward preserves lissity on each logarithmic stratum. If g is a saturated log perfectoid test object from Definition 4.19 and f' is the base change of f, then proper base change and strictness identify $g^*\mathrm{R}f_{\eta,*}\mathbb{K}$ with $\epsilon^*\mathrm{R}f'_{\text{proét},*}\mathbb{L}$ whenever the pullback of $\mathbb{K}$ to the source of f' is $\epsilon^*\mathbb{L}$.

Pullback compatibility of $\mathrm{T}_{\text{ét}}$ carries the opposite of the first adjunction to an adjunction with left adjoint $\mathrm{L}f_\eta^*$. Uniqueness of right adjoints therefore gives

$$\mathrm{T}_{\text{ét},\mathrm{Y}}(\mathrm{R}f_+\mathcal{E}[d]) \simeq \mathrm{R}f_{\eta,*}\mathrm{T}_{\text{ét},\mathrm{X}}(\mathcal{E}).$$

Since $\mathrm{T}_{\text{ét}}$ is contravariant and $d_{\mathrm{X}} = d_{\mathrm{Y}} + d$, this is the asserted commutative diagram. □

Appendix A. Log prismatic stacks

A.1. **Olsson's log prismatic stack.** In this appendix by formal stacks we mean (accessible) fpqc sheaves in groupoids on animated p-nilpotent rings.

Let Log be the p-completion of Olsson's stack of fine log structures [Ols03, Theorem 1.1]. It has an intrinsic Frobenius lift sending a prelog structure $\alpha : \mathcal{M} \to \mathcal{O}$ to the logification of $m \mapsto \alpha(m)^p$. The flat cover of [Ols03, Corollary 3.15] shows that Log is $\mathbf{Z}_p$-flat, so this Frobenius lift can be thought of as defining a δ-structure (one can make this precise by developing a theory of δ-stacks, which we do not pursue here). We let Log^{Δ} be its prismatization.

For an ordinary p-nilpotent ring R, let $\mathcal{M}_{\mathrm{R}}^{\mathrm{W}}$ denote the cofree $\delta_{\log}$-lift of a fine log structure $\mathcal{M}_{\mathrm{R}}$ on Spec R, and write $(\mathcal{M}_{\mathrm{R}}^{\mathrm{W}})^a$ for its associated log structure [Kos22, Remark 2.12, Proposition 2.14, and Corollary 2.15]. In the explicit cofree construction, $\mathcal{M}_{\mathrm{R}}^{\mathrm{W}} \simeq \mathcal{M}_{\mathrm{R}} \times \mathrm{R}^{\mathbf{N}}$ as sets. Every element over $1 \in \mathcal{M}_{\mathrm{R}}$ maps to a Witt vector with zeroth component 1, hence to a unit. The explicit monoid law therefore shows that logification removes the second factor. Thus an étale-local fine chart $\mathrm{P} \to \mathcal{M}_{\mathrm{R}}$ induces a chart of $(\mathcal{M}_{\mathrm{R}}^{\mathrm{W}})^a$, whose map to W(R) is

$m \mapsto [\alpha(m)]$; in particular, $(\mathcal{M}_{\mathrm{R}}^{\mathrm{W}})^a$ is fine. The formula of [BL22b, Construction 3.11]

$$\pi_{\mathrm{Log}}(\mathcal{M}_{\mathrm{R}}, \mathrm{I} \to \mathrm{W}(\mathrm{R})) = (\mathrm{I} \to \mathrm{W}(\mathrm{R}), (\mathcal{M}_{\mathrm{R}}^{\mathrm{W}})^a|_{\mathrm{Spec}(\mathrm{W}(\mathrm{R})/\mathrm{I})})$$

defines a natural transformation on ordinary rings; derived extension and fpqc hypersheafification give $\pi_{\mathrm{Log}} : \mathrm{Log} \times \mathbf{Z}_p^{\Delta} \to \mathrm{Log}^{\Delta}$. This morphism allows us to define the log prismatic stack, which one can think of as a "prismatic stack relative to a δ-stack".

Definition A.1. Let $(\mathrm{X}, \mathcal{M})$ be a bounded fs log formal scheme. Then there is a map $\mathrm{X} \to \mathrm{Log}$ induced by the log structure. We define $(\mathrm{X}, \mathcal{M})^{\Delta}$ to be the fiber product

$$\begin{tikzcd} (\mathrm{X}, \mathcal{M})^{\Delta} \arrow[r] \arrow[d] & \mathrm{X}^{\Delta} \arrow[d] \\ \mathrm{Log} \times \mathbf{Z}_p^{\Delta} \arrow[r, "\pi_{\mathrm{Log}}"] & \mathrm{Log}^{\Delta}. \end{tikzcd}$$

This is the log prismatic stack introduced by Olsson in [Ols23].

The general philosophy behind this definition is that log formal schemes can be thought of as formal schemes with maps $\mathrm{X} \to \mathrm{Log}$, and logarithmic cohomology theories $\mathrm{H}_{\log}(-)$ can often be produced by taking a relative version $\mathrm{H}(\mathrm{X}/\mathrm{Log})$ using the ordinary cohomology theory $\mathrm{H}(-)$. For example logarithmic de Rham cohomology can be recovered this way; we will show in this subsection the same is true for prismatic cohomology.

To make arguments easier, and still cover the general scope we need for applications, we work in the situation of Assumption 3.1 for a finite extension $K/\mathbf{Q}_p$, where all of our log formal schemes admit covers by logarithmic Breuil-Kisin prisms. For a framed log affine, the log q-de Rham prism is a logarithmic Breuil-Kisin atlas. We define them in general as follows.

Definition A.2. Let $(\mathrm{X}, \mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$. A logarithmic Breuil-Kisin prism for $(\mathrm{X}, \mathcal{M})$ is a log prism $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^a$ such that

$$(\mathrm{Spf}\,\mathrm{A}/\mathrm{I}, \mathrm{M}_{\mathrm{Spf}\,\mathrm{A}/\mathrm{I}}) \to (\mathrm{X}, \mathcal{M})$$

is a strict open immersion of log p-adic formal schemes.

A logarithmic Breuil-Kisin atlas for $(\mathrm{X}, \mathcal{M})$ means this is an isomorphism. We recall that an ordinary Breuil-Kisin prism for a p-adic formal scheme X is a bounded prism (A, I) such that $\mathrm{Spf}\,\mathrm{A}/\mathrm{I} \to \mathrm{X}$ is an open immersion. It is an atlas if this map is an isomorphism. If $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^a$ is an atlas supplied by Lemma A.3 and $(\mathrm{B}, (d), \mathcal{N})^a$ is a strict test prism, then [Kos22, Proposition 4.13] makes

$$\mathrm{B} \longrightarrow \mathrm{A} \coprod_{*} \mathrm{B}$$

completely faithfully flat. Thus the sheaf represented by the atlas covers the final object of the strict log prismatic site for the fpqc topology.

Lemma A.3. If $(\mathrm{X}, \mathcal{M})$ satisfies Assumption 3.1 over $\mathcal{O}_K$, it admits a covering family of logarithmic Breuil-Kisin prisms. Moreover if X is affine, we can choose a log prism so that

$$(\mathrm{Spf}\,\mathrm{A}/\mathrm{I}, \mathrm{M}_{\mathrm{Spf}\,\mathrm{A}/\mathrm{I}}) \to (\mathrm{X}, \mathcal{M})$$

is an isomorphism.

Proof. The construction is Zariski local on X. Let $(\mathrm{U}, \mathcal{M}_{\mathrm{U}}) = (\mathrm{Spf}\,\mathrm{R}, \mathcal{M}_{\mathrm{U}})$ be an affine strict open, put $\mathrm{S} = \mathrm{W}(k)[\![u]\!]$, and let $\mathrm{E}(u)$ be the Eisenstein polynomial of a uniformizer π. Log deformation theory lifts the given affine log smooth pair $(\mathrm{U}, \mathcal{M}_{\mathrm{U}})$ across the quotients $\mathrm{S}/(\mathrm{E}^m)$ to a (p, E)-complete affine fs log smooth pair $(\mathrm{Spf}\,\mathrm{A}, \mathcal{M}_{\mathrm{Spf}\,\mathrm{A}})$ over S satisfying

$$(\mathrm{Spf}\,\mathrm{A}/\mathrm{E}, \mathcal{M}_{\mathrm{Spf}\,\mathrm{A}/\mathrm{E}}) \simeq (\mathrm{U}, \mathcal{M}_{\mathrm{U}}).$$

The obstructions lie in positive Ext groups of the logarithmic cotangent complex, which is finite locally free in degree zero; they vanish because U is affine. Applying the corresponding obstruction theory for maps at each stage lifts absolute logarithmic Frobenius compatibly with the Frobenius $u \mapsto u^p$ of S.

It is clear $\mathrm{E}(u)$ is distinguished, so $(\mathrm{A}, (\mathrm{E}(u)), \mathcal{M}_{\mathrm{Spf}\,\mathrm{A}})^a$ is a bounded log prism with reduction $(\mathrm{U}, \mathcal{M}_{\mathrm{U}})$. Applying the construction to an affine cover gives a covering family of logarithmic Breuil-Kisin prisms. If X is affine, take $\mathrm{U} = \mathrm{X}$. □

After forgetting the δ-structure, the underlying pair is essentially determined by the argument of [GL25, Lemma 3.5].

A.2. **Quasicoherent sheaves on Olsson's log prismatic stack.** We show that quasicoherent sheaves on $(\mathrm{X}, \mathcal{M})^{\Delta}$ recover log prismatic crystals. This ends up being slightly more subtle than the argument for X^{Δ}, because we must generally avoid using Δ_{R} for R quasiregular semiperfect. The basic issue is that in the log setting these will generally not come with fine log structures, so they will not admit maps to Log.

We first show strict log prisms give points of Olsson's log prismatic stack.

Lemma A.4. Let $(\mathrm{X}, \mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$, and let $(\mathrm{A}, \mathrm{I}, \mathcal{N})^a$ be a bounded log prism with fine log structure and a strict map $(\mathrm{Spf}\,\mathrm{A}/\mathrm{I}, \mathcal{N}|_{\mathrm{Spf}\,\mathrm{A}/\mathrm{I}}) \longrightarrow (\mathrm{X}, \mathcal{M})$. Then there is a natural map

$$\rho_{\mathrm{X},\mathrm{A}} : \mathrm{Spf}\,\mathrm{A} \to (\mathrm{X}, \mathcal{M})^{\Delta}.$$

If $(\mathrm{A}, \mathrm{I}, \mathcal{N})^a$ is a logarithmic Breuil-Kisin atlas, then $\rho_{\mathrm{X},\mathrm{A}}$ is an fpqc epimorphism.

Proof. Construction 8.14 of [BL22b] gives $\mathrm{Spf}\,\mathrm{A} \to \mathrm{X}^{\Delta}$. By the fiber product definition of $(\mathrm{X}, \mathcal{M})^{\Delta}$, it remains to complete the diagram

$$\begin{tikzcd}
\mathrm{Spf}\,\mathrm{A} \arrow[r] \arrow[d, dashed] & \mathrm{X}^{\Delta} \arrow[d] \\
\mathbf{Z}_p^{\Delta} \times \mathrm{Log} \arrow[r, "\pi_{\mathrm{Log}}"] & \mathrm{Log}^{\Delta}
\end{tikzcd}$$

The log prism structure supplies the dashed arrow; unwinding definitions one may check the composites to Log^{Δ} agree so we get the desired map.

Now we check that this map is an fpqc epimorphism. For a log Breuil-Kisin atlas, after forgetting the log structure one has that $\rho_{\mathrm{A}} : \operatorname{Spf} \mathrm{A} \to \mathrm{X}^{\Delta}$ is an fpqc epimorphism. It follows

$$\operatorname{Spf} \mathrm{A} \times_{\mathrm{X}^{\Delta}} (\mathrm{X}, \mathcal{M})^{\Delta} \to (\mathrm{X}, \mathcal{M})^{\Delta}$$

is an fpqc epimorphism. By étale descent of X^{Δ}, the stack $(\mathrm{X}, \mathcal{M})^{\Delta}$ satisfies strict-étale descent. Thus when working locally we may choose a chart $\mathrm{P} = \mathbf{N}^r$ for the log structure on $\operatorname{Spf} \mathrm{A}$ and we may replace Log with a toric stack $\mathcal{A}_{\mathrm{P}} = [\operatorname{Spf} \mathbf{Z}_p[\mathrm{P}] / \operatorname{Spf} \mathbf{Z}_p[\mathrm{P}^{\mathrm{gp}}]]$ in the fiber product defining $(\mathrm{X}, \mathcal{M})^{\Delta}$. Lemma A.19 gives a fine log prism B_{P} and an fpqc epimorphism to the pullback of the toric atlas. Since $\operatorname{Spf} \mathrm{A} \to \mathrm{X}^{\Delta}$ and $\operatorname{Spf} \mathbf{Z}_p[\mathrm{P}] \to \mathcal{A}_{\mathrm{P}}$ are fpqc epimorphisms, one obtains fpqc epimorphisms

$$\operatorname{Spf} \mathrm{B}_{\mathrm{P}} \to \operatorname{Spf} \mathrm{A} \times_{\mathrm{X}^{\Delta}} (\mathrm{X}, \mathcal{M})^{\Delta} \to (\mathrm{X}, \mathcal{M})^{\Delta}.$$

We may check whether $\rho_{\mathrm{X},\mathrm{A}} : \operatorname{Spf} \mathrm{A} \to (\mathrm{X}, \mathcal{M})^{\Delta}$ is an fpqc epimorphism after base change along one, and indeed we obtain

$$\begin{tikzcd}
\operatorname{Spf}(\mathrm{A} \coprod_* \mathrm{B}_{\mathrm{P}}) \arrow[r] \arrow[d] & \operatorname{Spf} \mathrm{B}_{\mathrm{P}} \arrow[d] \\
\operatorname{Spf} \mathrm{A} \arrow[r, "\rho_{\mathrm{X},\mathrm{A}}"] & (\mathrm{X}, \mathcal{M})^{\Delta}
\end{tikzcd}$$

using Lemma A.5 (which only uses the first part of this lemma). Here $(\mathrm{A} \sqcup_* \mathrm{B}_{\mathrm{P}})$ is the coproduct in the log prismatic site $(\mathrm{X}, \mathcal{M})_{\Delta}$. But one can directly check that

$$\operatorname{Spf}(\mathrm{A} \coprod_* \mathrm{B}_{\mathrm{P}}) \to \operatorname{Spf} \mathrm{B}_{\mathrm{P}}$$

is faithfully flat, e.g. by [Kos22, Proposition 4.13]. This proves the claim. □

Lemma A.5. Let $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\operatorname{Spf} \mathrm{A}})^a$ be a logarithmic Breuil-Kisin atlas for $(\mathrm{X}, \mathcal{M})$, and let $(\mathrm{B}, (d), \mathcal{N})^a$ be an orientable bounded strict object of $(\mathrm{X}, \mathcal{M})_{\Delta}$. The coproduct $\mathrm{A} \coprod_* \mathrm{B}$ exists, and

$$\operatorname{Spf}(\mathrm{A} \coprod_* \mathrm{B}) \simeq \operatorname{Spf} \mathrm{A} \times_{(\mathrm{X}, \mathcal{M})^{\Delta}} \operatorname{Spf} \mathrm{B}.$$

If (A, I) is an ordinary Breuil-Kisin atlas for X and $(\mathrm{B}, (d))$ is an orientable bounded, possibly animated object of X_{Δ}, the analogous assertion holds without log structures.

Proof. In what follows we use $\overline{(-)}$ for reduction modulo a (log) prism's prismatic ideal.

Let D be the coproduct of the underlying prisms in $(\mathbf{Z}_p)_{\Delta}$, so Proposition 3.2.8 of [BL22a] gives

$$\operatorname{Spf} \mathrm{D} \simeq \operatorname{Spf} \mathrm{A} \times_{\mathbf{Z}_p^{\Delta}} \operatorname{Spf} \mathrm{B}.$$

Let D^{ex} be the completed exactification of the two pulled-back log structures. One may check that

$$\operatorname{Spf} \mathrm{D}^{\mathrm{ex}} \simeq \operatorname{Spf} \mathrm{A} \times_{\mathbf{Z}_p^{\Delta} \times \mathrm{Log}} \operatorname{Spf} \mathrm{B}.$$

In particular, D^{ex} is a bounded prism whose prismatic ideal is extended from D. The defining fiber product of the log prismatic stack gives

$$\mathrm{eq}\left(\mathrm{Spf}\,\mathrm{D} \rightrightarrows (\mathrm{X},\mathcal{M})^{\Delta} \times_{\mathbf{Z}_p^{\Delta}} \mathrm{Spf}\,\mathrm{D}\right) \simeq \mathrm{eq}\left(\mathrm{Spf}\,\mathrm{D}^{\mathrm{ex}} \rightrightarrows \mathrm{X}^{\Delta} \times_{\mathrm{Log}^{\Delta}} \mathrm{Spf}\,\mathrm{D}^{\mathrm{ex}}\right).$$

Here the two arrows in each equalizer are induced by the maps from Spf A and Spf B to $(\mathrm{X},\mathcal{M})^{\Delta}$. The first equalizer is the desired fiber product $\mathrm{Spf}\,\mathrm{A} \times_{(\mathrm{X},\mathcal{M})^{\Delta}} \mathrm{Spf}\,\mathrm{B}$. To compute the second, put $\mathrm{X}_{\overline{\mathrm{D}^{\mathrm{ex}}}} = \mathrm{X} \times_{\mathrm{Log}} \mathrm{Spf}\,\overline{\mathrm{D}^{\mathrm{ex}}}$ and consider

$$\begin{array}{ccccc} (\mathrm{X}_{\overline{\mathrm{D}^{\mathrm{ex}}}}/\mathrm{D}^{\mathrm{ex}})^{\Delta} & \longrightarrow & \mathrm{X}^{\Delta}_{\overline{\mathrm{D}^{\mathrm{ex}}}} & \longrightarrow & \mathrm{X}^{\Delta} \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spf}\,\mathrm{D}^{\mathrm{ex}} & \longrightarrow & (\mathrm{Spf}\,\overline{\mathrm{D}^{\mathrm{ex}}})^{\Delta} & \longrightarrow & \mathrm{Log}^{\Delta}. \end{array}$$

The left square is cartesian by [BL22b, Remark 8.11], and the right square is cartesian by finite limit preservation [BL22b, Remark 8.9]. Hence the outer square is cartesian, and

$$\mathrm{X}^{\Delta} \times_{\mathrm{Log}^{\Delta}} \mathrm{Spf}\,\mathrm{D}^{\mathrm{ex}} \simeq (\mathrm{X}_{\overline{\mathrm{D}^{\mathrm{ex}}}}/\mathrm{D}^{\mathrm{ex}})^{\Delta}.$$

Finite limit preservation identifies the second equalizer with $(\mathrm{Z}_{\log}/\mathrm{D}^{\mathrm{ex}})^{\Delta}$, where

$$\mathrm{Z}_{\log} = \mathrm{Spf}\,\overline{\mathrm{D}^{\mathrm{ex}}} \times_{\mathrm{Spf}\,\bar{\mathrm{A}} \times_{\mathrm{Log}} \mathrm{Spf}\,\bar{\mathrm{B}}} (\mathrm{Spf}\,\bar{\mathrm{A}} \times_{\mathrm{X}} \mathrm{Spf}\,\bar{\mathrm{B}}).$$

We have therefore shown

$$\mathrm{Spf}\,\mathrm{A} \times_{(\mathrm{X},\mathcal{M})^{\Delta}} \mathrm{Spf}\,\mathrm{B} \simeq (\mathrm{Z}_{\log}/\mathrm{D}^{\mathrm{ex}})^{\Delta}.$$

In this case $\mathrm{Z}_{\log}$ is cut out by a regular sequence, so [AKN26, Example 9.3 and Theorem 9.6] and [BL22b, Corollary 7.18] give

$$(\mathrm{Z}_{\log}/\mathrm{D}^{\mathrm{ex}})^{\Delta} \simeq \mathrm{Spf}\,\Delta_{\mathrm{Z}_{\log}/\mathrm{D}^{\mathrm{ex}}}.$$

The construction in the proof of [Kos22, Proposition 4.13], applied to A and B, identifies $\mathrm{A} \coprod_* \mathrm{B}$ with the log prism associated to the prelog prismatic envelope of the exactified surjection $\mathrm{D}^{\mathrm{ex}} \to \mathrm{Z}_{\log}$. By [Kos22, Propositions 3.6 and 3.7], this is the ordinary prismatic envelope with its associated log structure. Hence

$$\Delta_{\mathrm{Z}_{\log}/\mathrm{D}^{\mathrm{ex}}} \simeq \mathrm{A} \coprod_* \mathrm{B}.$$

This proves the logarithmic assertion. Omitting Log and the exactification gives the ordinary assertion; the relative prismatization construction also applies when B is animated. ☐

Proposition A.6. Let $(\mathrm{X},\mathcal{M}) = (\mathrm{Spf}\,\mathrm{R},\mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$, and let

$$\rho_{\mathrm{X},\mathrm{A}} : \mathrm{Spf}\,\mathrm{A} \to (\mathrm{X},\mathcal{M})^{\Delta}$$

be the fpqc epimorphism of Lemma A.4 associated with a logarithmic Breuil-Kisin atlas. Then the Čech nerve of Spf A identifies with $\mathrm{Spf}\,\mathrm{A}^{(n)}$ where $\mathrm{A}^{(n)}$ are the iterated self-coproducts of $(\mathrm{A},\mathrm{I},\mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^a$ in the category of log prisms. In particular

$$(\mathrm{X},\mathcal{M})^{\Delta} \simeq \left|\mathrm{Spf}\,\mathrm{A}^{(\bullet)}\right|$$

in formal stacks.

Proof. Lemma A.5, applied with $\mathrm{B} = \mathrm{A}$, identifies the degree one term of the Čech nerve with $\mathrm{Spf}(\mathrm{A} \coprod_* \mathrm{A})$. The higher terms are the iterated coproducts $\mathrm{Spf}\,\mathrm{A}^{(n)}$, for both the site-theoretic Čech nerve and the Čech nerve of $\rho_{\mathrm{X},\mathrm{A}} : \mathrm{Spf}\,\mathrm{A} \to (\mathrm{X}, \mathcal{M})^{\Delta}$. Using Lemma A.4, this is an fpqc epimorphism so the claim follows. □

Proposition A.7. Let $(\mathrm{X}, \mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$. There is a natural symmetric monoidal equivalence

$$\iota_{(\mathrm{X},\mathcal{M})} : \widehat{\mathrm{D}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}) \xrightarrow{\sim} \mathrm{D}_{\mathrm{qc}}((\mathrm{X}, \mathcal{M})^{\Delta}).$$

It is compatible with pullback along strict morphisms between pairs satisfying Assumption 3.1 and restricts to an equivalence on perfect objects. The same assertions hold after base change along an algebra object on $\mathbf{Z}_p^{\Delta}$.

Proof. We first note that the assertion can be checked strict-étale locally. Indeed for the crystal side by [Ino25, Lemma 5.9], crystals may be computed on the strict subsite; strict-étale locally on an affine, using a log Breuil-Kisin $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^a$ yields

$$\lim_{[n]\in\Delta} \widehat{\mathrm{D}}_{(p,\mathrm{I}^{(n)})}(\mathrm{A}^{(n)}) \simeq \widehat{\mathrm{D}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta})$$

where $\mathrm{A}^{(n)}$ is the logarithmic Čech nerve. The proof of Corollary A.16 gives strict-étale descent of the crystal side.

The same holds for the stack $(\mathrm{X}, \mathcal{M})^{\Delta}$ by étale descent for X^{Δ} in X [BL22b]. We may therefore assume $(\mathrm{X}, \mathcal{M}) = (\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ is affine and that its log prismatic stack is covered by Spf A for a log Breuil-Kisin prism, by Lemma A.4.

Now we can use Proposition A.6 to identify

$$\mathrm{D}_{\mathrm{qc}}((\mathrm{X}, \mathcal{M})^{\Delta}) \simeq \lim_{[n]\in\Delta} \mathrm{D}_{\mathrm{qc}}(\mathrm{Spf}\,\mathrm{A}^{(n)}) \simeq \widehat{\mathrm{D}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}).$$

Indeed, $\rho_{\mathrm{X},\mathrm{A}} : \mathrm{Spf}\,\mathrm{A} \to (\mathrm{X}, \mathcal{M})^{\Delta}$ is an fpqc epimorphism and we have identified its Čech nerve with $\mathrm{Spf}\,\mathrm{A}^{(n)}$, and thus D_{qc} satisfies descent for such maps. See the discussion after [Pen25, Definition 2.4] for more details. □

For a strict morphism f, we write $\mathrm{R}f_{\Delta,*}$ and $\mathrm{R}f_*$ for the right adjoints to pullback on the two categories of Proposition A.7, respectively.

Proposition A.8. Let $f : (\mathrm{X}, \mathcal{M}) \to (\mathrm{Y}, \mathcal{N})$ be a strict morphism between pairs satisfying Assumption 3.1 over $\mathcal{O}_K$. For every $\mathcal{E} \in \widehat{\mathrm{D}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta})$, there is a natural equivalence

$$\iota_{(\mathrm{Y},\mathcal{N})} \mathrm{R}f_{\Delta,*}\mathcal{E} \simeq \mathrm{R}f_*\iota_{(\mathrm{X},\mathcal{M})}\mathcal{E}.$$

Proof. Compatibility of Proposition A.7 with strict pullback gives a commutative diagram

$$\begin{tikzcd}
\widehat{\mathrm{D}}((\mathrm{Y},\mathcal{N})_{\Delta},\mathcal{O}_{\Delta}) \arrow[r, "\iota_{(\mathrm{Y},\mathcal{N})}"] \arrow[d, "f^*"] & \mathrm{D}_{\mathrm{qc}}((\mathrm{Y},\mathcal{N})^{\Delta}) \arrow[d, "(f^{\Delta})^*"] \\
\widehat{\mathrm{D}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}) \arrow[r, "\iota_{(\mathrm{X},\mathcal{M})}"] & \mathrm{D}_{\mathrm{qc}}((\mathrm{X},\mathcal{M})^{\Delta}).
\end{tikzcd}$$

Both vertical pullback functors preserve colimits. Taking right adjoints of this square gives the asserted equivalence. □

Remark A.9. Laurent F-crystals in [KY25, IKY26] use the saturated log prismatic site. By [Ino25, Lemma 5.9], the strict subsite is cofinal in the full log prismatic site. If $(\mathrm{X},\mathcal{M})$ is fs and saturated, a strict object has the same characteristic monoid as the pullback of $\overline{\mathcal{M}}$, and is therefore saturated. The strictification in [Ino25, Lemma 5.9] has this same characteristic monoid, so the strict subsite is also cofinal in the saturated log prismatic site. Thus the limits defining the crystal categories on the full and saturated sites agree. We suppress this distinction for $\widehat{\mathrm{D}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta})$ and its perfect and Laurent variants, but make no comparison between arbitrary sheaves on the underlying ringed topoi.

A.3. **Solid Laurent crystals.** In this subsection, we define ordinary and derived p-complete solid generalizations of

$$\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}).$$

We always use the analytic structure $\mathrm{Spa}(\mathrm{A},\mathrm{A})$.

Definition A.10. Let $(\mathrm{A},(d))$ be an orientable prism. Associated to the derived (p,d)-complete ring A is the analytic ring (A,A) in the sense of [AM24]. We define

$$\mathrm{D}_{\blacksquare}(\mathrm{A}[1/d]^{\wedge}_{p}) := \mathsf{Mod}_{\mathrm{A}[1/d]^{\wedge}_{p}}\mathrm{D}_{\blacksquare}((\mathrm{A},\mathrm{A})),$$

where $\mathrm{A}[1/d]^{\wedge}_{p} = \varprojlim_n (\mathrm{A}/^{\mathrm{L}}p^n)[1/d]$ is regarded as an algebra object of $\mathrm{D}_{\blacksquare}((\mathrm{A},\mathrm{A}))$. Equivalently, this is the solid category of the inverse limit of the analytic rings associated with the Huber pairs $((\mathrm{A}/^{\mathrm{L}}p^n)[1/d],\mathrm{A}/^{\mathrm{L}}p^n)$. We similarly put

$$\widehat{\mathrm{D}}_{\blacksquare}(\mathrm{A}[1/d]^{\wedge}_{p}) := \mathsf{Mod}_{\mathrm{A}[1/d]^{\wedge}_{p}}\widehat{\mathrm{D}}_{\blacksquare}((\mathrm{A},\mathrm{A})).$$

Here $\widehat{\mathrm{D}}_{\blacksquare}((\mathrm{A},\mathrm{A}))$ is the derived category of solid derived p-complete modules.
Then

$$\mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}) := \lim_{(\mathrm{A},(d),\mathrm{M}_{\mathrm{Spf\,A}})^a\in(\mathrm{X},\mathcal{M})_{\Delta}} \mathrm{D}_{\blacksquare}(\mathrm{A}[1/d]^{\wedge}_{p}).$$

Its derived p-complete variant is $\widehat{\mathrm{D}}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$.

The definition is independent of the generator d, and one may (by ind-Zariski localization) assume that the prismatic ideal is principal.

In what follows we repeatedly use the following formal fact: if $\mathcal{C} \simeq \varprojlim_i \mathcal{C}_i$ is a limit of presentable symmetric monoidal stable ∞-categories and an algebra $\mathrm{S} \in \mathcal{C}$ has images S_i, then

$$\mathrm{Mod}_{\mathrm{S}}(\mathcal{C}) \simeq \varprojlim_i \mathrm{Mod}_{\mathrm{S}_i}(\mathcal{C}_i).$$

Indeed, algebra objects and their module objects are computed in limits.

We define perfect Laurent crystals by dualizability:

$$\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}) := \mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})^{\mathrm{dual}}.$$

Lemma A.11. Let $(\mathrm{A},(d))$ be an orientable prism. There is a functorial equivalence

$$\mathrm{D}_{\blacksquare}(\mathrm{A}[1/d]^{\wedge}_{p})^{\mathrm{dual}} \simeq \varprojlim_{n} \mathsf{Perf}((\mathrm{A}/^{\mathrm{L}}p^{n})[1/d]).$$

If $\mathrm{A}[1/d]^{\wedge}_{p}$ is static with bounded p^{∞}-torsion, the right side is $\mathsf{Perf}(\mathrm{A}[1/d]^{\wedge}_{p})$.

Proof. This follows from [Kip26a, Lemma 3.7.8]. ☐

Consequently,

$$\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}) \simeq \varprojlim_{(\mathrm{A},(d),\mathrm{M}_{\mathrm{Spf\,A}})^{a}\in(\mathrm{X},\mathcal{M})_{\Delta}} \mathsf{Perf}((\mathrm{A})[1/d]^{\wedge}_{p}).$$

Lemma A.12. Let $\mathrm{A}\to\mathrm{B}$ be a faithfully flat map of connective animated rings. If $\pi_0\mathrm{B}$ is countably presented as a $\pi_0\mathrm{A}$-module, then $\mathrm{A}\to\mathrm{B}$ is descendable of index at most 2.

Proof. See [Mat16, Proposition 3.31 and the proof of Corollary 3.33]. ☐

Lemma A.13. Let $(\mathrm{X},\mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$, let $(\mathrm{A},\mathrm{I},\mathrm{M}_{\mathrm{Spf\,A}})^{a}$ be a logarithmic Breuil-Kisin atlas, and let $(\mathrm{B},(d),\mathcal{N})^{a}$ be an orientable bounded strict object of the same log prismatic site. Put

$$\mathrm{C} = \mathrm{A}\coprod_{*}\mathrm{B}.$$

Then C exists, and

$$\mathrm{B}\longrightarrow\mathrm{C}$$

is completely descendable for the (p,d)-adic topologies in the sense of [Kip26b, Definition 2.4.1.7], and the induced algebra is descendable in $\mathrm{D}_{\blacksquare}((\mathrm{B},\mathrm{B}))$.
Moreover, $\mathrm{Spa}(\mathrm{C},\mathrm{C})\to\mathrm{Spa}(\mathrm{B},\mathrm{B})$ is a proper descendable cover. The same assertions hold if (A,I) is an ordinary Breuil-Kisin atlas, $(\mathrm{B},(d))$ is an orientable bounded, possibly animated object of X_{Δ}, and $\mathrm{C}=\mathrm{A}\coprod_{*}\mathrm{B}$.

Proof. We prove the logarithmic assertion. In the ordinary case, the relative prismatization has the same countable cell presentation, and the argument applies to connective animated rings.

There are two main points in the argument. First, the reductions modulo (p^k,d^k) are descendable of index ≤ 2. Second, the nullhomotopies witnessing this need not lift compatibly to the derived adic limit. We show that the resulting $\varprojlim^1$ obstruction disappears after taking its tensor square, giving complete descendability of index ≤ 4.

Now we discuss the details. For the first point, we see Proposition 4.13 of [Kos22] constructs C by exactification followed by a log prismatic envelope. Propositions 3.11 and 4.13 of [Kos22] make $\mathrm{B} \to \mathrm{C}$ (p, d)-completely faithfully flat as well. The criterion of [BS22, §2.6, before Definition 2.42] therefore makes $\mathrm{B}/(p^n, d^k) \to \mathrm{C}/(p^n, d^k)$ flat. It is faithfully flat in the ordinary sense because (p, d) lies in the Jacobson radical (by nilpotence), therefore any maximal ideal contains (p, d). In particular to check surjectivity on mSpec we can work modulo (p, d) where we already know faithful flatness.

Because of our assumption on the log structure, exactification uses finitely many monoid algebra extensions and localizations, while the prismatic envelope used in the construction of C adjoins the iterated δ-operations on finitely many generators. Thus

$$\mathrm{C} \simeq (\mathrm{B}[x_1, x_2, \ldots]/(f_1, f_2, \ldots))^{\wedge}_{(p,d)}.$$

If Mon is the countable set of monomials in the x_i, the algebra before completion has the module presentation

$$\mathrm{B}^{(\mathbf{N}\times\mathrm{Mon})} \longrightarrow \mathrm{B}^{(\mathrm{Mon})} \longrightarrow \mathrm{B}[x_1, x_2, \ldots]/(f_1, f_2, \ldots) \longrightarrow 0.$$

The first map sends the basis vector indexed by (i, m) to mf_i. As we work modulo (p^n, d^k) we may ignore the completion; hence $\pi_0(\mathrm{C}/(p^n, d^k))$ is countably presented over $\pi_0(\mathrm{B}/(p^n, d^k))$, and Lemma A.12 proves the first assertion.

Rigidity of prismatic ideals ([BS22, Lemma 3.5]) makes $\mathrm{B} \to \mathrm{C}$ adic. Let $\mathrm{J} = \mathrm{fib}(\mathrm{B} \to \mathrm{C})$ in the derived (p, d)-complete category. The assertion that the map is descendable of index ≤ 2 modulo (p^k, d^k) says that the canonical map $u : \mathrm{J}\widehat{\otimes}_{\mathrm{B}}\mathrm{J} \to \mathrm{B}$ becomes null modulo (p^k, d^k) for every k. If $\mathrm{J}_k = \mathrm{J} \otimes^{\mathrm{L}}_{\mathrm{B}} \mathrm{B}/(p^k, d^k)$, derived completeness gives

$$\mathrm{Map}_{\mathrm{B}}(\mathrm{J}\widehat{\otimes}_{\mathrm{B}}\mathrm{J}, \mathrm{B}) \simeq \mathrm{R}\varprojlim_k \mathrm{Map}_{\mathrm{B}/(p^k,d^k)}(\mathrm{J}_k^{\otimes 2}, \mathrm{B}/(p^k, d^k)).$$

Write X_k for the mapping spectrum on the right. The Milnor exact sequence gives

$$0 \longrightarrow \varprojlim_k{}^1 \pi_1\mathrm{X}_k \longrightarrow \pi_0\mathrm{R}\varprojlim_k \mathrm{X}_k \longrightarrow \varprojlim_k \pi_0\mathrm{X}_k \longrightarrow 0.$$

The reduction of u is null for every k, so its image in $\varprojlim_k \pi_0\mathrm{X}_k$ vanishes. Hence $[u]$ lies in $\mathrm{Fil}^1 = \varprojlim^1_k \pi_1\mathrm{X}_k$. Tensoring maps gives compatible pairings

$$\mathrm{X}_k \otimes \mathrm{X}_k \longrightarrow \mathrm{Map}_{\mathrm{B}/(p^k,d^k)}(\mathrm{J}_k^{\otimes 4}, \mathrm{B}/(p^k, d^k)).$$

The induced pairing of derived limit spectral sequences carries $\mathrm{Fil}^a \otimes \mathrm{Fil}^b$ into Fil^{a+b}. Therefore

$$[u] \in \mathrm{Fil}^1 \Longrightarrow [u\widehat{\otimes}u] \in \mathrm{Fil}^2 = 0,$$

since $\varprojlim^s_k = 0$ for $s \geq 2$ on a countable tower. Thus the fourth tensor power of the fiber map $\mathrm{J} \to \mathrm{B}$ is null. The tensor nilpotence criterion of [Mat18, Equation (6)] makes C descendable in the derived (p, d)-complete category. This is complete descendability in the sense of [Kip26b, Definition 2.4.1.7].

The lax symmetric monoidal functor from derived (p, d)-complete B-modules to $\mathrm{D}_{\blacksquare}((\mathrm{B}, \mathrm{B}))$ used in the proof of [Kip26b, Proposition 2.5.1.2] carries this tensor nilpotence witness to the induced analytic algebra.

It remains to prove properness. Locally after exactification, the construction of [Kos22, Propositions 3.11 and 4.13] has the form $\mathrm{C} \simeq \mathrm{D}\{x_1/d, \dots, x_m/d\}_\delta^\wedge$ with $\mathrm{D}/(d, x_1, \dots, x_m) \simeq \mathrm{B}/d$. Modulo (p, d), the standard triangular relations for this prismatic envelope give a monic equation of degree p for each successive δ-generator. The exactification ratios have prescribed images in B/d, so $\pi_0(\mathrm{B}/(p, d)) \to \pi_0(\mathrm{C}/(p, d))$ is integral. Since p and d are topologically nilpotent, [Kip26a, Example 2.1.13] makes $\mathrm{Spa}(\mathrm{C}, \mathrm{C}) \to \mathrm{Spa}(\mathrm{B}, \mathrm{B})$ proper. Its algebra is descendable by the first part of the proof, so it is a cover by [Kip26a, Example 2.1.12]. In the possibly animated ordinary case, the argument applies to π_0. □

Theorem A.14. Let $(\mathrm{X}, \mathcal{M})$ be affine and satisfy Assumption 3.1 over $\mathcal{O}_K$, and let $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^a$ be a logarithmic Breuil-Kisin atlas. If $(\mathrm{A}^{(m)}, \mathrm{I}, \mathrm{M}^{(m)})^a$ is the degree-m term of its Čech nerve in the strict log prismatic site, then

$$\mathrm{D}_\blacksquare((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}_\blacksquare(\mathrm{A}^{(m)}[1/\mathrm{I}]_p^\wedge).$$

The analogous natural functor for derived p-complete solid categories is also an equivalence:

$$\widehat{\mathrm{D}}_\blacksquare((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge) \simeq \varprojlim_{[m]\in\Delta} \widehat{\mathrm{D}}_\blacksquare(\mathrm{A}^{(m)}[1/\mathrm{I}]_p^\wedge).$$

Proof. By [Ino25, Lemma 5.9], it suffices to work on the strict log prismatic subsite, where orientable objects form a Zariski basis. For an orientable strict test prism $(\mathrm{B}, (d), \mathcal{N})^a$, [Kos22, Proposition 4.13] identifies the pullback of the atlas with

$$\mathrm{B} \longrightarrow \mathrm{A} \coprod_{*} \mathrm{B}.$$

The complete descendability result of Lemma A.13 gives

$$\mathrm{D}_\blacksquare(\mathrm{B}[1/d]_p^\wedge) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}_\blacksquare((\mathrm{A}^{(m)} \coprod_{*} \mathrm{B})[1/d]_p^\wedge)$$

because forming module objects commutes with limits in symmetric monoidal presentable categories. This shows the assertion that $(\mathrm{B}, (d), \mathrm{M}_{\mathrm{Spf}\,\mathrm{B}})^a \mapsto \mathrm{D}_\blacksquare(\mathrm{B}[1/d]_p^\wedge)$ satisfies descent for $h_{\mathrm{A}} \to *$ in the log prismatic topos of $(\mathrm{X}, \mathcal{M})_{\Delta}$; here we use h for the Yoneda embedding on a log prism. Indeed, it suffices to check this on all representable sheaves, which we have just done. Thus, taking global sections one obtains

$$\mathrm{D}_\blacksquare((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}_\blacksquare(\mathrm{A}^{(m)}[1/\mathrm{I}]_p^\wedge).$$

The same formula holds with $\widehat{\mathrm{D}}_\blacksquare$ by a similar argument. □

Lemma A.15. Let $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ be a framed log affine over $\mathrm{Spf}\,\mathrm{W}(k)[\zeta_p]$. Let

$$((\mathrm{A}^+)^{(m)}, [p]_q, \mathcal{M}^{(m)})^a$$

be the degree m term of the log q-de Rham Čech nerve in $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}$, and put

$$\mathrm{A}^{(m)} = (\mathrm{A}^+)^{(m)}[1/[p]_q]_p^\wedge.$$

Then
$$D_\blacksquare((\operatorname{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]^\wedge_p) \simeq \varprojlim_{[m]\in\Delta} D_\blacksquare(A^{(m)}).$$
Moreover, the Γ action on the degree zero term gives a conservative evaluation functor to $D_\blacksquare(A)^{h\Gamma}$. The same holds for $\widehat{D}_\blacksquare$.

Proof. The log q-de Rham prism is a logarithmic Breuil-Kisin atlas, so Theorem A.14 gives the asserted Čech descriptions in both categories. The Γ action arises from log prism automorphisms, and the crystal property therefore gives the action on degree zero evaluation. The projection from either Čech limit to its degree zero term is conservative. $\square$

Corollary A.16. On log formal schemes satisfying Assumption 3.1 over $\mathcal{O}_K$, the assignment
$$(X, \mathcal{M}) \longmapsto D_\blacksquare((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]^\wedge_p)$$
is a stack for the strict-étale topology. The same holds for $\widehat{D}_\blacksquare((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]^\wedge_p)$ and $\mathsf{Perf}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]^\wedge_p)$.

Proof. By Zariski descent, we may assume that $(X, \mathcal{M})$ is affine and that $U \to X$ is an affine strict-étale cover. Let $U^{(\bullet)}$ be its Čech nerve. Choose a logarithmic Breuil-Kisin atlas for $(X, \mathcal{M})$. Strict-étale localization and the unique completely étale lifting property give compatible logarithmic Breuil-Kisin atlases for the $U^{(n)}$ and their higher Čech nerves in the log prismatic site; see [Kos22, Lemma 4.12]. Write $A_X^{(m)}$ and $A_{U^{(n)}}^{(m)}$ for the Laurent coefficient rings in degree m of their prismatic Čech nerves. They form the (augmented) bicosimplicial diagram

$$\begin{tikzcd}
D_\blacksquare((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]^\wedge_p) \arrow[r] \arrow[d] & D_\blacksquare(A_X^{(0)}) \arrow[r, shift left] \arrow[r, shift right] \arrow[d] & D_\blacksquare(A_X^{(1)}) \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] \arrow[d] & \cdots \\
D_\blacksquare((U, \mathcal{M}|_U)_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]^\wedge_p) \arrow[r] \arrow[d, shift left] \arrow[d, shift right] & D_\blacksquare(A_U^{(0)}) \arrow[r, shift left] \arrow[r, shift right] \arrow[d, shift left] \arrow[d, shift right] & D_\blacksquare(A_U^{(1)}) \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] \arrow[d, shift left] \arrow[d, shift right] & \cdots \\
D_\blacksquare((U^{(1)}, \mathcal{M}|_{U^{(1)}})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]^\wedge_p) \arrow[r] \arrow[d, shift left=2] \arrow[d] \arrow[d, shift right=2] & D_\blacksquare(A_{U^{(1)}}^{(0)}) \arrow[r, shift left] \arrow[r, shift right] \arrow[d, shift left=2] \arrow[d] \arrow[d, shift right=2] & D_\blacksquare(A_{U^{(1)}}^{(1)}) \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] \arrow[d, shift left=2] \arrow[d] \arrow[d, shift right=2] & \cdots \\
\vdots & \vdots & \vdots &
\end{tikzcd}$$

For fixed n, Theorem A.14 identifies the limit of the nth row with the Laurent crystal category of $U^{(n)}$. For fixed m, compatibility with strict-étale localization identifies the mth column with the Čech nerve of a completely faithfully étale map. Such maps are completely descendable, so the limit of this column is $D_\blacksquare(A_X^{(m)})$. Since limits commute with limits, we obtain

$$\begin{aligned}
\varprojlim_{[n]\in\Delta} \mathrm{D}_{\blacksquare}((\mathrm{U}^{(n)}, \mathcal{M}|_{\mathrm{U}^{(n)}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}) &\simeq \varprojlim_{[n]\in\Delta} \varprojlim_{[m]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{A}_{\mathrm{U}^{(n)}}^{(m)}) \\
&\simeq \varprojlim_{[m]\in\Delta} \varprojlim_{[n]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{A}_{\mathrm{U}^{(n)}}^{(m)}) \\
&\simeq \varprojlim_{[m]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{A}_{\mathrm{X}}^{(m)}) \\
&\simeq \mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}).
\end{aligned}$$

All equivalences are induced by the augmentations, so their composite is the strict-étale descent equivalence. The same bicosimplicial argument applies to $\widehat{\mathrm{D}}_{\blacksquare}$: the rows are computed by Theorem A.14, while descent for the columns follows modulo p^n from ordinary solid descent and then by taking the inverse limit over n.

The descent equivalence is symmetric monoidal, and dualizable objects are detected termwise in limits. It therefore restricts to the asserted descent equivalence for $\mathsf{Perf}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$. □

The canonical t-structure on $\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$ is defined by taking the connective objects to be the crystals whose evaluations on every prism are connective, and taking the coconnective objects to be the right orthogonal of the connective objects shifted by one. Extension of scalars preserves connective objects, so this defines an accessible t-structure.

Corollary A.17. Let $(\mathrm{X}, \mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$. The canonical t-structure on $\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$ restricts to $\mathsf{Perf}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$.

Proof. The assertion is strict-étale local by Corollary A.16, so choose a framed log q-de Rham atlas and put $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$. This prism covers the final object by Theorem A.14, so its evaluation functor detects perfectness. If $\mathcal{E}$ is perfect, Lemma A.11 identifies $\mathcal{E}(\mathrm{A})$ with an ordinary perfect A-complex. Since A is regular Noetherian, its truncations are perfect. Hence the truncations of $\mathcal{E}$ are perfect as well. □

We denote its heart by $\mathsf{Coh}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$. Since the t-structure is defined by termwise truncation, Corollary A.16 gives strict-étale descent for its heart.

A.4. **Analytifications.** We compare Breuil-Kisin presentations with the analytification of prismatic stacks.

Proposition A.18. Let X be an affine p-adic formal scheme which is flat and lci over $\mathbf{Z}_p$, and let (A, I) be an ordinary Breuil-Kisin atlas. The induced map

$$\mathrm{Spa}(\mathrm{A}, \mathrm{A}) \longrightarrow \mathrm{X}^{\Delta, \blacksquare}$$

is a representable proper effective epimorphism. If $\mathrm{A}^{(n)}$ denotes the degree-n iterated self-coproduct in the category of prisms over X, then

$$\mathrm{X}^{\Delta, \blacksquare} \simeq \left|\mathrm{Spa}(\mathrm{A}^{(\bullet)}, \mathrm{A}^{(\bullet)})\right|.$$

Proof. Construction 3.7.1 of [Kip26a] gives an effective presentation of $X^{\Delta,\blacksquare}$ by the prismatizations of a semiperfectoid Čech presentation of X. After a Zariski refinement, its affine terms are $\mathrm{Spa}(B, B)$ for orientable bounded, possibly animated prisms B over X. Lemma A.5, left exactness of éid sheafification, and finite limit preservation of $(-)^{\blacksquare}$ identify the pullback of $\mathrm{Spa}(A, A)$ with $\mathrm{Spa}(A \coprod_* B, A \coprod_* B)$. Lemma A.13 makes its projection to $\mathrm{Spa}(B, B)$ a proper descendable cover. Representability is local on the target, as is properness by [Kip26a, Remark 2.3.1]; effective epimorphisms are local on the target in any ∞-topos. This proves the first assertion. Applying Lemma A.5 iteratively with $B = A$ identifies the Čech terms with $\mathrm{Spa}(A^{(n)}, A^{(n)})$. The resulting Čech nerve realizes to $X^{\Delta,\blacksquare}$ because $\mathrm{Spa}(A, A) \to X^{\Delta,\blacksquare}$ is an effective epimorphism. □

For a fine monoid P, let $\mathcal{A}_P = [\mathrm{Spf}\,\mathbf{Z}_p[P]/\,\mathrm{Spf}\,\mathbf{Z}_p[P^{gp}]]$ be its Artin cone. Its toric atlas is $\mathrm{Spf}\,\mathbf{Z}_p[P] \to \mathcal{A}_P$. The universal log structure induces an étale map $\mathcal{A}_P \to \mathrm{Log}$ by [Ols03, Corollary 5.24].

Lemma A.19. Suppose $(X, \mathcal{M})$ has a strict chart by $P = \mathbf{N}^r$, and let $(A, I, M_{\mathrm{Spf}\,A})^a$ be a logarithmic Breuil-Kisin atlas carrying a lift of this chart. The formal stack

$$\mathrm{Spf}\,A \times_{\mathcal{A}_P^{\Delta}} (\mathbf{Z}_p^{\Delta} \times \mathrm{Spf}\,\mathbf{Z}_p[P])$$

admits an fpqc cover by an orientable bounded fine log prism $(B_P, I, \mathcal{N}_P)^a$. After solidification the resulting map is a proper effective epimorphism. Moreover, $(\mathrm{Spf}(B_P/I), \mathcal{N}_P|_{\mathrm{Spf}(B_P/I)}) \to (X, \mathcal{M})$ is strict.

Proof. Put

$$X_P = \mathrm{Spf}(A/I) \times_{\mathcal{A}_P} \mathrm{Spf}\,\mathbf{Z}_p[P].$$

If $\widetilde{e}_1, \ldots, \widetilde{e}_r$ are the lifted chart generators, then the chosen lift trivializes the pulled-back torus torsor and gives

$$X_P \simeq \mathrm{Spf}\left((A/I)[u_1^{\pm 1}, \ldots, u_r^{\pm 1}]_p^{\wedge}\right).$$

Set

$$B_P = A[u_1^{\pm 1}, \ldots, u_r^{\pm 1}]_{(p,I)}^{\wedge}$$

with $\varphi(u_i) = u_i^p$ and chart $e_i \mapsto u_i \widetilde{e}_i$. This is an orientable bounded fine log prism with reduction X_P. Its projection to $(X, \mathcal{M})$ is strict.

Relative base change [BL22b, Remark 8.11] gives

$$\mathrm{Spf}\,A \times_{\mathcal{A}_P^{\Delta}} (\mathbf{Z}_p^{\Delta} \times \mathrm{Spf}\,\mathbf{Z}_p[P]) \simeq (X_P/A)^{\Delta}.$$

By the relative form of Lemma A.5, pullback along an orientable bounded prism $C \to (X_P/A)^{\Delta}$ gives

$$\mathrm{Spf}(B_P \coprod_* C) \longrightarrow \mathrm{Spf}\,C.$$

This is completely faithfully flat by [Kos22, Proposition 4.13]. The fpqc local prism factorization in the proof of [BL22b, Theorem 6.5 and Remark 7.23] proves the first assertion. After solidification these pullbacks are proper descendable covers by Lemma A.13. The presentation of [Kip26a, Construction 3.7.1] then proves the second assertion. □

Proposition A.20. Let $(\mathrm{X}, \mathcal{M})$ be affine and satisfy Assumption 3.1 over $\mathcal{O}_K$, and let $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}})^a$ be a logarithmic Breuil-Kisin atlas. The induced map

$$\mathrm{Spa}(\mathrm{A}, \mathrm{A}) \longrightarrow (\mathrm{X}, \mathcal{M})^{\triangle,\blacksquare}$$

is a representable proper effective epimorphism. If $\mathrm{A}^{(n)}$ denotes the degree-n iterated self-coproduct in $(\mathrm{X}, \mathcal{M})_{\triangle}$, then

$$(\mathrm{X}, \mathcal{M})^{\triangle,\blacksquare} \simeq \left|\mathrm{Spa}(\mathrm{A}^{(\bullet)}, \mathrm{A}^{(\bullet)})\right|.$$

Proof. Formation of $\mathrm{X}^{\triangle,\blacksquare}$ preserves étale covers by [Kip26a, Construction 3.7.1], so by the construction of $(\mathrm{X}, \mathcal{M})^{\triangle,\blacksquare}$ the assertions may be checked strict-étale locally on $(\mathrm{X}, \mathcal{M})$ for a Breuil-Kisin atlas. After further strict-étale localization, choose a chart by $\mathrm{P} = \mathbf{N}^r$ for $\mathrm{M}_{\mathrm{Spf\,A}}$; its reduction is a chart for $\mathcal{M}$ on $\mathrm{X} = \mathrm{Spf\,A}/\mathrm{I}$. This induces a map $\mathrm{Spf\,A} \to \mathcal{A}_{\mathrm{P}}^{\triangle} \to \mathrm{Log}^{\triangle}$. By Definition A.1, the fiber product $\mathrm{Spf\,A} \times_{\mathrm{X}^{\triangle}} (\mathrm{X}, \mathcal{M})^{\triangle}$ is $\mathrm{Spf\,A} \times_{\mathrm{Log}^{\triangle}} (\mathbf{Z}_p^{\triangle} \times \mathrm{Log})$. Since $\mathcal{A}_{\mathrm{P}} \to \mathrm{Log}$ is étale, the relative base change formula in [BL22b, Remark 8.11] gives

$$\mathrm{Spf\,A} \times_{\mathrm{Log}^{\triangle}} (\mathbf{Z}_p^{\triangle} \times \mathrm{Log}) \simeq \mathrm{Spf\,A} \times_{\mathcal{A}_{\mathrm{P}}^{\triangle}} (\mathbf{Z}_p^{\triangle} \times \mathcal{A}_{\mathrm{P}}).$$

The toric atlas $\mathrm{Spf}\,\mathbf{Z}_p[\mathrm{P}] \to \mathcal{A}_{\mathrm{P}}$ and Lemma A.19 give an effective epimorphism

$$\mathrm{Spa}(\mathrm{B}_{\mathrm{P}}, \mathrm{B}_{\mathrm{P}}) \longrightarrow \mathrm{Spa}(\mathrm{A}, \mathrm{A}) \times_{\mathrm{X}^{\triangle,\blacksquare}} (\mathrm{X}, \mathcal{M})^{\triangle,\blacksquare}.$$

The projection from the fiber product on the right to $(\mathrm{X}, \mathcal{M})^{\triangle,\blacksquare}$ is the base change of $\mathrm{Spa}(\mathrm{A}, \mathrm{A}) \to \mathrm{X}^{\triangle,\blacksquare}$, so it is an effective epimorphism by Proposition A.18. Thus the composite from $\mathrm{Spa}(\mathrm{B}_{\mathrm{P}}, \mathrm{B}_{\mathrm{P}})$ covers $(\mathrm{X}, \mathcal{M})^{\triangle,\blacksquare}$. After pullback along this composite, the map from $\mathrm{Spa}(\mathrm{A}, \mathrm{A})$ becomes

$$\mathrm{Spa}(\mathrm{A} \coprod_{*} \mathrm{B}_{\mathrm{P}}, \mathrm{A} \coprod_{*} \mathrm{B}_{\mathrm{P}}) \longrightarrow \mathrm{Spa}(\mathrm{B}_{\mathrm{P}}, \mathrm{B}_{\mathrm{P}}).$$

This is a proper descendable cover by Lemma A.13. Representability is local on the target, as is properness by [Kip26a, Remark 2.3.1]; the covering property is local in any ∞-topos. This proves the first assertion. Lemma A.5, left exactness of éid sheafification, and finite limit preservation of $(-)^{\blacksquare}$ identify the degree-n Čech term with $\mathrm{Spa}(\mathrm{A}^{(n)}, \mathrm{A}^{(n)})$. Since the map from $\mathrm{Spa}(\mathrm{A}, \mathrm{A})$ is an effective epimorphism, its Čech nerve realizes to $(\mathrm{X}, \mathcal{M})^{\triangle,\blacksquare}$. □

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